भारत सरकार
Government of India
जल शक्ति मंत्रालय,
Ministry of Jal Shakti,
जल संसाधन, नदी विकास और गंगा संरक्षण विभाग,
Department of Water Resources,
River Development and
Ganga Rejuvenation

केंद्रीय भूमि जल बोर्ड
Central Ground Water Board

वार्षिक रिपोर्ट

ANNUAL REPORT

2022-23

भारत सरकार
GOVERNMENT OF INDIA
जल शक्ति मंत्रालय,
MINISTRY OF JAL SHAKTI
जल संसाधन, नदी विकास और गंगा संरक्षण विभाग
DEPARTMENT OF WATER RESOURCES, RIVER DEVELOPMENT & GANGA REJUVENATION
केंद्रीय भूमि जल बोर्ड
CENTRAL GROUND WATER BOARD

ANNUAL REPORT 2022 - 23

Faridabad
March 2024

डॉ. सुनील कुमार अम्बष्ट
अध्यक्ष
Dr. Sunil Kumar Ambast
Chairman

सत्यमेव जयते

भारत सरकार
जल शक्ति मंत्रालय
जल संसाधन,
नदी विकास और गंगा संरक्षण विभाग
केन्द्रीय भूमि जल बोर्ड
Government of India
Ministry of Jal Shakti
Department of Water Resources,
River Development & Ganga Rejuvention
Central Ground Water Board

Message

Central Ground Water Board is a multidisciplinary Scientific Organization with a vision of "Sustainable Development and Management of Ground Water Resources of the Country".

The activities of Central Ground Water Board are aligned with the objective of sustainable ground water management. Aquifer Mapping studies, Ground water exploration, Ground water Resource assessment, Artificial recharge studies, Ground water level and Ground water quality monitoring, ground water regulation etc. are the major activities of CGWB.

The Annual Report 2022-23 is a compilation of major activities and achievements of the Board during the period 1st April 2022 to 31st March 2023. The year 2022-23 has been a landmark year for CGWB as the first phase of NAQUIM studies was completed in March 2023. Starting from 2012, the entire mappable area of 25 lakh sq km has been covered under the NAQUIM studies. Further, all the pending NAQUIM reports of previous years were also issued in 2022-23. From 2022-23, we started do assessment of dynamic ground water resources on an annual basis. We also covered 1 lakh sq km for high resolution aquifer mapping using heli-borne TEM surveys. All the envisaged targets for exploratory drilling, geophysical studies, trainings, outreach activities, short term investigations etc. were also met.

I have no doubt that this Annual Report will be useful in creating awareness on activities and achievements of the Board. The Report will also serve as reference document for years to come.

I appreciate the efforts of officers of CGWB in bringing out this Annual Report.

Dr. Sunil Kumar Ambast

Faridabad, March 4, 2024

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AAP 2022-23 के लिए कार्यकारी सारांश

- केंद्रीय भूजल बोर्ड (सीजीडब्ल्यूबी) भविष्य के भूजल विकास के लिए क्षेत्रों की पहचान करने के लिए जलभृतों को चित्रित करने और उनके मापदंडों को निर्धारित करने के लिए भूजल अन्वेषण करता है। 76 ड्रिलिंग रिगों के बेड़े का उपयोग करके इन-हाउस ड्रिलिंग कार्यों के माध्यम से 2022-23 के दौरान, सीजीडब्ल्यूबी ने 837 कुओं (240 खोजपूर्ण कुएं, 92 अवलोकन कुएं, 505 पीज़ोमीटर) का निर्माण किया। आदिवासी क्षेत्रों में 34 और सूखा प्रभावित क्षेत्रों में 28 कुओं का निर्माण किया गया। 190 कुएँ कठोर चट्टान में, 39 जलोढ़ में और 11 बोल्डरी संरचनाओं में बनाए गए थे। हरियाणा के यमुनानगर जिले के टोपरा खुर्द में 41 एलपीएस के उच्चतम डिस्चार्ज के साथ राज्यों में 133 उच्च उपज वाले कुओं का निर्माण किया गया है।

- वर्ष 2022-23 के लिए भूजल स्तर की निगरानी में, वर्ष के दौरान 25000 कुओं की निगरानी की गई। उथले और गहरे जलभृत डेटा का विश्लेषण पिछले दशक की तुलना में 50% से अधिक मॉनिटर किए गए कुओं में जल स्तर में वृद्धि का संकेत देता है। हालाँकि, 35-55% कुओं में जल स्तर में गिरावट भी देखी गई है। प्रीमूनसून अवधि में, अरुणाचल प्रदेश, असम, कर्नाटक, केरल, ओडिशा, टीएन और पश्चिम बंगाल के कुछ हिस्सों में 2 मीटर से कम उथले पानी का स्तर देखा गया। उत्तर-पश्चिम भारत में जल स्तर की गहराई अधिक (10-40 मीटर) थी। गहरे जलभृतों में, जल स्तर अधिकतर 5-20 मीटर के बीच था। आंध्र प्रदेश, छत्तीसगढ़, गुजरात, हरियाणा, कर्नाटक, केरल, एमपी, महाराष्ट्र के कुछ हिस्सों में 40 मीटर से अधिक गहरा स्तर देखा गया।

- सीजीडब्ल्यूबी का अध्ययन पानी की कमी वाले क्षेत्रों में जलभृत प्रणालियों के मानचित्रण और प्रबंधन पर केंद्रित है, जिसमें उत्तर-पश्चिम भारत के शुष्क क्षेत्रों में एनजीआरआई के सहयोग से उच्च रिज़ॉल्यूशन जलभृत मानचित्रण भी शामिल है। पोखरण, चूरू, पाली-जालौर और श्री गंगानगर में पैलियो-चैनलों का सीमांकन किया गया है। राजस्थान में सर्वेक्षण इकाइयाँ, और हरियाणा में कुरूक्षेत्र सर्वेक्षण इकाई है। ट्यूबवेलों के निर्माण के लिए कुल 1292 उपयुक्त स्थलों की पहचान की गई है। कृत्रिम पुनर्भरण संरचनाओं के निर्माण के लिए 1029 संभावित स्थानों की पहचान की गई है। ग्रामीण विकास योजनाओं के लिए अभिसरण योजना के साथ-साथ राजस्थान के पानी की कमी वाले क्षेत्रों लिए भूजल पुनर्भरण संवर्धन और वर्षा जल संचयन कार्य शामिल हैं।

- NAQUIM को 2012 में भूजल प्रबंधन और विनियमन योजना योजना के एक भाग के रूप में शुरू किया गया था। NAQUIM के तहत अध्ययन वर्ष के दौरान पूरा किया गया, पूरे देश के ~33 लाख किमी2 में से, 25 लाख किमी2 का संपूर्ण मानचित्रण योग्य क्षेत्र मार्च 2023 तक पूरा कर लिया गया है। असम, पंजाब (केंद्रशासित प्रदेश) राज्यों में 6 राष्ट्रीय स्तर की कार्यशालाएँ आयोजित की गईं । चंडीगढ़, कर्नाटक, राजस्थान, बिहार और महाराष्ट्र को विभिन्न स्तरों पर राज्यों/केंद्रशासित प्रदेशों में NAQUIM निष्कर्षों का प्रसार किया गया और भूजल प्रबंधन हस्तक्षेपों को लागू करने के लिए जिला अधिकारियों (DM/DC) के साथ NAQUIM निष्कर्षों को साझा किया गया।

- पीआईबी परियोजना के तहत प्रस्तावित गतिविधियों को मूल रूप से योजना के घटक । (एक्विफर मैपिंग और प्रबंधन कार्यक्रम, NAQUIM) के तहत 2021-26 की अवधि के लिए भूजल प्रबंधन और विनियमन योजना के हिस्से के रूप में प्रस्तावित किया गया था। संपूर्ण परियोजना अवधि यानी 2022-23 से 2025-26 के लिए परियोजना की कुल अनुमानित लागत 805 करोड़ रुपये है। वर्ष के दौरान पीआईबी के तहत कार्यों के लिए निविदा प्रक्रिया शुरू कर दी गई है।

- भारत के गतिशील भूजल संसाधनों के 2022 के आकलन के अनुसार, जो केंद्रीय भूजल बोर्ड (सीजीडब्ल्यूबी) और राज्य भूजल विभागों द्वारा संयुक्त रूप से आयोजित किया गया था। भारत के लिए कुल वार्षिक भूजल पुनर्भरण 437.60 बिलियन क्यूबिक मीटर (बीसीएम) होने का अनुमान है। मुख्य स्रोत 241.35 बीसीएम मानसून वर्षा है, जो कुल पुनर्भरण का 55% है। कुल वार्षिक निकालने योग्य भूजल संसाधन 398.08 बीसीएम है। इसमें से, 239.16 बीसीएम वर्तमान में सालाना निकाला जाता है, जिसमें 87% (208.49 बीसीएम) सिंचाई के लिए उपयोग किया जाता है।

- निष्कर्षण के चरण के आधार पर, राज्यों/केंद्रशासित प्रदेशों में मूल्यांकन इकाइयों को सुरक्षित (67%), अर्ध-महत्वपूर्ण (12%), गंभीर (4%) या अति-शोषित (14%) के रूप में वर्गीकृत किया गया है। मूल्यांकन इकाइयों के संदर्भ में , भूजल निष्कर्षण का स्तर हरियाणा, पंजाब, राजस्थान, दादरा और नगर हवेली और दमन और दीव में सबसे अधिक है, जहां निष्कर्षण 100% निकालने योग्य संसाधनों से अधिक है। 2020 के आकलन की तुलना में, कुल वार्षिक पुनर्भरण में 1.45 बीसीएम की मामूली वृद्धि हुई है और 0.46 बीसीएम द्वारा निकाले जाने योग्य संसाधन। भूजल दोहन 244.92 बीसीएम से मामूली कम होकर 239.16 बीसीएम हो गया है।
- राजीव गांधी राष्ट्रीय भूजल प्रशिक्षण एवं अनुसंधान संस्थान (आरजीएनजीडब्ल्यूटीआरआई) ने 60 टियर-I, 19 टियर-II और 50 टियर-III प्रशिक्षण आयोजित किए हैं। संस्थान द्वारा अन्य संगठनों के

सहयोग से विभिन्न अनुसंधान और विकास परियोजनाएं शुरू की गईं। संस्थान की त्रैमासिक पत्रिका "भुजलन्यूज" में छह तकनीकी पेपर प्रकाशित हुए। संस्थान सीजीडब्ल्यूए दिशानिर्देशों के अनुसार भूजल पेशेवरों की मान्यता में शामिल है। 2022-23 में मान्यता के दो दौर आयोजित किए गए।

- पीएमकेएसवाई का लक्ष्य छोटे और सीमांत किसानों की आय बढ़ाने के लिए सुनिश्चित सिंचाई प्रदान करना है। इसे पर्याप्त रूप से उपलब्ध भूजल का उपयोग करके सिंचाई कवरेज बढ़ाने के लिए 2015 में लॉन्च किया गया था। इसके अंतर्गत आने वाले प्रमुख राज्य असम, अरुणाचल प्रदेश, नागालैंड, त्रिपुरा, मणिपुर, मिजोरम, तमिलनाडु, गुजरात, उत्तर प्रदेश और उत्तराखंड हैं। सुनिश्चित सिंचाई के लिए भूजल के उपयोग के लिए पीएमकेएसवाई-एचकेकेपी-जीडब्ल्यू योजना के तहत, 13 परियोजनाओं को मंजूरी दी गई है। 10 राज्य राज्यों में मार्च 2023 तक, 29,695 कुओं का निर्माण किया गया, 77,964 हेक्टेयर कमांड क्षेत्र बनाया गया और 67,285 सीमांत किसानों को लाभ हुआ।

- वर्ष 2022-23 में, विभिन्न राज्य सरकारों और वायु सेना, डीआरडीओ आदि जैसे रक्षा संगठनों को तकनीकी सहायता के एक भाग के रूप में, देश के विभिन्न हिस्सों में 127 अल्पकालिक जल आपूर्ति जांच (STWSI) की गईं। वर्ष के दौरान केंद्रीय भूजल बोर्ड के अधिकारियों द्वारा 405 तकनीकी रिपोर्ट प्रकाशित की गई हैं। सीजीडब्ल्यूबी के अधिकारियों द्वारा राष्ट्रीय पत्रिकाओं में 37 तकनीकी पत्र और अंतर्राष्ट्रीय पत्रिकाओं में 8 तकनीकी पत्र प्रकाशित/प्रस्तुत किए गए हैं। वर्ष 2022-23 में 46 विशेष अध्ययन शुरू/जारी किये गये हैं। केंद्रीय भूजल बोर्ड द्वारा 7 राष्ट्रीय सहयोग और 8 अंतर्राष्ट्रीय सहयोग किए गए/जारी रखे गए हैं।

- आज़ादी का अमृत महोत्सव समारोह के हिस्से के रूप में, NAQUIM परिणामों और पानी से संबंधित पहलुओं के बारे में जागरूकता बढ़ाने के लिए कई सार्वजनिक संपर्क कार्यक्रम आयोजित किए गए। 2022-23 में, 252 ऐसे कार्यक्रम आयोजित किए गए जिनमें 10,035 महिलाओं सहित 25,723 लोगों की भागीदारी थी। 2018 से संचयी रूप से, 1,351 सार्वजनिक संपर्क कार्यक्रम आयोजित किए गए हैं, जिनमें 44,110 महिलाओं सहित 118,213 लोगों तक पहुंच बनाई गई है।

EXECUTIVE SUMMARY FOR AAP 2022-23

- Central Ground Water Board (CGWB) carries out ground water exploration to delineate aquifers and determine their parameters to identify areas for future ground water development.During 2022-23, CGWB constructed 837 wells (240 exploratory wells, 92 observation wells, 505 piezometers) through in-house drilling operations using a fleet of 76 drilling rigs.34 wells were constructed in tribal areas and 28 in drought affected areas. 190 wells were constructed in hard rock, 39 in alluvium and 11 in bouldary formations. 133 high yielding wells across states have been constructed with highest discharge of 41 lps at Topra Khurd in Yamunanagar district of Haryana. Since its inception.

- In ground water level monitoring for year 2022-23, 25000 Wells monitored during the year. Analysis of shallow and deeper aquifer data indicates rise in water level in more than 50% of the monitored wells compared to the previous decade. However, water level declines are also seen in 35-55% of wells. In premoonsoon period, shallow water levels of less than 2m were observed in parts of Arunachal Pradesh, Assam, Karnataka, Kerala, Odisha, TN and West Bengal. Depth to water levels were deeper (10-40m) in northwest India.In deeper aquifers, water levels mostly ranged from 5-20m. Levels deeper than 40m were seen in parts of Andhra Pradesh, Chhatisgarh, Gujarat, Haryana, Karnataka, Kerala, MP, Maharashtra among others.

- Studies are focused on mapping and managing aquifer systems in water-stressed regions, including high resolution aquifer mapping in collaboration with NGRI in arid regions of northwest India.Paleo-channels have been demarcated in Pokhran, Churu, Pali-Jalore, and Sri Ganganagar Survey Units in Rajasthan, and the Kurukshetra survey unit in Haryana. A total of 1292 suitable sites have been identified for the construction of tubewells. 1029 feasible locations have been identified for the construction of artificial recharge structures. Groundwater recharge augmentation and rainwater harvesting works are included for water-scarce regions of Rajasthan along with convergence planning for rural development schemes.

- NAQUIM was initiated in 2012 as a part of the Ground Water Management and Regulation plan scheme. The studies under NAQUIM were completed during the year, Out of ~33 lakh km^2 of the entire country, entire mappable area of 25 lakh km^2has been completed by March 2023. 6 National level workshops were organized in the States of Assam, Punjab (UT of Chandigarh), Karnataka, Rajasthan, Bihar and Maharashtra to disseminate the NAQUIM

findings to States / UTs at various levels and also shared NAQUIM findings with district authorities (DM/DC) for implementing groundwater management interventions.

- The activities proposed under the PIB project were originally proposed as part of the Ground Water Management and Regulation Scheme for the period 2021-26 under Component I (Aquifer Mapping and Management Program, NAQUIM) of the Scheme. Total estimated cost of the project is Rs 805 Cr for the entire project period i.e. 2022-23 to 2025-26. The tendering process for the works under PIB has been initiated during the year.

- As per the 2022 assessment of India's dynamic ground water resources, which was conducted jointly by the Central Ground Water Board (CGWB) and state ground water departments.The total annual ground water recharge for India is estimated at 437.60 billion cubic meters (bcm). The main source is monsoon rainfall at 241.35 bcm, comprising 55% of total recharge.The total annual extractable ground water resource is 398.08 bcm. Of this, 239.16 bcm is currently extracted annually, with 87% (208.49 bcm) used for irrigation.

- Based on the stage of extraction, assessment units across states/UTs are categorized as safe (67%), semi-critical (12%), critical (4%) or over-exploited (14%).In terms of assessment units, the stage of ground water extraction is highest in Haryana, Punjab, Rajasthan, Dadra & Nagar Haveli and Daman & Diu, where extraction exceeds 100% of extractable resources.Compared to the 2020 assessment, total annual recharge has increased slightly by 1.45 bcm and extractable resources by 0.46 bcm. Ground water extraction has marginally reduced from 244.92 bcm to 239.16 bcm.

- Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI), has conducted 60 Tier I, 19 Tier II and 50 Tier III trainings.Various research and development projects were undertaken by the institute in collaboration with other organizations. Six technical papers were published in the institute's quarterly journal "BhujalNews."The institute is involved in accreditation of groundwater professionals as per CGWA guidelines. Two rounds of accreditation were conducted in 2022-23.

- PMKSY aims to provide assured irrigation to enhance incomes of small and marginal farmers. It was launched in 2015 to extend irrigation coverage by utilizing groundwater where sufficiently available. Key States covered under this are Assam, Arunachal Pradesh, Nagaland, Tripura, Manipur, Mizoram, Tamil Nadu, Gujarat, Uttar Pradesh and Uttarakhand.Under the PMKSY-HKKP-GW scheme for utilizing groundwater for assured irrigation, 13 projects have been sanctioned in 10 states. As of March 2023, 29,695 wells constructed, 77,964 Ha command area created, and 67,285 marginal farmers have benefited.

- In year 2022-23, as a part of Technical Assistance to various State Govts and Defense organizations as Air force, DRDO etc, 127 short term water supply investigations (STWSI) were carried out in different parts of the country. 405 Technical Reports have been published during the year from officers of Central Ground Water Board.37 Technical papers to National Journals and 8 technical papers have been published/submitted to International Journals by the officers of CGWB. 46 Special studies have been taken up/continued in the year 2022-23. 7 National collaborations and 8 International collaborations have been done/continued by Central Ground Water Board.

- As part of the Azadi Ka Amrit Mahotsava celebrations, several public interaction programs were organized to raise awareness about NAQUIM outcomes and water-related aspects. In 2022-23, 252 such programs were held with participation of 25,723 people, including 10,035 women. Cumulatively since 2018, 1,351 public interaction programs have been conducted, reaching out to 118,213 people, including 44,110 women.

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CHAPTER 1 CENTRAL GROUND WATER BOARD

1. INTRODUCTION

Central Ground Water Board (CGWB) is a scientific department under Department of Water Resources, River Development & Ganga Rejuvination, Ministry of Jal Shakti, Government of India and the national apex organization for all aspects of ground water surveys, exploration, development and management. Central Ground Water Board is also discharging the functions as Central Ground Water Authority (CGWA) to regulate and control the development and management of ground water in the country since 1997. The Central Ground Water Board was formed in 1970 by renaming erstwhile Exploratory Tube well Organization. Subsequently, Ground Water Division of Geological Survey of India got merged with it in 1972. The Board was placed under various ministries in the past and since 1982, it is an organization under Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, Government of India.

1.1 VISION

Sustainable Development and Managemement of Ground Water Resources of the Country.

1.2 MANDATE

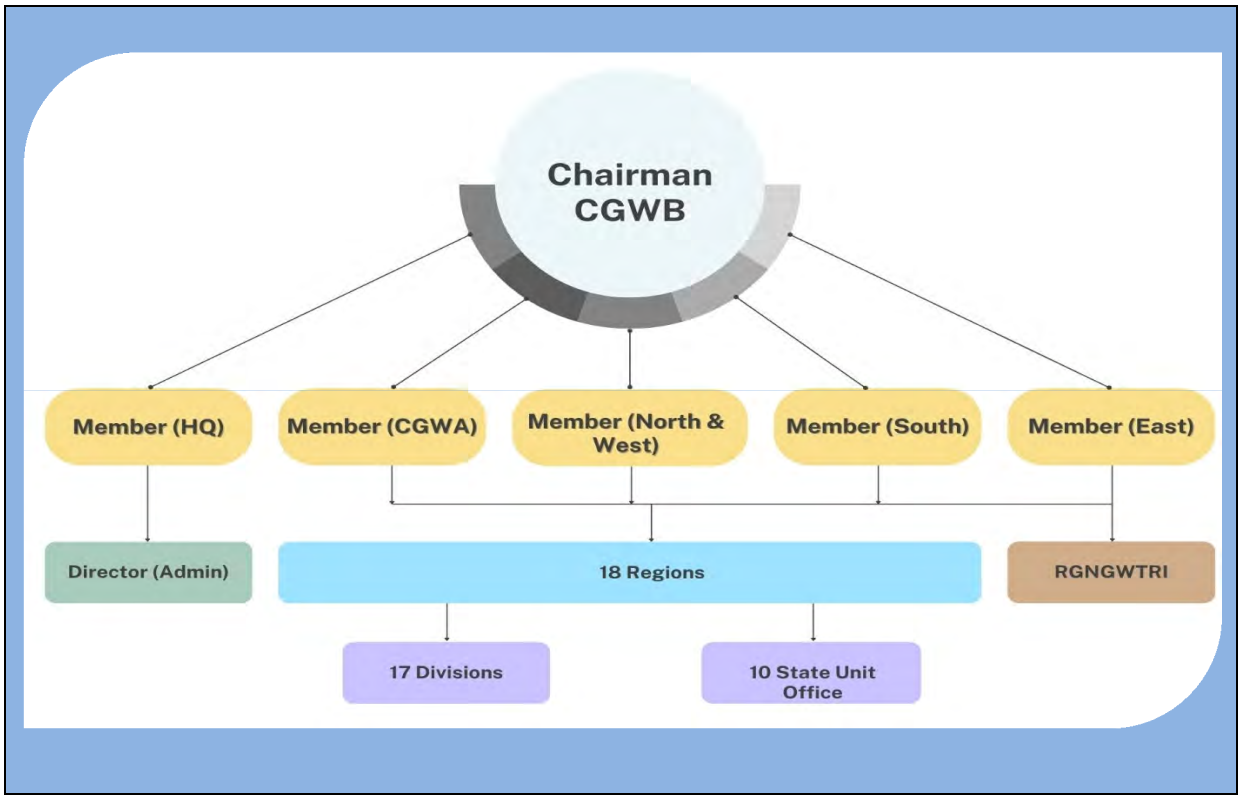
CGWB is a multidisciplinary Scientific Organization with a mandate “To develop and disseminate technologies, monitor and implement National policies for the scientific and sustainable development and management of India's Ground Water resources, including its exploration, assessment, conservation, augmentation, protection from pollution and distribution, based on principles of economic and ecological efficiency and equity”

1.3 ORGANIZATIONAL SET UP

Central Ground Water Board (CGWB) operates as a subordinate office of Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti Government of India. CGWB is a multi-disciplinary scientific organization consisting of hydrogeologists, geophysicists, chemists, hydrologists, hydrometeorologists and engineers. CGWB has about 600 scientists, 150 engineers and 3,250 supporting staff (technical, administrative and ministerial).

The Board is headed by the Chairman and has five Members who look after different regional offices and also perform other specified functions. CGWB also has five permanent members representing Central Water Commission, Ministry of Jal Shakti, Ministry of Environment, Forest& Climate Change and Oil and Natural Gas Commission (ONGC). *(Website:https://cgwb.gov.in)*

Organizational setup of Central Ground Water Board is shown in the following figure:



Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) was established during IXth Five Year Plan at Raipur as a training wing of Central Ground Water Board, Ministry of Jal Shakti, Department of Water Resources, River Development and Ganga Rejuvenation · Government of India is running continuously since 1997. The Institute created as training wing of Central Ground Water Board, Ministry of Jal Shakti, Department of Water Resources, River Development and Ganga Rejuvenation · Government of India in 12th Plan Period is presently a plan scheme of Human Resource Development (HRD)/ Capacity Building.

Rajiv Gandhi National Ground Water Training and Research Institute is envisaged to function as "Centre of Excellence" in training and research in the core and applied field of ground water.

1.4 OFFICES OF CENTRAL GROUND WATER BOARD

CGWB has its headquarters at Bhujal Bhawan, Faridabad, Haryana; the Central Ground Water Authority (CGWA) operates from its office in New Delhi. Field activities are performed through 18 Regional Offices, 10 State Unit Offices and 17 Divisional Offices located in various part of the Country. The Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI)

located in Raipur, Chhattisgarh, serves as the training arm of CGWB. Locations of the offices are given below:

Table 1.1: Regions & Regional Offices covering the States

S. No.	Name of Office	Location	Jurisdiction
Central Headquarters			
1	CGWB, CHQ	Faridabad	Entire Country
Central Ground Water Authority			
2	CGWA	New Delhi	Entire Country
Rajiv Gandhi National Ground Water Training and Research Institute			
3	RGNGWTRI	Raipur	Entire Country
North West Himalayan Region			
4	North West Himalayan Region	Jammu	U.T. of Jammu & Kashmir and Ladakh
5	Division-VII	Jammu	U.T. of Jammu & Kashmir and Ladakh
North Himalayan Region			
6	North Himalayan Region	Dharamshala	Himachal Pradesh
7	Division-XVII	Dharamshala	Himachal Pradesh
North Western Region			
8	North Western Region	Chandigarh	Punjab, Haryana, Delhi & U.T. of Chandigarh
9	State Unit Office	Delhi	Delhi
10	Division-II	Ambala	Punjab, Haryana, NCT of Delhi & U.T. of Chandigarh
Uttarakhand Region			
11	Uttarakhand Region	Dehradun	Uttarakhand
Northern Region			
12	Northern Region	Lucknow	Uttar Pradesh
13	Division-III	Varanasi	Parts of Uttar Pradesh
14	Division-XVI	Bareilly	Part of Uttar Pradesh & Uttarkhand
15	State Unit Office	Prayagraj	Part of Uttar Pradesh
Western Region			
16	Western Region	Jaipur	Parts of Rajasthan
17	Division-XI	Jodhpur	Rajasthan
18	State Unit Office	Jodhpur	Part Rajasthan
Mid-Eastern Region			
19	Mid-Eastern Region	Patna	Bihar, Jharkhand
20	Division-V	Ranchi	Bihar and Jharkhand
21	State Unit Office	Ranchi	Jharkhand

S. No.	Name of Office	Location	Jurisdiction
Eastern Region			
22	Eastern Region	Kolkata	West Bengal, Sikkim, U.T. of Andaman & Nicobar Islands
23	Division-XV	Kolkata	West Bengal, Sikkim, U.T. of Andaman & Nicobar Islands
North Eastern Region			
24	North Eastern Region	Guwahati	Assam, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya & Tripura
25	Division-VII	Guwahati	Assam, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya & Tripura
26	State Unit Office	Shillong	Meghalaya
27	State Unit Office	Agartala	Tripura
28	State Unit Office	Itanagar	Arunachal Pradesh
West Central Region			
29	West Central Region	Ahmedabad	Gujarat, U.T of Daman & Diu and U.T. of Dadra & Nagar Haveli
30	Division-I	Ahmedabad	Gujarat, Daman & Diu and & U.T. of Dadra & Nagar Haveli
North Central Region			
31	North Central Region	Bhopal	Madhya Pradesh
32	Division-XII	Bhopal	Madhya Pradesh
North Central Chhattisgarh Region			
33	North Central Chhattisgarh Region	Raipur	Chhattisgarh
34	Division-XIII	Raipur	Chhattisgarh
South Eastern Region			
35	South Eastern Region	Bhubaneswar	Odisha
36	Division-X	Bhubaneswar	Odisha
Central Region			
37	Central Region	Nagar	Maharashtra
38	Division-VI	Nagpur	Maharashtra
39	State Unit Office	Pune	Parts of Maharashtra
Southern Region			
40	Southern Region	Hyderabad	Andhra Pradesh & Telangana
41	Division-IX	Hyderabad	Andhra Pradesh & Telangana

S. No.	Name of Office	Location	Jurisdiction
42	State Unit Office	Visakhapatnam	Parts of Andhra Pradesh
South Western Region			
43	South Western Region	Bengaluru	Karnataka & Goa
44	Division-XIV	Bengaluru	Karnataka, Goa and Parts of Kerala
45	State Unit Office	Belagavi	Parts of Karnataka and Goa
South East Coastal Region			
46	South East Coastal Region	Chennai	Tamil Nadu and U.T. of Puducherry
47	Division-IV	Chennai	Tamil Nadu, U.T. of Puducherry and Parts of Kerala
Kerala Region			
48	Kerala Region	Thiruvananthapuram	Kerala and U.T. of Lakshadweep

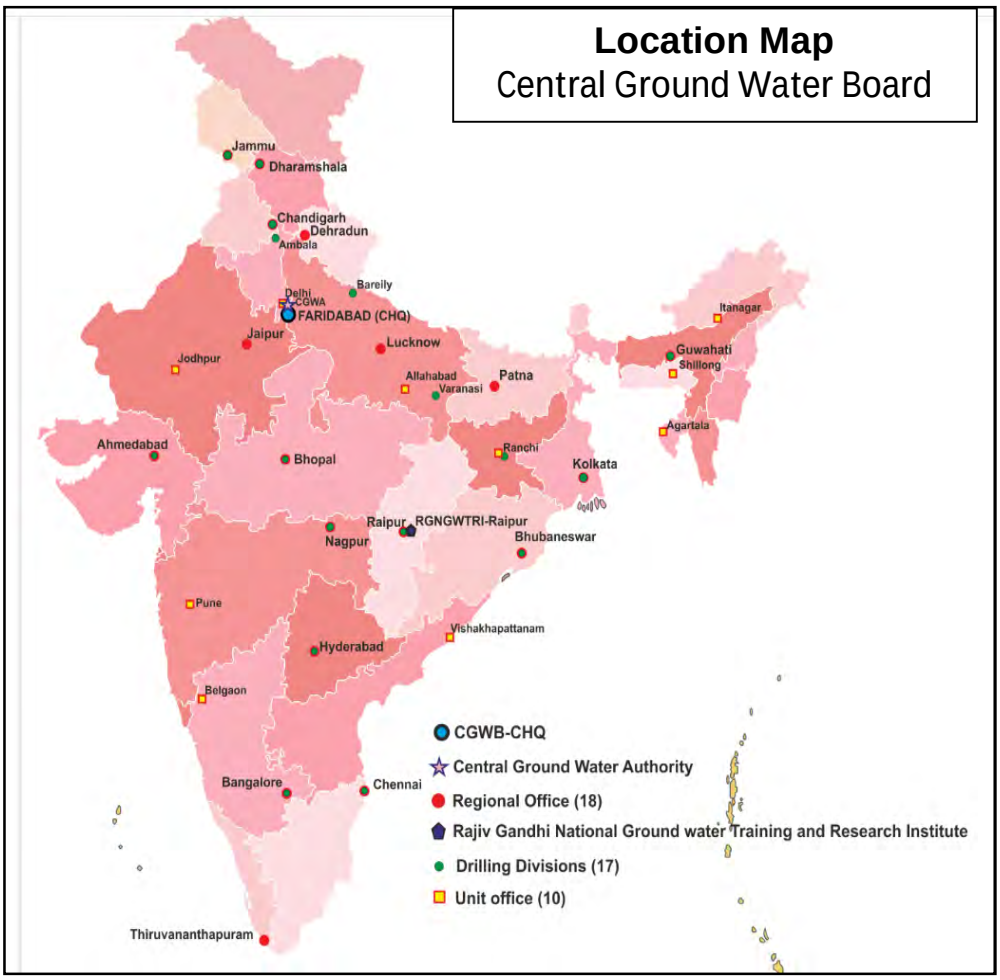


Fig 1: Location and Jurisdiction of the CGWB offices

1.5 MAJOR SCHEMES AND ACTIVITIES OF CENTRAL GROUND WATER BOARD

1.5.1 Ground Water Management & Regulation Scheme (GWMR)

Government of India is implementing the Ground Water management and Regulation scheme. It is a Central Sector Scheme, which deals exclusively with ground water and is being implemented by CGWB, DoWR, RD and GR, Ministry of Jal Shakti. The GWM&R Scheme is being continued since 2007-08 and has been approved for continuation during the period 2021-2026.

Ground Water Management and Regulation scheme is a continuing Central Sector Scheme, which is being implemented since 2007-08 by Central Ground Water Board (CGWB). To ensure continuity of the activities, the scheme has been approved for implementation till 31st March 2026 and includes activities like monitoring, assessment and regulation of groundwater resources and strengthening of infrastructure for technological upgradation.

One of the major activities under the scheme is National Aquifer Mapping & Management (NAQUIM) Programme, under which it was targeted to cover ~25 lakh sq km of mappable area and it has been covered by 31st March 2023. In addition to above other activities like ground water level & quality monitoring, assessment of dynamic ground water resources, regulation and control of ground water withdrawal, demonstrative recharge projects etc. have also been implemented.

A) **National Aquifer Mapping and Management Program (NAQUIM):** NAQUIM which is being implemented by CGWB, envisages mapping of aquifers (water bearing formations), their characterization and development of Aquifer Management Plans to facilitate sustainable management of Ground Water Resources. NAQUIM was initiated in 2012 as a part of the Ground Water Management and Regulation plan scheme. Out of ~33 lakh km^2 of the entire country, a mappable area of ~25 lakh km^2 has been identified and has been covered under this programme by March 2023.

B) **Ground water level monitoring:** Ground water level monitoring is one of the key activities of Central Ground Water Board. The primary objective of ground water monitoring is to record the response of various natural and anthropogenic stresses on the groundwater regime which impacts the recharge and discharge parameters. At present, CGWB has a network of nearly 25000 ground water observation wells throughout the country. Ground water levels are measured four times a year during the months of January, March/April/May, August and November.

C) **Groundwater quality studies:** Central Ground Water Board has 16 Regional Chemical Laboratories to carry out chemical analysis of water samples. The Chemical laboratories are well equipped with sophisticated instruments like ICP-MS, Atomic Absorption Spectrophotometer (AAS) etc. Ground water sampling for quality monitoring is done once a year during the premonsoon period. In addition

to it ground water samples are also collected and analysed as a part of other scientific studies. CGWB has also taken up sampling and analysis of ground water samples for URANIUM content across the country.

D) **Geophysical Studies:** Geophysical studies are non-invasive techniques and provide indirect information about disposition and characteristics of aquifers. CGWB has inhouse facility for various geophysical studies. Outputs of geophysical studies are used mostly in aquifer mapping studies and other case specific investigations.

E) **High Resolution mapping of Aquifer through Heli-borne Geophysical Studies**: Heli-borne Transient Elecromagnetic (Heli-TEM) studies in conjunction with the classical ground water surveys and exploration have the advantage of faster spatial coverage. Central Ground Water Board, Ministry of Jal Shakti and CSIR-NGRI, Hyderabad have signed a Memorandum of Agreement for High Resolution Aquifer Mapping and Management in Arid Regions of North Western India, by using advanced Heli-borne Geophysical Surveys and other scientific studies in parts of the States of Rajasthan, Gujarat, Haryana, Punjab and Himachal Pradesh, covering an area of 3.88 lac sq/km under the National Aquifer Mapping Programme (NAQUIM) with financial outlay of Rs. 54 Cr for Phase-I. The heliborne survey data has been obtained through a total of 40313 flight line km covering an area of 97165 sq.km in Rajasthan, Gujarat and Haryana states.

F) **Exploratory Drilling:** Ground Water Exploration is one of the core activities of Central Ground Water Board. Drilling aided ground water exploration provides direct information about disposition and characteristics of the aquifers. Every year around 700 wells are drilled by CGWB for ground water exploration through in-house resources. CGWB has a fleet of 76 operational drilling rigs and has capability to construct wells in various types of terrain in the country. In view of enhanced requirements of ground water exploration under the NAQUIM programme, CGWB has also taken up exploratory drilling through outsourcing.

G) **Aquifer Rejuvenation and Water Conservation:** CGWB implemented innovative Aquifer Rejuvenation interventions. During 2021-23, CGWB has taken up artificial recharge for groundwater augmentation in certain water-stressed areas of Jaisalmer, Jodhpur & Sikar districts of Rajasthan.

H) **Assessment of Dynamic Ground Water Resources of India:** Periodic Assessment of dynamic ground water resources is done jointly by CGWB and the respective state governments. The latest assessment for the year 2022 has been carried out jointly by the Central Ground Water Board and State/UT Nodal/Ground Water Departments based on the methodology recommended by Ground Water Estimation Committee -2015 (GEC-2015). The entire assessment has been carried out using the GIS based web portal 'India- Groundwater Resource Estimation System (IN-GRES) which has been developed by CGWB in association with IIT-

Hyderabad. IN-GRES provides a common and standardized platform for GW Resource Assessment for the entire country (Central and State Governments).

I) **Outreach Programs:** Public Interaction programs (PIPs) were initiated in the year 2018 as a medium to disseminate findings of NAQUIM studies at grassroots level. The participants are sensitized on various aspects of water conservation, ground water management and findings of the NAQUIM study.

J) **CGWA- Regulation and Control of Ground Water Extraction:** Central Ground Water Authority (CGWA) has been constituted under Section 3(3) of the Environment Protection Act, 1986. CGWA has been entrusted with the responsibility of regulating and controlling ground water development and management in the country and issuing necessary directives for the purpose. To promote ease of doing business, CGWA has put in place a web-based system NOCAP for processing of applications and issuance of NOCs. New Guidelines for regulation and control of ground water extraction have been issued in September 2020 and revised in March 2023 by Central Ground Water Authority.

1.5.2 Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI)

Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTR), Raipur caters to training and research in the field of ground water in the country. Since 2012 (XII Plan) RGNGWTRI has been implementing a three tiered (Tier I- National Level, Tier II- State/ District Tier -III: Block Level) training programme. It also occasionally carries out training programmes related to ground water management for foreign nationals and institutes. Other important activities of RGI include conducting exams for accreditation of ground water consultants and conducting research on various aspects of ground water.

1.5.3 PMKSY- Har Khet Ko Pani- Ground Water (PMKSY-HKKP-GW)

It is a centrally sponsored scheme launched by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, which envisages to provide irrigation facility for Small and Marginal Farmers in areas having sufficient potential for future development of ground water. The Scheme is being implemented with an objective to support goal of Hon'ble Prime Minister of India, for doubling the farmer incomes. The scheme was approved for 2015-20, but keeping in view various requirements to implement the scheme, guidelines were revised during May 2019 and scheme has been re-launched in July 2019. So far 13 projects have been implemented in 10 states namely Assam, Arunachal Pradesh, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, Uttarakhand. The total cost of these 13 ongoing projects are Rs. 978.00 Crore and Central Assistance is Rs. 827.15 Crore. Out of 12completed projects, three projects have completed during the year 2022-23.

1.5.4 National Hydrology Project (NHP)

National Hydrology Project (NHP) is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of Department of Water Resources, RD&GR, Ministry of Jal Shakti,

Government of India with a total outlay of Rs.3679.7674 Crore (Rs. 3,640 Crore for NHP and Rs.39.7674 Crore for establishment of National Water Informatics Centre (NWIC), a repository of Nation-wide Water Resources data for a period of 08 years extending from 2016-17 to 2023-24.

CGWB is one of the implementing agencies of the NHP. Major activity traken up under NHP in CGWB include installation of Digital Water Level Recorders with telemetry systems.

CHAPTER 2 GROUND WATER SCENARIO OF INDIA

2.1 INTRODUCTION

India is one of the largest users of ground water in various sectors like agriculture, domestic and industrial. Potential use of Ground water irrigation has been increased at a very rapid pace since 1970s. Ground water sustainability and its availability play a vital role in ensuring livelihood as well as food security across the country, especially in (UP, Panjab, Haryana, WB, etc.) where economies depend on agriculture. Rural agriculture and drinking water sector are dependent partly or wholly on groundwater through pumping and/or spring or river due to facile access. However, urban populations highly rely on groundwater due to unreliable and deficient quality of water supplies from other sources. Unplanned Intensive irrigation, domestic and industrial use of Groundwater created serious problems for scientific community in planning management of groundwater. Saltwater encroachment near coast, depletion in water tables, water logging, drying of aquifers, pollution of groundwater, and salinity, etc. are major consequences of overexploitation. Deterioration in groundwater quality by various causes is another serious issue. Despite the cumulative effort of different stakeholders, 20% of groundwater blocks are semi-critical, critical, or overexploited. Moreover, aquifers depletion in the most populated and economically productive areas of the country encourages us to understand and create an overall scenario of Indian aquifer system.

CGWB, an apex organisation in the world working under Ground water, keep an India-focus ground water management and regulation for the reason that it is one of the pioneer organisations to Develop and disseminate technologies, and monitor and implement national policies for the scientific and sustainable development and management of India's ground water resources, including their exploration, assessment, conservation, augmentation, protection from pollution and distribution, based on principles of economic and ecological efficiency and equity.

Combined together, it can be understood that various issues of ground water are going to reduce the fresh water availability for irrigation, domestic and industrial uses. Realizing this, central Ground water board instigate several protective and legislative measures on ground water quality, aquifer mapping and ground water regime monitoring to overcome the groundwater management-related problems.In this chapter, CGWB attempt to unpack myriad mystery of Indian aquifer system using water table, ground water resource, and aquifer disposition

Ground water regime monitoring is one of the key activities of CGWB to generate information on ground water level/ quality through representative sampling. Ground water levels are being monitored throughout the Country four times in a year by Central Ground Water Board through

a network of around 25000 monitoring wells during the months of January, Premonsoon (March/April/ May), August and Postmonsoon. The ground water regime monitoring was started in the year1969 by Central Ground Water Board. At present CGWB has a network of 22339 ground water observation wells, out of which 16190 observation wells are dugwells and 6149 are piezometers. The water level / piezometric head data collected from these observations were entered into the National database and are analysed for obtaining background information of ground water regime and changes on regional scale.

2.2 Aquifer system of India

Aquifer System is an outcome of the extensive hydrogeological studies carried out by Central Ground Water Board (CGWB)include Hydrogeological information collected by CGWB through its survey, investigations and exploration were collated, analyzed and integrated to define the extent and characteristics of aquifer systems.

Based on the hydrogeological characteristics, the entire country has been classified into 14 Principal Aquifer Systems and 42 major aquifers. Alluvium is the major aquifer system which covers maximum area of around 31% of the entire country and is present in Uttar Pradesh, Bihar, West Bengal, Assam, Odisha and Rajasthan.The sandstone aquifer covers around 8% area in the country and found in Chhattisgarh, Andhra Pradesh, Madhya Pradesh, Gujarat, Karnataka andRajasthan. The rest of the country is covered with the other formations that cover around 60% of the area. Among these, Basalt aquifer covers maximum of around 17% area of the country and mainly spread over Maharashtra, Madhya Pradesh, Gujarat, Rajasthan and Karnataka. Shale aquifer accounts for around 7% of area in the country and is available mostly in Chhattisgarh, Andhra Pradesh, Madhya Pradesh, Rajasthan and in the north-eastern states as well as in the Himalayan terrain. Limestone aquifer covers a very small area of around 2% in the country and mainly available inthe states of Chhattisgarh, Andhra Pradesh, Karnataka, and Gujarat and in the Himalayan states.Around 20% of the area of the country is covered by Banded Gneissic Complex (BGC) and Gneiss aquifers which are available almost in all the peninsular states as well as the Himalayan states.The rest 15% of the entire area is covered by aquifers namely; Schist, Granite, Quartzite, Charnockite, Khondalite, Lat- erites and Intrusive.

2.3 Ground Water level scenario

Ground Water Year Book is prepared annually by CGWB depicting changes in ground water regime of the country through different seasons. It is an effort to obtain information on ground water levels through representative monitoring wells. The important attributes of ground water regime monitoring are ground water level.

Ground water levels are being measured by Central Ground Water Board four times a year during January, March/April/ May, August and November to obtain information on ground water levels through representative monitoring wells. The factor affecting the ground water regime can be either natural or anthropogenic such as rainfall, evapotranspiration, pumpage from the aquifer, recharge due to irrigation systems and other practices like waste disposal etc.

A network of 25437 observation wells, as on March 2023, located all over the country is being monitored.

2.4 Ground Water Resources scenario

1.	**TotalAnnualGroundWaterRecharge**	**437.60 bcm**
2.	Annual Extractable Ground Water Resources	398.08 bcm
3.	Annual Ground Water Extraction	239.16 bcm
4.	Stage of Ground Water Extraction	60.08 %

Table 2.1: Categorization of Assessment Units as per GWRA-2022

Sl.No	Category	Number of Assessment Units		Recharge worthy Area		Annual Extractable Ground Water Resource	
		Number	%	inlakh sq.km	%	(inbcm)	%
1	Safe	4780	67	16.18	66	291.88	**73**
2	SemiCritical	885	12	3.03	12	47	**12**
3	Critical	260	4	0.77	3	13.02	**3**
4	Over-Exploited	1006	14	4.30	17	46.05	**12**
5	Saline	158	2	0.4	2	NA	**NA**
	TOTAL	**7089**		**24.69**		**398.08**	

As per Dynamic Ground Water Resource Estimation of the Country as on 2022, the total annual ground water recharge has been assessed as 437.60 bcm. Keeping an allocation for natural discharge, the annual extractable ground water resource works out as 398.08 bcm. The total annual ground water extraction (as in 2022) has been assessed as 239.16 bcm. The average stage of ground water extraction for the country as a whole works out to be about 60.08 %. The extraction of ground water for various uses in different parts of the country is not uniform. Out of the total 7089 assessment units (Blocks/ Districts/ Mandals/Talukas/Firkas) in the country 1006 units in various States (14 %) have been categorized as 'Over- Exploited' indicating ground water extraction exceeding the annually replenishable ground water recharge. A total of 260 (4 %) assessment units have been categorized as 'Critical', where the stage of ground water extraction is between 90-100 % of annual extractable resources available. There are 885 'Semi-Critical' units (12 %), where the stage of ground water extraction is between 70 % and 90% and 4780 (67 %) assessment units have been categorized as 'Safe' where the stage of Ground water extraction is less than 70 %.

Apart from this, there are 158 assessment units (2 %), which have been categorized as 'Saline' as major part of the ground water in phreatic aquifers is brackish or saline. Similarly out of 24.69 lakh sq km recharge worthy area of the country, 4.30 lakh sq km (17 %) are under 'Over-

Exploited', 0.77 lakh sq km (3 %) are under 'Critical', 3.03 lakh sq km (12 %) are under 'Semi-Critical', 16.18 lakh sq km (66 %) are under 'Safe' and 0.4 lakh sq km (2 %) are under 'Saline' category assessment units. Out of 398.08 bcm of Total Annual Extractable Resources of the country, 46.05 bcm (12 %) are under 'Over-Exploited', 13.02 bcm (3 %) are under 'Critical', 47 bcm (12 %) are under 'Semi-Critical', 291.88 bcm (73 %) are under 'Safe' category assessment units.

CHAPTER 3 GROUND WATER EXPLORATION

3.1 INTRODUCTION

Ground Water Exploration aided by drilling is one of the major activities of the Board. It is aimed at delineation of aquifers in different hydrogeological setups and determination of their hydraulic parameters. The exploratory drilling operations have enabled demarcation of aquifers both in lateral and vertical extensions and evaluation of various aquifer parameters, designing of suitable structures and assessment of their yield potential in various hydrogeological settings. These studies have helped in identifying areas worthy for future ground water development. Ground Water exploration contributes to a large extent in guiding the States to implement ground water development schemes.

Groundwater exploration is being carried out by the Board through its fleet of 76 drilling rigs (23 Direct Rotary, 51 Down the Hole and 02 Dual Rotary). During the year 2022-23, Central Ground Water Board under its in-house Ground Water Exploration programme constructed 837 wells (EW- 240, OW- 92, PZ- 505). The region/division/state wise deployment of rigs during AAP 2022-23 is mentioned in table 3.1 and the division-wise details of pumping test units is mentioned in table 3.2.

Priority was accorded for exploration in Over Exploited/ Critical/ Semi-Critical/Drought prone areas and areas affected with ground water pollution etc. The statement showing State, Division & Region wise distribution of boreholes drilled/ completed during 2022-23 is presented in Table 3.3& 3.4(a) & (b). Out of 240 exploratory wells, 190 wells were constructed in hard rock, 39 wells in alluvium and 11 well in bouldary formations (Table 3.6) including 34 wells constructed in Tribal areas and 28 wells constructed in drought area (Table 3.7) of the country.

The Board since inception has drilled 45762 bore holes (including 9023 bore holes through outsourcing) as on 31.03.2023 to identify worthy areas for ground water development in the country.

The statement showing State-wise distribution of boreholes drilled/completed till March, 2023 in the country is presented in Table 3.5.

Table 3.1 Region/Division/State wise Deployment of Rigs during AAP 2022-23

Region	Division	State	Number Of Rigs			
			DR	DTH	DUR	T
NWHR,Jammu	VIII,Jammu	Jammu & Kashmir	0	2	1	**3**
NWR, Chandigarh	II,Ambala	Punjab	2	0	0	**2**
		Haryana	1	0	0	**1**
WR,Jaipur	XI,Jodhpur	Rajasthan	2	3	0	**5**
WCR,Ahmedabad	I,Ahmedabad	Gujarat	4	2	0	**6**
NCR,Bhopal	XII,Bhopal	Madhya Pradesh		4		**4**
NCCR,Raipur	XIII,Raipur	Chhattisgarh		4		**4**
CR,Nagpur	VI,Nagpur	Maharashtra		5		**5**
NR,Lucknow	III,Varanasi	Eastern U.P	3	2	0	**5**
	XVI,Bareilly	Western U.P	3	0	0	**3**
MER,Patna	V,Ranchi	Bihar	1	1	0	**2**
		Jharkhand	3	0	0	**3**
ER,Kolkatta	XV,Kolkata	West Bengal	3	2		**5**
NER,Guwahati	VII,Guwahati	Assam	2	2		**4**
SER,Bhubaneswar	X,Bhubaneswar	Orissa	1	4		**5**
SR,Hyderabad	IX, Hyderabad	Andhra Pradesh		4		**4**
		Telangana		1		**1**
SWR,Bangalore	XIV,Bangalore	Karnataka		4		**4**
SECR,Chennai	IV,Chennai	Tamil Nadu	1	4		**5**
KR,Trivendrum	IV,Chennai	Kerala		1		**1**
	XIV, Bangalore			1		**1**
UR,Dehradun	XVI,Bareilly	Uttarakhand			1	**1**
NHR,Dharamshala	XVII,Dharamshala	Himachal Pradesh		2		**2**
		Total	**23**	**51**	**2**	**76**
Abbreviation : DR : Direct Rotary	**DUR :** Dual Rotary	**DTH :** Down The Hole				

Table 3.2 Status of Division Wise Pumping Units of CGWB

Division	No. of Pumping Units
I.Ahmedabad	2
II.Ambala	1
III.Varanasi	2
IV.Chennai	0
V.Ranchi	2
VI.Nagpur	2
VII.Guwahati	1
VIII.Jammu	1
IX.Hyderabad	1
X.Bhubneshwar	2
XI.Jodhpur	2
XII.Bhopal	1
XIII.Raipur	1
XIV. Bangalore	2
XV.Kolkata	2
XVI.Bareilly	2
XVII. Dharamshala	1

Table 3.3 State wise Wells constructed by CGWB during the Year 2022-23 (till 31th Mar., 2023)

STATE	TENTATIVE TARGET 2022-23				CUMULATIVE ACHIEVEMENT 2022-23 (SINCE APRIL 2022)			
	EW	OW	PZ	T	EW	OW	PZ	T
West Bengal	3	2	25	**30**	5	4	21	30
Bihar	6	0	4	**10**	4	1	5	10
Jharkhand	15	0	18	**33**	9	5	20	34
Assam	9	0	9	**18**	11	0	7	18
Arunachal Pradesh			4	**4**				**0**
Meghalaya				**0**				**0**
Manipur				**0**				**0**
Nagaland				**0**				**0**
Tripura				**0**				**0**
Odisha	17	8	24	**49**	18	12	24	54
Chhatisgarh	10	5	30	**45**	12	3	33	48
Uttar Pradesh	18	0	21	**39**	19	10	15	44
Uttar Pradesh	9	0	6	**15**	5	8	2	15
Uttarakhand	1	1	0	**2**	1	0	1	2
New Delhi			7	**7**	0	0	4	4
Punjab			10	**10**	0	0	10	10
Haryana			10	**10**	1	1	6	8
Chandigarh				**0**				**0**
Jammu & Kashmir	0	0	15	**15**	2	0	13	15
Himachal Pradesh	0	0	14	**14**	0	0	14	14

STATE	TENTATIVE TARGET 2022-23				CUMULATIVE ACHIEVEMENT 2022-23 (SINCE APRIL 2022)			
	EW	OW	PZ	T	EW	OW	PZ	T
Rajasthan	28	7	30	**65**	32	8	25	65
Gujarat	12	8	20	**40**	12	8	21	41
Daman & Div				**0**				**0**
Madhya Pradesh	14	6	30	**50**	18	3	30	51
Maharashtra	32	6	35	**73**	32	7	45	84
Dadra & Nagar Haveli				**0**				**0**
Andhra Pradesh	28	12	20	**60**	45	17	35	97
Telangana	0	0	25	**25**	0	0	33	33
Tamil Nadu			83	**83**	2	0	87	89
Puducherry (UTP)				**0**				**0**
Kerala	0	0	12	**12**	0	0	10	10
Karnataka	12	4	26	**42**	12	4	36	52
Kerala	0	0	10	**10**	0	1	8	9
Total	**214**	**59**	**488**	**761**	**240**	**92**	**505**	**837**

Abbreviation used

EW-Exploratory Well **OW-Observation Well** **PZ-Piezometers**

Table3.4 (a) Division- wise wells constructed by CGWB during 2022-23

DIVISION	TARGET 2022-23				CUMULATIVE ACHIEVEMENT 2022-23 (SINCE APRIL 2022)				Achievement (%)
	EW	OW	PZ	T	EW	OW	PZ	T	
I.Ahmedabad	12	8	20	**40**	12	8	21	**41**	**102.50**
II.Ambala	0	0	27	**27**	1	1	20	**22**	**81.48**
III.Varanasi	18	0	21	**39**	19	10	15	**44**	**112.82**
IV.Chennai	0	0	95	**95**	2	0	97	**99**	**104.21**
V.Ranchi	21	0	22	**43**	13	6	25	**44**	**102.33**
VI.Nagpur	32	6	35	**73**	32	7	45	**84**	**115.07**
VII.Guwahati	9	0	13	**22**	11	0	7	**18**	**81.82**
VIII.Jammu	0	0	15	**15**	2	0	13	**15**	**100.00**
IX.Hyderabad	28	12	45	**85**	45	17	68	**130**	**152.94**
X. Bhubneshwar	17	8	24	**49**	18	12	24	**54**	**110.20**
XI.Jodhpur	28	7	30	**65**	32	8	25	**65**	**100.00**
XII.Bhopal	14	6	30	**50**	18	3	30	**51**	**102.00**
XIII.Raipur	10	5	30	**45**	12	3	33	**48**	**106.67**
XIV. Bangalore	12	4	36	**52**	12	5	44	**61**	**117.31**
XV.Kolkata	3	2	25	**30**	5	4	21	**30**	**100.00**
XVI.Bareilly	10	1	6	**17**	6	8	3	**17**	**100.00**
XVII. Dharamshala	0	0	14	**14**	0	0	14	**14**	**100.00**
TOTAL	**214**	**59**	**488**	**761**	**240**	**92**	**505**	**837**	**109.99%**

Table 3.4 (b). Region-wise wells constructed by CGWB during 2022-23

REGION	TENTATIVE TARGET 2022-23				ACHIEVEMENT 2022-23 (01.04.22 TO 31.03.2023)				% ACHIEVEMENT
	EW	OW	PZ	T	EW	OW	PZ	T	
NWHR. Jammu	0	0	15	**15**	2	0	13	**15**	**100.00**
NWR. Chandigarh	0	0	27	**27**	1	1	20	**22**	**81.48**
WR. Jaipur	28	7	30	**65**	32	8	25	**65**	**100.00**
WCR. Ahmedabad	12	8	20	**40**	12	8	21	**41**	**102.50**
NCR. Bhopal	14	6	30	**50**	18	3	30	**51**	**102.00**
NCCR. Raipur	10	5	30	**45**	12	3	33	**48**	**106.67**
CR. Nagpur	32	6	35	**73**	32	7	45	**84**	**115.07**
NR. Lucknow	27	0	27	**54**	24	18	17	**59**	**109.25**
MER. Patna	21	0	22	**43**	13	6	25	**44**	**102.32**
ER. Kolkata	3	2	25	**30**	5	4	21	**30**	**100.00**
NER. Guwahati	9	0	13	**22**	11	0	7	**18**	**81.82**
SER. Bhubaneshwar	17	8	24	**49**	18	12	24	**54**	**110.20**
SR. Hyderabad	28	12	45	**85**	45	17	68	**130**	**152.94**
SWR. Bangalore	12	4	26	**42**	12	4	36	**52**	**123.81**
SECR. Chennai	0	0	83	**83**	2	0	87	**89**	**107.23**
KR. Trivandrum	0	0	22	**22**	0	1	18	**19**	**86.36**
UR. Dehradun	1	1	0	**2**	1	0	1	**2**	**100.00**
NHR. Dharamsala	0	0	14	**14**	0	0	14	**14**	**100.00**
TOTAL	**214**	**59**	**488**	**761**	**240**	**92**	**505**	**837**	**109.99**

Abbreviation used

EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table 3.5: Status of Cumulative Boreholes Drilled by CGWB till 31st March 2023 in States & UTs

S No.	State /UT	EW	OW	PZ	T	EW	OW	PZ	SH	DW	Total	TOTAL (I+II)
		(I) Through Outsourcing (Contractual)				(II) Through Departmental Rigs						
A.	**STATES**											
1	Andhra Pradesh	456	122	0	**578**	969	442	342	9	4	**1766**	**2344**
2	Arunachal Pradesh	0	0	0	**0**	46	10	0	1	1	**58**	**58**
3	Assam	17	20	0	**37**	466	214	72	16	42	**810**	**847**
4	Bihar	151	39	0	**190**	326	196	80	10	514	**1126**	**1316**
5	Chhattisgarh	414	15	105	**534**	898	288	194	0	28	**1408**	**1942**
6	Goa	13	1	0	**14**	58	18	14	0	31	**121**	**135**
7	Gujarat	392	116	0	**508**	1188	554	522	27	255	**2546**	**3054**
8	Haryana	75	55	80	**210**	411	274	245	23	170	**1123**	**1333**
9	Himachal Pradesh	0	0	0	**0**	272	49	24	1	0	**346**	**346**
10	Jammu & Kashmir	21	0	0	**21**	503	124	50	8	114	**799**	**820**
11	Jharkhand	275	28	0	**303**	505	254	69	4	71	**903**	**1206**
12	Karnataka	804	146	0	**950**	1643	742	405	7	5	**2802**	**3752**
13	Kerala	10	0	0	**10**	613	214	249	16	13	**1105**	**1115**
14	Madhya Pradesh	725	57	80	**862**	1451	750	206	8	149	**2564**	**3426**
15	Maharashtra	92	2	88	**182**	1824	578	212	2	166	**2782**	**2964**
16	Manipur	0	0	0	**0**	29	14	1	0	2	**46**	**46**
17	Meghalaya	0	0	0	**0**	120	40	2	2	8	**172**	**172**
18	Mizoram	0	0	0	**0**	3	3	0	0	0	**6**	**6**
19	Nagaland	0	0	0	**0**	15	6	1	0	3	**25**	**25**
20	Orissa	575	40	67	**682**	1769	453	175	21	191	**2609**	**3291**
21	Punjab	121	105	0	**226**	232	215	123	20	14	**604**	**830**
22	Rajasthan	809	195	0	**1004**	1520	566	598	93	591	**3368**	**4372**
23	Sikkim	0	0	0	**0**	31	9	0	0	0	**40**	**40**
24	Tamil Nadu	425	206	179	**810**	1429	451	418	13	93	**2404**	**3214**
25	Tripura	0	0	0	**0**	64	31	1	5	22	**123**	**123**
26	Telangana	373	78	0	**451**	796	534	567	5	27	**1929**	**2380**
27	Uttarakhand	28	12	0	**40**	76	7	4	1	129	**217**	**257**
28	Uttar Pradesh	694	274	0	**968**	1156	786	227	40	501	**2710**	**3678**
29	West Bengal	263	80	100	**443**	607	310	198	12	82	**1209**	**1652**
TOTAL(A)		**6733**	**1591**	**699**	**9023**	**19020**	**8132**	**4999**	**344**	**3226**	**35721**	**44744**

S No.	State /UT	EW	OW	PZ	T	EW	OW	PZ	SH	DW	Total	TOTAL (I+II)
		(I) Through Outsourcing (Contractual)				(II) Through Departmental Rigs						
B.	**UNION TERRITORIES**											
1	Andaman & Nicobar	0	0	0	**0**	46	13	0	1	0	**60**	**60**
2	Chandigarh	0	0	0	**0**	9	17	14	2	15	**57**	**57**
3	Dadra &NagarHaveli	0	0	0	**0**	14	1	0	0	0	**15**	**15**
4	Delhi	0	0	0	**0**	149	64	196	13	380	**802**	**802**
5	Daman & Diu	0	0	0	**0**	0	0	7	0	0	**7**	**7**
6	Pondicherry	0	0	0	**0**	30	20	8	5	14	**77**	**77**
TOTAL(B)		**0**	**0**	**0**	**0**	**248**	**115**	**225**	**21**	**409**	**1018**	**1018**
GRAND TOTAL(A+B)		**6733**	**1591**	**699**	**9023**	**19268**	**8247**	**5224**	**365**	**3635**	**36739**	**45762**

Abbreviation used

EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table 3.6 Division / State / Formation wise Achievement during 2022-23 (as on 31.03.2022)

DIVISION	STATE/ UT	HARD ROCK				ALLUVIUM				BOULDARY				TOTAL			
		EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T
I.AHMEDABAD	Gujarat	6	2	20	**28**	6	6	1	**13**	0	0	0	**0**	**12**	**8**	**21**	**41**
II.AMBALA	Haryana	0	0	0	**0**	1	1	6	**8**	0	0	0	**0**	1	1	6	**8**
	Punjab	0	0	0	**0**	0	0	14	**14**	0	0	0	**0**	0	0	14	**14**
	Delhi	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**
III.VARANASI	Uttar Pradesh	12	4	12	**28**	7	6	3	**16**	0	0	0	**0**	19	10	15	**44**
IV.CHENNAI	Tamil Nadu	1	0	83	**84**	1	0	4	**5**	0	0	0	**0**	2	0	87	**89**
	Kerala	0	0	10	**10**	0	0	0	**0**	0	0	0	**0**	0	0	10	**10**
V.RANCHI	Bihar	0	0	0	**0**	4	1	5	**10**	0	0	0	**0**	4	1	5	**10**
	Jharkhand	9	5	20	**34**	0	0	0	**0**	0	0	0	**0**	9	5	20	**34**
VI.NAGPUR	Maharashtra	32	7	45	**84**	0	0	0	**0**	0	0	0	**0**	32	7	45	**84**
VII.GUWAHATI	Assam	0	0	0	**0**	3	0	2	**5**	8	0	5	**13**	11	0	7	**18**
	Arunachal Pradesh	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**
	Meghalaya	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**
	Tripura	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**
VIII.JAMMU	Jammu & Kashmir	0	0	0	**0**	0	0	0	**0**	2	0	13	**15**	2	0	13	**15**

DIVISION	STATE/ UT	HARD ROCK				ALLUVIUM				BOULDARY				TOTAL			
		EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T
IX.HYDERABAD	Andhra Pradesh	45	17	35	**97**	0	0	0	**0**	0	0	0	**0**	45	17	35	**97**
	Telangana	0	0	33	**0**	0	0	0	**0**	0	0	0	**0**	0	0	33	**33**
X.BHUBANESWAR	Orissa	15	10	24	**49**	3	2	0	**5**	0	0	0	**0**	18	12	24	**54**
XI.JODHPUR	Rajasthan	25	4	25	**54**	7	4	0	**11**	0	0	0	**0**	32	8	25	**65**
XII.BHOPAL	Madhya Pradesh	18	3	30	**51**	0	0	0	**0**	0	0	0	**0**	18	3	30	**51**
XIII.RAIPUR	Chattisgarh	12	3	33	**48**	0	0	0	**0**	0	0	0	**0**	12	3	33	**48**
XIV.BANGALORE	Karnataka	12	4	36	**52**	0	0	0	**0**	0	0	0	**0**	12	4	36	**52**
	Kerala	0	1	8	**9**	0	0	0	**0**	0	0	0	**0**	0	1	8	**9**
XV.KOLKATTA	West Bengal	3	2	13	**18**	2	2	8	**12**	0	0	0	**0**	5	4	21	**30**
XVI.BAREILLY	Uttarkhand	0	0	0	**0**	0	0	0	**0**	1	0	1	**2**	1	0	1	**2**
	Uttar Pradesh	0	0	0	**0**	5	8	2	**15**	0	0	0	**0**	5	8	2	**15**
XVII.DHARAMSHALA	Himachal Pradesh	0	0	0	**0**	0	0	0	**0**	0	0	14	**14**	0	0	14	**14**
	TOTAL	**190**	**62**	**427**	**679**	**39**	**30**	**45**	**114**	**11**	**0**	**33**	**44**	**240**	**92**	**505**	**837**

Abbreviation usedEW-Exploratory Well OW-Observation Well PZ-Piezometers

Table 3.7: Division/ State /Head wise Achievement in Normal/ Tribal/ Drought areas during 2022-23 (as on 31.03.2023)

DIVISION	STATE/ UT	NORMAL				TRIBAL				DROUGHT				TOTAL			
		EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T
I.AHMEDABAD	Gujarat	12	08	21	**41**	0	0	0	**0**	0	0	0	**0**	12	08	21	41
II.AMBALA	Haryana	1	1	6	**8**	0	0	0	**0**	0	0	0	**0**	1	1	6	8
	Punjab	0	0	14	**14**	0	0	0	**0**	0	0	0	**0**	0	0	14	14
	Delhi	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	0
III.VARANASI	Uttar Pradesh	7	6	3	**16**	0	0	0	**0**	12	4	12	**28**	19	10	15	44
IV.CHENNAI	Tamil Nadu	2	0	87	**89**	0	0	0	**0**	0	0	0	**0**	2	0	87	89
	Kerala	0	0	10	**10**	0	0	0	**0**	0	0	0	**0**	0	0	10	10
V.RANCHI	Bihar	4	1	5	**10**	0	0	0	**0**	0	0	0	**0**	4	1	5	10
	Jharkhand	0	0	0	**0**	9	5	20	**34**	0	0	0	**0**	9	5	20	34
VI.NAGPUR	Maharashtra	32	7	45	**84**	0	0	0	**0**	0	0	0	**0**	32	7	45	84
VII.GUWAHATI	Assam	8	0	5	**13**	3	0	2	**5**	0	0	0	**0**	11	0	7	18
	Arunachal Pradesh	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	0
	Meghalaya	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	0
	Tripura	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	0
VIII.JAMMU	Jammu & Kashmir	2	0	13	**15**	0	0	0	**0**	0	0	0	**0**	2	0	13	15
IX.HYDERABAD	Andhra Pradesh	45	17	35	**97**	0	0	0	**0**	0	0	0	**0**	45	17	35	97
	Telangana	0	0	33	**33**	0	0	0	**0**	0	0	0	**0**	0	0	33	33
X.BHUBANESWAR	Orissa	8	5	16	**29**	10	7	8	**25**	0	0	0	**0**	18	12	24	54
XI.JODHPUR	Rajasthan	32	8	25	**65**	0	0	0	**0**	0	0	0	**0**	32	8	25	65
XII.BHOPAL	Madhya Pradesh	18	3	30	**51**	0	0	0	**0**	0	0	0	**0**	18	3	30	51
XIII.RAIPUR	Chattisgarh	0	0	10	**10**	12	3	23	**38**	0	0	0	**0**	12	3	33	48
XIV.BANGALORE	Karnataka	12	4	36	**52**	0	0	0	**0**	0	0	0	**0**	12	4	36	52
	Kerala	0	1	8	**9**	0	0	0	**0**	0	0	0	**0**	0	1	8	9
XV.KOLKATTA	West Bengal	5	4	21	**30**	0	0	0	**0**	0	0	0	**0**	5	4	21	30
XVI.BAREILLY	Uttarkhand	1	0	1	**2**	0	0	0	**0**	0	0	0	**0**	1	0	1	2
	Uttar Pradesh	5	8	2	**15**	0	0	0	**0**	0	0	0	**0**	5	8	2	15
XVII.DHARAMSHALA	Himachal Pradesh	0	0	14	**14**	0	0	0	**0**	0	0	0	**0**	0	0	14	14
	TOTAL	**194**	**73**	**440**	**723**	**34**	**15**	**53**	**102**	**12**	**4**	**12**	**28**	**240**	**92**	**505**	**837**

Abbreviation usedEW-Exploratory Well OW-Observation Well PZ-Piezometers

3.2 Well Development/Pumping Test

Once constructed, a tube-well is developed to increase its specific capacity to prevent sand rushing into the well and to obtain maximum well life. Thereafter, pumping tests are conducted for evaluating aquifer parameters i.e. Transmissivity, storage co-efficient and well parameters *viz*. specific capacity and well efficiency, with a view to evolve efficient design for tube wells, assessment of yield capabilities and spacing criteria for tube wells. Total of 176 tubewells were developed and tested during the year 2022-23 (Table 3.8).

Table 3.8 Status of Development & Pumping Test of Wells by CGWB

DIVISION	STATE	No. of Wells developed and tested during the year 2022-23	Balance No. of wells to be tested (Backlog)
I.Ahmedabad	Gujarat	0	**52**
II.Ambala	Haryana	1	**6**
	Punjab	2	**18**
	Delhi	0	**0**
	Chandigarh	0	**2**
III.Varanasi	Uttar Pradesh	7	**117**
IV.Chennai	Tamilnadu	2	**12**
	Kerala	0	**1**
V.Ranchi	Bihar	2	**16**
	Jharkhand	9	**54**
VI.Nagpur	Maharashtra	17	**21**
VII.Guwahati	Assam	10	**42**
	Meghalya	0	**20**
	Tripura	0	**2**
	Arunachal Pradesh	0	**7**
VIII.Jammu	Jammu & Kashmir	8	**10**
IX.Hyderabad	Andhra Pradesh	21	**1**
	Telangana	2	**0**
X.Bhubneshwar	Orissa	15	**8**
XI.Jodhpur	Rajasthan	24	**21**
XII.Bhopal	Madhya Pradesh	2	**20**

XIII.Raipur	Chhattisgarh	8	**28**
XIV. Bangalore	Karnatka	15	**6**
	Kerala	5	**1**
XV.Kolkata	West Bangal	14	**35**
XVI.Bareilly	Uttar Pradesh	11	**34**
	Uttranchal	0	**5**
XVII. Dharamshala	Himachal Pradesh	1	**0**
	TOTAL	**176**	**539**

3.3 Handing over of Exploratory Wells

As per the directions, all the offices of CGWB are directed that from year 2022-23 the wells drilled under GWMR scheme are not to be handed over and have to be included in Monitoring network.

Till March 2023, a total of 19268 exploration wells have been drilled by the Board. Out of a total of 15280 successful wells having adequate ground water discharge, 14532 exploratory wells have been handed over to the State agencies upto 31.03.2022. The status of handing over of exploratory wells drilled by Central Ground Water Board to the State Government is indicated in Table3.9.

Table 3.9 Status of Handing over of Wells drilled by CGWB

S.No.	States	Total Wells drilled (EW) upto 31.03.2023	No. of Successful Wells (EW) upto 31.03.2023	No. of Wells Handed Over to the user agencies upto 31.03.2022
1	Andhra Pradesh	969	665	618
2	Arunachal Pradesh	46	41	41
3	Assam	466	438	421
4	Bihar	326	284	257
5	Chhattisgarh	898	820	783
6	Goa	58	49	49
7	Gujrat	1188	835	808
8	Haryana	411	237	221
9	Himachal Pradesh	272	231	224
10	Jammu & Kashmir	503	379	345
11	Jharkhand	505	377	298
12	Karnataka	1643	1406	1341
13	Kerala	613	453	444
14	Madhya Pradesh	1451	1070	1032
15	Maharashtra	1824	1559	1494
16	Manipur	29	21	21
17	Meghalaya	120	116	116

18	Mizoram	3	3	3
19	Nagaland	15	9	9
20	Orissa	1769	1668	1639
21	Punjab	232	205	189
22	Rajasthan	1520	1117	1068
23	Sikkim	31	10	10
24	Tamil Nadu	1429	1118	1093
25	Tripura	64	62	62
26	Telangana	796	613	607
27	Uttrakhand	76	67	37
28	Uttar Pradesh	1156	945	864
29	West Bengal	607	309	292
TOTAL(A)		**19020**	**15107**	**14386**
1	Andaman & Nicobar	46	12	10
2	Chandigarh	9	9	7
3	Dadra &NagarHaveli	14	8	8
4	Delhi	149	131	108
	Daman & Diu	0	0	0
5	Pondicherry	30	13	13
Total(B)		**248**	**173**	**146**
Total(A+B)		**19268**	**15280**	**14532**

3.4. High Yielding Wells

Ground Water Exploration is carried to study the sub-surface hydrogeological set up of the aquifer to evaluate aquifer parameters. During 2022-23, CGWB under its scientific exploratory drilling programmehasexplored high yielding aquifers in various parts of the Country based on hydrogeological studies coupled with remote sensing and geophysical techniques. Such studies will help in identifying similar ground water sources in other parts of the state having similar hydrogeological conditions and in guiding the State agencies to adopt the follow up action with regard to ground water development for drinking water supply and meeting other demands. High Yielding Wells explored during 2022-23 are indicated in Table 3.9 below. During the year 2022-23, the Board has constructed 133 high yielding wells as detailed below:

Table 3.9 State wise details of high yielding wells constructed during the year 2022-23

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
1	**NER**	Bokulia	Karbi Anglong	26.060	93.200	3.3
2	**NER**	Aolaguri	Bongaigaon	26.420	90.780	4.73
3	**NER**	Srijan Gram	Bongaigaon	26.380	90.720	35.08

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
4	NER	Dhing	Nagaon	26.470	92.470	7.986
5	NER	Bilashpur	Bongaigaon	26.410	90.410	10.41
6	NWR	Topra Khurd	Yamunanagar	30.132	77.192	41
7	Kerala Region (KR)	Parakkalayi	Kasargod	12.366	75.160	8.3
8	Kerala Region (KR)	Nettanige	Kasargod	12.606	75.175	15.1
9	Kerala REgion (KR)	Perla- Enmakaje	Kasargod	12.647	75.107	11
10	Kerala REgion (KR)	Cheripady	Kasargod	12.439	75.160	5.5
11	Kerala REgion (KR)	Edathode	Kasargod	12.374	75.231	4.3
12	Kerala REgion (KR)	Meenja	Kasargod	12.713	74.953	4.3
13	Kerala REgion (KR)	Vorkady	Kasargod	12.752	74.948	9.2
14	SECR,Chennai	Piuliyankudi	Thenkasi	9.168	77.393	6.8
15	SECR,Chennai	Alangulam	Thenkasi	8.867	77.493	12
16	SECR,Chennai	Golasanahalli	Dharmapuri	12.372	78.058	4.8
17	SECR,Chennai	Anchetty	Krishnagiri	12.354	77.720	4.8
18	SECR,Chennai	Maniyathahalli,	Dharmapuri	11.994	78.021	4.8
19	SECR,Chennai	K.Vetrapatty	Dharmapuri	12.123	78.502	12
20	NR Lucknow	Bhadeicha	Hardoi	27.329	80.113	41
21	NR Lucknow	Chittampur	Mirzapur	25.033	82.757	25.8
22	NR Lucknow	Babauna	Lakhimpur Khiri	27.838	80.505	31.13
23	NR Lucknow	Gharthaniya	Lakhimpur Khiri	28.038	80.454	26.01
24	NR Lucknow	Baharchath	Mirzapur	24.908	82.564	4.6
25	NR Lucknow	Garhwa	Mirzapur	24.846	82.879	4.6
26	NR Lucknow	Khadipur	Ayodhya	26.678	82.257	33
27	WR Jaipur	Bengati Khurd	Jodhpur	27.049	72.171	4.36
28	WR Jaipur	Maheson ki Dhani	Jodhpur	26.931	71.674	5
29	WR Jaipur	Jhedasar	Hanumangarh	29.020	74.390	12.6
30	SER Bhubaneswar	Bhatli	Bargarh	21.473	83.540	3.35
31	SER Bhubaneswar	Kunjamara	Sambalpur	21.247	84.666	5.5

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
32	SER Bhubaneswar	Salebhatta	Sambalpur	21.292	84.668	10.36
33	SER Bhubaneswar	Salebhatta	Sambalpur	21.292	84.683	14.13
34	SER Bhubaneswar	Rairakhol	Sambalpur	21.075	84.354	3.35
35	SER Bhubaneswar	Phulkusum	Sambalpur	21.257	84.369	11.58
36	SER Bhubaneswar	Phulkusum	Sambalpur	21.257	84.369	18.86
37	SER Bhubaneswar	Sahajbahal	Sambalpur	21.232	84.308	6.88
38	SER Bhubaneswar	Sahajbahal	Sambalpur	21.232	84.308	4.36
39	SER Bhubaneswar	Pandakimal	Sambalpur	20.984	84.321	4.36
40	SER Bhubaneswar	Maliberua	JHARSUGUDA	21.961	84.170	4.364
41	SER Bhubaneswar	Jharmal	JHARSUGUDA	21.908	84.144	4
42	SER Bhubaneswar	Badasalepalli	SUNDARGARH	22.007	84.230	4.37
43	SER Bhubaneswar	Kendujiani	Mayurbhanj	21.631	86.111	4.4
44	SER Bhubaneswar	Patbil	Mayurbhanj	21.697	86.091	8.5
45	SER Bhubaneswar	Katara	Jagatsinghpur	20.233	86.272	8
46	SER Bhubaneswar	Alasudha	Jagatsinghpur	20.256	86.218	33
47	SER Bhubaneswar	Alasudha	Jagatsinghpur	20.256	86.218	8
48	SER Bhubaneswar	Tarikund	Jagatsinghpur	20.307	86.128	4.5
49	SER Bhubaneswar	Tarikund	Jagatsinghpur	20.307	86.128	4.5
50	NCCR, Raipur	Khusuropali	Gariyaband	20.841	82.211	5.1
51	NCCR, Raipur	Pond	Gariyaband	20.766	81.945	10.16
52	NCCR, Raipur	Pond	Gariyaband	20.766	81.945	10.16
53	NCCR, Raipur	Baghhara	Kabirdham	22.156	81.489	6.1

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
54	NCCR, Raipur	Jamunia	Kabirdham	21.780	81.232	6.1
55	NCCR, Raipur	Maharajpur	Kabirdham	21.942	81.188	3.2
56	NCCR, Raipur	Chandaini	Kabirdham	21.908	81.170	34.48
57	NCCR, Raipur	Rawabhata	Raipur	21.328	81.633	8.4
58	SWR,Bangalore	Krishnaraja Pura/ Magge EW	Mysuru	11.974	76.261	4.58
59	SWR,Bangalore	Krishnaraja Pura/ Magge OW	Mysuru	11.974	76.261	3.78
60	SWR,Bangalore	Dhummavad EW	Dharwad	15.314	74.007	4.22
61	SWR,Bangalore	Dhummavad OW	Dharwad	15.314	74.007	3.8
62	SWR,Bangalore	Govt Primary Higher School, K.B.Thanda PZ	Vijayanagara	14.899	76.252	3.41
63	SWR,Bangalore	Govt PU College Kottur PZ	Vijayanagara	14.822	76.227	3.28
64	SWR,Bangalore	Govt Middle Primary School, Bhogenagarkoppa EW	Dharwad	15.209	75.060	10
65	SWR,Bangalore	Govt. Higher Primary School, Chilugodu PZ	Vijaynagara	15.128	76.153	6.47
66	SWR,Bangalore	Govt First Grade College KalghatgiDastikoppa EW	Dharwad	15.198	74.991	5.5
67	SWR,Bangalore	Govt Higher Primary School, Sasvihalli PZ	Vijayanagara	14.729	76.123	4.28
68	SR Hyderabad	Mamidipalli	Srikakulam	18.862	84.547	4.2
69	SR Hyderabad	Mamidipalli	Srikakulam	18.862	84.547	4.2
70	SR Hyderabad	Chakalakonda	Nellore	14.957	79.510	3.3
71	SR Hyderabad	Besi Ramachandrapuram	Srikakulam	18.918	84.550	8.19
72	SR Hyderabad	Somalaregada	Nellore	14.858	79.367	4.8
73	SR Hyderabad	Somalaregada	Nellore	14.858	79.367	6.2
74	SR Hyderabad	Rapur	Nellore	14.199	79.562	11.7
75	SR Hyderabad	Jirravaripalem	Nellore	14.835	79.675	6.7
76	SR Hyderabad	Rapur	Nellore	14.199	79.562	11.7
77	SR Hyderabad	Bejji	Srikakulam	18.611	83.688	4.2

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
78	SR Hyderabad	Jirravaripalem	Nellore	14.835	79.675	8.19
79	SR Hyderabad	Bejji	Srikakulam	18.611	83.688	6.7
80	SR Hyderabad	Thimmareddipalli	Nellore	15.048	79.429	6.7
81	SR Hyderabad	Thimmareddipalli	Nellore	15.048	79.429	3.6
82	SR Hyderabad	Chinna Annaluru	Nellore	14.915	79.688	8.19
83	SR Hyderabad	Chinna Annaluru	Nellore	14.915	79.688	6.7
84	SR Hyderabad	Ravivalasa	Srikakulam	18.574	84.244	6.7
85	SR Hyderabad	Gottigundala (EW-I)	Nellore	14.633	79.275	6.7
86	SR Hyderabad	Gottigundala (OW-I)	Nellore	14.633	79.275	4.2
87	SR Hyderabad	Gottigundala (EW-II)	Nellore	14.633	79.275	5.4
88	SR Hyderabad	Gottigundala (OW-II)	Nellore	14.633	79.275	3.5
89	SR Hyderabad	Chunchuluru	Nellore	14.633	79.275	8.19
90	SR Hyderabad	Chunchuluru	Nellore	14.633	79.275	8.19
91	SR Hyderabad	Bathulapalli	Nellore	14.519	79.706	4.2
92	SR Hyderabad	Gareepally	Karimnagar	18.498	79.249	4.2
93	SR Hyderabad	Tatti Annaram (office)	Rangareddy	17.367	78.585	4.2
94	SR Hyderabad	Tatti Annaram (office)	Rangareddy	17.367	78.585	4.2
95	ER, Kolkata	Ghughuridanga Lower Primary School	Murshidabad	23.237	88.195	15.26
96	ER, Kolkata	Kallabera Junior High School	Purulia	22.863	86.570	4.36
97	ER, Kolkata	Bansdiha Primary School	Bankura	22.826	86.648	5.5
98	ER, Kolkata	Dhabani Madhyamik Shiksha Kendra	Purulia	22.912	86.549	12.0
99	ER, Kolkata	Laxminarayanpur (Dhaboni) PHED	Birbhum	24.273	87.719	10.13
100	WCR, Ahmedabad	MANPUR	SURENDRANAGAR	23° 0'50.33"N	71°39'11.00"E	8.94
101	WCR, Ahmedabad	KHERVA	SURENDRANAGAR	22°59'52.31"N	71°45'6.90"E	9.35
102	WCR, Ahmedabad	KARELI	SURAT	21° 8'49.30"N	73° 2'30.29"E	23.31
103	WCR, Ahmedabad	KARELI	SURAT	21° 8'49.30"N	73° 2'30.29"E	22.75

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
104	WCR, Ahmedabad	KOTHIYA KHAD	ANAND	22°18'57.52"N	72°57'19.14"E	24
105	WCR, Ahmedabad	KOTHIYA KHAD	ANAND	22°18'57.52"N	72°57'19.14"E	24
106	WCR, Ahmedabad	CHOKARI	VADODARA	22°13'16.14"N	72°55'51.79"E	18
107	WCR, Ahmedabad	KANPAR	BHAVNAGAR	21°58'25.53"N	71°52'48.63"E	3.08
108	WCR, Ahmedabad	KANPAR	BHAVNAGAR	21°58'25.53"N	71°52'48.63"E	4.41
109	WCR, Ahmedabad	PIPRALA	BHAVNAGAR	21°22'44.66"N	72° 5'40.79"E	3.5
110	WCR, Ahmedabad	JAMWADI	RAJKOT	21°55'37.04"N	70°46'3.09"E	9.5
111	WCR, Ahmedabad	MADHVIPUR	RAJKOT	22° 3'24.92"N	71°17'19.80"E	4.45
112	WCR, Ahmedabad	MOTI KANTADI	PANCHMAHAL	22°49'6.43"N	73°25'33.86"E	4
113	WCR, Ahmedabad	PIPALI	MORBI	22°51'19.85"N	70°51'58.13"E	11.66
114	CR, Nagpur	Sakharpa_EW	Ratnagiri	16.991	73.725	7.76
115	CR, Nagpur	Sakharpa_OW	Ratnagiri	16.991	73.725	7.76
116	CR, Nagpur	Kalambani Bk.	Ratnagiri	17.710	73.416	19.66
122	CR, Nagpur	Agave_EW	Ratnagiri	16.886	73.437	7.76
123	CR, Nagpur	Agave_OW	Ratnagiri	16.886	73.437	7.76
125	CR, Nagpur	Kalasawali_EW	Ratnagiri	16.747	73.538	14.88
126	CR, Nagpur	Kalasawali_OW	Ratnagiri	16.747	73.538	12.18
118	CR, Nagpur	Pendhari	Sindhudurg	16.477	73.507	3.77
119	CR, Nagpur	Phanasgaon (VittalaDevi)	Sindhudurg	16.466	73.584	3.17
120	CR, Nagpur	Achirne	Sindhudurg	16.446	73.761	25.05
121	CR, Nagpur	Achara_EW	Sindhudurg	16.210	73.470	3.17
124	CR, Nagpur	Achara_OW	Sindhudurg	16.200	73.699	3.17
117	CR, Nagpur	Osargaon	Sindhudurg	16.200	73.699	3.17
127	CR, Nagpur	Chikhala_Pz	Buldhana	20.445	76.083	3.17
128	CR, Nagpur	Shekapur_Pz	Buldhana	20.460	75.997	7.76
129	NCR, Bhopal	Baghraji	Jabalpur	23.324	80.256	17
130	NCR, Bhopal	Ghat pipariya	Jabalpur	23.037	79.821	16.6
131	NCR, Bhopal	Talad	Jabalpur	23.542	80.042	18.3

Sr no	Region	Name of Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
132	**NCR, Bhopal**	Silondi	Jabalpur	23.501	80.128	8.6
133	**MER, Patna**	Kissan Inter College, Korma	Gaya	24.846	84.975	4.36
134	**MER, Patna**	Plus 2 high School Bahulahaar	Gaya	24.516	84.652	5.54
135	**MER, Patna**	Plus 2 high School Bahulahaar	Gaya	24.516	84.652	4.92
136	**MER, Patna**	PDS Plus 2 H/School,Brahmpur	Darbhanga	26.315	85.739	43.1
137	**MER, Patna**	PDS Plus 2 H/School,Brahmpur	Darbhanga	26.315	85.739	12.5
138	**MER, Patna**	UM School,Rampura	Darbhanga	26.171	85.740	14
139	**MER, Patna**	UM School,Rampura	Darbhanga	26.171	85.740	13
140	**CR, Nagpur**	Raipur_Pz	Buldhana	20.387	76.147	7.76
141	**CR, Nagpur**	DeulgaonMahi_Pz	Buldhana	20.095	76.191	7.76
142	**NWHR, Jammu**	Hasplot, Thana Mandi	Rajouri	33.536	74.363	7
143	**NWHR, Jammu**	Heera Nagar	Kathua	32.457	75.268	Artisian well of 15 lps discharge with Pz head of 12 magl
144	**NWHR, Jammu**	GGM Science College	Jammu	32.723	74.852	5.65

High discharge well at **Bastwamata** (MCD-24), in Balesar block of Jodhpur District with 372lpm discharge.	High discharge well at Govt. Higher Primary School, **Chilugodu**, HB Halli taluk, Vijayanagaradistrict, Karnataka State with 24lps discharge
High yielding well at **Achara**, Malwan block, Sindhudurg district, with 4.43 lps discharge	High yielding well at **GottiGundala**, block Kondapuram,district Nellore of Andhra Pradesh with 20 lps discharge

Fig.3.4: High Yielding Well from States

CHAPTER 4 GEOPHYSICAL STUDIES

4.0 Introduction

Geophysical investigations are used for exploration of groundwater and in delineating the underground structures which control the occurrence, distribution and movement of ground water. Application of geophysical techniques for ground water investigations on regular basis commenced in CGWB during the seventies. The Board has made extensive use of both the surface and the subsurface (well logging) geophysical techniques in the search of groundwater and proper construction of water wells. The findings of the geophysical studies, as a practice, are combined with the hydrogeological and geomorphologic investigations to place them on firm footing. The techniques have become an integral part of the ground water exploration programme.

4.1 Borehole Geophysics

The Borehole geophysics is used in groundwater to obtain information pertaining to lithology, fractures, permeability, porosity and water quality so as to delineate subsurface disposition of aquifers. Borehole-geophysical logging determines the character and thickness of the different lithological/Hydro-geologic units in drilled pilot boreholes. Saline / brackish water bearing aquifers are present in different parts of India. Fresh water bearing aquifers are often intervened by the saline water aquifers. Such information is essential for proper placement of casing and screens in water-supply wells and for characterizing and remediation of problems related to ground-water salinity. The proper positioning and condition of casing and screen pipes in a well can be rapidly evaluated with geophysical borehole logging.

Surface geophysical surveys specially the traditional Electrical Resistivity survey in soft and hard rock formations are commonly employed in CGWB to delineate the ground water bearing zones/structures, pin-pointing sites for construction of boreholes and providing inputs for formulating proposals for constructing artificial recharge structures. Geophysical survey has also been conducted for delineating the bedrock topography and sandy horizon of non – perennial channel.

4.2 Transient-Electromagnetic Techniques (TEM)

In recent times, Transient-Electromagnetic techniques (TEM) are also being used for identifying the sub-surface layers parameters in term of Resistivity and thickness as is done through Electrical resistivity surveys. TEM surveys, however, takes less time in comparison to the conventional Electrical resitivitysurvey. Imaging Resistivity 2-D survey at present is conducted by CGWB by Multi Electrodes Resistivity Meter. Various other techniques like Self Potential, Induced Polarization, Mise-a-la-masse of electrical method, refraction seismic, electromagnetic – the Horizontal Loop, Very Low Frequency (VLF) *and* magnetic were also conducted by CGWB in past. The Heliborne Survey has also been incorporated through pilot project of Aquifer mapping and identification of palaeo-channel in parts of Prayagraj and Kaushambi district in collaboration with CSIR-NGRI. Apart from these, resistivity survey (VES) were also carried out during the year for short-term water supplyinvestigation on request of other Government organization and Public Sector Undertakings.

Regionwise details of availability of Geophysical instruments in CGWB are presented below in table 4.1

Table 4.1: Regionwise details of availability of Geophysical instruments in CGWB

Sl. No.	Region	Geophysical Logger	Resistivity Meter	EM
1	ER-Kolkata	Uptron (18Y) Logger u-1	1.CRM Auto –C (7 y) 2. Res Meter(2 y)	TEM – 1 (New)
2	KR- Trivandrum	Nil	1. Terrameter (21 Y) 2. CRM-500 (15Y)	TEM – 1 (New)
3	NCCR- Raipur	Nil	SSR-MP-ATS (adv) (7Y)	TEM – 1 (New)
4	NCR - Bhopal	Nil	1. DDR-4 MP (20Y) 2. SSR-MP1 (18Y)	
5	WCR- Ahmadabad	Uptron (25 y) Logger	1.SSR-1 (7Y) 2.DDR-4 (25Y)	
6	NWHR- Jammu	Uptron (22 y) Logger	1.DDR-4 (18 Y) 2. Terrameter (18Y)	
7	NHR- Dharmshala	Nil		
8	UR - Dehradun	Nil	1.Terrameter SAS 300 (21Y)	
9	NER-Guwahati	Uptron (>27Y) Logger (+Gamma)	1. SSR-MP1 (13Y) 2. Terrameter (18 Y)	
10	CR- Nagpur	1.Uptron (27Y) Logger	1.Terrameter (19Y)	

Sl. No.	Region	Geophysical Logger	Resistivity Meter	EM
		(+ Gamma), 2. Digital (25Y)Portable Logger (+ Gamma, Flow meter &Temp.) 3.Terrameter spot Logger (part of Terrameter)	2.CRM – 20 (15Y)	
11	MER - Patna	Uptron (> 20 Y) Logger	1. CRM 500 – 40 W (Ranchi) (15Y) 2. CRM-Auto C – 40 W (5Y)	
12	SER - Bhubaneswar	Uptron (> 15 y) Logger	1. CRM 20 (5Y)	
14	SR - Hyderabad	1.Upron (3no and 25 y) Logger (+Gamma)	1. DDR- 4–80 W (21Y) 2. SSR- MP1–80W (7Y) 3. DDR- 3 – 80 W (7Y) 4. Terrameter with manual logger (22Y) 5. Terrameter LS (6Y) 6. ERT System Terrameter LS–2D (6)	
15	SWR- Bangalore	Uptron (20Y) Logger(+Gamma)	1. SSR-MP-ATS - 100W – 2 no. (2Y) 2. Res Meter (5Y)	
16	SECR-Chennai	Uptron (19 Y) Logger	1. DDR – 3= (7Y) 2. CRM – 500- (15Y) 3. Adv WTS – 2 No (5Y)	
17	WR - Jaipur	Uptron (27Y) Logger (+Gamma), u-13	1. DDR- 4 – 80 W (19 Y) 2. CRM-Auto-C – 40 W (7 Y)	
18	NR - Lucknow	Uptron (21Y) Logger(+Gamma), u-14	1.Adv. Terrameter LS – 2D (6.5Y) 2.Adv Syscal R2 – 250	

Sl. No.	Region	Geophysical Logger	Resistivity Meter	EM
			W (25 Y) 3. ERT system 4.Terrameter LS – 2D	
	Total Logger (17 No.), two spot logger with resistivity meter	1. Uptron - 14, (+8 gamma) (8 working) 2. OYO potable-1 (+gamma, flow & Temp). (Not working) 3. SAS log 200 – 2 (part of Terrameter), One is working	1. 33 working , 8 Repairable 2. Imaging System-2 no 3. TEM- 9 no	

Table 4.2 Region wise progress of Geophysical Activities for AAP 2022-23

Sr. No.	State/UT	Region	Annual Target VES	Number of VES As on 31st March 2023	Number of Loggings carried out As on 31st March 2023	Number of Profilings carried out As on 31st March 2023	Achievement TEM As on 31st March2023
1	Madhya Pradesh	NCR	**225**	234	0	0	416
2	Jammu and Kashmir	NWHR	**50**	35	0	0	0
3	Ladakh	NWHR	**25**	39	0	0	0
4	Gujarat	WCR	**50**	52	0	0	0
5	Rajasthan	WR	**75**	75	19	19	270
6	Uttar Pradesh	NR	**95**	77	9	21	456
7	Delhi	SUO	**0**	15	0	0	16
8	Uttarakhand	UR	**20**	70	0	0	0
9	Andaman and Nicobar Islands	ER	**25**	32	0	0	0
10	Sikkim	ER	**10**	0	0	0	0
70	West Bengal	ER	**15**	53	2	2	198
12	Bihar	MER	**25**	19	0	0	122
13	Jharkhand	MER	**25**	39	0	0	0
14	Chhattisgarh	NCCR	**20**	65	0	11	366
15	NER	NER	**50**	36	0	0	0
16	Odisha	SER	**150**	315	0	0	0
17	Maharashtra	CR	**75**	127	0	0	0
18	Andhra Pradesh	SR	**150**	381	0	0	190
19	Telangana	SR	**15**	54	0	0	111
20	Goa	SWR	**0**	5	0	0	117
21	Karnataka	SWR	**75**	88	0	0	300
	Total		**1175**	**1811**	**30**	**53**	**2562**

4.3 Heliborn Survey - High Resolution Aquifer Mapping And Management In Arid Regions Of North-Western India

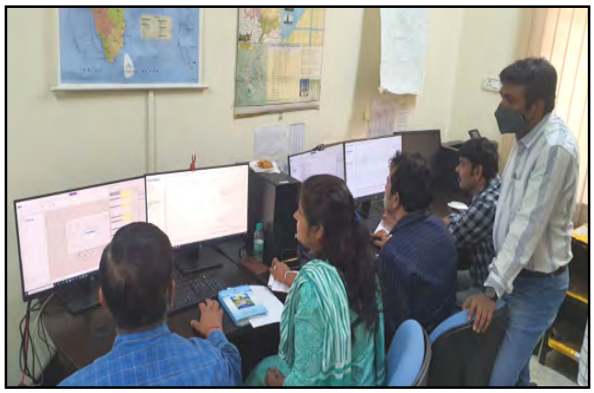
Central Ground Water Board, Ministry of Jal Shakti and CSIR-NGRI, Hyderabad have signed a Memorandum of Agreement for High Resolution Aquifer Mapping and Management in Arid Regions of North Western India, by using advanced Heliborne Geophysical Surveys and other scientific studies in parts of the States of Rajasthan, Gujarat, Haryana, Punjab and Himachal Pradesh, covering an area of 3.88 lakh sq. km under the National Aquifer Mapping Programme (NAQUIM) proposed to be covered in two phases with financial outlay of Rs. 200 crores. The Phase I of Heliborne Surveys was proposed for an area of 1.01 lakh sq. km, out of which 19,020 sq. km was priority areas having 2.5 Km flight spacing, covering parts of Sikar, Jaisalmer and Jodhpur districts in Rajasthan, Kurukshetra & Yamuna Nagar district in Haryana and remaining areas are normal area to be covered with flight spacing of 5 Km.

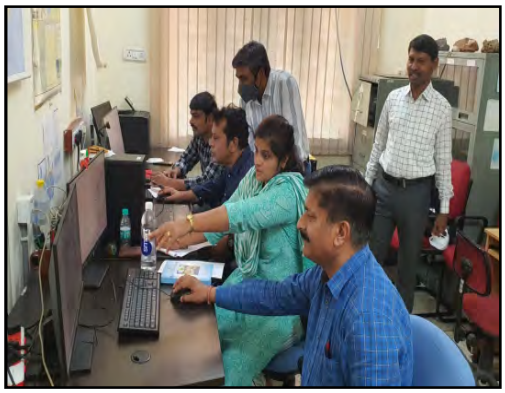
During Phase I heliborne Electromagnetic data has been obtained through a total traverse of 40,313 flight Line Km covering an area of 97,165 sq. km in parts of Rajasthan, Gujrat and Haryana States. Phase I survey area was divided into 8 survey units. These units were distributed as follows: 6 survey units in Rajasthan, namely Sri Ganganagar, Churu, Srimadhopur, Pokhran, Bilara, and Pali-Jalore; 1 survey unit in Haryana named as Kurukshetra; and 1 survey unit in Gujarat, named as Rajkot. In addition, ground geophysical surveys comprising of 86 Line Km Electrical Resistivity Tomography (ERT) surveys, 108 nos. of Ground Transient Electromagnetic (G-TEM) and 16 Line Km 2D Electrical Vector Resistivity Imaging (EVRI) have also been carried out in the survey area for heliborne data correlation and validation. The studies aimed at finding aquifer system with relatively fresh and saline zones, aquifer geometry of principal aquifer with demarcation of de-saturated and saturated aquifers, identification of paleochannels network and their linkage with the aquifer systems in the area, groundwater potential worthy sites for withdrawal of groundwater and water conservation structures through artificial or Managed Aquifer Recharge (MAR). Remaining area of 2.93 lakh. sq. km is envisaged to be covered under Pahse-II for 2023-25.

The general outcomes for all the survey area covered under Phase-I are as follows:

- Information on sub-surface disposition of various litho-units/fracture zones down to the depth up to 500 m in favourable conditions has been obtained.
- Demarcated saline/freshwater bearing zones in the study area.
- Delineated saturated/unsaturated zones within the identified aquifers.
- 3D geophysical model, geophysical thematic maps at horizontal and vertical plains have been created.

- Paleochannels have been demarcated in Pokhran, Churu, Pali-Jalore, and Sri Ganganagar Survey Units in Rajasthan, and the Kurukshetra survey unit in Haryana.
- A total of 1292 suitable sites have been identified for the construction of tubewells.
- 1029 feasible locations have been identified for the construction of artificial recharge structures.





Hands on training to CGWB Scientists on "Data Processing on Aarhus Workbench" at NGRI, Hyderabad, from 18 to 29-April-2022.

CHAPTER 5 GROUND WATER LEVEL MONITORING

5.1 Introduction to Ground Water Level Monitoring

Ground water regime monitoring is one of the key activities of Central Ground Water Board (CGWB). Monitoring of ground water regime is an effort to obtain information on ground water level and chemical quality through representative sampling. The primary objective of ground water monitoring by, is to record the response of various natural and anthropogenic stress on the groundwater regime which impacts the recharge and discharge parameters with reference to geology, climate, physiography, land use pattern and hydrologic characteristics on a regional scale. The key applications of the groundwater data acquired during regime monitoring in the country are :

The data is used for volumetric estimation of groundwater resources.

- To assess the impact of groundwater recharge and draft on long term basis.
- To categories area into overexploited, critical, semi-critical and safe based on long term water level trend analysis.
- In deciding the depth of water well drilling as well as depth of lowering the pumps.
- Help in design, implement and monitor the effectiveness of groundwater management, protection and conservation program
- To identify groundwater quality affected areas for taking remedial measures.
- To plan groundwater recharge interventions and study the impact in time and space.

The ground water levels are measured manually four times a year during the months of January, March/April/ May, August and November coinciding with crop season and onset of monsoon as well as to capture deepest and shallowest water level in the hydrological year. The ground water regime monitoring started in the year 1969 by Central Ground Water Board. Ground water samples are collected once a year during the month of March/April/May to obtain background information of ground water quality changes on regional scale.

At present, CGWB has a network of 25437 (Dug Wells: 16004, Piezometers: 8578, Hand Pump: 790, spring: 65) ground water observation wells throughout the Country.

Table 5.1: State/Ut -Wise Status of Groundwater Monitoring Wells

S. No	Name of the State/UT	Number of GW Monitoring Stations (March 2023)				
		Dug Well	Piezometer	Handpump	Spring	Total
1	Andhra Pradesh	633	701	0	0	1334
2	Arunachal Pradesh	26	5	0	0	31
3	Assam	355	80	0	0	435
4	Bihar	800	90	0	0	890
5	Chhattisgarh	1041	246	0	0	1287
6	Delhi	22	107	0	0	129
7	Goa	83	45	0	0	128
8	Gujarat	836	434	0	0	1270
9	Haryana	488	809	0	0	1297
10	Himachal Pradesh	138	32	0	24	194
11	Jammu & Kashmir	287	92	0	0	379
12	Jharkhand	419	96	0	0	515
13	Karnataka	1366	780	0	0	2146
14	Kerala	1351	259	0	0	1610
15	Madhya Pradesh	1178	434	0	0	1612
16	Maharashtra	1658	273	0	0	1931
17	Manipur	2	0	0	0	2
18	Meghalaya	60	14	0	0	74
19	Mizoram	3				3
20	Nagaland	12	1	0	0	13
21	Odisha	1483	235	0	0	1718
22	Punjab	180	990	0	0	1170
23	Rajasthan	642	709	0	0	1351
24	Sikkim	0	4	0	0	4
25	Tamil Nadu	794	587	0	0	1381
26	Telangana	257	950	0	0	1207
27	Tripura	100	18	0	0	118
28	Uttar Pradesh	838	356	0	0	1194
29	Uttarakhand	40	12	132	41	225
30	West Bengal	754	185	658	0	1597

S. No	Name of the State/UT	Number of GW Monitoring Stations (March 2023)				
		Dug Well	Piezometer	Handpump	Spring	Total
UTs						
1	Andaman & Nicobar	110	2	0	0	112
2	Chandigarh	1	22	0	0	23
3	Dadra & Nagar Haveli	17	0	0	0	17
4	Daman & Diu	21	3	0	0	24
5	Pondicherry	9	7	0	0	16
	Total	**16004**	**8578**	**790**	**65**	**25437**

The database thus generated forms the basis for planning the ground water development and management programme. This data is used for assessment of ground water resources and changes in the regime consequent to various development and management activities

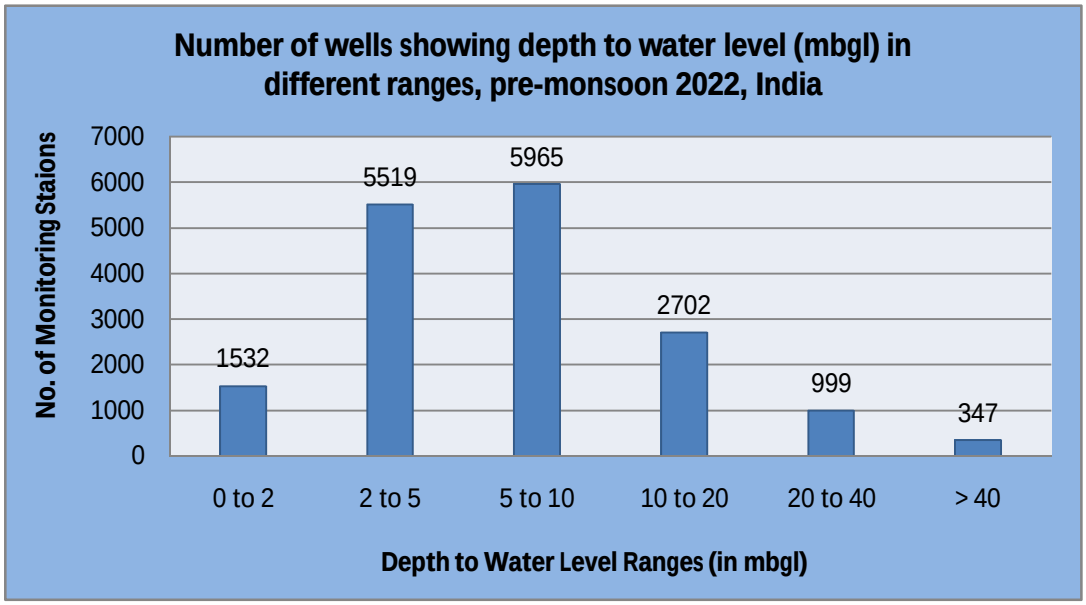
5.2 Groundwater Regime in Unconfined Aquifer

5.2.1 Depth To Water Level in Unconfined Aquifer Pre-Monsoon 2022

The groundwater level data for pre-monsoon 2022 indicates that out of the total 17064 wells analysed, 1532 (9%) wells are showing water level less than 2 m bgl (metres below ground level), 5519 (32.3%) wells are showing water level in the depth range of 2 to 5 m bgl, 5965 (35.0 %) wells are showing water level in the depth range of 5 to10 m bgl, 2702 (15.8%) wells are showing water level in the depth range of 10 to 20 m bgl, 999 (5.9%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 347 (2%) wells are showing water level more than 40 m bgl. The maximum depth to water level of 145.50 m bgl is observed in Bikaner district of Rajasthan whereas the minimum is less than 1 m bgl **(Plate I, Annexure I)**.

Groundwater level data of pre-monsoon 2022 for the country reveals that the general depth to water level of the country ranges from 5 to 10 m bgl.Very shallow water level of less than 2 m bgl is observed in few states, such as Andhra Pradesh, Assam, Karnataka, Kerala, Odisha, Tamil Nadu and West Bengal in small patches. Groundwater level in the range of 2 to 5 m bgl is seen in Assam,Uttar Pradesh, Bihar, Coastal parts of Odisha, few pockets in Andhra Pradesh, Gujarat and Maharashtra. Major part of the country shows water level in the range 5 to 10 m bgl, especially in the states of Madhya Pradesh, Uttar Pradesh, Bihar,

Jharkhand, West Bengal, Odisha, Chhattisgarh, Maharashtra, Gujarat, Tamil Nadu, Telangana and Karnataka. In major parts of north-western and western



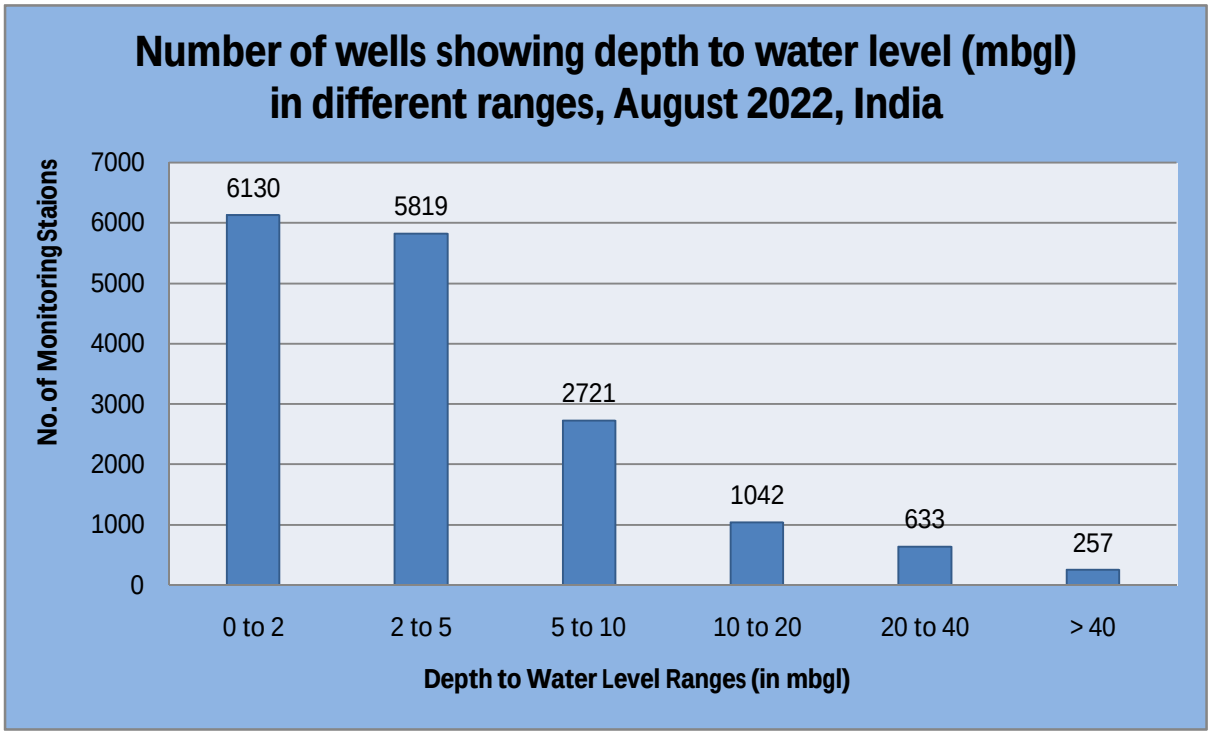
states, especially in the states of Delhi, Haryana, Punjab and Rajasthan, depth to water level is generally deeper and ranges from about 20 to more than 40 m bgl.

5.2.2 Depth to Water Level – August 2022

The groundwater level data for August 2022**(*Annexure-II*)** indicates that, out of the 16602wells analysed, 6130 (36.9 %) wells are showing water level less than 2 m bgl (metres below ground level), 5819(35.0%) wells are showing water level in the depth range of 2-5 m bgl, 2719 (16.4%) wells are showing water level in the depth range of 5-10 m bgl, 1042 (6.3%) wells are showing water level in the depth range of 10-20 m bgl, 633 (3.8%) wells are showing water level in the depth range of 20-40 m bgl and the remaining 257(1.5%) wells are showing water level more than 40 m bgl,The map of depth to water levels during August 2022 is shown in **Plate-II**.

The general depth to water level of the country ranges from 0 to 5 m bgl. Almost 72% of the wells analysed show water level in the range of 0 to 5 m bgl.Very shallow water level of less than 2 m bgl is observed in considerable number of wells in almost all the states,except Chandigarh, Punjab, Delhi and Tamil Nadu.Water level in the range of 2 to 5 m bgl, is observed in almost all states. Punjab, where less than 15 % of wells have water level of 2 to 5 mbgl.In major parts of north-western and western states, depth to water level is generally deeper and ranges fromabout 10 to 40 m bgl. In parts Delhi, Haryana, Punjab and Rajasthan, water level of more than 40 m bgl is also recorded.The maximum depth to water level of 130.6 m bgl is observed at Khara in Jodhpur district of Rajasthan,whereas the minimum is

0.01 m bglat places in Andhra Pradesh, Karnatka, Maharashtra, Gujarat and Madhya Pradesh.

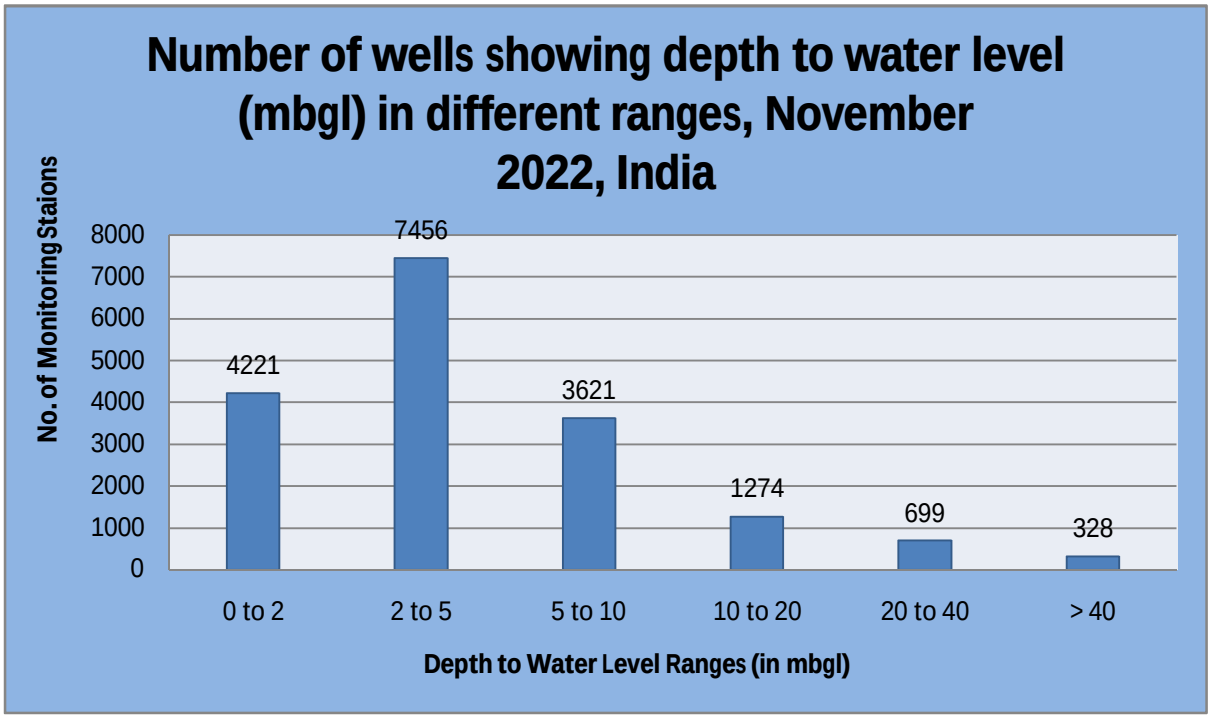


5.2.3 Depth to Water Level – November 2022 (Post-Monsoon)

The groundwater level data for November 2022 indicates that out of the total 17599 wells analysed, 4221 (24.0 %) wells are showing water level less than 2 m bgl,7456(42.4%) wells are showing water level in the depth range of 2 to 5 m bgl, 3621 (20.6%) wells are showing water level in the depth range of 5 to 10 m bgl, 1273 (7.2%) wells are showing water level in the depth range of 10 to 20 m bgl, 699 (4.0%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 329(1.9%) wells are showing water level more than 40 m bgl.In general, depth to water level ranges from 2 to 5 m bgl as observed at about 66.4% of the monitoring stations. Very shallow water level of less than 2 m bgl is observed in all the states, except Chandigarh.

Groundwater level in the range of 2 to 5 m bgl is predominant in the entire Country, as major part of the country shows water level in the range 2 to 5 m bgl. In parts of north-western and western states, especially in the states/UTs of Chandigarh, Delhi, Haryana, Punjab and Rajasthan, depth to water level is generally deeper and ranges from about 10m bgl to more than 40 m bgl.

The peninsular part of country recorded a water level in the range of 2 to 10 m bgl. The maximum depth to water level of 124.5 m bgl is observed in Jodhpur district of Rajasthan whereas the minimum is less than 1 m bgl**(Plate-III, *Annexure- III*)**.

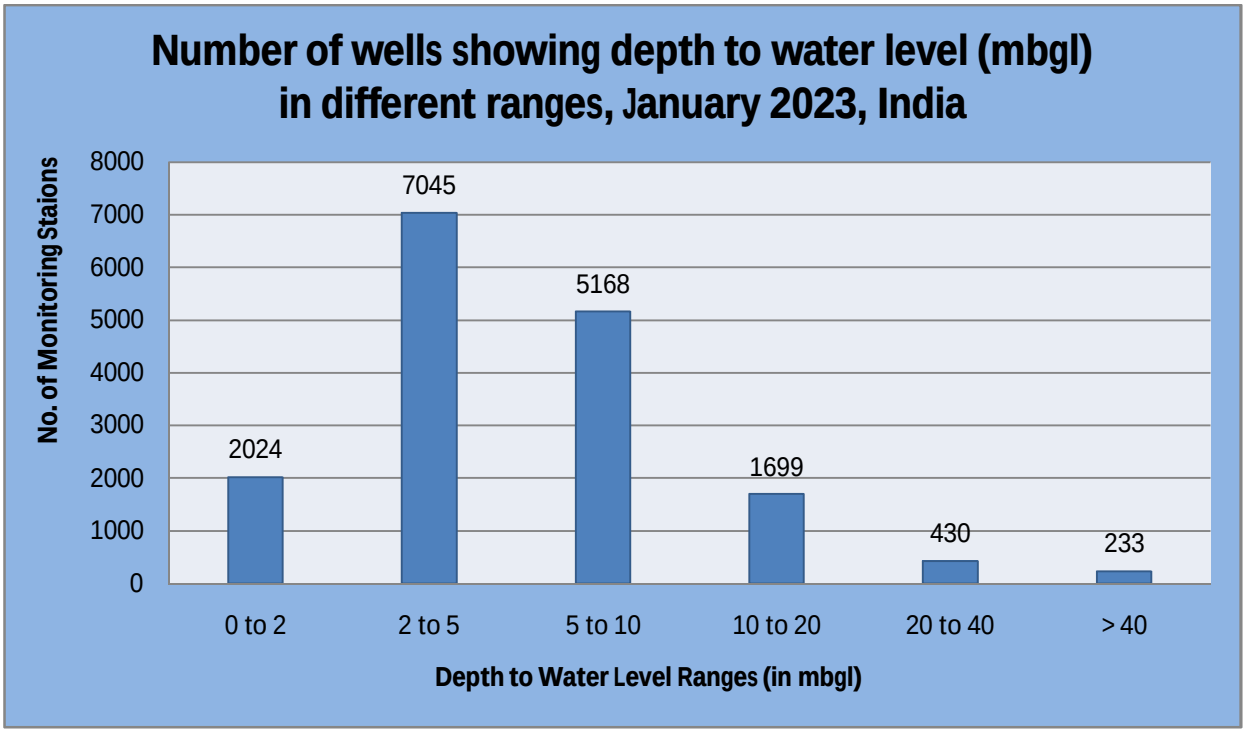


5.2.3 Depth to Water Level– January 2023

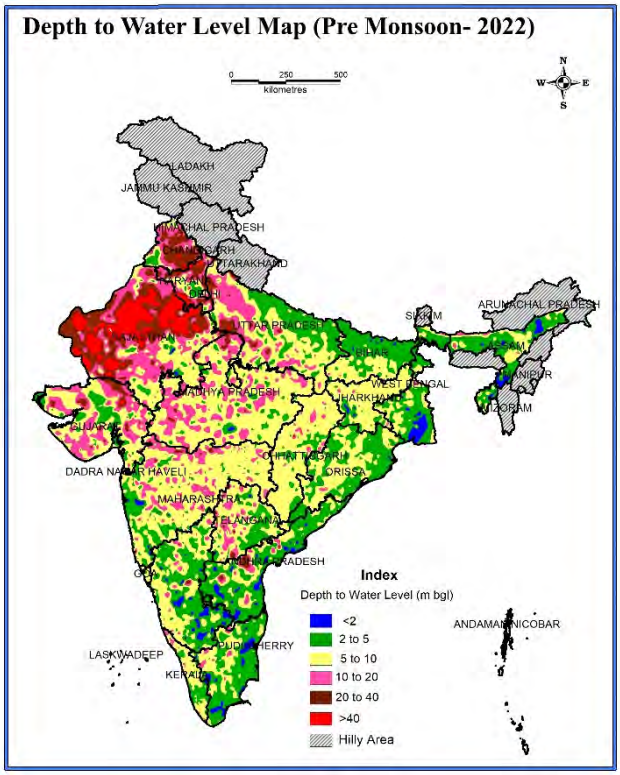
The groundwater level data for January 2023 indicates that out of the total 16599 wells analysed, 2024 (12.2%) wells are showing water level less than 2 m bgl (metres below ground level), 7045 (42.4%) wells are showing water level in the depth range of 2 to 5 m bgl, 5168 (31.1%) wells are showing water level in the depth range of 5 to 10 m bgl, 1699 (10.2%) wells are showing water level in the depth range of 10 to 20 m bgl, 430 (2.6%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 233 (1.4%) wells are showing water level more than 40 m bgl. The maximum depth to water level of 146.11 m bgl is observed in Jodhpur district of Rajasthan, whereas the minimum is less than 1 m bgl. **(Plate-IV, *Annexure-IV*)**.

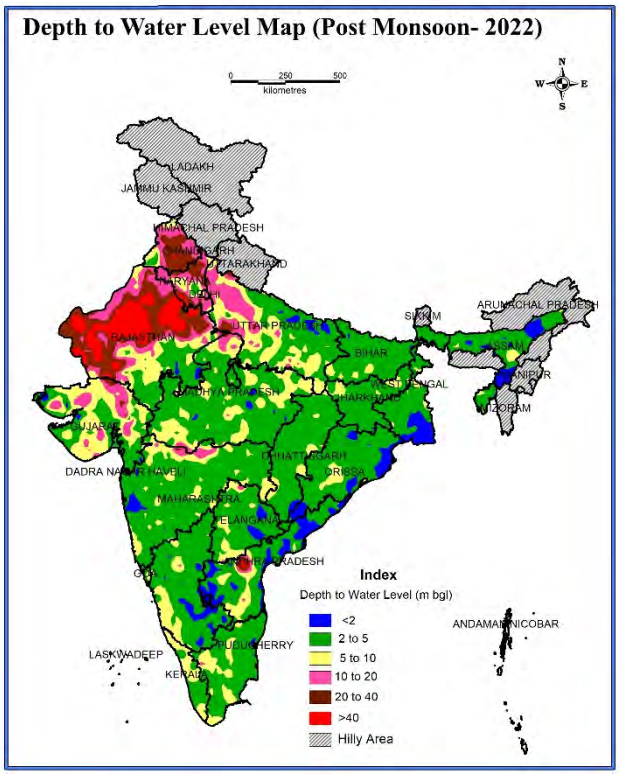
The depth to water level map of January 2023 for the country indicates that in general depth to water level ranges from 2 to 10 m bgl as observed at about 86% of themonitoring stations. Shallow water level of less than 2 m bgl is observed in very small and isolated pockets in the states of Assam, Odisha, Andhra Pradesh, Tamil Nadu, Konkan areas of Maharashtra and northern parts of Uttar Pradesh and Bihar.Water level in the range of 5 to 10 m bgl is prominent in all the states of Central and western parts of Karnataka and Kerala in Southern India. In major parts of north-western states depth to water level generally

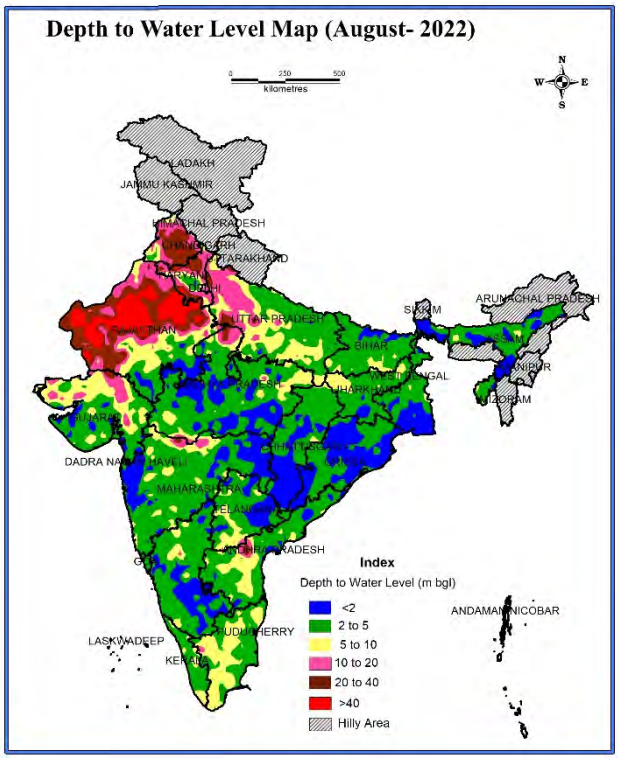
ranges from 10 to 40 m bgl. Water level of more than 40 m bgl is also prevalent in the north western part of the country. In some parts of Haryana, Punjab and Delhi and major parts of Rajasthan, water level of more than 40 m bgl is recorded.

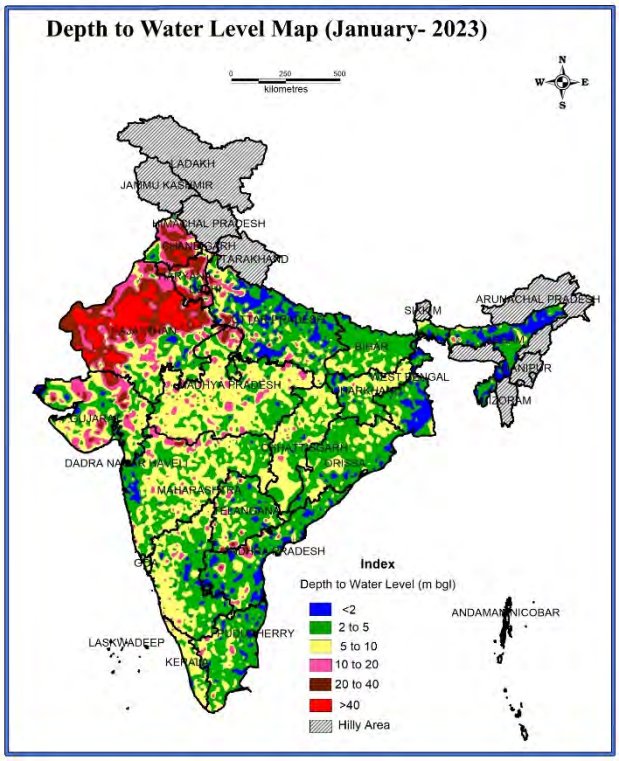


DEPTH TO WATER LEVEL AT A GLANCE









Annexure – I

State-wise Depth to Water Level Distribution of Percentage of Observation Wells Pre-Monsoon 2022 in Unconfined Aquifer

Sr. No.	State/UT Name	No of well analysed	DTWL, mbgl		No./Percentage of wells showing depth to water level (mbgl) in the range of											
			Min	Max	0 to 2		2 to 5		5 to 10		10 to 20		20 to 40		> 40	
					No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
1	Andhra Pradesh	714	0.01	43.28	142	19.9	357	50.0	170	23.8	33	4.6	8	1.1	4	0.6
2	Arunachal Pradesh	24	0.15	12.19	5	20.8	8	33.3	9	37.5	2	8.3	0	0.0	0	0.0
3	Assam	279	0.17	22.98	58	20.8	161	57.7	48	17.2	10	3.6	2	0.7	0	0.0
4	Bihar	585	0.35	14.08	42	7.2	330	56.4	202	34.5	11	1.9	0	0.0	0	0.0
5	Chhattisgarh	873	0.35	34.00	25	2.9	241	27.6	478	54.8	123	14.1	6	0.7	0	0.0
6	Goa	77	1.50	15.97	1	1.3	35	45.5	31	40.3	10	13.0	0	0.0	0	0.0
7	Gujarat	701	0.42	52.42	23	3.3	155	22.1	241	34.4	212	30.2	65	9.3	5	0.7
8	Haryana	990	0.65	98.00	35	3.5	193	19.5	165	16.7	240	24.2	258	26.1	99	10.0
9	Himachal Pradesh	83	0.82	29.79	6	7.2	30	36.1	22	26.5	19	22.9	6	7.2	0	0.0
10	Jharkhand	318	0.45	14.55	13	4.1	81	25.5	193	60.7	31	9.7	0	0.0	0	0.0
11	Karnataka	1383	0.04	32.26	307	22.2	470	34.0	437	31.6	166	12.0	3	0.2	0	0.0
12	Kerala	1389	0.16	51.70	150	10.8	421	30.3	582	41.9	223	16.1	11	0.8	2	0.1
13	Madhya Pradesh	1203	0.45	44.62	15	1.2	182	15.1	607	50.5	366	30.4	30	2.5	3	0.2
14	Maharashtra	1679	0.10	36.50	66	3.9	378	22.5	836	49.8	365	21.7	34	2.0	0	0.0
15	Meghalaya	48	0.58	9.37	8	16.7	30	62.5	10	20.8	0	0.0	0	0.0	0	0.0
16	Nagaland	11	1.45	18.04	1	9.1	1	9.1	6	54.5	3	27.3	0	0.0	0	0.0
17	Odisha	1314	0.15	15.70	143	10.9	671	51.1	472	35.9	28	2.1	0	0.0	0	0.0
18	Punjab	752	1.20	68.20	12	1.6	91	12.1	124	16.5	194	25.8	303	40.3	28	3.7
19	Rajasthan	945	0.25	145.50	17	1.8	97	10.3	214	22.6	254	26.9	182	19.3	181	19.2
20	Tamil Nadu	875	0.14	48.80	160	18.3	418	47.8	232	26.5	59	6.7	4	0.5	2	0.2
21	Telangana	526	0.02	69.70	30	5.7	175	33.3	210	39.9	94	17.9	14	2.7	3	0.6
22	Tripura	94	0.75	28.35	14	14.9	52	55.3	23	24.5	3	3.2	2	2.1	0	0.0
23	Uttar Pradesh	812	0.42	44.46	16	2.0	336	41.4	290	35.7	139	17.1	29	3.6	2	0.2

24	Uttarakhand	165	0.53	76.17	3	1.8	36	21.8	39	23.6	52	31.5	20	12.1	15	9.1
25	West Bengal	711	0.35	20.18	115	16.2	360	50.6	216	30.4	19	2.7	1	0.1	0	0.0
26	Andaman and Nicobar	110	0.01	5.90	86	78.2	21	19.1	3	2.7	0	0.0	0	0.0	0	0.0
27	Chandigarh	8	3.02	21.53	0	0.0	4	50.0	2	25.0	1	12.5	1	12.5	0	0.0
28	Daman & Diu and Dadra & Nagar Haveli	9	3.15	9.15	0	0.0	3	33.3	6	66.7	0	0.0	0	0.0	0	0.0
29	Delhi	117	1.25	53.34	5	4.3	35	29.9	35	29.9	27	23.1	12	10.3	3	2.6
30	Jammu and Kashmir	260	0.10	36.58	33	12.7	139	53.5	62	23.8	18	6.9	8	3.1	0	0.0
31	Puducherry	9	1.84	3.29	1	11.1	8	88.9	0	0.0	0	0.0	0	0.0	0	0.0
	Total	**17064**	**0.01**	**145.50**	**1532**	**9.0**	**5519**	**32.3**	**5965**	**35.0**	**2702**	**15.8**	**999**	**5.9**	**347**	**2.0**

Annexure – II

State-wise Depth to Water Level Distribution of Percentage of Observation Wells August 2022 in Unconfined Aquifer

Sr. No.	State/UT Name	No of well analysed	DTWL, mbgl		No./Percentage of wells showing depth to water level (mbgl) in the range of											
			Min	Max	0 to 2		2 to 5		5 to 10		10 to 20		20 to 40		> 40	
					No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
1	Andhra Pradesh	747	0.01	99.50	298	39.9	273	36.5	125	16.7	39	5.2	10	1.3	2	0.3
2	Arunachal Pradesh	25	0.10	13.04	11	44.0	10	40.0	3	12.0	1	4.0	0	0.0	0	0.0
3	Assam	295	0.10	23.47	161	54.6	110	37.3	19	6.4	4	1.4	1	0.3	0	0.0
4	Bihar	628	0.06	12.56	145	23.1	391	62.3	87	13.9	5	0.8	0	0.0	0	0.0
5	Chhattisgarh	965	0.01	31.00	492	51.0	363	37.6	95	9.8	12	1.2	3	0.3	0	0.0
6	Goa	80	0.56	16.75	22	27.5	39	48.8	16	20.0	3	3.8	0	0.0	0	0.0
7	Gujarat	680	0.01	60.70	241	35.4	241	35.4	111	16.3	65	9.6	19	2.8	3	0.4
8	Haryana	266	0.04	88.92	45	16.9	43	16.2	48	18.0	67	25.2	49	18.4	14	5.3
9	Himachal Pradesh	88	0.08	36.92	27	30.7	27	30.7	17	19.3	11	12.5	6	6.8	0	0.0
10	Jharkhand	364	0.09	12.10	105	28.8	165	45.3	88	24.2	6	1.6	0	0.0	0	0.0
11	Karnataka	1346	0.01	26.00	655	48.7	462	34.3	203	15.1	23	1.7	3	0.2	0	0.0
12	Kerala	1411	0.01	52.20	400	28.3	486	34.4	436	30.9	79	5.6	8	0.6	2	0.1

Sr. No.	State/UT Name	No of well analysed	DTWL, mbgl		No./Percentage of wells showing depth to water level (mbgl) in the range of											
			Min	Max	0 to 2		2 to 5		5 to 10		10 to 20		20 to 40		> 40	
					No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
13	Madhya Pradesh	1288	0.01	38.92	591	45.9	443	34.4	173	13.4	65	5.0	16	1.2	0	0.0
14	Maharashtra	1691	0.01	55.27	755	44.6	651	38.5	194	11.5	72	4.3	15	0.9	4	0.2
15	Meghalaya	50	0.05	6.64	27	54.0	22	44.0	1	2.0	0	0.0	0	0.0	0	0.0
16	Nagaland	10	0.41	9.95	2	20.0	4	40.0	4	40.0	0	0.0	0	0.0	0	0.0
17	Odisha	1390	0.02	8.75	944	67.9	401	28.8	45	3.2	0	0.0	0	0.0	0	0.0
18	Punjab	618	0.13	51.20	30	4.9	58	9.4	83	13.4	140	22.7	268	43.4	39	6.3
19	Rajasthan	978	0.02	130.60	148	15.1	178	18.2	153	15.6	169	17.3	154	15.7	176	18.0
20	Tamil Nadu	891	0.36	48.00	66	7.4	429	48.1	323	36.3	67	7.5	4	0.4	2	0.2
21	Telangana	587	0.02	49.55	255	43.4	199	33.9	99	16.9	27	4.6	6	1.0	1	0.2
22	Tripura	89	0.08	27.77	29	32.6	45	50.6	10	11.2	3	3.4	2	2.2	0	0.0
23	Uttar Pradesh	801	0.05	44.11	166	20.7	318	39.7	192	24.0	100	12.5	23	2.9	2	0.2
24	Uttarakhand	160	0.24	93.90	13	8.1	36	22.5	47	29.4	34	21.3	21	13.1	9	5.6
25	West Bengal	735	0.03	19.77	376	51.2	268	36.5	79	10.7	12	1.6	0	0.0	0	0.0
26	Chandigarh	11	2.55	33.08	0	0.0	3	27.3	3	27.3	2	18.2	3	27.3	0	0.0
27	Daman & Diu and Dadra & Nagar Haveli	5	1.00	5.83	2	40.0	1	20.0	2	40.0	0	0.0	0	0.0	0	0.0
28	Delhi	117	0.43	52.68	11	9.4	28	23.9	37	31.6	24	20.5	14	12.0	3	2.6
29	Jammu and Kashmir	277	0.31	34.06	113	40.8	118	42.6	26	9.4	12	4.3	8	2.9	0	0.0
30	Puducherry	9	2.31	6.55	0	0.0	7	77.8	2	22.2	0	0.0	0	0.0	0	0.0
	Total	**16602**	**0.01**	**130.60**	**6130**	**36.9**	**5819**	**35.0**	**2721**	**16.4**	**1042**	**6.3**	**633**	**3.8**	**257**	**1.5**

Annexure-III

State-wise Depth to Water Level Distribution of Percentage of Observation Wells Post-Monsoon 2022 in Unconfined Aquifer

Sr. No.	State/UT Name	No of well analysed	DTWL, mbgl		No./Percentage of wells showing depth to water level (mbgl) in the range of											
			Min	Max	0 to 2		2 to 5		5 to 10		10 to 20		20 to 40		> 40	
					No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
1	Andhra Pradesh	740	0.01	37.62	370	50.0	233	31.5	100	13.5	30	4.1	7	0.9	0	0.0
2	Arunachal Pradesh	25	0.10	10.93	9	36.0	8	32.0	7	28.0	1	4.0	0	0.0	0	0.0
3	Assam	299	0.01	19.03	146	48.8	125	41.8	21	7.0	7	2.3	0	0.0	0	0.0
4	Bihar	664	0.33	11.30	104	15.7	458	69.0	99	14.9	3	0.5	0	0.0	0	0.0
5	Chhattisgarh	1024	0.20	40.00	234	22.9	619	60.4	157	15.3	13	1.3	1	0.1	0	0.0
6	Goa	83	0.15	17.24	7	8.4	37	44.6	30	36.1	9	10.8	0	0.0	0	0.0
7	Gujarat	711	0.10	61.42	121	17.0	274	38.5	201	28.3	81	11.4	30	4.2	4	0.6
8	Haryana	945	0.04	97.00	85	9.0	177	18.7	133	14.1	204	21.6	240	25.4	106	11.2
9	Himachal Pradesh	84	0.54	36.25	16	19.0	29	34.5	15	17.9	19	22.6	5	6.0	0	0.0
10	Jharkhand	392	0.40	16.10	47	12.0	244	62.2	96	24.5	5	1.3	0	0.0	0	0.0
11	Karnataka	1402	0.01	27.30	461	32.9	540	38.5	350	25.0	49	3.5	2	0.1	0	0.0
12	Kerala	1424	0.05	52.82	280	19.7	456	32.0	544	38.2	130	9.1	13	0.9	1	0.1
13	Madhya Pradesh	1225	0.02	37.67	228	18.6	585	47.8	323	26.4	79	6.4	10	0.8	0	0.0
14	Maharashtra	1724	0.10	55.00	394	22.9	850	49.3	402	23.3	63	3.7	13	0.8	2	0.1
15	Meghalaya	50	0.07	6.73	26	52.0	23	46.0	1	2.0	0	0.0	0	0.0	0	0.0
16	Nagaland	13	0.45	10.04	3	23.1	3	23.1	6	46.2	1	7.7	0	0.0	0	0.0
17	Odisha	1415	0.03	11.60	570	40.3	709	50.1	133	9.4	3	0.2	0	0.0	0	0.0
18	Punjab	502	0.79	48.79	19	3.8	75	14.9	79	15.7	143	28.5	168	33.5	18	3.6
19	Rajasthan	964	0.06	124.50	73	7.6	235	24.4	164	17.0	174	18.0	145	15.0	173	17.9
20	Tamil Nadu	896	0.52	47.52	190	21.2	463	51.7	197	22.0	44	4.9	1	0.1	1	0.1
21	Telangana	551	0.01	62.20	192	34.8	228	41.4	101	18.3	25	4.5	2	0.4	3	0.5
22	Tripura	98	0.54	26.38	26	26.5	56	57.1	11	11.2	4	4.1	1	1.0	0	0.0
23	Uttar Pradesh	904	0.15	44.14	218	24.1	367	40.6	196	21.7	101	11.2	18	2.0	4	0.4
24	Uttarakhand	152	0.20	68.10	17	11.2	44	28.9	36	23.7	27	17.8	18	11.8	10	6.6
25	West Bengal	733	0.05	16.55	193	26.3	402	54.8	124	16.9	14	1.9	0	0.0	0	0.0
26	Andaman and Nicobar	109	0.04	4.90	94	86.2	15	13.8	0	0.0	0	0.0	0	0.0	0	0.0
27	Chandigarh	13	2.51	49.01	0	0.0	4	30.8	2	15.4	3	23.1	3	23.1	1	7.7
28	Daman & Diu and Dadra & Nagar Haveli	22	1.45	7.80	2	9.1	16	72.7	4	18.2	0	0.0	0	0.0	0	0.0
29	Delhi	116	0.62	51.69	9	7.8	28	24.1	43	37.1	22	19.0	11	9.5	3	2.6
30	Jammu and Kashmir	310	0.20	88.88	86	27.7	146	47.1	45	14.5	20	6.5	11	3.5	2	0.6
31	Puducherry	9	1.16	6.04	1	11.1	7	77.8	1	11.1	0	0.0	0	0.0	0	0.0
	Total	**17599**	**0.01**	**124.50**	**4221**	**24.0**	**7456**	**42.4**	**3621**	**20.6**	**1274**	**7.2**	**699**	**4.0**	**328**	**1.9**

State-wise Depth to Water Level Distribution of Percentage of Observation Wells January 2023 Annexure IV

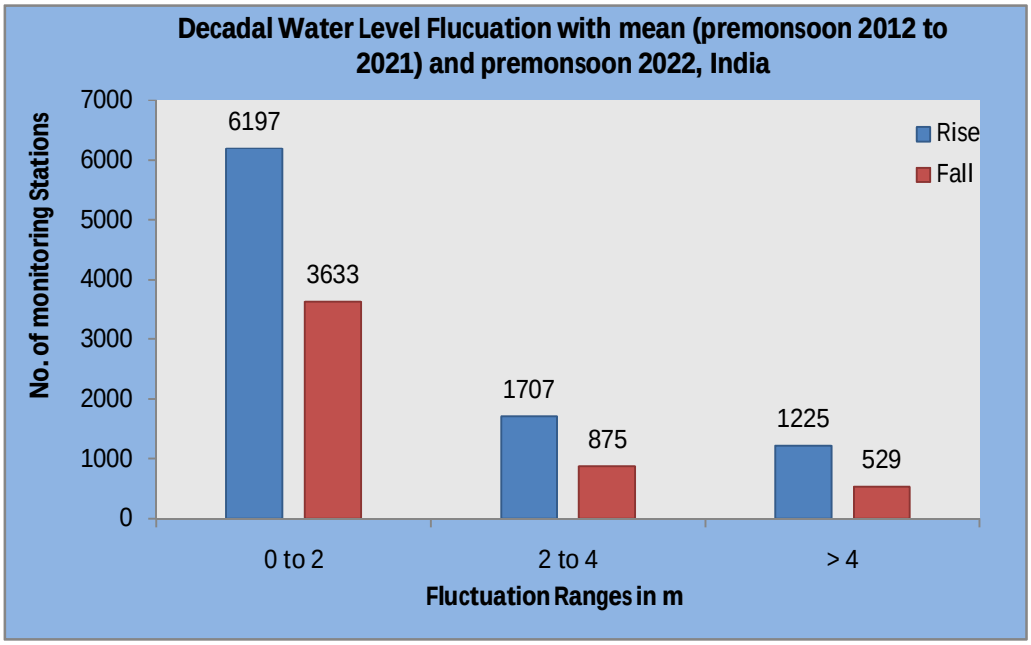
Sr. No.	State/UT Name	No of well analysed	DTWL, mbgl		No./Percentage of wells showing depth to water level (mbgl) in the range of											
			Min	Max	0 to 2		2 to 5		5 to 10		10 to 20		20 to 40		> 40	
					No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
1	Andhra Pradesh	782	0.01	69.82	283	36.2	337	43.1	110	14.1	38	4.9	13	1.7	1	0.1
2	Arunachal Pradesh	26	0.05	11.47	6	23.1	8	30.8	9	34.6	3	11.5	0	0.0	0	0.0
3	Assam	299	0.15	20.34	69	23.1	176	58.9	45	15.1	8	2.7	1	0.3	0	0.0
4	Bihar	789	0.42	15.70	49	6.2	515	65.3	214	27.1	11	1.4	0	0.0	0	0.0
5	Chhattisgarh	1018	0.30	50.00	74	7.3	513	50.4	380	37.3	47	4.6	3	0.3	1	0.1
6	Goa	83	1.60	16.31	3	3.6	37	44.6	29	34.9	14	16.9	0	0.0	0	0.0
7	Gujarat	721	0.33	61.81	77	10.7	235	32.6	243	33.7	128	17.8	35	4.9	3	0.4
8	Haryana	268	0.09	88.92	37	13.8	52	19.4	50	18.7	71	26.5	42	15.7	16	6.0
9	Himachal Pradesh	81	0.29	38.21	10	12.3	30	37.0	20	24.7	16	19.8	5	6.2	0	0.0
10	Jharkhand	399	0.26	15.60	24	6.0	153	38.3	212	53.1	10	2.5	0	0.0	0	0.0
11	Karnataka	1365	0.01	23.50	95	7.0	539	39.5	547	40.1	182	13.3	2	0.1	0	0.0
12	Kerala	1399	0.01	53.00	168	12.0	427	30.5	600	42.9	189	13.5	12	0.9	3	0.2
13	Madhya Pradesh	1183	0.20	39.32	70	5.9	375	31.7	518	43.8	206	17.4	14	1.2	0	0.0
14	Maharashtra	1659	0.10	56.90	161	9.7	719	43.3	610	36.8	154	9.3	13	0.8	2	0.1
15	Meghalaya	49	0.20	8.27	10	20.4	29	59.2	10	20.4	0	0.0	0	0.0	0	0.0
16	Nagaland	13	1.87	12.04	1	7.7	4	30.8	6	46.2	2	15.4	0	0.0	0	0.0
17	Odisha	1408	0.05	16.50	278	19.7	778	55.3	337	23.9	15	1.1	0	0.0	0	0.0
18	Punjab	219	0.57	47.15	13	5.9	42	19.2	38	17.4	71	32.4	48	21.9	7	3.2
19	Rajasthan	958	0.15	146.11	52	5.4	175	18.3	209	21.8	194	20.3	152	15.9	176	18.4
20	Tamil Nadu	897	0.33	46.90	112	12.5	503	56.1	218	24.3	60	6.7	3	0.3	1	0.1
21	Telangana	591	0.16	59.90	121	20.5	247	41.8	172	29.1	42	7.1	7	1.2	2	0.3
22	Tripura	97	0.89	26.70	18	18.6	57	58.8	16	16.5	5	5.2	1	1.0	0	0.0
23	Uttar Pradesh	999	0.35	44.36	105	10.5	474	47.4	259	25.9	130	13.0	28	2.8	3	0.3
24	Uttarakhand	164	0.08	70.78	14	8.5	46	28.0	34	20.7	35	21.3	21	12.8	14	8.5
25	West Bengal	738	0.05	22.21	124	16.8	410	55.6	179	24.3	24	3.3	1	0.1	0	0.0
26	Chandigarh	12	2.74	48.08	0	0.0	4	33.3	2	16.7	2	16.7	3	25.0	1	8.3
27	Daman & Diu and Dadra & Nagar Haveli	5	5.49	7.34	0	0.0	0	0.0	5	100.0	0	0.0	0	0.0	0	0.0
28	Delhi	115	0.96	55.49	7	6.1	30	26.1	43	37.4	21	18.3	12	10.4	2	1.7
29	Jammu and Kashmir	253	0.12	89.00	42	16.6	123	48.6	52	20.6	21	8.3	14	5.5	1	0.4
30	Puducherry	9	1.60	6.30	1	11.1	7	77.8	1	11.1	0	0.0	0	0.0	0	0.0
	Total	**16599**	**0.01**	**146.11**	**2024**	**12.2**	**7045**	**42.4**	**5168**	**31.1**	**1699**	**10.2**	**430**	**2.6**	**233**	**1.4**

3.2.4 Decadal Water Level Fluctuation

Water Level Fluctuation with Decadal Mean (Pre-Monsoon 2012 to 2021) to Pre-Monsoon 2022

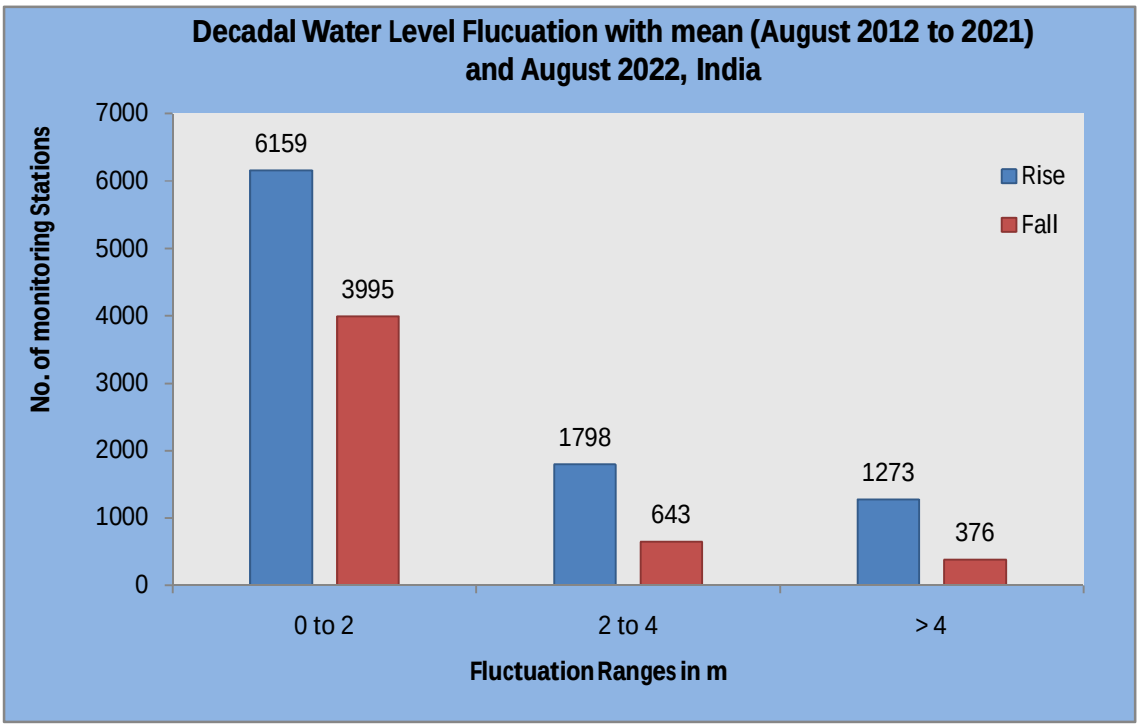
A comparison of depth to water level of pre-monsoon 2022 with decadal mean of pre-monsoon (2012-2021) indicate that 9129 (64.3%) of wells are showing rise in water level, out of which 43.7% (6197) wells are showing rise of less than 2 m. About 12.0% (1707) wells are showing rise in water level in the range of 2 to 4 m and about 8.6% (1225) wells are showing rise in water level in the range of more than 4 m. 5037 (35.5%) wells are showing decline in water level, out of which 25.6% wells are showing decline in water in the range of 0 to 2 m. 6.2% wells are showing decline in water level in 2 to 4 m range and remaining 3.7% (529) are in the range of more than 4 m. A comparison of depth to water level of pre-monsoon 2022 with decadal mean of pre-monsoon (2012-2021) indicates that 64.3% of the analysed wells show rise in water level whereas 35.5% wells show decline in water level. Less than 0.5% well show no change. Rise and decline in water level are primarily in the 0 to 2 m range. More than 50% of the wells analysed have registered rise in groundwater level, mostly in the range of 0 to 4 m, in all the states/UTs.

Decline of more than 4 m has also been observed in pockets in the states/UTs of Chhattisgarh, Delhi, Gujarat, Haryana, Maharashtra, Punjab, Rajasthan, Telangana and Uttarakhand **(Plate-XX, *Annexure-XII*).**



Water Level Fluctuation with Decadal Mean (August 2012 to AUGUST 2021) to AUGUST 2022

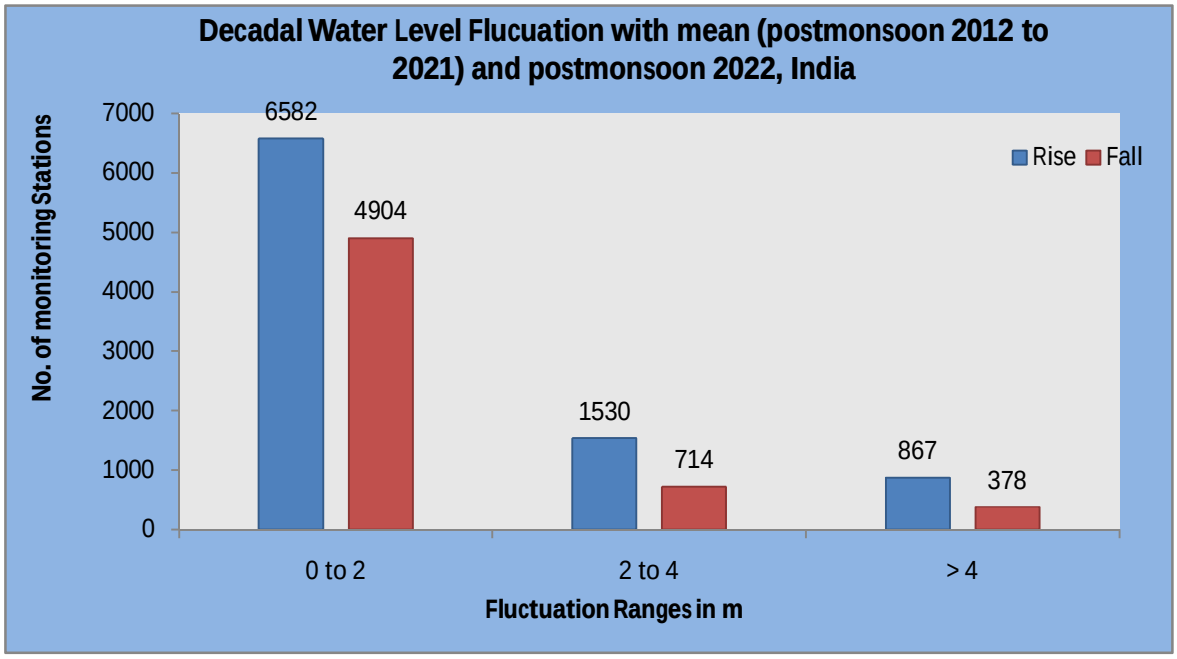
A comparison of depth to water level of August 2022 with decadal mean of August (2012-2021) indicate that about 64.7% (9230) of wells are showing rise in water level, out of which 43.2% (6159) wells are showing rise of less than 2 m. About 12.6% (1798) wells are showing rise in water in the range of 2 to 4 m and about 8.9% (1273) wells are showing rise in water level in the range of more than 4 m. About 35% (5014) wells are showing decline in water level, out of which 28% (3995) wells are showing decline in water in the range of 0 to 2 m. 4.5% (643) wells are showing decline in water level in 2 to 4 m range and remaining 2.6% (376) are in the range of more than 4 m. A comparison of depth to water level of August 2022 with decadal mean of August (2012-2021) indicates that 43.2% of the wells analysed have registered rise in groundwater level, mostly in the range of 0 to 2 m, in the states/UTs namely Arunachal Pradesh, Assam, Bihar, Chandigarh, Dadra Nagar Haveli, Delhi, Haryana, Kerala, Meghalaya, Pondicherry, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh and West Bengal. Decline of more than 4 m has also been observed in pockets in the states/UTs of Chandigarh, Chhattisgarh, Delhi, Gujarat, Haryana, Punjab, Rajasthan, Tamil Nadu, Telangana, Uttar Pradesh, Uttarakhand and West Bengal**(Plate XXI, *Annexure XIII*)**.



Water Level Fluctuation with Decadal Mean (POST- MONSOON-2012 to POST-MONSOON-2021) to post-monsoon-2022

A comparison of depth to water level of post-monsoon 2022 with decadal mean of post-monsoon (2012-2021) indicate that, out of 14986 wells analysed, 8979 (59.9%) of wells are showing rise in water level, out of which 43.9% (6582) wells are showing rise of less than 2 m. About 10.2% (1530) wells are showing rise in water level in the range of 2 to 4 m and about 5.8% (867) wells are showing rise in water level in the range of more than 4 m. About 5996 (40.0%) wells showing decline in water level, 32.2% (4904) wells are showing decline in water in the range of 0 to 2 m. 4.8%(714) wells are showing decline in water level in 2 to 4 m range and remaining 2.5% (378) are in the range of more than 4 m **(Plate XXII, *Annexure-XIV*)**.

A comparison of depth to water level of post-monsoon 2022 with decadal mean of post-monsoon(2012-2021) indicates that 43.9% of the wells have registered rise in groundwaterlevel, mostly in the range of 0 to 2 m. Rise in water level is widespread in southern India comprising of Maharashtra, Telangana, Andhra Pradesh, Gujarat, Bihar, Jharkhand, eastern Uttar Pradesh, Tamil Nadu, Karnataka and Kerala.Fall in water level within 0 to 2 m range is observed in Madhya Pradesh, Maharashtra, Goa, Chhattisgarh, Odisha and parts of NE states in patches. Decline of more than 4 m has also been observed in pockets in the states/UTs of Chandigarh, Delhi, Gujarat, Haryana, Punjab, Rajasthan and western Uttar Pradesh.



Water Level Fluctuation with Decadal Mean (Jan 2013 to Jan 2022) to Jan 2023

A comparison of depth to water level of January 2023 with decadal mean of January (2013-2022) indicates that out of 14283 well analysed, 8073 (about 56.5%) of wells are showing rise in water level, out of which 41.8% (5966) wells are showing rise of less than 2 m. About 9.1% (1305) wells are showing rise in water level in the range of 2 to 4 m and only 5.6% (802) wells are showing rise in the range of more than 4 m. Total6201 (about 43.3%) wells showing decline in water level, 35.1% (5019) wells are showing decline in water in the range of 0 to 2 m. 5.6% (802) wells are showing decline in water level in 2 to 4 m range and remaining 2.9% (410) are in the range of more than 4 m.Rise in water level is observed in major portions of peninsular India in 0 to 4 m range and morethan 4 m is seen in southern states Telangana, Andhra Pradesh, Keral and Tamil Nadu. Decline in water level is observed in north eastern states Arunachal Pradesh, Tripura, Assam. In rest of India fall in water level is recorded in Bihar, Chandigarh, Goa, Punjab, Karnataka, Odisha and Rajasthan **(Plate XXIII, *Annexure-XV*)**.

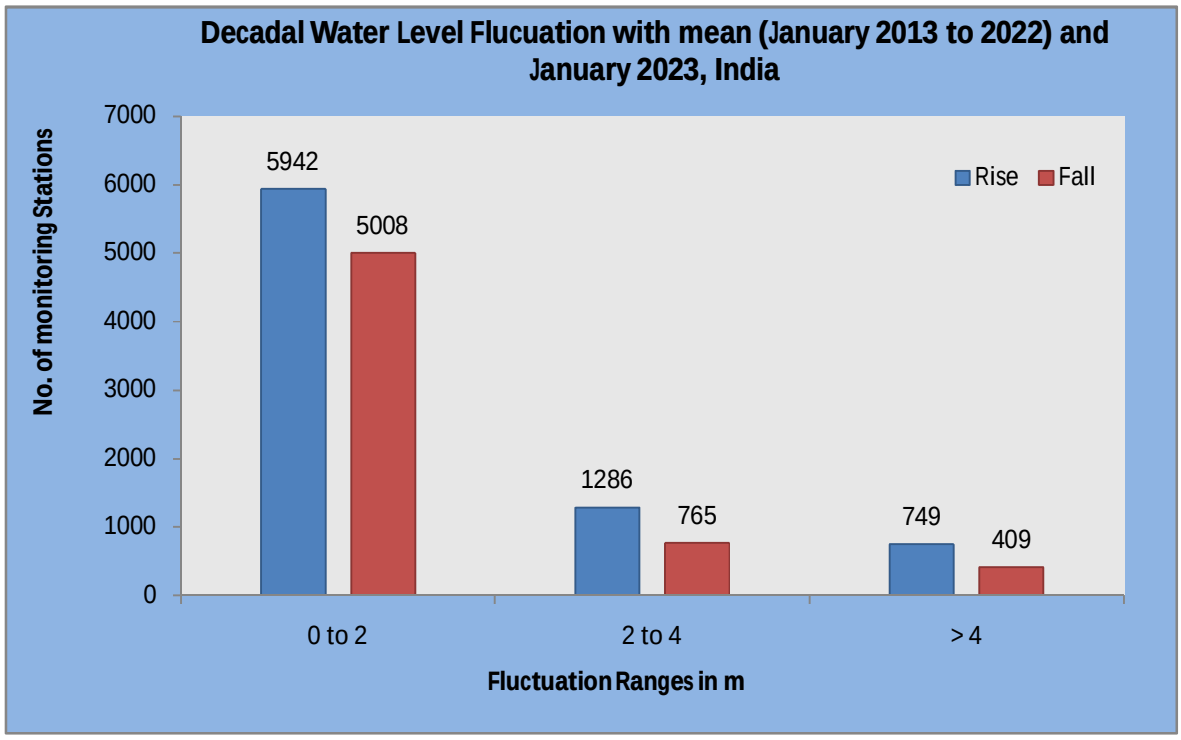
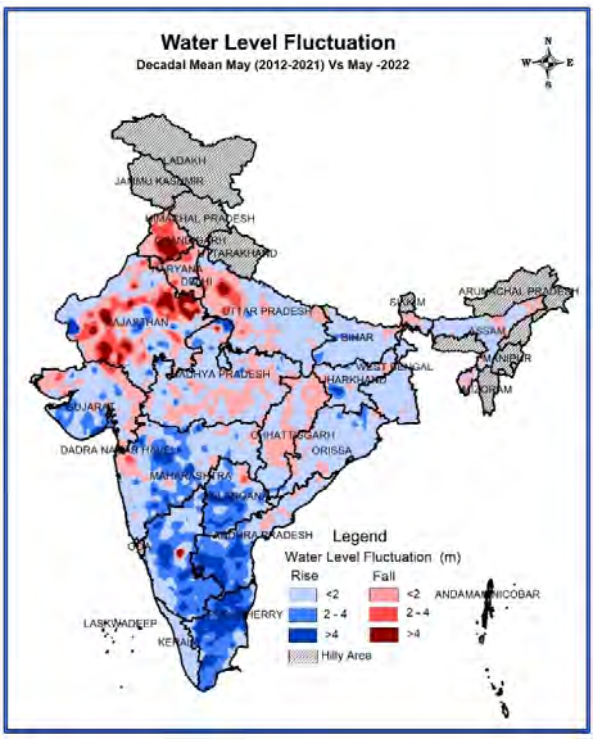
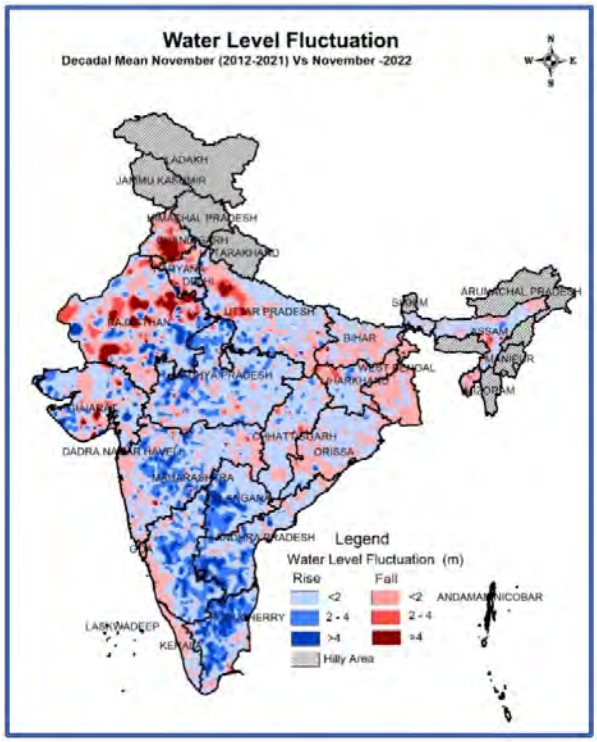
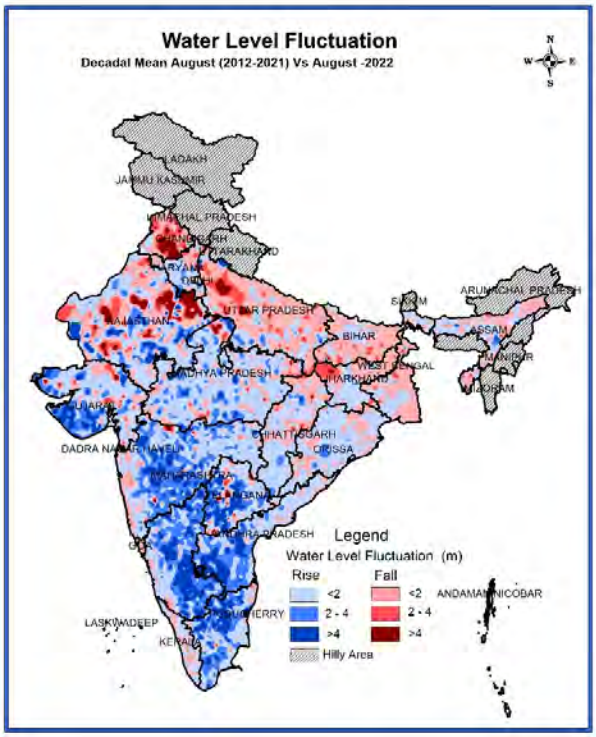
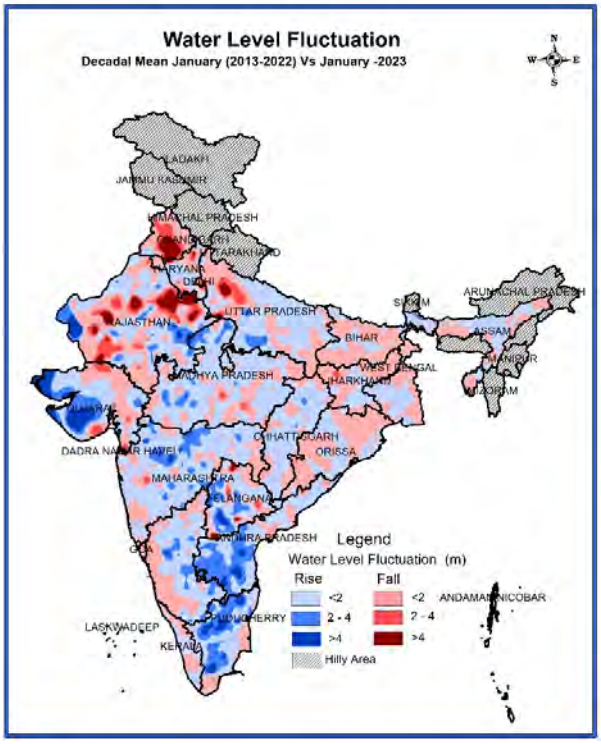


PLATE – XIX









Decadal mean water level flucuation at a glance **PLATE – XX**

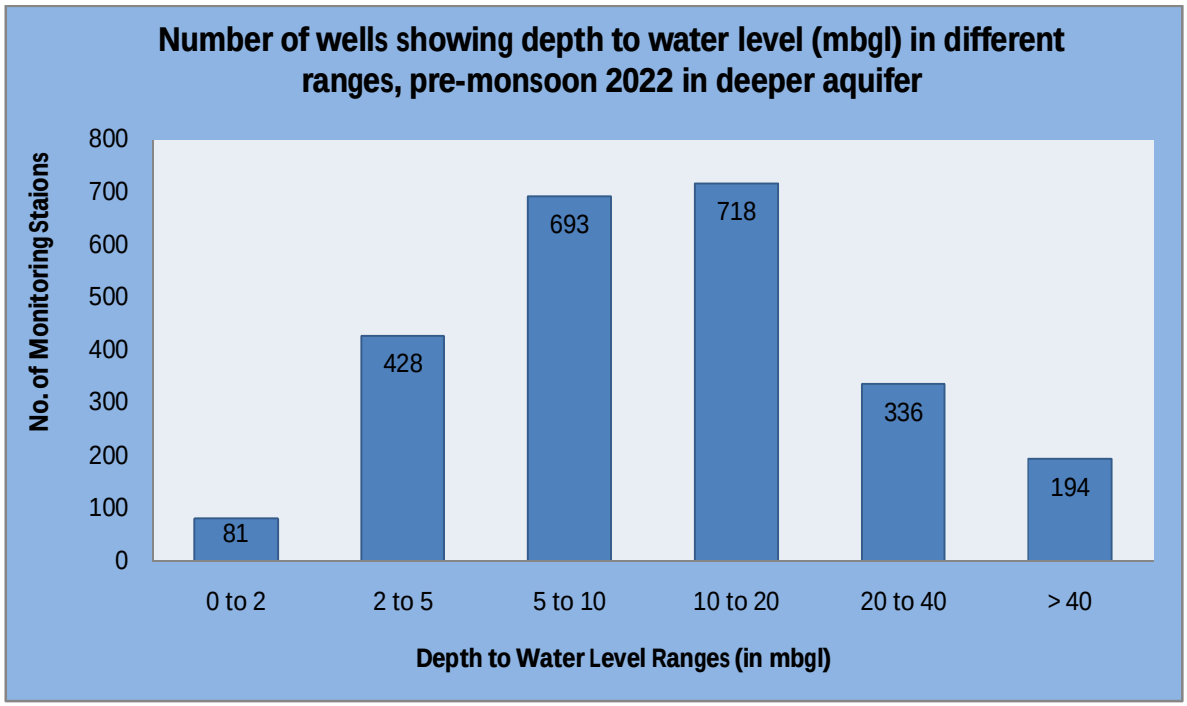
3.3 Groundwater Regime In Deeper Aquifer

3.3.1 Depth To Water Level In Deeper Aquifer

Depth to Water Level – Pre-monsoon 2022 in deeper aquifer

The groundwater level data ***(Annexure-XVI)*** of deeper aquifers for pre-monsoon 2022 indicates that out of the total 2450 wells analysed, 81 (3.3%) wells are showing water level less than 2 m bgl, 428(17.5%) wells are showing water level in the depth range of 2 to 5 m bgl, 693(28.3%) wells are showing water level in the depth range of 5 to 10 m bgl, 718 (29.3%) wells are showing water level in the depth range of 10 to 20 m bgl,336 (13.7%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 194 (7.9%) wells are showing water level more than 40 m bgl**.**

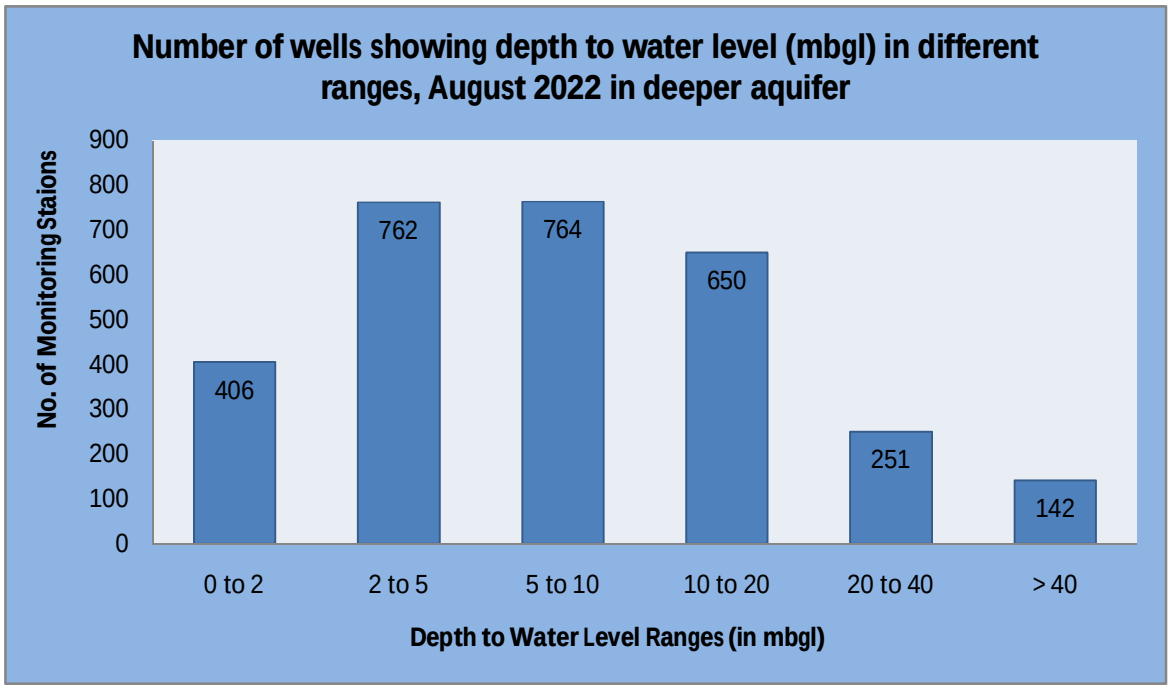
Thus, the general range of water level is between 5 to 20m. From the analysis of the data, it's also revealed that deeper piezometric level of more than 40 m is observed in Andhra Pradesh, Chhattisgarh, Gujarat, Haryana, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Punjab, Tamil Nadu, Telangana, Uttar Pradesh, West Bengal, Delhi and Puducherry.



Depth to Water Level – August 2022 in deeper aquifer

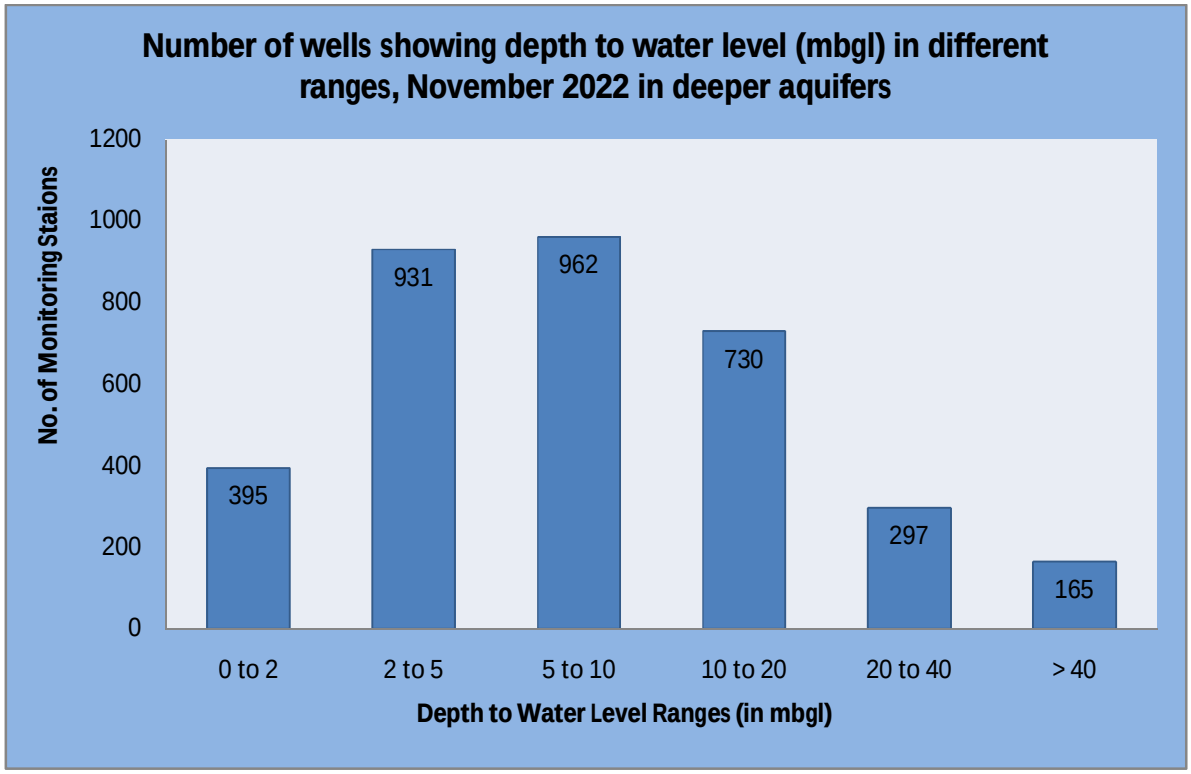
The groundwater level data***(Annexure-XVII)*** of deeper aquifers for August 2022 indicates that out of the total 2975 wells analysed, 406 (13.6%) wells are showing water level less than 2 m bgl, 762(25.6%) wells are showing water level in the depth range of 2 to 5 m bgl, 764 (25.7%) wells are showing water level in the depth range of 5 to 10 m bgl, 650 (21.8%) wells are showing water level in the depth range of 10 to 20 m bgl,251 (8.4%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 142 (4.8%) wells are showing water level more than 40 m bgl**.**

Thus, the general range of water level is between 2 to 10m. From the analysis of the data, it's also revealed that deeper piezometric level of 20 to 40 m is observed in Andhra Pradesh, Gujarat, Haryana, Karnataka, Madhya Pradesh, Punjab, Tamil Nadu, West Bengal, Chandigarh, Delhi and Puducherry. Deeper piezometric level of more than 40 m is observed Andhra Pradesh, Gujarat, Haryana, Punjab, Tamil Nadu, Uttarakhand, Chandigarh and Delhi.



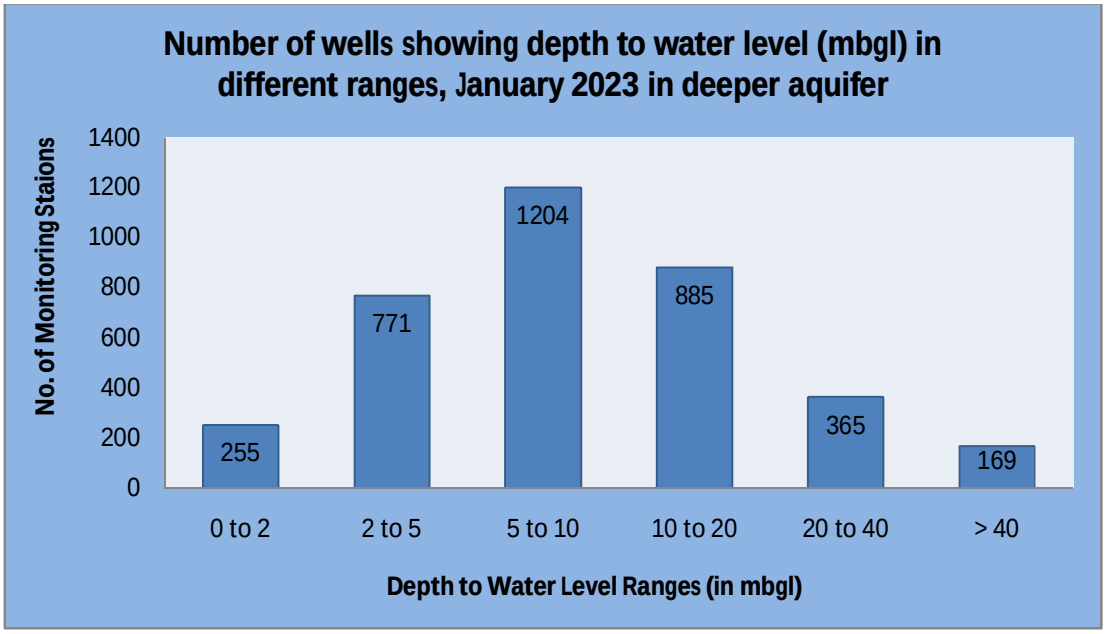
Depth to Water Level – november 2022 in deeper aquifer

The groundwater level data***(Annexure-XVIII)*** of deeper aquifers for November 2022, indicates that out of the total 3480 wells analysed, 395(11.4%) wells are showing water level less than 2 m bgl,931(26.8%) wells are showing water level in the depth range of 2 to 5 m bgl, 962 (27.6%) wells are showing water level in the depth range of 5 to 10 m bgl, 730 (21.0%) wells are showing water level in the depth range of 10 to 20 m bgl,297 (8.5%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 165 (4.7%) wells are showing water level more than 40 m bgl**.** Thus, the general range of water level is between 2 to 10m. From the analysis of the data, it's also revealed that deeper piezometric level of 20 to 40 m is observed inAndhra Pradesh, Chhattisgarh, Gujarat, Haryana, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab, Tamil Nadu, Telangana, Uttar Pradesh, West Bengal, Chandigarh, Delhi and Puducherry;and more than 40m in Andhra Pradesh, Chhattisgarh, Gujarat, Haryana, Karnataka, Maharashtra, Punjab, Tamil Nadu, Telangana, Uttarakhand, Chandigarh and Delhi



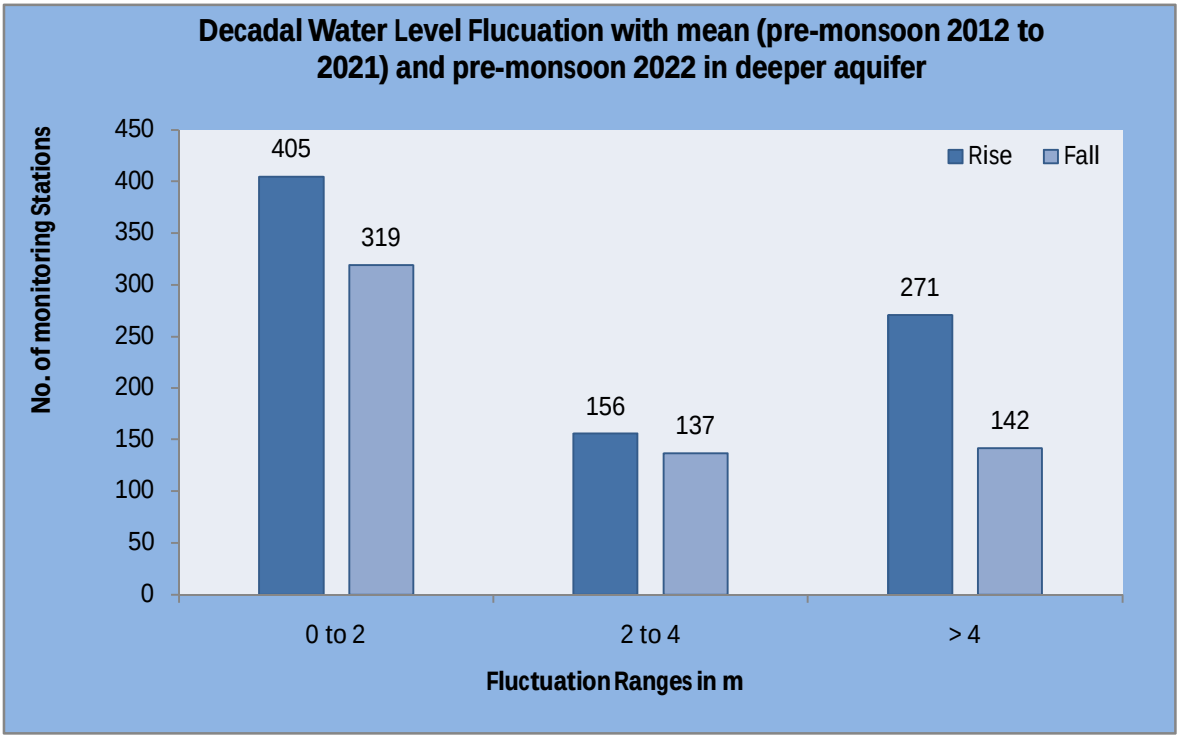
Depth to Water Level – january 2023 in deeper aquifer

The groundwater level data ***(Annexure-XIX)*** of deeper aquifers for January 2023, indicates that out of the total 3649 wells analysed, 255 (7.0%) wells are showing water level less than 2 m bgl,771(21.1%) wells are showing water level in the depth range of 2 to 5 m bgl, 1204 (33.0%) wells are showing water level in the depth range of 5 to 10 m bgl, 885 (24.3%) wells are showing water level in the depth range of 10 to 20 m bgl,365 (10.0%) wells are showing water level in the depth range of 20 to 40 m bgl and the remaining 169 (4.6%) wells are showing water level more than 40 m bgl**.** Thus, the general range of water level is between 2 to 10m. From the analysis of the data, it's also revealed that deeper piezometric level of 20 to 40 m is observed in Andhra Pradesh, Chhattisgarh, Gujarat, Haryana, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Odisha, Punjab, Tamil Nadu, Telangana, Uttar Pradesh, West Bengal, Chandigarh, Delhi and Puducherry and more than 40 m in Andhra Pradesh, Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Tamil Nadu, Telangana, Uttarakhand, Chandigarh and Delhi.



3.3.4 Decadal Mean Water Level Fluctuation In Deeper Aquifer

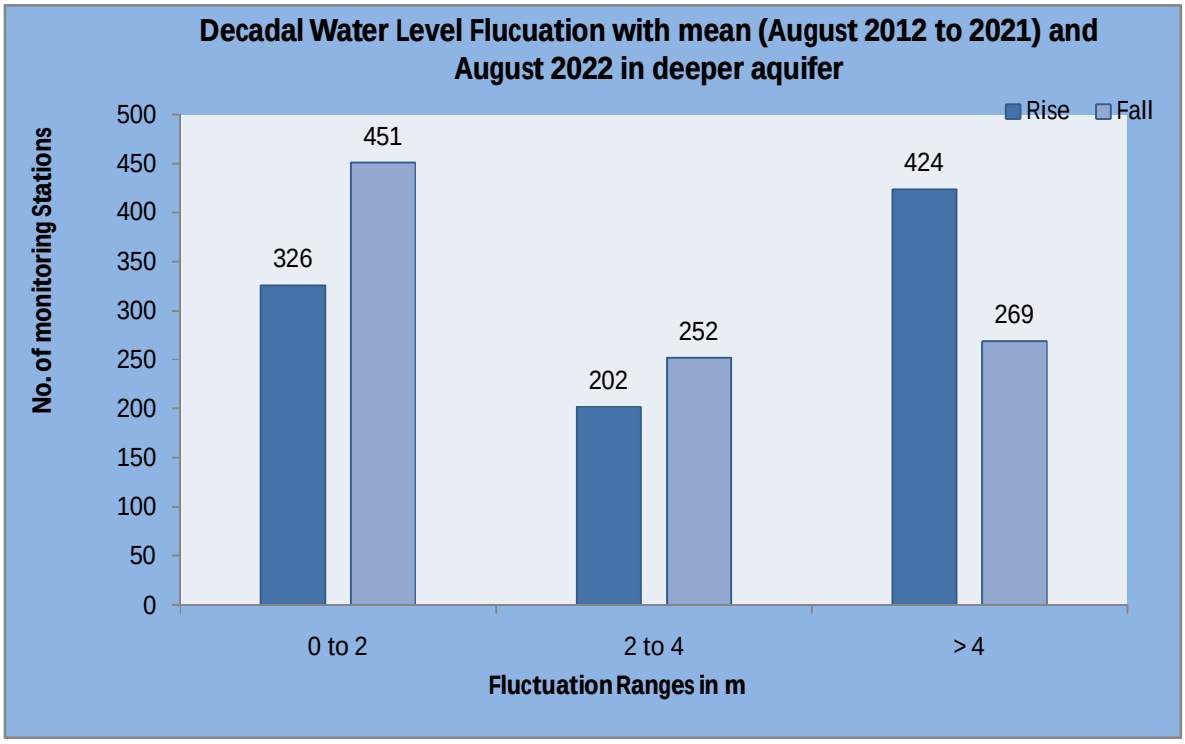
Water Level Fluctuation with Decadal Mean (pre-monsoon-2012 topre-monsoon-2021) to pre-monsoon2022 IN DEEPER AQUIFER



A comparison of depth to water level of pre-monsoon2022 with decadal mean of pre-monsoon (2012-2021) indicates that, out of total 1430wells analysed***(Annexure-XXVII)***,832(58.2%) of wells are showing rise in water level, among which 28.3% (405) wells are showing rise of less than 2 m. About 10.9% (156) wells are showing rise in water level in the range of 2 to 4 m and only 19.0% (271) wells are showing rise in the range of more than 4 m. Among the598(41.8%) wells showing decline in water level, 22.3% (319) wells are showing decline in water in the range of 0 to 2 m. 9.6% (137) wells are showing decline in water level in 2 to 4 m range and remaining 9.9% (142) are in the range of more than 4 m. Rise in water level in majority of wells observed in Andhra Pradesh, Assam, Chhattisgarh, Gujarat, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Tamil Nadu, Telangana, West Bengal, Andaman and Nicobar and Delhi. Similarly fall in significant number of wells is observed in Chhattisgarh, Goa, Gujarat, Haryana, Kerala, Madhya Pradesh, Odisha, Punjab, Tamil Nadu, West Bengal and Puducherry. Rise of more than 4 m is observed in Andhra Pradesh, Chhattisgarh, Gujarat, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Telangana and Delhi.

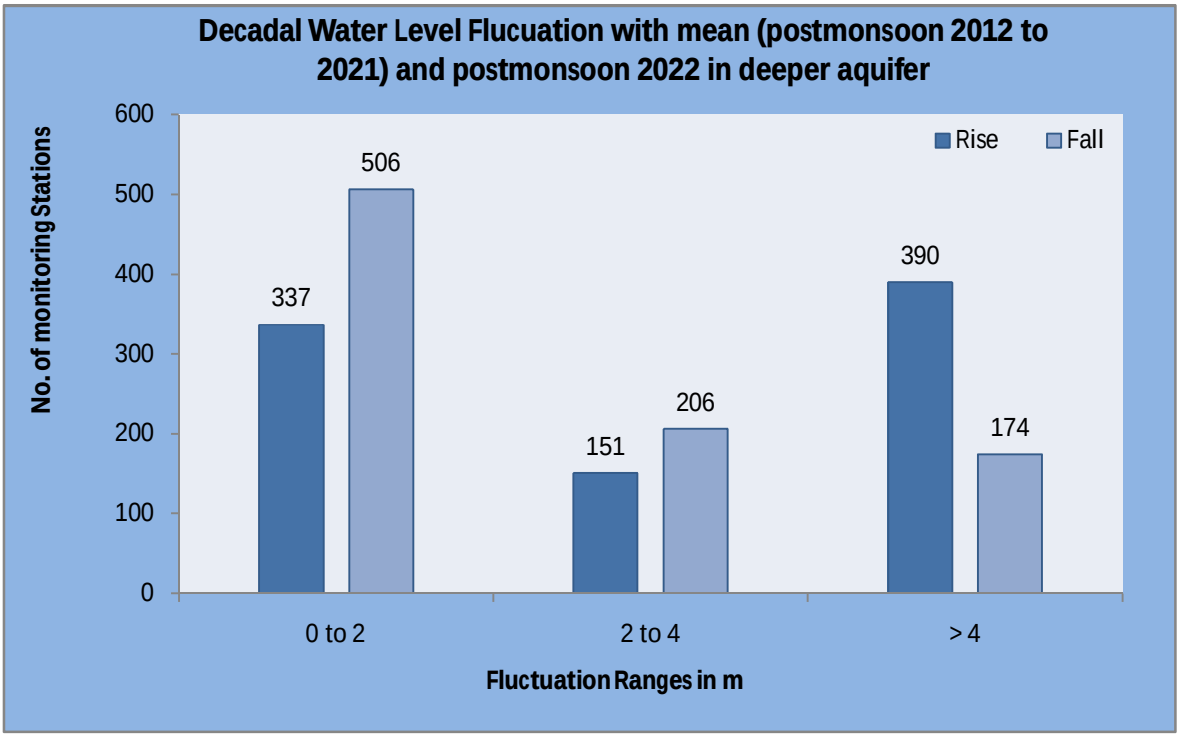
Water Level Fluctuation with Decadal Mean (august-2012 to august-2021) to august 2022 in deeper aquifer

A comparison of depth to water level of August2022 with decadal mean of August (2012-2021) indicates that, out of total 1924 wells analysed***(Annexure-XXVIII)***,952(49.5%) of wells are showing rise in water level, among which 16.9% (326) wells are showing rise of less than 2 m. About 10.5% (202) wells are showing rise in water level in the range of 2 to 4 m and 22.0% (424) wells are showing rise in the range of more than 4 m. Among the972(50.5%) wells showing decline in water level, 23.4% (451) wells are showing decline in water in the range of 0 to 2 m. 13.1% (252) wells are showing decline in water level in 2 to 4 m range and remaining 14.0% (269) are in the range of more than 4 m. Rise in water level in majority of wells observed in Andhra Pradesh, Assam, Gujarat, Karnataka, Kerala, Madhya Pradesh, Odisha, Tamil Nadu, Delhi and Puducherry. Similarly fall in significant number of wells is observed in Chandigarh, Chhattisgarh, Gujarat, Haryana, Kerala, Odisha, Punjab, Telangana and West Bengal.



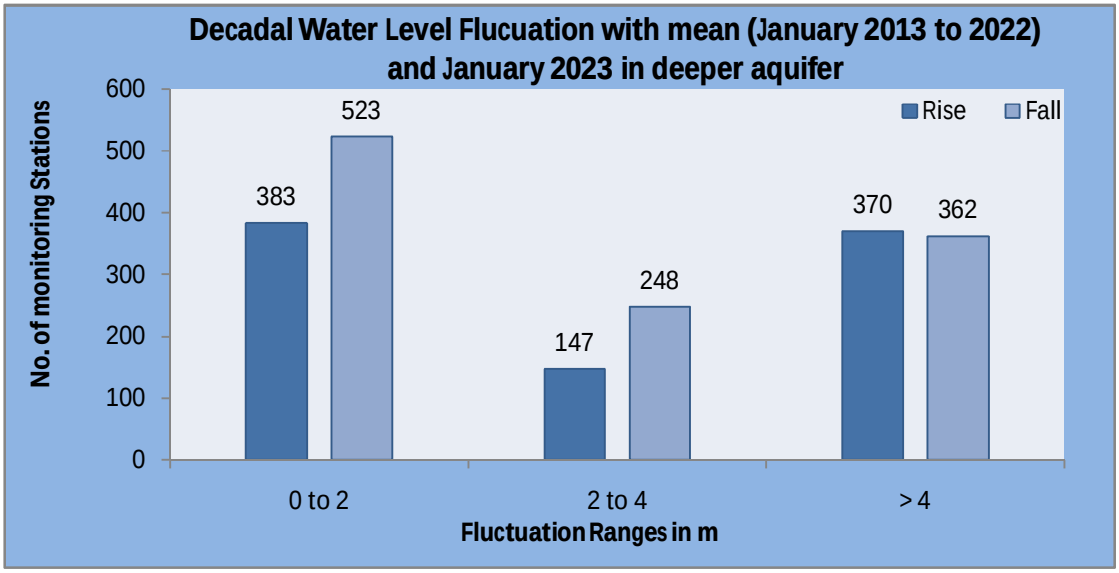
Water Level Fluctuation with Decadal Mean (NOVEMBER2012 to NOVEMBER2021) to NOVEMBER2022 In Deeper Aquifer

A comparison of depth to water level of November2022 with decadal mean of November (2012-2021) indicates that, out of total 1764 wells analysed***(Annexure-XXIX)***, 878 (49.8%) of wells are showing rise in water level, among which 19.1% (337) wells are showing rise of less than 2 m. About 8.6% (151) wells are showing rise in water level in the range of 2 to 4 m and only 22.1% (390) wells are showing rise in the range of more than 4 m. Among the886(52.2%) wells showing decline in water level, 28.7% (506) wells are showing decline in water in the range of 0 to 2 m. 11.7% (206) wells are showing decline in water level in 2 to 4 m range and remaining 9.9% (174) are in the range of more than 4 m. Rise in water level in majority of wells observed in Andhra Pradesh, Assam, Gujarat, Karnataka, Madhya Pradesh, Maharashtra, Odisha, Tamil Nadu, Telangana, Chandigarh, Delhi and Puducherry.Similarly fall in significant number of wells is observed in Chhattisgarh, Goa, Gujarat, Haryana, Kerala, Punjab, Tripura and West Bengal.



Water Level Fluctuation with Decadal Mean (JANUARY-2013 to JANUARY-2022) to JANUARY-2023 In Deeper Aquifer

A comparison of depth to water level of January 2023 with decadal mean of January (2013-2022) indicates that, out of total 2033 wells analysed **(Annexure-XXX)**, 900(44.3%) of wells are showing rise in water level, out of which 18.8% (383) wells are showing rise of less than 2 m. About 7.2% (147) wells are showing rise in water level in the range of 2 to 4 m and 18.2% (370) wells are showing rise in the range of more than 4 m. Among the1133 (55.7%) wells showing decline in water level, 25.7% (532) wells are showing decline in water in the range of 0 to 2 m. 12.2% (248) wells are showing decline in water level in 2 to 4 m range and remaining 17.8% (362) are in the range of more than 4 m.



Rise in water level in majority of wells observed in Andhra Pradesh, Assam, Chhattisgarh, Gujarat, Karnataka, Madhya Pradesh, Maharashtra, Tamil Nadu, Telangana, Delhi and Puducherry. Similarly fall in significant number of wells is observed in Chhattisgarh, Goa, Haryana, Kerala, Meghalaya, Odisha, Punjab, Telangana, West Bengal and Chandigarh.

CHAPTER 6 NATIONAL AQUIFER MAPPING & MANAGEMENT

6.1 Objectives

The National Aquifer Mapping and Management Programme (NAQUIM) was initiated and implemented by Central Ground Water Board under the scheme of Ground Water Management & Regulation during XII plan (2012-17). The Programme was designed as a follow-up of the recommendations of four Working Groups of the erstwhile Planning Commission namely "Sustainable Ground Water Development and Management", "Minor Irrigation and Watershed Development" , "Urban and Industrial Water Supply and Sanitation "and "Data Base Development and Management".

With an aim to support effective management of ground water resources in the country, the NAQUIM programme was taken up with the following objectives

1. Delineation of aquifers in terms of their horizontal and vertical extent.
2. Characterization of aquifers in terms of their hydrogeological roperties, resource availability and quality
3. Preparation of aquifer maps
4. Developing management strategies to ensure sustainability of ground water resources.

   NAQUIM coverage

6.2 Introduction

- NAQUIM envisages mapping of aquifers (water bearing formations), their characterization and development of Aquifer Management Plans to facilitate sustainable management of Ground Water Resources. NAQUIM was initiated in 2012 as a part of the Ground Water Management and Regulation plan scheme. Out of ~33 lakh km^2 of the entire country, a mappable area of ~25 lakh km^2 was identified and has been covered under this programme by March 2023.
- NAQUIM findings are being shared with States / UTs at various levels for implementing groundwater management interventions. The findings of NAQUIM studies are being utilized by many agencies especially the State government agencies involved in ground water management and water supply.
- Public interaction Programmes (PIP) is also being conducted in various parts of the country to promote outreach on ground water management.

- Total Area: **33 lakh sq km**
- Total Mappable Area: **~ 25 lakh sq km**
- Target for Coverage till 31st March 2023: **~ 25 lakh sq km**
- Achievement: :**~ 25 lakh sq km**
- Total States/UTs :**36**
- States/UTs covered: **36**

Table 6.1 : States/UTs covered under NAQUIM

STATES/UTs covered prior to 22-23	STATES/UTs covered in 2022-23
Chandigarh UT	Andaman & Nicobar UT
Dadra & Nagar Haveli and Daman and Diu	Andhra Pradesh
Delhi	Arunachal Pradesh
Goa	Assam
Haryana	Bihar
Himachal Pradesh	Chhattisgarh
Jammu & Kashmir UT	Gujarat
Kerala	Jharkhand
Lakshdweep UT	Karnataka
Ladakh UT	Madhya Pradesh
Manipur	Maharashtra
Meghalaya	Odisha
Mizoram	Rajasthan
Nagaland	Sikkim
Pudducherry UT	Uttar Pradesh

STATES/UTs covered prior to 22-23	STATES/UTs covered in 2022-23
Punjab	West Bengal
Tamil Nadu	--
Telangana	--
Tripura	--
Uttarakhand	--

Table 6.2: NAQUIM coverage till March 2023 (in Sq. Km.)

S.No.	State/UT	Total Area	Area targeted for coverage	Coverage till March 2023
1	Andaman & Nicobar UT	8,249	1,774	1,774
2	Andhra Pradesh	1,63,900	1,41,784	1,41,784
3	Arunachal Pradesh	83,743	4,703	4,703
4	Assam	78,438	61,826	61,826
5	Bihar	94,163	90,567	90,567
6	Chandigarh UT	115	115	115
7	Chhattisgarh	1,36,034	96,000	96,000
8	Dadra & Nagar Haveli,	602	602	602
9	Daman & Diu UT	1,483	1,483	1,483
10	Goa	3,702	3,702	3,702
11	Gujarat	1,96,024	1,60,978	1,60,978
12	Haryana	44,212	44,179	44,179
13	Himachal Pradesh	55,673	8,020	8,020
14	Jammu & Kashmir UT	1,67,396	9,506	9,506
15	Jharkhand	79,714	76,705	76,705
16	Karnataka	1,91,808	1,91,791	1,91,791
17	Kerala	38,863	28,088	28,088
18	Lakshadweep UT	32	32	32
19	Ladakh UT	54,840	963	963
20	Madhya Pradesh	3,08,000	2,69,349	2,69,349
21	Maharashtra	3,07,713	2,59,914	2,59,914
22	Manipur	22,327	2,559	2,559
23	Meghalaya	22,429	10,645	10,645

S.No.	State/UT	Total Area	Area targeted for coverage	Coverage till March 2023
24	Mizoram	21,081	700	700
25	Nagaland	16,579	910	910
26	Odisha	1,55,707	1,19,636	1,19,636
27	Pudducherry UT	479	454	454
28	Punjab	50,368	50,368	50,368
29	Rajasthan	3,42,239	3,34,152	3,34,152
30	Sikkim	7,096	1,496	1,496
31	Tamil Nadu	1,30,058	1,05,829	1,05,829
32	Telangana	1,11,940	1,04,824	1,04,824
33	Tripura	10,492	6,757	6,757
34	Uttar Pradesh	2,46,387	2,40,649	2,40,649
35	Uttarakhand	53,484	11,430	11,430
36	West Bengal	88,752	71,947	71,947
	Total	**3294105**	**2514437**	**2514437**

CHAPTER 7 GROUND WATER QUALITY

7.1 Introduction

Central Ground Water Board has 16 Regional Chemical Laboratories to carry out chemical analysis of ground water samples. The Chemical laboratories are well equipped with sophisticated instruments like Inductive Coupled Plasma Mass-Spectrometer (ICP-MS), Atomic Absorption Spectrophotometer (AAS) etc. Ground water sampling for quality monitoring is done once a year during the pre-monsoon period. In addition to it ground water samples are also collected and analyzed as a part of other scientific studies. CGWB has also taken up sampling and analysis of ground water samples for URANIUM content across the country. So far nearly 55883 groundwater samples have been analyzed by CGWB which included 27896 for Basic parameters, 27619 for Heavy Metals with Uranium and 368 for Total Organic Carbon.The details of water samples analysed by different chemical laboratories during 2022-23 are presented in table (i).

7.2 Ground Water Quality Studies

Central Ground Water Board has 16 Regional Chemical Laboratories to carry out Chemical Analysis of major and minor inorganic constituents in water samples. The Chemical laboratories are well equipped to carryout Basic parameters, Trace Metaland Toxic elements determinations using sophisticated instruments like Atomic Absorption Spectrophotometer (AAS), Ion Chromatograph, Fluorimeter (Uranium Analyser), Digital PC based UV–VIS Spectrophotometer, Ion Meter, Flame Photometer, Conductivity meter, pHmeter, and Nephelometer. The laboratories are also provided with Electronic Mono-pan and Top loading Balances, Ultra -Pure Water System, Deionizer, Double Distillation Plant, Hot AirOven, Water Bath, Magnetic Stirrer and Hot Plates.The Chemical Laboratory at Lucknow, Chandigarh, Hyderabad and Guwahati are additionally equipped with InductiveCoupled Plasma Spectrometer (ICPS) for sequential analysis of multiple toxic elements withhigh accuracy.Total Organic Carbon (TOC) analyser is installed in the Regional ChemicalLaboratory at Chandigarh, Lucknow, Chennai and Kolkata. The Chemical Laboratories atChandigarh, Lucknow, Nagpur and Bengaluru are also equipped with Liquid Scintillation Counter (LSC) to undertake the analysis of radiation.

The chemical analysis data generated by these laboratories is utilized for evaluating the groundwater quality in compliance withNational Standards (BIS 2012), to study the impact of

anthropogenic activities on groundwater quality, demarcate areas of water quality deterioration and assess the point and non-point sources of ground water pollution so as to take necessary action for management ofground waterresources.

7.3 Location of Chemical Labs of CGWB

16 chemical laboratories- Jammu, Chandigarh, Jaipur, Ahmadabad, Bhopal, Raipur, Nagpur, Lucknow, Patna, Kolkata, Guwahati, Bhubaneswar, Hyderabad,Bangalore,Chennai and Trivandrum.

7.4 Number of Labs NABL accredited

10 Labs NABL Accredited- Chandigarh, Jaipur, Bhopal, Raipur, Nagpur, Lucknow, Kolkata, Guwahati, Bhubaneswar and Hyderabad.

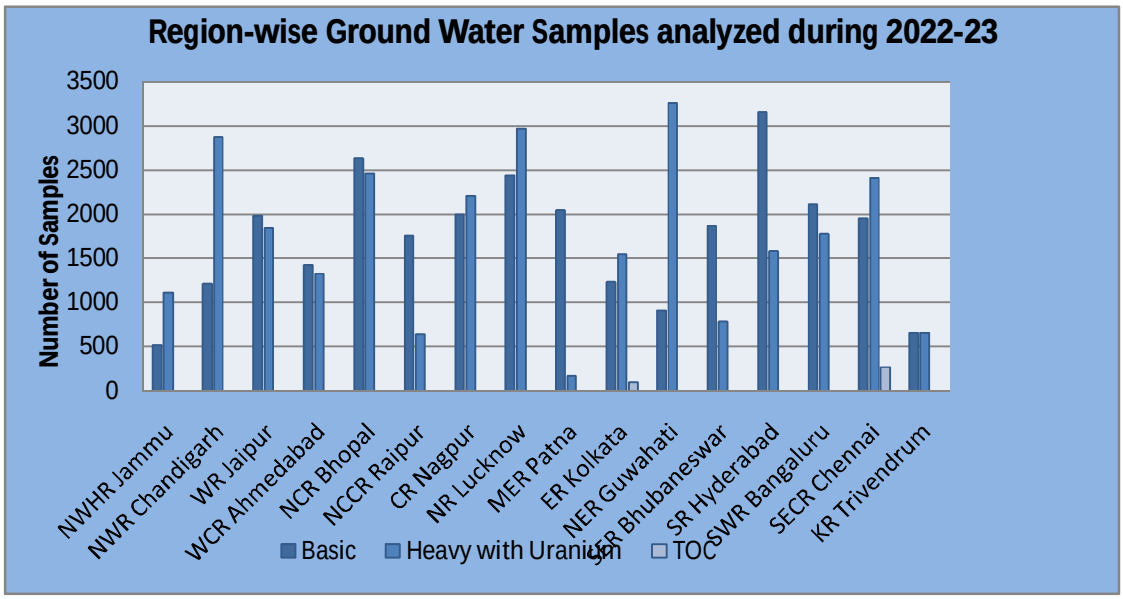
Table 7.1 Status Of Instruments In Regional Laboratory Of Cgwb

S. No.	Instrument/ Equipment	No. of Instruments/Equipment in 16 Regional
1	pH Meter	1-2 in each Laboratory
2	EC Meter	1-2 in each Laboratory
3	Flame- Photometer	1-2 in each Laboratory
4	UV/Visible-Spectro-photometer	1 in each Laboratory
5	Atomic Absorption Spectro-	1 in each Laboratory
6	Ion Chromatograph (IC)	1 in each Laboratory
7	Ion meter	1 in each Laboratory
8	Inductive Coupled Plasma	1 each in Lucknow, Chandigarh, Hyderabad
9	Total Organic Carbon (TOC)	1 each in Lucknow, Chandigarh, Kolkata &
10	Liquid Scintillation Counter	1 each in Lucknow, Chandigarh, Bengaluru,
11	Ultra-Water Purification System	1 in each Laboratory
12	Fluorimeter	1 in each Laboratory

Table7.2 Region-wise Ground Water Samples analyzed during 2022-23

S. No.	Regional Offices	Number of Analyzed Samples		
		Basic	Heavy Metals	TOC
1.	NWHR, Jammu	515	1117	-
2.	NWR, Chandigarh	1212	2875	-
3.	WR, Jaipur	1980	1842	-
4.	WCR, Ahmadabad	1423	1324	-
5.	NCR, Bhopal	2631	2466	-
6.	NCCR, Raipur	1756	644	-
7.	CR, Nagpur	1995	2206	-
8.	NR, Lucknow	2439	2973	-
9.	MER, Patna	2049	166	-
10.	ER, Kolkata	1236	1542	101
11.	NER, Guwahati	913	3257	-
12.	SER, Bhubaneswar	1871	781	-
13.	SR, Hyderabad	3152	1583	-
14.	SWR, Bangalore	2113	1777	-
15.	SECR, Chennai	1953	2410	267
16.	KR, Trivandrum	658	656	-
	Total	27,896	27,619	368
Grand Total(Basic + Heavy + TOC)		**55,883**		

Table7.3



CHAPTER 8 ASSESSMENT OF DYNAMIC GROUND WATER RESOURCES

8.0 Introduction

The Dynamic ground water resources (as in 2022) of the entire country have been assessed jointly by CGWB and State Ground Water Departments under the supervision of the State level Committees. The dynamic ground water resources are also known as Annual Ground Water Recharge, since it gets recharged every year from rainfall and other sources (secondary sources) such as applied irrigation water, surface water bodies, water conservation structures, etc. Methodology adopted for the assessment has been outlined in Chapter-2 of this report. This section provides a summary of the Ground water Resources Assessment 2022 (GWRA-2022) made for the country.

As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction are assessed for each assessment unit (block/taluka/mandal/tehsils/firka etc.). The 'Stage of Ground Water Extraction' is then computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe (<= 70 %), Semi-Critical (>70 % and <=90 %), Critical (>90 % and <=100%) and Over-Exploited (>100 %).

A Central Level Expert Group (CLEG) was constituted by the Department of Water Resources, RD & GR, Ministry of Jal Shakti, Government of India for over-all supervision of the re-assessment of ground water resources in the entire country with reference year 2022. The groundwater resources assessment at the State/UT Levels was carried out jointly by Central Ground Water Board and State Nodal/Ground Water Departments under the guidance of respective State Level Committees (SLC) with over-all supervision of CLEG.The National Compilation on Dynamic Ground Water Resources of India, 2022 has been published in November, 2022(http://cgwb.gov.in/documents/2022-11-11-GWRA%202022.pdf).Ground Water Resource Assessment -2022 (GWRA-2022) has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad ((http://ingres.iith.ac.in).

8.1. Dynamic Fresh Ground Water Resources

As per the 2022 assessment of Dynamic Ground Water Resources, the Total Annual Ground Water Recharge for the entire country has been assessed as 437.60 billion cubic meter (bcm) and Total natural discharges works out to be 36.85 bcm. Hence, Annual Extractable Ground Water Resources for the entire country is 398.08 bcm.

Major source of ground water recharge is the monsoon rainfall, which is 241.35 bcm and about 55 % of the total annual ground water recharge. The contribution in Annual Ground Water Recharge from rainfall during monsoon season is more than 70% in the states/UT of Goa, Gujarat, Jharkhand, Kerala, Madhya Pradesh, Manipur, Meghalaya, Mizoram, Rajasthan and Daman & Diu. The overall contribution of rainfall (both monsoon & non-monsoon) recharge to country's total annual ground water recharge is 61 % and the share of recharge from 'Other sources' viz. canal seepage, return flow from irrigation, recharge from tanks, ponds and water conservation structures taken together is 39 %.

Annual Ground Water Recharge is significantly high in the Indus-Ganga-Brahmaputra alluvial belt in the North, East and North East India covering the states of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal and valley areas of North Eastern States, where rainfall is plenty and thick piles of unconsolidated alluvial formations are conducive for recharge. Annual Ground Water Recharge in these regions varies from 0.25 to more than 0.5 m. The coastal alluvial belt particularly Eastern Coast also has relatively high annual ground water recharge, in the range 0.25 to more than 0.5 m. In western India, particularly Rajasthan and parts of northern Gujarat that have arid climate, the annual ground water recharge is scanty, mostly up to 0.025 m. Similarly, in major parts of the southern peninsula covered with hard rock terrains, annual ground water recharge mostly ranges from 0.10 to 0.15 m. This is primarily because of comparatively low infiltration and storage capacity of the rock formations prevailing in the region. The remaining part of Central India is mostly characterized by moderate recharge in the range of 0.10 to 0.25 m.

The overall estimate of Annual Ground Water Recharge for the entire country shows an increase of 1.45 bcm in the present assessment as compared to the last assessment i.e. 2020. The Annual Extractable Ground Water Resources shows an increase of 0.46 bcm.

8.2. Groundwater Extraction

The assessment of ground water extraction is carried out considering the Minor Irrigation Census data and sample surveys carried out by the State Ground Water Departments. The Total Annual Ground Water Extraction of the entire country for the year 2022 has been estimated as 239.16 bcm. Agriculture sector is the predominant consumer of ground water resources. About

87 % of total annual ground water extraction i.e. 208.49 bcm is for irrigation use. Only 30.69 bcm is for Domestic & Industrial use, which is about 13 % of the total extraction. In the states of, Delhi, Goa, Kerala, Jammu & Kashmir, Ladakh, Manipur, Meghalaya, Mizoram, Nagaland, Tripura, Andaman & Nicobar Island and Chandigarh the ground water extraction for domestic uses is more than 40 %.

8.3. Stage of Groundwater Extraction

The overall stage of groundwater extraction in the country is 60.08 %. The stage of ground water extraction is very high in the states of Haryana, Punjab, Rajasthan, Dadra & Nagar Haveli and Daman & Diu where it is more than 100%, which implies that in these states the annual ground water consumption is more than annual extractable ground water resources. In the states of Delhi, Tamil Nadu, Uttar Pradesh, Karnataka and UTs of Chandigarh, Lakshadweep and Puducherry, the stage of ground water extraction is between 60-100%. In rest of the states, the stage of ground water extraction is below 60 %.

8.4. Categorization of Assessment Units

Out of the total 7089 assessment units (Blocks/ Taluks/ Mandals/ Districts/Firkas/Valleys), 1006 has been categorized as 'Over-exploited', 260 as 'Critical', 885 as 'Semi-critical', and 4780 units as 'Safe'. There are 158 assessment units, which are completely saline. 124 new assessments are included in the current assessment of 2022. The percentage of Over-exploited and Critical administrative units more than 25% of the total units are in Delhi, Haryana, Karnataka, Punjab, Rajasthan, Tamil Nadu, Dadra& Nagar Haveli and Daman & Diu.

Similarly out of 24.69 lakh sq km recharge worthy area of the country, 4.30 lakh sq km (17 %) are under 'Over-Exploited', 0.77 lakh sq km (3 %) are under 'Critical', 3.03 lakh sq km (12 %) are under 'Semi-Critical', 16.18 lakh sq km (66 %) are under 'Safe' and 0.4 lakh sq km (2 %) are under 'Saline' category assessment units. Out of 398.08 bcm of Total Annual Extractable Resources of the country, 46.05 bcm (12%) are under 'Over-Exploited', 13.02 bcm (3%) are under 'Critical', 47 bcm (12%) are under 'Semi-Critical', 291.88 bcm (73 %) are under 'Safe' category assessment units.

8.5. Comparison with previous Assessment

In comparison to 2020 assessment, the total numbers of assessment units in the country have increased from 6965 to 7089 with major contribution (in increase) from the State of West Bengal, Karnataka,Rajasthan, Uttar Pradesh, Telangana, Gujarat and Jharkand. The total annual ground water recharge has increased from 436 to 437.6 bcm, where major increase is noticed in the States of Bihar, Telangana, Andhra Pradesh, Tamilnadu, Arunachal Pradesh, Odisha and Gujarat. The changes are attributed mainly to changes in recharge from 'Other Sources'.

Accordingly, the annual extractable resource of GW Resource Assessment, 2022 on comparison GW Resource Assessment, 2022 also shows a increase from 397.6 to 398.08 bcm. The ground water extraction has marginally decreased from 244.92 to 239.16 bcm. The overall stage of groundwater extraction has marginally decreased from 61.6 % to 60.08 %.

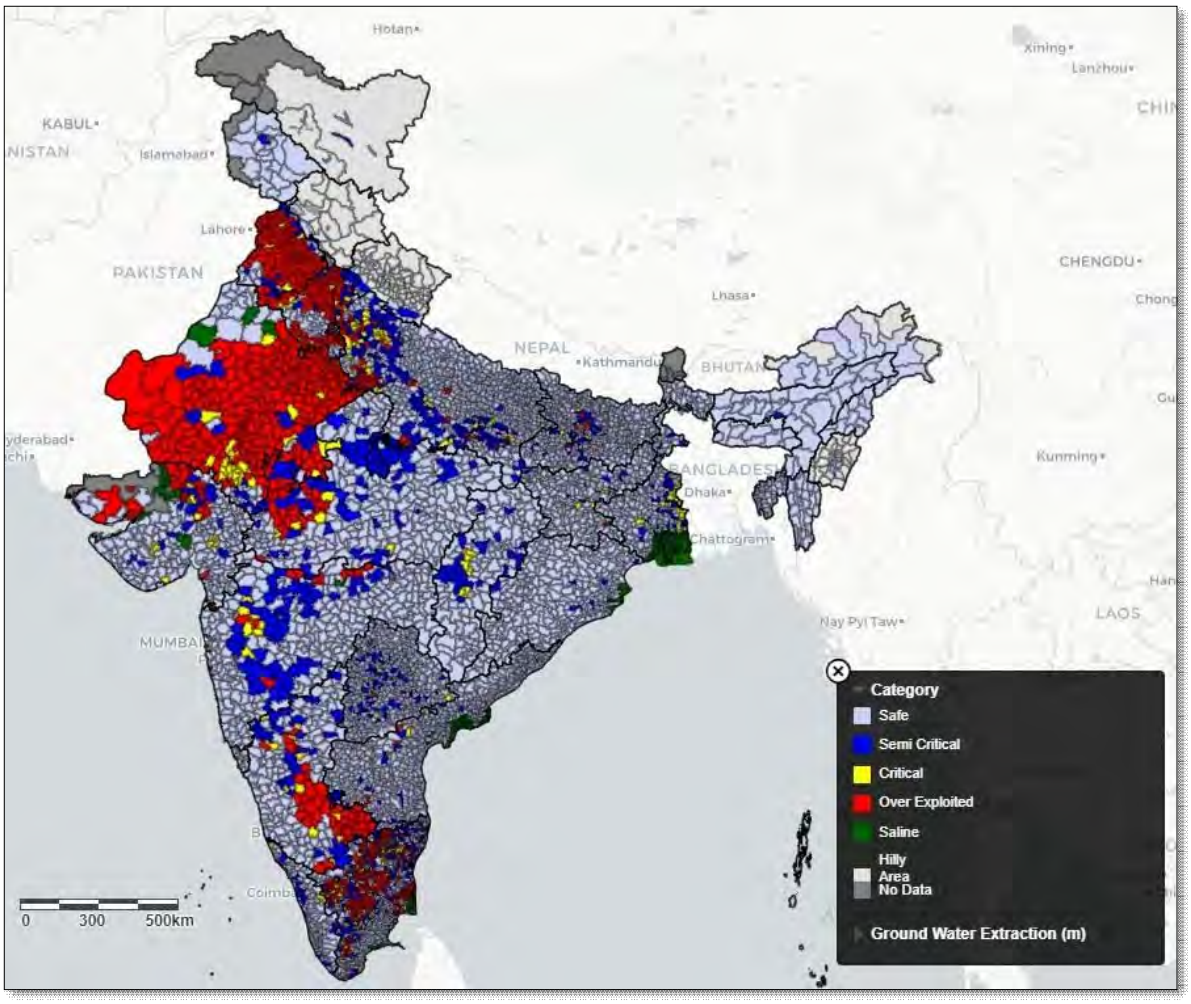


Figure 8.1: Categorization of Assessment Units

Table 8.1: State-wise Ground Water Resources of India, 2022

STATE-WISE GROUND WATER RESOURCES OF INDIA, 2022

(inbcm)

S.No.	States / Union Territories	Ground Water Recharge					Total Natural Discharges	Annual Extractable Ground Water Resource	Current Annual Ground Water Extraction				Annual GW Allocation for Domestic Use as on 2025	Net Ground Water Availability for future use	Stage of Ground Water Extraction (%)
		Monsoon Season		Non-monsoon Season		Total Annual Ground Water Recharge			Irrigation	Industrial	Domestic	Total			
		Recharge from rainfall	Recharge from other sources	Rechargef rom rainfall	Recharge from other sources										
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Andhra Pradesh	9.14	9.41	0.91	7.77	27.23	1.36	25.86	6.46	0.16	0.83	7.45	**1.09**	18.54	**28.81**
2	Arunachal Pradesh	1.96	0.94	1.06	0.56	4.52	0.41	4.07	0.02	0.01	0.01	0.03	0.01	4.03	**0.79**
3	Assam	17.92	1.15	6.52	0.94	26.53	2.56	21.4	2.06	0.01	0.58	2.65	0.62	18.71	**12.38**
4	Bihar	19.94	7.07	1.14	5	33.15	3.1	30.04	10.01	0.35	3.14	13.5	3.41	16.76	**44.94**
5	Chhattisgarh	8.08	1.8	0.15	2.01	12.04	1.04	11.01	4.62	0.11	0.73	5.46	0.83	5.56	**49.58**
6	Delhi	0.1388	0.0895	0.0094	0.1728	0.4105	0.0411	0.3695	0.0904	0.0007	0.2716	0.3627	0.2878	0.0288	**98.1612**
7	Goa	0.35	0.02	0	0.04	0.41	0.08	0.33	0.026	0.004	0.048	0.078	0.05	0.25	**23.63**
8	Gujarat	19	2.63	0	4.83	26.46	1.88	24.58	12.1	0.16	0.82	13.09	1.04	12.18	**53.23**
9	Haryana	3.15	2.79	0.70	2.83	9.48	0.87	8.61	10.30	0.60	0.65	11.54	0.66	1.04	**134.14**
10	HimachalPradesh	0.6	0.14	0.14	0.15	1.03	0.09	0.94	0.18	0.05	0.12	0.35	0.12	0.59	**37.56**
11	Jharkhand	4.92	0.45	0.48	0.36	6.21	0.51	5.69	0.93	0.21	0.65	1.78	0.65	3.92	**31.35**
12	Karnataka	8.83	4.29	1.19	3.43	17.74	1.70	16.04	10.01	0.13	1.09	11.22	1.17	6.34	**69.93**
13	Kerala	4.25	0.15	0.47	0.87	5.74	0.54	5.19	1.17	0.01	1.55	2.73	2.2	2.18	**52.56**
14	MadhyaPradesh	26.87	1.56	0.11	6.69	35.23	2.66	32.58	17.39	0.17	1.69	19.25	1.88	14.21	**59.1**
15	Maharashtra	20.72	2.43	0.54	8.6	32.29	1.84	30.45	15.29	0.003	1.35	16.65	1.35	14.38	**54.68**
16	Manipur	0.4	0	0.11	0.01	0.52	0.05	0.47	0.02	0.0002	0.02	0.04	0.02	0.43	**7.95**
17	Meghalaya	1.29	0.01	0.42	0	1.72	0.17	1.51	0.003	0.0007	0.05	0.05	0.06	1.45	**3.55**
18	Mizoram	0.19	0	0.03	0	0.22	0.02	0.2	0.000	0.00	0.01	0.01	0.01	0.19	**3.96**
19	Nagaland	0.36	0.33	0.08	0.02	0.79	0.08	0.71	0.002	0.00002	0.02	0.02	0.02	0.69	**2.89**
20	Odisha	10.44	2.82	1.81	2.72	17.79	1.44	16.34	5.83	0.16	1.24	7.23	1.37	9.03	**44.25**
21	Punjab	4.67	9.09	0.72	4.46	18.94	1.87	17.07	26.69	0.16	1.17	28.02	1.19	1.57	**165.99**
22	Rajasthan	8.71	0.62	0.20	2.61	12.13	1.17	10.96	14.18	0.14	2.23	16.56	2.28	0.87	**151.07**

23	**Sikkim**	0.1712	0.0039	0.0956	0.0005	0.2712	0.0271	0.2441	0.0089	0.0022	0.0036	0.0147	0.0038	0.2291	**6.04**
24	**TamilNadu**	7.42	9.76	1.33	2.59	21.11	2.04	19.09	13.68	0.18	0.57	14.43	1.36	6.42	**75.59**
25	**Telangana**	7.19	6.66	0.98	6.44	21.27	2.02	19.25	7.257	0.154	0.596	8	3.82	11.23	**41.6**
26	**Tripura**	0.81	0.06	0.22	0.22	1.31	0.25	1.06	0.02	0.0007	0.08	0.10	0.09	0.96	**9.70**
27	**UttarPradesh**	35.44	13.96	0.82	21.23	71.45	6.13	65.3	40.72	0.41	5.01	46.14	5.48	19.99	**70.66**
28	**Uttarakhand**	1.28	0.31	0.1	0.32	2.01	0.16	1.86	0.63	0.12	0.15	0.89	0.15	0.96	**48.04**
29	**WestBengal**	15.46	1.65	3.04	3.46	23.61	2.19	21.42	8.38	0.14	1.54	10.07	1.76	11.29	**47.01**
30	**Andamanand Nicobar**	0.2979	0.0002	0.3203	0.0001	0.6185	0.0618	0.5566	0.0001	0.001	0.0065	0.0075	0.0069	0.5486	**1.35**
31	**Chandigarh**	0.01	0.01	0.00	0.03	0.05	0.01	0.05	0.01	0.002	0.03	0.04	0.03	0.01	**80.99**
32	**Dadra&NagarHaveli**	0.06	0.01	0.003	0.02	0.09	0.01	0.08	0.01	0.09	0.01	0.11	0.02	0.01	**133.2**
	Daman& Diu	0.037	0.001	0.000	0.001	0.038	0.002	0.036	0.003	0.055	0.000	0.057	0.016	0.000	**157.927**
33	**JammuandKashmir**	1.16	1.94	1.15	0.64	4.90	0.46	4.44	0.31	0.05	0.71	1.07	0.73	3.35	**24.18**
34	**Ladakh**	0.01	0.05	0.02	0	0.08	0.01	0.07	0.00037	0.000200	0.03	0.03	0.03	0.04	**41.36**
35	**Lakshadweep**	0.01	0	0	0	0.01	0.01	0.01	0	0.00	0	0	0	0	**61.6**
36	**Puducherry**	0.06	0.09	0.01	0.04	0.21	0.02	0.19	0.08	0.01	0.05	0.13	0.05	0.05	**69.17**
	Grand Total	**241.35**	**82.30**	**24.88**	**89.07**	**437.60**	**36.85**	**398.08**	**208.49**	**3.64**	**27.05**	**239.16**	**33.86**	**188.03**	**60.08**

Table 8.2: State/UT-Wise Categorization of Assessment Units, as per GWRA-2022

CATEGORIZATION OF BLOCKS/MANDALS/TALUKAS IN INDIA(2022)												
S. No.	State/Union Territories	Total No.of Assessed Units	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
	States		Nos.	%	Nos.	%	Nos.	%	Nos.	%	Nos.	%
1	AndhraPradesh	667	598	89.7	19	2.8	5	0.7	6	0.9	39	**5.85**
2	ArunachalPradesh	11	11	100.00								
3	Assam	28	27	96.43	1	3.57						
4	Bihar	535	469	87.66	46	8.60	12	2.24	8	1.50		
5	Chhattisgarh	146	116	79.45	24	16.44	6	4.11				
6	Delhi	34	4	11.76	8	23.53	7	20.59	15	44.12		
7	Goa	12	12	100.00								
8	Gujarat	252	189	75.00	20	7.94	7	2.78	23	9.13	13	**5.16**
9	Haryana	143	36	25.17	9	6.29	10	6.99	88	61.54		
10	HimachalPradesh	10	10	100.00								
11	Jharkhand	263	241	91.63	11	4.18	6	2.28	5	1.90		
12	Karnataka	234	139	59.40	35	14.96	11	4.70	49	20.94		
13	Kerala	152	122	80.26	27	17.76	3	1.97				
14	MadhyaPradesh	317	226	71.29	60	18.93	5	1.58	26	8.20		
15	Maharashtra	353	272	77.05	62	17.56	7	1.98	11	3.12	1	**0.28**
16	Manipur	9	9	100.00								
17	Meghalaya	12	12	100.00								
18	Mizoram	26	26	100.00								
19	Nagaland	11	11	100.00								
20	Odisha	314	300	95.54	8	2.55					6	**1.91**
21	Punjab	153	17	11.11	15	9.80	4	2.61	117	76.47		
22	Rajasthan	302	38	12.58	20	6.62	22	7.28	219	72.52	3	**0.99**
23	Sikkim	6	6	100.00								
24	TamilNadu	1166	463	39.71	231	19.81	78	6.69	360	30.87	34	**2.92**
25	Telangana	594	494	83.00	80	13.60	7	1.20	13	2.20		
26	Tripura	59	59	100.00								
27	UttarPradesh	836	557	66.63	169	20.22	47	5.62	63	7.54		
28	Uttarakhand	18	14	77.78	4	22.22						
29	WestBengal	345	232	67.25	31	8.99	22	6.38			60	**17.39**

30	**Andamanand Nicobar**	36	35	97.22							1	**2.78**
31	**Chandigarh**	1			1	100.00						
32	**Dadra &NagarHaveli**	1							1	100.00		
	Daman &Diu	2							2	100.00		
33	**Jammu andKashmir**	20	19	95.00	1	5.00						
34	**Ladakh**	8	7	87.50	1	12.50						
35	**Lakshadweep**	9	7	77.78	2	22.22						
36	**Puducherry**	4	2	50.00			1	25.00			1	**25.00**
	Grand Total	**7089**	**4780**	**67.43**	**885**	**12.48**	**260**	**3.67**	**1006**	**14.19**	**158**	2.23

Note-

Blocks- Bihar, Chhatisgarh, Haryana,Jharkhand, Kerala,MadhyaPradesh,Manipur, Mizoram, Odisha,Punjab, Rajasthan, Tripura,UttarPradesh,Uttarakhand,West Bengal

Taluks-Goa,Gujarat,Karnataka, Maharashtra

Mandals- AndhraPradesh,Telangana

District-Arunachal Pradesh,Assam, Meghalaya,Nagaland,Sikkim, Dadra& Nagar Haveli,Daman& Diu,Jammu&Kashmir

Valley- HimachalPradesh,Ladakh

Islands- Andaman& Nicobar,Lakshadweep

Firka- TamilNadu

Region-Puducherry

UT-Chandigarh

Tehsil- Delhi

9.1 Convergence with MGNERGS in Water Stressed areas

CGWB is providing technical guidance in site selection & design of artificial recharge structures and capacity building of MGNREGS officials in selected 09 Blocks of 08 States as a pilot study (Table1). The work has been taken up under the MGNREGS fund, so far the work is completed in Andhra Pradesh, Telangana & Tamil Nadu (one block each) and is under progress in the rest of the blocks of the states.

The latest progress is annexed at Annexure-1.

Table 9.1: The details of water stressed areas taken up under MGNREGS

S.No.	State	District	Block
1.	Andhra Pradesh	Prakasam	Peddaravedu
2.	Telangana	Nallagonda	Munugode
3.	Karnataka	Chickballalpur	Gowribidanur
4.	Tamil Nadu	Krishnagiri	Mathur
5.	Rajasthan	Bhilwara	Madalgarh
		Sirohi	Reodhar
6.	Haryana	Kurukshetra	Babain
7.	Madhya Pradesh	Ujjain	Badnagar
8.	Maharashtra	Amravati	Warud

The glimpse of photographs of artificial recharge structures constructed in convergence with MGNERGS is shown in Fig.1and Fig.2:

Fig.9.1: Multi Arch Check Dam (MACD) at Gowribidanur Block, Chickballalpur District, Karnataka

Dug well recharge structure *Countour Bunding*

Fig.9.2: Artificial Recharge Structures at Warud Block, Amravati District, Maharashtra

9.2 Implementation of Master Plan for Artificial Recharge 2020:

CGWB has prepared a conceptual document on "Master Plan for Artificial Recharge to Ground Water in India" in 2020 to work out the feasibility of various structures for the different terrain conditions of the country and estimated costs, providing a broad outline of the project and expected investments which include artificial recharge in both rural and urban areas. A total 14175144 number of structures at an estimated cost of Rs 1.33 lakh Cr have been envisaged for rainwater harvesting and artificial recharge in the country.

One district in each state has been identified for the implementation of Master Plan for Artificial Recharge-2020 on pilot basis. The list of identified districts in each States/UTs is shown in Table 2:

Table 9.2: The list of districts identified for implementation of Master Plan for Artificial recharge 2020

S. No.	State/UTs	District
1	Andhra Pradesh	Prakasam
2	Arunachal Pradesh	Papum Pare
3	Assam	Kamrup (Metro) &Kamrup (Rural)
4	Bihar	Gaya
5	Chhattisgarh	Bemetara
6	Chandigarh	Chandigarh
7	Delhi	North Delhi
8	Goa	South Goa
9	Gujarat	Gandhinagar
10	Haryana	Yamuna Nagar
11	Himachal Pradesh	Solan
12	Jharkhand	Ranchi
13	Karnataka	Chitradurga
14	Kerala	Kasaragod
15	Madhya Pradesh	Chhatarpur
16	Maharashtra	Nagpur (Katol &Narkhed blocks)
17	Manipur	Imphal East
18	Meghalaya	West Garo Hills
19	Mizoram	Aizwal
20	Nagaland	Dimapur
21	Odisha	Nuapada
22	Punjab	Sangrur
23	Puducherry	Puducherry
24	Rajasthan	Ajmer
25	Sikkim	South District(Renamed as Namchi District)
26	Tamil Nadu	Virudhunagar
27	Telangana	Vikarabad
28	Tripura	Khowai
29	Uttarakhand	Haridwar
30	Uttar Pradesh	Jhansi
31	West Bengal	Purulia
32	U.T of Andaman & Nicobar	South Andaman
33	UT of D & N and D & D	Dadra & Nagar Haveli
34	UT of Jammu & Kashmir	Jammu
35	U.T. of Lakshadweep	Lakshadweep
36	UT of Ladakh	Leh

The implementation of Master Plan for Artificial recharge has been initiated in 20 States and 03 UT which are Arunachal Pradesh, Bihar, Chhattisgarh, Delhi, Gujarat, Haryana, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab, Rajasthan, Sikkim, Tamil Nadu, Tripura, Uttar Pradesh, Uttarakhand, West Bengal, UT of Andaman and Nicobar, Chandigarh, Ladakh. The progress is annexed at annexure-II. The glimpse of some photographs showing implementation of Artificial Structures of Master Plan 2020 is shown in Fig.3 to Fig.8:

Fig.9.3: Artificial Recharge Structures at Nuapada block of Nuapada district, Odisha

Fig.9.4: Artificial Recharge Structures at Yamunanagar district, Haryana

Fig.9.5: Artificial Recharge Structures at Kamrup Rural and Urban District, Assam

Fig.9.6: Artificial Recharge Structures at Jammu District, Jammu & Kashmir

Fig.9.7: Artificial Recharge Structures at ChitradurgaDistrict,Karnataka

Fig.9.8: Artificial Recharge Structures at Virudhunagar district, Tamil Nadu

9.3 Groundwater Augmentation through Artificial Recharge in water-stressed areas of Rajasthan (2021- continuing)

During 2021-22, CGWB has taken up artificial recharge for groundwater augmentation in certain water-stressed areas of Jaisalmer, Jodhpur & Sikar districts of Rajasthan.

Phase-I: Construction of Concrete Gravity Dam at Bastawa Mata (Block Balesar) andZoned Earth Fill Dam with Clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan with the objectives of artificial recharge to groundwater aquifer systems. The cost of the project is about Rs. 34.63 Cr, which includes Defect Liability Period of 01 Year after construction and 03 years of O&M by WAPCOS. The work is under progress.

Phase-II: Under this phase, it is proposed to construct Stone Masonary Check Dams (MCD), Anicuts, Concrete Check Dams (CCD & Recharge shafts) at **101** sites in parts of Balesar, Shekhla, Lohawat, Tinwari and Phalodi blocks of Jodhpur district and Sankra block of Jaisalmer and Shrimadhopur block of Sikar districts of Rajasthan. The district wise details of the Nos. of structures are to be constructed is given in Table 3. The details of structures constructed so far is provided in Table 4.

Table 9.3: The districts wise details of the nos. of AR structures are to be constructed under Phase II

District	Block	Types of WHS & Nos.				No. of Sites
		MCD	Anicut	CCD	RS	
Jodhpur	Balesar	49	08	-	-	57
Jodhpur	Shekhla	06	03	-	-	09
	Lohawat	10	07	-	-	17
	Tinwari	02	04	-	-	06
	Phalodi	-	01	-	-	01
Jaisalmer	Sankra	-	-	04	-	04
Sikar	Shri Madhopur	02	02	-	03	07
Total		**69**	**25**	**4**	**3**	**101**

Table9.4: The details of the nos. of structures constructed under Phase II

Description	Packages								Total
	I		**II**				**III**	**IV**	
	Jodhpur		**Jodhpur**				**Jaisalmer**	**Sikar**	
	Balesar	**Sekhala**	**Tinwari**	**Sekhala**	**Phalodi**	**Lohawat**	**Sankra**	**Srimadhopur**	
Proposed Structures	56	1	6	9	1	17	4	7	101
Work Completed	42	1	6	9	0	16	-	6	80

Phase III (Extended Phase-II):

Due to deterioration in ground water resources in Rajasthan, the Artificial recharge Project under Phase –II has been extended as Phase III (Extended Phase-II) in Jodhpur, Jaisalmer, Sikar, Jhunjhunu and Alwar districts of Rajasthan. The estimated cost of the project is 61.82 Cr. The district wise details of the Nos. of structures are to be constructed is given in Table 5:

Table 9.5: The district wise details of the Nos. of AR structures are to be constructed under Phase II (extended)

Description	Jodhpur District	Jaisalmer District	Alwar District	Jhunjhunu District	Barmer District	TOTAL
Proposed Structures	22	(15 Sites) 24	01	02	04	53

9.4 Artificial Recharge structures in Kala Amb Valley & surrounding area in Sirmaur District, Himachal Pradesh:

Kala Amb Valley in Simaur district falls under ground water dependent area and the ground water is mainly utilized for drinking and industrial development. It has been observed during the past decade that the large withdrawal of ground water in the industrial sector has resulted in depletion of ground water level in the area. It has also been observed that during the monsoons, runoff quickly flows out of the area with less natural recharge to ground water due to sloppy areas. As surplus runoff volume in streams and rivulets are available, appropriate structures across these streams at selective locations may facilitate the harvesting of water which in turn enhances the ground water recharge in downstream areas. Keeping in view of above, a total number of 09 Check dams and 19 Recharge wells have been proposed for construction to augment ground water resources. The proposal has been submitted to Ministry for approval.

9.5 District Recharge Plan Maps:

It has been decided to prepare "District Recharge Plans" for the 255 water stressed district of the country which were identified during Jal Shakti Abhiyan (JSA) 2019 and the same need to be shared to the State Govt dept/implementing agencies and District authorities. The District Recharge Plan will be prepared on 1:50,000 and printable on A0 size. The District Recharge Plan will comprise of the following layers.

- Admin Boundary.
- Location of villages.
- Watershed Boundary.
- Drainage layer with water bodies.
- Major Aquifer units.
- Important location with base layer (major roads).
- Area suitable for Artificial Recharge.
- Area suitable for different artificial recharge structures.
- Tentative location of the artificial recharge structures.

The list of all 255 District Recharge Plan maps prepared by CGWB are mentioned in the table 6:

Table 9.6: The details of the state wise District Recharge Plan

Name of the State	Total number of water stressed district
Andaman & Nicobar	1
Andhra Pradesh	9
Arunachal Pradesh	1
Assam	1
Bihar	12
Chhattisgarh	2
Dadra & Nagar Havelli	1
Daman & Diu	1
Delhi	10
Goa	1
Gujarat	5
Haryana	19
Himachal Pradesh	4
Jammu & Kashmir	1
Jharkhand	2
Karnataka	18
Kerala	2
Lakshadweep	1
Madhya Pradesh	11
Maharashtra	8
Manipur	1
Meghalaya	1
Mizoram	1
Nagaland	1
Odisha	1
Puducherry	1
Punjab	20
Rajasthan	29
Sikkim	1
Tamil Nadu	27
Telangana	24
Tripura	1
Uttar Pradesh	35
Uttarakhand	1
West Bengal	1
Grand total	**255**

Annexure-I

Work carried out under MGNREGS during March - 2023

State	**Andhra Pradesh**	**Telangana**	**Karnataka**	**Tamil Nadu**	**Rajasthan**	**Rajasthan**	**Haryana**	**Madhya Pradesh**	**Maharashtra**
District	**Prakasam**	**Nallagonda**	**Chickballalpur**	**Krishnagiri**	**Bhilwara**	**Sirohi**	**Kurukshetra**	**Ujjain**	**Amravati**
Block	**Peddaravedu**	**Munugode**	**Gowribidanur**	**Mathur**	**Madalgarh**	**Reodhar**	**Babain**	**Badnagar**	**Warud**
Commencement of Execution of Work	April 2018	April 2018	19.06.2022	18.07.2022	19.04.2022	19.04.2022	18.09.2022	18.03.2022	18.12.2022
No. of Structure Envisaged	1253	2174	103	20	56	126	15	436	1798
No. of Structure Completed	1253	1713	33	20	47	81	9	353	422
No. of Structure In Progress	Nil	Nil	7	nil	1	16	1	83	20
Geo-tagging (Completed /in progress)	Completed	Completed	completed	Completed	Completed	Completed	In-progress	Completed	In-progress
Initiation of Impact Assessment Study (Yes/No)	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Status of the Project	Completed	Completed	Under Progress	Completed	Under Progress	Under Progress	Under Progress	Under Progress	Under Progress

Type of Structures constructed	• Farm Pond/Dugout Ponds : 1034 • Recharge structures to Borewells : 3 • Check Dams : 34 • Desilting of MI Tanks : 23 • Roof top Rain Water Harvesting and Recharge Structures : 04 • Staggered trenches : 11 • Boundary trench with 1m depth : 144	• Farm Pond-426 • Recharge Shafts with Bore wells-138 • Check Dams-02 • Soak Pits-1065 • PTS-04 • Community Soak Pits-18 • Desiltaion of MI Tanks-60	• Multi Arch Check Dam-33	• Check Dams-04 • Nala Bund-04 • Recharge Wells and De siltation of MI Tanks-12	• Recharge Shaft in Ponds-47	• Recharge Shaft in Ponds-81		• Check Dam -26 • Check Dam Nala Trenching – 109 • New Pond-13 • Nala Trenching - 112 , • Nala Trenching Culvert Nirman – 02 • Percolation Tank – 06 • Stop Dam-07 • Stop Dam Cum Culvert Nala Trenching- 04 • Sankrin Pond – 31 • Pond West Weir Renovation – 06 • West Weir Nala Trenching-02 • Pond Renovation – 31 • West Weir Renovation-02 • Check Dam Renovation - 02	Well Recharge-102 • Soak Pits-234,LBS-2, • Gabion-27 • Farm Pond-39 • CCT-1 • Nala Kholikaran-17
Remarks						All 126 sites released by CGWB. As per the informatio n received from State Govt. only	As per latest data received from ABPO Babain, Total 15 No. of ponds are selected under MNREGA		

						97 structures will be constructed.	instead of 16 Ponds. in Addition to this total 51 Nos. of recharge structures (recharge Pit =26 and Soakage Pits= 25) are completed.		

Annexure –II

Implementation of Master Plan 2020						
Sr. No.	**Name of the State**	**District Identified for Master Plan 2020 Implementation**	**State agency identified for Master plan implementation**	**Type and number of structures proposed in Master Plan 2020 of the identified district**	**Actual implementation of Master Plan in identified district (Type and Number of structures constructed)**	**Remarks**
1	Arunachal Pradesh	Papum Pare	Water Resource Department	FP-5000,CD-10000,GS-5000,ECB-200000,ST-11000,RTRWH-4516	194 nos. of check dams constructed.	
2	Bihar	Gaya	Minor Water Resources Department, GoB	PT(H)-37,GP-592,CB&T(h)-592, cd-28, NB-101. CB&T(m)-807, RS-1613, PT(M)-76, DS(Tank)-2941, Inj.wells-3921, Revival of Ahar Payne-169, RTRWH-2500, Revivalof UWB-9	Revival of Ahaar: 213 Revival of Pyne: 148 Check Dam: 2 Renovation of Ponds:75 Construction of Soakpit:8	

3	Chhattisgarh	Bemetara	Water Resource Department (WRD), Chhattisgarh	RS/GHRW-5287,PT-705,GP/CB/GS-4028, NB/CB/KT Bhandara-2350,RTRWH-1380	Ponds/ Tanks-31 RTRWH structures-51. 2 Check dams-02 Anicuts -11 Soak pit/ Recharge Shaft -72 New Talab and Government Ponds-160 Irrigation Nalas/Canals-12.	
4	Chandigarh	Chandigarh			Rooftop Rain Water Harvesting Structures (RTRWH) constructed: -516 nos.	1. Only RTRWH has been suggested under the master Plan and the same has been implemented by the concerned authority. 2. Roof top rain water Harvesting mandatory under by-laws
5	Delhi	North Delhi	Irrigation and Flood Control Board and Delhi Jal board , Govt. of NCT Delhi	RS&RT-456, RTRWH-7000	Percolation pond-01 Work on 11 no. of ponds yet to be taken up as the funds are not available with IFC & D	
6	Gujarat	Gandhinagar	GWRDC , Municipal Corporation, Gandhinagar ,Water Resources Department, Rural Development Department	RTRWH-15792,Recharge shaft-159, Recharge through defunct tubewells-174	Total 634 Structures proposed in 223 WSP 1)Rain Water Harvesting Structure, Excavation of pond, Nalah Bund -58 2)Work order for 15 recharge tubewells costing Rs 2.53 crore issued and work in progress.	Work orders for 21 recharge tubewells costing Rs 3.59 crore issued and work in progress. Out of these 12 recharge tubewells drilling including pipe lowering are completed. It is planned to complete works by 30/06/2023

7	Haryana	Yamuna Nagar	SLNA, Rural Development & Forest Department	CD-10, RS-100, FP-17236,RTRWH-13665	RTRWH-78 Soakage pits-513	In Haryana, DPRs for Artificial recharge structures for Ambala, Kurukshetra, Yamunanagar and Panchkula districts have been submitted to CHQ for financial approval
8	Himachal Pradesh	Solan	BaddiBarotiwalaNalagarh Development Authority (BBNDA)	Gabion-3412, CD/NB/CP-134, SSD-23, CD+SSD-76, Modification pond and tank-248, RS-45,Inj Well-10, RTRWH-100	Check Dam -03 RWH structures-02.	Total 1037 Lakh amount has been sanctioned and in which 600 lakh amount has been released (100 lakh to JSV, Nalagarh and 500 lakh to JSV, Baddi)
9	Karnataka	Chitradurga	SLNA, Ground Water Directorate	CD-4143, PT-802, SSD,CD cumSSD-29	Check Dam - 214, Percolation Tank - 4, Earthen Check Dam– 2243.	
10	Kerala	Kasaragod	Ground Water Department, Kerala	Percolation Pond-300, CB-15000, GB-40000, NB-5250, SSD-20. RTRWH-14027	57 ground water conservation/AR/other works has been undertaken as on date. 38 Check dam/VCB and remaining works such as renovation of existing AR structures, canal network and allied works.	

11	Madhya Pradesh	Chhatarpur	Water resource Department (WRD), Govt of Madhya Pradesh	Total 9105 number of structure proposed (Percolation tank, check dam, recharge shaft, nala bund/ cement plug, Farm pond/ village pond)	Total 719 structures constructed (Percolation tank, Check Dam, Farm Pond, Nala bund/ Cement plug/ Gully plug) Implementation scheme- MGNREGA, Rajiv Gandhi Watershed Mission	
12	Maharashtra	Nagpur (Katol &Narkhed blocks)	Groundwater Surveys & Development Agency (GSDA), Govt of Maharashtra	PT:280, CD: 755, RTRWH:350774, RS: 27, URH: 650	Recharge shafts (RS)-618 RTRWH -15 Soak pit- 25 Nala deepening-04 Gabbian structures-32 KT weir -03 Storage CNB-15.	Recharge shafts under ABHY by GSDA. 68 Recharge shaft, 1UGB, 2 nala deepening, under JJM by Rural Water Supply (ZP). 20 RTRWH, 10 soak pit, 5 nala deepening, under MREGS/15th Finance by Gram Panchayat. 03 KT Weir, 10 soak pit, 5 nala deepening, under MREGS/15th Finance by Gram Panchayat. 3 KT Weir and 4 Storage CNB under River rejuvenation/mahamandal and CM Water Conservation Scheme by Soil and water Conservation department.

13	Odisha	Nuapada	Directorate of Ground Water Development, Dept Of Water Resources	CD-6, PT-3, NB/CB-6, ST Conversion to PT- 3SSD-6, RTRWH-78	Percolation tanks: 05. Renovation of existing tanks: 01 Sub-surface dykes: 04. Concrete Nallah bund: 02 Recharge shaft with RCC well: 05 Recharge shafts with filtration chamber in newly constructed/existing tanks/ponds: 20	
14	Punjab	Sangrur	Department of Soil and Water Conservation, Punjab	RS-3408, FP-35170, RTRWH-31887, IW-2674	soak pits-82 Rain water harvesting structures-20	Under recent development, work on 129 recharge schemes (Injection wells) at the outlet of canals is expected to start soon all over Punjab. Funds have been allocated and tendering complete. Apart from this, approval and funds have been allocated for construction of RWH structures at 886 Rural water supply schemes under JJM.
15	Rajasthan	Ajmer	Ground Water Department (GWD), Govt of Rajathan	CD-831,FP-4547, PT-1629, CAT (Catchment Area Treatment under which plantation, staggered trench and continuous countour trench)-13741hacteres, anicut-422, MST- 24, MPT- 3615,	Farm ponds 32 been completed in 2021-22. In 2022-23 -Farm pond 30 nos. has been proposed by Agriculture Dept.	

16	Sikkim	South District(Renamed as Namchi District)	Department of Water Resources and River Development, Govt. of Sikkim	SpringshedDevelopment : 450 Gabions/Nala Bunds / Check Dams: 50 Nala Plug / Sub-surface Dyke : 100 RTRWH : 500 RWH along slopes with impermeable membrane : 41	Pond : 480 Trenches : 8000	
17	Tamil Nadu	Virudhunagar	State Ground & Surface Water Resources Data Centre, Water Resources Department, Govt. of Tamil Nadu	CD/NB-244, RS/BW-2604, OTHERS (FP/RP)-244, PP-81, RT-10kms	75 recharge shafts constructed during 2022-23.	During the meeting held on 19.04.2023 with Addl. Chief Secretary (WRD) and Chief Engineers of Water Resources Department , the District Recharge Plan prepared by CGWB was presented. ACS has asked to CGWB to share Recharge Plan for all the 12 delta districts of Tamil Nadu for taking up the works before monsoon.
18	Tripura	Khowai	Public Works Department (PWD) (WR), Govt. of Tripura	Gabion-35,CB-14,CD-21, Modification of village tanks-35,RTRWH-5,753	• RWHS- 179 and Renovation of Water bodies/tanks-03 • Currently the State Govt has under taken tendering work for 300 Recharge Trench, which is likely to be completed soon.	Currently the State Govt has under taken tendering work for 300 Recharge Trench, which is likely to be completed soon. Tendering at Finalisation Stage

19	Uttarakhand	Haridwar	Minor Irrigation, Uttarakhand	RTRWH-400, CD-100, PT-10, CT-225	Amrit Sarowar- 78 Recharge Pits- 115 Rooftop Rainwater Harvesting- 10 Ponds Renovated-82 Recharge Shaft- 20 Farm Pond- 18 Micro Irrigation (Sub surface drainage)- 633	
20	Uttar Pradesh	Jhansi	Minor Irrigation Dept., Govt of Uttar Pradesh	CD/NB/CP-158, DW/TW/RS-158, Pond-263, PT-47	Check Dam- 38. Ponds-38 RTRWH- 65 Nala Bunds-15 Contour Bund/ Trench-09	
21	West Bengal	Purulia District	State Water Investigation Directorate, WRI & DD, Govt. of West Bengal	CD-177, RTRWH-75, PT in CT&MC-58, G/CB-445, SSD-222, DWR-45	Check Dam : 129 SFMIS : 250 Water Detention Structure : 238 Water Harvesting Tank: 01.	

22	U.T of Andaman & Nicobar	South Andaman District	Andaman Public Works Depertment	CD-350, PT-1000, G/CB-150, SSD-100, Spring Shed Development with catchment area treatment and Microwatershed management- 170	RTRWH Structures: 439. Balance 414 likely to be completed within December 2023	UT Administration planned to construct and retrofit 853 existing Govt Buildings with RTRWH. It is about 1/3rd of the proposed target under master Plan from U/T Fund. Fund being limited, work is undertaken in phased manner. Few ground verifications done. One Officer from ER already visiting the UT for co-ordination
23	UT of Ladakh	Leh District	I&FC Department, UT of Ladakh	G/NB/SS Bhandara-46, Rev & Res. Pond-31, Diversion of flows from Perennial Nalas/Springs-15, Artificial Glaciers-200	Check dams- 08 Zings (Ponds)-08	

10.0 Technical Assistance (SHORT TERM WATER SUPPLY INVESTIGATIONS)

Table 10.1 List of short-term investigations carried out region wise

	Region	Title of short term water supply investigation	Agency for which the study was taken up
1	NWR, Chandigarh	1. Feasibility study for construction of tubewell for Provision of Swimming Pool at Jalandhar Cantt. (W), Punjab. Report is prepared by Sh. Manish Shrivastav (AHG)	Jalandhar Cantt., Punjab
2	NWR, Chandigarh	2. Feasibility study of Rain water Harvesting and Artificial Recharge at Children School, Military Engineering Services, Ambala Cantt. Haryana. Report is prepared by Sh. Aditya Sharma (AHM)	Military Engineering Services, Ambala Cantt., Haryana
3	NWR, Chandigarh	3. Feasibility study of Rain water Harvesting and Artificial Recharge for Constr of 256 DUs at Military Engineering Services, Ambala Cantt. Haryana. Report is prepared by Sh. Aditya Sharma (AHM)	Military Engineering Services, Ambala Cantt., Haryana
4	NWR, Chandigarh	4. Feasibility study of Rain water Harvesting and Artificial Recharge for DEFI OTM ACCN of HQ2 Corps Under GE (P), Military Engineering Services, Ambala Cantt. Haryana. Report is prepared by Sh. Aditya Sharma (AHM)	Military Engineering Services, Ambala Cantt., Haryana
5	NWR, Chandigarh	5. Feasibility study of Rain water Harvesting and Artificial Recharge at KLP ACCN for HQ40 ARTY DIVISION, Military Engineering Services, Ambala Cantt. Haryana. Report is prepared by Sh. Aditya Sharma (AHM)	Military Engineering Services, Ambala Cantt., Haryana
6	NWR, Chandigarh	6. Feasibility study survey for Water Strata at (AF) Chandigarh 12 Wing Area, Military Engineering Services, Air force Chandigarh. Report is prepared by Sh. Gyanendra Rai (STA) (HG).	Military Engineering Services, Air force Chandigarh
7	NWR, Chandigarh	7. Feasibility study for construction of tubewell at Air Force Station, Highgrounds, Chandigarh.Report is prepared by Sh. Manish Shrivastav (AHG) .	Air Force Station, Chandigarh
8	NWR, Chandigarh	8. Feasibility study for construction of tubewell at Pump House No. 3(Bhagat Road) & Pump House No. 11, Raina Park, Jalandar Cantt West, Jalandhar.Report is prepared by Sh. Manish Shrivastav (AHG).	Jalandhar Cantt West, Jalandhar

	Region	Title of short term water supply investigation	Agency for which the study was taken up
9	NWR, Chandigarh	9. Feasibility study for drilling of borewell for Pump House No. 25 (MAP BI Line) at Jalandhar Cantt West, Jalandhar. Report is prepared by Sh. Manish Shrivastav (AHG).	Jalandhar Cantt West, Jalandhar
10	NWR, Chandigarh	10. Feasibility study for construction of tubewell in at ALG Ground Area, Ferozpur, Punjab. Report is prepared by Sh. Gyanendra Rai (STA).	
11	NWR, Chandigarh	11. Feasibility study for construction of tubewell at OTM ACCN Zone, Jalandhar Cantt East.Report is prepared by Sh. Manish Shrivastav (AHG).	Jalandhar Cantt East
12	NWR, Chandigarh	12. Feasibility study for construction of new borewell (02 nos.) at 36 field Ammunation Depot, Bathinda Military Station, Bathinda, Punjab Report is prepared by Sh. Ayush Kesharwani (Scientist-B).	Bathinda Military Station, Bathinda
13	NWR, Chandigarh	13. Feasibility study for construction of New borewell (03 nos.) at Airforce Station,Bhisiana, Bathinda, Punjab Report is prepared by Sh. Ayush Kesharwani (Scientist-B).	Airforce Station, Bathinda
14	NWR, Chandigarh	14. Feasibility study for construction of New borewell (02 nos.) at Hisar Military Station,Village Bhojraj, Dist. Hisar, Haryana. Report is prepared by Sh. Ayush Kesharwani (Scientist-B)	Hisar Military Station
15	NWR, Chandigarh	15. Feasibility study for construction of new borewell (02Nos.) at Air Force Station, Adampur, Jalandhar, Punjab. Report was prepared by Sh. Manish Shrivastav (AHG).	Air Force Station, Adampur, Jalandhar, Punjab.
16	NWR, Chandigarh	16. Feasibility of Constructing a Tubewell for Military Station Sangrur Cantt. Punjab. Report was prepared by Sh. Gyanendra Rai (STA-HG)	Military Station Sangrur Cantt. Punjab
17	NWR, Chandigarh	17. Feasibility Study for Construction of Tubewells in Battalion HQ BSF 73 Bn SBF, Ajnala, Amritsar, Punjab. Report was prepared by Naima Akhtar (Scientist-B)	HQ BSF 73 Bn SBF, Ajnala, Amritsar, Punjab
18	NWR, Chandigarh	18. Feasibility Study for construction of tubewells in BOP Wadhai Cheema, Ajnala, Amritsar, Punjab. Report was prepared by Naima Akhtar (Scientist – C)	Border Out-Post (BOP) Wadhai Cheema, Ajnala, Amritsar, Punjab
19	NWR, Chandigarh	19. Feasibility study of Rain water Harvesting and Artificial Recharge at KLP Accn for 371 Comp Arty Bde at Ambala Cantt, Haryana. Report is prepared by Sh. Manish Shrivastav (AHG)	Ambala Cantt. Haryana
20	NWR, Chandigarh	20. Feasibility study of Rain water Harvesting and Artificial Recharge at KLP Accn for ONE AD REGT (L/70) Phase-I at Ambala Cantt, Haryana, Haryana. Report is prepared by Sh. Manish Shrivastav (AHG)	Ambala Cantt. Haryana

	Region	Title of short term water supply investigation	Agency for which the study was taken up
21	NWR, Chandigarh	21. Feasibility study report of 02 nos. of borewell at MD ACCN (280DUs) under EVI Ambala Cantt. Haryana. Report is prepared by Sh. Manish Shrivastav (AHG)	Ambala Cantt. Haryana
22	NWR, Chandigarh	22. Feasibility Study report of a new borewell at BOP Bhariyal, Block-Dinanagar, Dist.- Gurdaspur, Punjab. Report is prepared by Sh. Ayush Kesherwani (Scientist-B)	Border Out-Post (BOP) Bhariyal, Gurdaspur, Punjab
23	NCR Bhopal	1. Short term report of Indian Institute of soil science (ICAR), Nabi Bagh, Berasia Road, Bhopal	Bhopal
24	NCR Bhopal	2. Construction of Rainwater harvesting structure in the Air force station, Amla, Betul	Air force station, Amla Betul
25	NCCR, Raipur	1. Geophysical investigations of GC-CRPF, campusatvillageBhilai, Arang, Raipur,Chhattisgarh by Dr.A.K.Sinha, Scientist-C (GP) &Sh.Nageswara Rao Elisela, AGP	Bhilai, Arang, Raipur,Chhattisgarh
26	NCCR, Raipur	2. Geophysical investigations in parts of Bilha and Masturi Blocks of Bilaspur District, ChhattisgarhDr. A.K. Sinha, Scientist-C (GP) & Sh. Nageswara Rao Elisela, AGP	Bilha and Masturi Blocks of Bilaspur District, Chhattisgarh
27	NHR Dharamshala	Investigation for Army area Pooh, Kinnaur, H.P.	Army
28	NHR Dharamshala	Investigation for Army area Karcham, Kinnaur	Army
29	NHR Dharamshala	Investigation for Army area Samdhu, Kinnaur	Army
30	NHR Dharamshala	Investigation for Army area Dogri, Kinnaur	Army
31	NHR Dharamshala	Investigation for R.W.H in Army area Subathu, Solan, H.P.	Army
32	UR Dehradun	Report on The Feasibility for Construction of Abstraction Structure for Ground Water at Nayalgarh Village, Srinagar District Pauri-Garhwal, Uttarakhand	Gram Pradhan, Nayalgarh village, Srinagar Grahwal Uttarakhand
33	UR Dehradun	Feasibility report for the Construction of Tubewells at Military Station Hempur, Kashipur Block, District Udham Singh Nagar, Uttarakhand	Hempur Military Station, Kashipur
34	UR Dehradun	Feasibility for construction/implementation of AR structures in Roorkee Cantonment	Garisson Engineer, Military Engineering Services, Roorkee
35	UR Dehradun	Feasibility report for the construction of tubewell at Ghunghora-BirpurGarhi Cantonment Area	Garisson Engineer, Military Engineering Services

	Region	Title of short term water supply investigation	Agency for which the study was taken up
36	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of BOP Budhwar,Jammu	BSF,Jammu
37	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of Bakkal,nearchenab railway bridge,Reasi district	Chief Engineer,konkan Railway
38	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of Punzgam,Zangli and Kalaroos,Dist.Kupwara	Garrison Engineer,874 Engr.
39	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of Ascon Node(dassalGujran),Dist.Rajouri	Garrison Engineer,862 Engr.
40	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of Palma GarrisonAsconNode(dassalGujran),Dist.Rajouri	Assisstant Garrison Engineer,CIF.
41	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at tawiriver,Dist.Jammu	Jammu Smart City LTD.
42	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Wazurcantonment,Dist.Anantnag	Wazurcantonment,Dist.Anantnag
43	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Central University,Dist.jammu	Central University,Dist.jammu
44	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at JindrahMillatryStation,Dist.Udhampur	JindrahMillatryStation,Dist.Udhampur
45	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Patti Mansar area,Dist.samba	Forest department,Jammu
46	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Government Polytechnic,Chanderkote,Dist.Ramban	PHE Department UT of J&K.
47	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Government Polytechnic,KV,KashtigarhChanderkote,Dist.Doda	Asstt.Commissioner,Doda, UT of J&K.
48	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at 31 Rajput in Chanigund and 8 Dogra,Kanasar,	GarrissionEngineer,kargil
49	NWHR Jammu	Feasibility report of construction of Rainwater harvesting and Artificial Recharge at Kaluchak and Ratnuchak Millitry Station	Kaluchak and Ratnuchak Millitry Station, Jammu

	Region	Title of short term water supply investigation	Agency for which the study was taken up
50	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Palma Garrison, District Rajouri, UT of J&K	Garrission Engineer, Palma, Rajouri
51	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells at Bulla Chak,Tegowali and Nikowal villages of Suchetgarh, District Jammu, UT of J&K	Executive Engineer, TubewellIrigation Division, I&FC, UT of J&K
52	NWHR Jammu	Resistivty survey for construction of tubewell at IIM Campus, District Jammu, UT of J&K	Chief Administrative Officer, IIM Jammu
53	NWHR Jammu	Hydrogeological and Geophysical investigation for water supply/feasibility report of construction of tubewells at Ring Road Area, Kote Bhalwal,Dist.Jammu	Jammu Development Authority
54	NWHR Jammu	Resistivty survey for construction of tubewell at Talla Nagara, Patyari&Talatand, District Rajouri, UT of J&K	PHE Department UT of J&K.
55	NWHR Jammu	Hydrogeological & Geophysical Investigation For Water Supply / Feasibility Report Of Gerrison (North), Akhnoor, District Jammu (J&K-UT).	Garrission Engineer
56	NWHR Jammu	Hydrogeological & Geophysical Investigation For Water Supply / Feasibility Report Of KharianGerrison Miran Sahib District-Jammu (J&K-UT).	Garrission Engineer
57	NWHR Jammu	Feasibiliy report for construction of tubewell at CPS Muthi, District Jammu	PHE Department UT of J&K.
58	NWHR Jammu	Feasibiliy report for construction of tubewell at Peerkho, District Jammu	PHE Department UT of J&K.
59	NWHR Jammu	Feasibiliy report for construction of tubewell at Tubewell No 11,NearRamdada Hotel, Chand Nagar, District Jammu	PHE Department UT of J&K.
60	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of construction of tubewells atTatnihal&Bankoot area of Banihal ,DistrictRamban	Additional Chief Manager, Elec. USBRL Project
61	SUO, Delhi	Hydrogeological Investigation report for construction of one bore well at Air Force Station, Arjangarh (Near UG OPS room) under 503 SU Campus	Delhi Cantonment
62	SUO, Delhi	Assessment of Rain Water Harvesting Structures at 27 Equipment Depot, Air Force Station	Delhi Cantonment

	Region	Title of short term water supply investigation	Agency for which the study was taken up
63	SUO, Delhi	Hydrogeological Investigation report for construction of 04 tube wells at Aerospace Museum, Air Force Station, Palam, Delhi Cantonment, Delhi	Delhi Cantonment
64	NR Lucknow	Site selectionn for construction of roof top rain water harvesting structure for provn of 06 jco's, lucknow cantonment area, lucknow, uttarpradesh	Lucknow Cantonment
65	NR Lucknow	Water availbility , feasibility study at Kanpur cantonment	Kanpur Cantonment
66	NR Lucknow	Provision of New Borewell at Prabudh Vihar , devi Khera, Lucknow	Lucknow Cantonment
67	NR Lucknow	Provision of KLP ACCN for 01 X INF BN of HQ 71 MTN BDE at Lucknow	Lucknow Cantonment
68	NR Lucknow	Site selectionn for construction of roof top rain water harvesting structure for hq 71 mtndbelucknow cantonment area, lucknow, uttarpradesh	Lucknow Cantonment
69	NR Lucknow	Supply of data of Ground Water Strata for estimation of Bore well at Head Post office compound Bareilly	Head Post Office, Bareilly
70	NR Lucknow	Design of RWHS for proposed building at Telibagh,Lucknow	BSNL Office Lucknow
71	NER Guwahati	NOTE ON HYDROGEOLOGICAL INVESTIGATION AT THE PREMISES OF AEGCL, KAHILPARA, GUWAHATI, ASSAM	AEGCL , Govt. of Assam Undertaking
72	NER Guwahati	REPORT ON HYDROGEOLOGICAL INVESTIGATION IN THE PREMISES OF 5 AFSB (AGCHIA), AIR FORCE, BORJHAR, KAMRUP (M), ASSAM	Air Force, GoI
73	NER Guwahati	NOTE ON HYDROGEOLOGICAL INVESTIGATION AT THE PREMISES OF BSF, GEETA NAGAR, GUWAHATI, ASSAM	BSF, GoI
74	NER Guwahati	FEASIBILITY REPORT FOR CONSTRUCTION OF A DEEP TUBE WELL IN THE CAMPUS OF MLA HOSTEL, DISPUR, KAMRUP (M) DISTRICT, ASSAM	PHED, Govt. of Assam
75	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT MSME-DI, BAMUNIMAIDAM, GUWAHATI ASSAM	MSME-DI, GoI

	Region	Title of short term water supply investigation	Agency for which the study was taken up
76	NER Guwahati	A NOTE ON HYDROGEOLOGICAL INVESTIGATION IN THE PREMISES OF THE OFFICE CUM RESIDENTIAL QUARTERS INCLUDING TRANSIT ACCOMMODATION AT SPECIAL BUREAU, KHANAPARA, GUWAHATI, KAMRUP(M), ASSAM	Special Bureau, GoI
77	NER Guwahati	FEASIBILITY NOTE FOR CONSTRUCTION OF GROUNDWATER ABSTRACTION STRUCTURE (DEEP BORE WELL) IN THE PREMISES OF SHIISHU SHIKSHA SAMITY, PRASHANTA PATH, OPPOSITE DOOR DARSHAN KENDRA, R.G. BARUAH ROAD, GUWAHATI-24, KAMRUP (M), ASSAM	Shishu Shiksha Samity, A registered society under Govt. of Assam
78	NER Guwahati	REPORT ON SHORT TERM WATER SUPPY INVESTIGATION AT NARENGI MILITARY STATION, GUWAHATI, ASSAM	Indian Army, GoI
79	NER Guwahati	NOTE ON HYDROGEOLOGICAL INVESTIGATION AT THE PREMISES OF ASSAM RIFLES TRANSIT CAMP, KHANAPARA, GUWAHATI	Assam Rifles, GoI
80	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT CRPF GROUP CENTRE, AMERIGOG, 9TH MILE, KHANAPARA, GUWAHATI, ASSAM	CRPF, GoI
81	NER Guwahati	NOTE ON HYDROGEOLOGICAL INVESTIGATION AT THE PREMISES OF POSOCO CAMPUS, KAHILIPARA, GUWAHATI	POSOCO, GoI undertaking
82	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT PRAGJYOTI BHAWAN, SIB, VIP ROAD, HENGRABARI, GUWAHATI	SIB Govt. of India
83	NER Guwahati	REPORT ON HYDROGEOLOGICAL INVESTIGATION AT NARANGI MILITARY STATION, KAMRUP (M) DISTRICT, ASSAM	Indian Army GoI
84	NER Guwahati	REPORT ON SHORT-TERM INVESTIGATION FOR EVALUATING FEASIBILITY FOR CONSTRUCTION OF A BORE WELL IN THE PREMISES AIZAWL MILITARY STATION	Indian Army, GoI
85	NER Guwahati	REPORT ONHYDROGEOLOGICAL INVESTIGATION FOR CONSTRUCTION OF A DEEP TUBE WELL IN THE B. K. HOUSE COMPLEX, SPECIAL BUREAU, AGARTALA, WEST TRIPURA DISTRICT, TRIPURA	Special Breau, GoI
86	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT COLLEGE OF VETERINARY SCIENCES & ANIMAL HUSBANDRY, CENTRAL AGRICULTURAL UNIVERSITY, JALUKIE, NAGALAND	Central Agricultural University, Jalukie-797110, Nagaland. GoI

	Region	Title of short term water supply investigation	Agency for which the study was taken up
87	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT NATUANACHA AND BHITORKHOLA, RANGIA, ASSAM	PHED, Govt. of Assam
88	NER Guwahati	REPORT ON HYDROGEOLOGICAL INVESTIGATION AT THE PREMISES ASSAM URBAN WATER SUPPLY AND SEWERAGE BOARD (AUWSASB) JORHAT DIVISION, GARMUR ROAD, JORHAT, ASSAM	Assam Urban Water Supply and Sewerage Board (AUWSASB)
89	NER Guwahati	NOTE ON HYDROGEOLOGICAL INVESTIGATION AT THE PREMISES OF THE COLLEGE OF VETERINARY SCIENCE, KHANAPARA, GUWAHATI, KAMRUP (M) DISTRICT, ASSAM	College of Veterinary Science, Khanapara
90	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT STATE PUBLIC HEALTH LABORATORY, BAMUNIMAIDAM, GUWAHATI ASSAM	PHED, Govt. of Assam
91	NER Guwahati	A NOTE ON HYDROGEOLOGICAL INVESTIGATION AT MOUNTAIN SHADOW AREA IN THE PREMISES OF AIR FORCE, BORJHAR, KAMRUP (M), ASSAM	Air Force, GoI
92	NER Guwahati	A NOTE ON HYDROGEOLOGICAL INVESTIGATION IN THE PREMISES OF THE OFFICE CUM HOSTELS AT INDIAN INSTITUTE OF HANDLOOM TECHNOLOGY, KHANAPARA, GUWAHATI, KAMRUP(M), ASSAM	IIHT, Khanapara, GOI
93	NER Guwahati	REPORT ON SHORT TERM WATER SUPPLY INVESTIGATION AT INDO TIBETAN BORDER POLICE FORCE (ITBP), BASAR, LEPA RADA (EARSTWHILE WEST SIANG) DISTRICT, ARUNACHAL PRADESH	49th BN, ITBP, Basar
94	NER Guwahati	STWSI at HQ Eastern Air Command Upper Shillong	Garrison Engineer (I) Air Force Upper Shillong
95	NER Guwahati	STWSI at Umroi Military Station, Barapani, Ri Bhoi, Meghalaya	Garrison Engineer, Umroi, Military Station
96	NER Guwahati	STWSI at U Lumsohpetbneng, Ri Bhoi, Meghalaya	PHED, Govt. of Meghalaya
97	NER Guwahati	STWSI at Sohmynting, West Jaintia Hills, Meghalaya	Deputy Commissioner (Planning), Jowai, West Jaintia Hills
98	NER Guwahati	STWSI at Laskein, West Jaintia Hills, Meghalaya	Deputy Commissioner (Planning), Jowai, West Jaintia Hills
99	WCR, Ahmedabad	Short term Geophysical Resistivity (VES) investigation at KVK-CAZRI, Kukma, Bhuj, Kachchh district	KVK-CAZRI, Kukma, Bhuj, Kachchh
100	WR, Jaipur	Hydrogeological Surveys for Provision of Rain Water Harvesting at Bhakrota and Jaipur Military Station, District: Jaipur, Rajasthan.	Indian Army

	Region	Title of short term water supply investigation	Agency for which the study was taken up
101	WR, Jaipur	Resistivity Surveys at Adarsh Vidya Mandir Shankar Vidyapeeth, Mount Abu, District: Sirohi, Rajasthan.	VIP reference MoJS.
102	SR Hyderabad	Report on Short Term Water Supply Investigations at ITBP Campus,Anadapuram,Visakhapatnam	Central Public Works Department, Government of India
103	SR Hyderabad	Water Supply Investigation at Facility of Electronics (FEL), MCEME Campus, Ministry of Defence,Trimulgherry Mandal ,Secunderabad,MedchalMalkajgiriDistrict,Telangana state.	MCEME, Ministry Of Defence
104	SR Hyderabad	Feasibilty Report for Digging of Borewell at 654 EME Battalion, Ramakrishnapuram, Ministry of Defence,Trimulgherry mandal, MedchalMalkajgiri, Telangana state.	654 EME Battalion, , Ministry of Defence,
105	SR Hyderabad	Report on Feasibility for Bore well site at RashtapatiNilayam,Bollaram, SecunderabadMandal,Telangana state	Rashtrapati Nilayam, Government of India
106	SR Hyderabad	Water Supply Investigation for Augmenting Ground Water Supply to NIMSME Area,Yousufguda,Hyderabad.	National Institute for Micro,Small and Medium Enterprises (NIMSME)
107	SECR, Chennai	Short term Ground water investigation at ESI Hospital and Staff quarters at Ayanavaram campus	ESI Hospital, Ayanavaram
108	SECR, Chennai	Short term investigation for providing Bore well facility at Head Post office premises, Kancheepuram	Head Post Office, Kancheepuram
109	SECR, Chennai	Short term investigation for digging of Bore well at ESI Dispensary campus, Choolai	ESI Dispensary, Choolai
110	SECR, Chennai	Short term investigation at Naval Air Force station ,Parundu, Ramanathapuram	Air Force Station, Ministry of Defence
111	SECR, Chennai	Short term investigation for digging of BW at Indian Coast Guard Air station near Tuticorin Airport	Indian Coast Guard
112	KR Kerala	Short term hydrogeological investigation on feasibility of roof top rainwater harvesting structure for INS Gaurda and Kataribg at Kichi, Ernakulam district	Indian Navy
113	SWR, Bangalore	Hydrogeological Investigation for constructing Borewells for Water Supply at DRDO, Kolar, Kolar district, Karnataka	DRDO, Kolar
114	SWR, Bangalore	Water Supply Investigation for Feasibility of Drilling Borewell at ASC Centre (South) 2 ATC Bangalore, Bangalore South Taluk, Bangalore Urban District.	ASC Centre (South) 2 ATC Bangalore, Bangalore South Taluk, Bangalore Urban District.

	Region	Title of short term water supply investigation	Agency for which the study was taken up
115	SWR, Bangalore	Water Supply Investigation for Feasibility of Drilling Borewell at HQGNA Gomantak (Naval Area) & NOFRA-II, Vasco, Goa State (AAP: 2021-2022)	HQGNA Gomantak (Naval Area) & NOFRA-II, Vasco, Goa State
116	SWR, Bangalore	Water Supply Investigation for Feasibility of drilling Borewell at DRDO Township, Phase-1, CV Raman Nagar, Bangalore North Taluk, Bangalore Urban District, Karnataka (AAP: 2022-2023)	CV Raman Nagar, Bangalore North Taluk, Bangalore Urban District
117	SWR, Bangalore	Water Supply Investigation for Feasibility of drilling Borewell at DRDO Township, Phase-2, CV Raman Nagar, Bangalore North Taluk, Bangalore Urban District, Karnataka (AAP: 2022-2023)	CV Raman Nagar, Bangalore North Taluk, Bangalore Urban District
118	SWR, Bangalore	Water Supply Investigation for Feasibility of Drilling Borewell at DRDO Site-2, Rajanukunte, Yelahanka, Bangalore North Taluk, Bangalore Urban District, Karnataka (AAP: 2022-2023)	Yelahanka, Bangalore North Taluk, Bangalore Urban District
119	SWR, Bangalore	Water Supply Investigation for Feasibility of Drilling Borewell at Airmen Training School (ATS), Tilakwadi, Belagavi, Karnataka State (AAP: 2022-2023)	Tilakwadi, Belagavi
120	SWR, Bangalore	Water Supply Investigation for Feasibility of Drilling Borewell at Head Quarter Naval Technical Group (HQNTG) Campus, Vimanapura, Bangalore Urban District, Karnataka (AAP: 2022-2023)	Vimanapura, Bangalore Urban District
121	ER, Kolkata	Short Term Water Supply Investigation for Airforce Station, Hasimara- prepared by Dr. Indranil Roy, Sc-D(HG)	MES, Siliguri, West Bengal
122	ER, Kolkata	Short Term Water Supply Investigation for Airforce Station, Bagdogra - prepared by Dr. Indranil Roy, Sc-D(HG)	MES, Siliguri, West Bengal
123	ER, Kolkata	Short Term Water Supply Investigation for Airforce Station, Kalaikunda - prepared by Shri Arpan Banerjee, Sc-C(HG)	MES, Kalaikunda, West Bengal
124	ER, Kolkata	Short Term Water Supply Investigation for Coast Gurad Andaman, Port Blair - prepared by Shri Sujit Sarkar, Sc-D(GP)	Coast Guard, Port Blair, A & N Islands
125	ER, Kolkata	Short Term Water Supply Investigation for Military Station, Burdang - prepared by Shri Anirvan Choudhury, Sc-D(HG)	MES, Gangtok, Sikkim
126	ER, Kolkata	Short Term Water Supply Investigation for Military Station, Panagarh - prepared by Shri Sujit Sarkar, Sc-D(GP)	MES, Panagarh, West Bengal
127	ER, Kolkata	Short Term Water Supply Investigation for Military Station, Salugara - prepared by Shri Sandip Bhowal, AHG	MES, Siliguri, West Bengal

Table 11.1 Regionwise List of Reports published in 2022-23

S.No	Region	Title of Report Published	Authors of Report
1	NWR, Chandigarh	Haryana State Hydrochemical Report	Smt. Balinder P. Singh (Sc.-D), Sh. Kiran G. Lale(STA Chemical), Sh. Kuldeep Gopal Bhartariya (Sc.-B), Sh. Rishi Raj (Assistant Chemist)
2	NWR, Chandigarh	Punjab State Hydrochemical Report	Smt. Balinder P. Singh (Sc.-D), Sh. Rishi Raj (Assistant Chemist), Sh. Kuldeep Gopal Bhartariya (Sc.-B), Sh. Kiran G. Lale(STA Chemical)
3	NWR, Chandigarh	Chandigarh UT Hydrochemical Report	Smt. Balinder P. Singh (Sc.-D), Sh. Rishi Raj (Assistant Chemist)
4	NWR, Chandigarh	Haryana State Hydrogeological Report	Smt. Amandeep Kaur, Sc. C
5	NWR, Chandigarh	Punjab State Hydrogeological Report	Sh. Rakesh Rana Sc-D Ms. Naima Akhtar Sc-C
6	NWR, Chandigarh	Chandigarh UT Hydrogeological Report	Sh. Dinesh Tewari, Sc. E Dr. Kriti Mishra, Sc. B
7	NWR, Chandigarh	Haryana State Geophysical Report	Mrs. Savita , AGP
8	NWR, Chandigarh	Punjab State Geophysical Report	Sh. Sunil Kumar, STA-HG
9	NWR, Chandigarh	Chandigarh UT Geophysical Report	Sh. Shailesh Bhatnagar, Sc-D
10	NWR, Chandigarh	Punjab State Exploration Report	Ms. Naima Akhtar Sc-C
11	NWR, Chandigarh	Chandigarh UT Exploration Report	Sh. Manish Shrivastav AHG

SNo	Region	Title of Report Published	Authors of Report
12	NWR, Chandigarh	Ground Water Year Book of Haryana State	Ms. Amandeep Kaur (Sc.B, HG), Ms. Gargi Walia (Sc. B, HG), Sh. Aditya Sharma, (Sc. B, Hydrometerologist) Sh. Kiran Lale (STA, Chemical)
13	NWR, Chandigarh	Ground Water Year Book of Punjab State and Chandigarh UT	Ms. Amandeep Kaur (Sc.B, HG), Ms. Gargi Walia (Sc. B, HG), Sh. Aditya Sharma, (Sc. B, Hydrometerologist) Sh. Rishi Raj (Assistant Chemist)
14	NWR, Chandigarh	Ground Water Resources Estimation Report 2022 of Haryana State	CGWB, NWR, Chandigarh Ground Water Cell, Irrigation & Water Resource Department, Haryana
15	NWR, Chandigarh	Ground Water Resources Estimation Report 2022 of Chandigarh UT	Chandigarh Administration & Municipal Corporation, Chandigarh and CGWB, NWR, Chandigarh
16	NWR, Chandigarh	Ground Water Resources Estimation Report 2022 of Punjab State	Ground Water Management Circle, Water Resources Department, Punjab S.A.S Nagar and CGWB, NWR, Chandigarh
21	NCCR, Raipur	Ground Water Resources Estimation Report 2022 of Chhattisgarh State	Sh. Uddeshya Kumar, Sc-C, Sh. B. Abhishek, Sc-C, Sh. Mukesh Anand, Sc-B &Ms.Gurpreet Kour, Sc-B
22	NCCR, Raipur	Chhattisgarh Exploration Report	Sh. Uddeshya Kumar, Sc-C
23	NCCR, Raipur	Ground Water Year Book of Chhattisgarh State	Ms. Priyanka B.Sonbarse, Sc-C & Ms. Sweta Mohanty, AHG
24	NHR Dharamshala	Ground Water Year Book of Himachal Pradesh 2022	Smt. Rachna Bhatti, Scientist-D, Sh. Prasant Kumar Singh, STA (Hg)
25	NHR Dharamshala	Ground Water Resources Estimation Report 2022 of Himachal Pradesh	Sh. Sanjay Pandey, Scientist-B, Sh. Prasant Kumar Singh, STA (Hg)
26	NHR Dharamshala	Himachal Pradesh State Hydrogeological Report	Sh. Sanjay Pandey, Scientist-B, Sh. Manohar Kumar, AHG

SNo	Region	Title of Report Published	Authors of Report
27	NHR Dharamshala	Himachal Pradesh State Exploration Report	Sh. Sanjay Pandey, Scientist-B, Sh. Devinder Kumar, AHG
29	UR Dehradun	Groundwater Year Book of Uttarakhand State 2021-22	Ms. Anjali Kushwaha, Scientist- B, Sh. Mohd. Arshad, AHG
30	UR Dehradun	Ground Water Resources Estimation Report 2022 of Uttarakhand State	Ms. Chandreyee De, Scientist-C
31	UR Dehradun	Hydrochemical report of Uttarakhand State	Ms. Anjali Kushwaha, Scientist- B
32	UR Dehradun	Uttarakhand State Exploration Report	Sh. Debojyoti Mondal, Scientist-B
34	NWHR Jammu	Ground Water Year Book 2022	Sh. Rayees Ahmad Pir, AHG & Sh. K.S. Rawat, Sc-C (Chemical)
35	NWHR Jammu	Ground Water Resources of UT of J&K (As on 31.03.2022)	Sh. Rayees Ahmad Pir, AHG & Sh. Naresh Singh Barti, AHG
36	NWHR Jammu	Ground Water Resources of UT of Ladakh (As on 31.03.2022)	Sh. Parvej Ahmad, Sc-B & Sh. Kush Kumar, STA(Hg)
37	NWHR Jammu	Ground Water Exploration Report of UT of J&K	Sh. M.L. Angurala, Sc-D, Sh. Mohd Abid Khan, AHG & Sh. Nikhil Kumar, AHG
38	NWHR Jammu	Ground Water Exploration Report of UT of Ladakh	Sh. M.L. Angurala, Sc-D, Sh. Mohd Abid Khan, AHG & Sh. Nikhil Kumar, AHG
39	NWHR Jammu	State Hydrochemical Report of UT of J&K	Sh. K.S. Rawat, Sc-C, Sh. Nimay Kashyap, ACH and Sh. Partha Pratim Mondal, STA(Ch)
40	NWHR Jammu	State Hydrochemical Report of UT of J&K	Sh. K.S. Rawat, Sc-C, Sh. Nimay Kashyap, ACH and Sh. Partha Pratim Mondal, STA(Ch)
41	NWHR Jammu	State Geophysical Report of UT of J&K	Sh. Gulshan Kumar, STA(GP)
42	NWHR Jammu	State Geophysical Report of UT of Ladakh	Sh. Gulshan Kumar, STA(GP)
43	NWHR Jammu	State Hydrogeological Report of UT of J&K	Sh. Rayees Ahmad Pir, AHG

SNo	Region	Title of Report Published	Authors of Report
44	NWHR Jammu	State Hydrogeological Report of UT of Ladakh	Sh. Rayees Ahmad Pir, AHG
45	NWHR Jammu	Ground Water Quality in Baribrahman-Samba Industrial Cluster	Sh. Naresh Singh Barti, AHG & Sh. Nimay Kashyap, ACH
46	NWHR Jammu	Hydrochemical Studies in Phyand Valley, Leh	Sh. K.S. Rawat, Sc-C, Sh. Sh. Parvej Ahmad, Sc-B, and Sh. Partha Pratim Mondal, STA(Ch)
47	SUO, Delhi	Ground Water Resources Estimation for the year 2021-22	Sh.S.K.Mohiddin, Sc-C, Sh.Saidul Haq, Sc-D, Ms.Shilpi Gupta, Sc-C, Sh.V.Praveen Kumar STA(HG)
48	SUO, Delhi	Ground Water Year Book for the year 2021-22	Sh.S.K.Mohiddin, Sc-C, Sh.Faisal Abrar, AHG, Sh.V.Praveen Kumar STA(HG)
52	NR Lucknow	Aquifer Mapping and Management plan of of Sitapur district	Sh Vidya Bhooshan, AHG
53	NR Lucknow	Aquifer Mapping and Management plan of of Sultanpur district	Sh Vikash Kumar, AHG
56	NR Lucknow	Aquifer Mapping and Management plan of of Lakhimpur Kheri district	Sh Rahul Kumar, AHG
57	NR Lucknow	Aquifer Mapping and Management plan of of Kushinagar district	Sh Mohd. Arshad, AHG, UR Dehradun
58	NR Lucknow	Aquifer Mapping and Management plan of ofKasganj district	ShParvej Ahmed, Sc-B, NWHR Jammu
59	NR Lucknow	Aquifer Mapping and Management plan of of Gonda district	ShDebojyoti Mondal, Sc-B, UR Dehradun
60	NR Lucknow	Aquifer Mapping and Management plan of ofDeoria district	ShEbadur Rehman, Sc-C

SNo	Region	Title of Report Published	Authors of Report
61	NR Lucknow	Aquifer Mapping and Management plan of ofBarabanki district	Smt Richa Gautam, Sc-B
62	NR Lucknow	Aquifer Mapping and Management plan of of Amethi district	Ms. Kriti Mishra Sc-B NWR Chandigarh
63	NR Lucknow	Ground Water Resources Estimation Report 2022 of Uttar Pradesh State	Shri Sujatro Ray Chowdhuri, Sc-C (Hg), Dr.Fakhre Alam, Sc-B (Hg) and Ms.Aditi Singh, AHG
64	NR Lucknow	Ground Water Year Book	Dr.R.K.Prashad, Sc-C (HM) Dr. S. K. Srivastava Sc D(Chem)
65	NR Lucknow	State Geophysical Report	Dr. Shashi Kant Singh, Sc-C (Gp) and Shri Anirudh Singh, AGP
66	NR Lucknow	State Exploration Report	ShS.K.Swaroop, Sc-D (Hg) and Sh Tejas Y Mankikar Sc C(HG)
67	NR Lucknow	State Hydrochemical Report	Dr S.K.Srivastava, Sc D, Smt. Kiran Singh, Sc C, Shri Karam Singh, Sc C, Dr Supriya Singh, Sc B and Dr Madhavai Rajak, Assist Chemist.
68	NER Guwahati	Ground Water Year Book of NER	Ms Mophi Mili, Sc - C, Ms. Dakshina Rabha, Sc -D.
69	NER Guwahati	BDR KarichukPhulani, Majuli district, Assam	Sh. Ebadur Rohman, Sc.-C , Sh. Rajat Gupta, AHG
70	NER Guwahati	BDR of KakorikataMajuli District, Assam	Sh. Ebadur Rohman, Sc.-C Sh. Rajat Gupta, AHG
71	NER Guwahati	BDR of Khalkuri, Binnakandi, Cachar Assam	Ms. Rinku Rani Das, Sc.-B
72	NER Guwahati	BDR of Silchar Polytechnic, Silchar, Cachar Assam	Ms. Rinku Rani Das, Sc.-B
73	NER Guwahati	BDR of Adi Ningsoo Lohit (Undivided), AP	Sh. R.K. Kalita Sc.-D & Sh. D.J. Khound, Sc.-D
74	NER Guwahati	BDR of Wingko EW Lohit (Undivided), AP	Sh. R.K. Kalita Sc.-D & Sh. D.J. Khound, Sc.-D
75	NER Guwahati	BDR of Dengaon, karbi Anglong, Assam	Sh. N.K. Jatav, Sc.-C
76	NER Guwahati	BDR of Dillai, Karbi Anglong,	Ms.WonjanoMorhui, Sc.-C

SNo	Region	Title of Report Published	Authors of Report
		Assam	
77	NER Guwahati	BDR of KV Diphu, Karbi Anglong, Assam	Sh. N.K. Jatav, Sc.-C
78	NER Guwahati	BDR of Mirdan, Karbi Anglong, Assam	Sh. N.K. Jatav, Sc.-C
79	NER Guwahati	BDR of Phuloni, Krbi Anglong, Assam	Sh. N.K. Jatav, Sc.-C
80	NER Guwahati	BDR of Phutoni, Karbi Anglong, Assam	Sh. N.K. Jatav, Sc.-C
81	NER Guwahati	BDR of Taralangso	Sh. N.K. Jatav, Sc.-C
82	NER Guwahati	BDR of Moranhat	Sh. D. Khanikar, Sc-D
83	NER Guwahati	BDR of Nagajanka	Sh. D. Khanikar, Sc-D
84	NER Guwahati	Aquifer Mapping and Management plan of Kamrup district, Assam	Sh. R. K. Kalita, Sc.D, Ms. Mophi Mili, Sc.C
85	NER Guwahati	Aquifer Mapping and Management plan of Golaghat district, Assam	Sh. HimangshuKachari, Sc.-B
86	NER Guwahati	Aquifer Mapping and Management plan of Dhemaji district, Assam	Sh. Ebadur Rahman, Sc-C
87	NER Guwahati	Aquifer Mapping and Management plan of Changlang district, AP	Sh. Rajat Gupta, AHG
88	NER Guwahati	Aquifer Mapping and Management plan of Tirap district, AP	Sh. Rajat Gupta, AHG
89	NER Guwahati	Aquifer Mapping and Management plan of Nagaon district, Assam	Ms. AnenuoPienyu, Sc.-C, Ms. Rinku Rani Das, Sc.-B
90	NER Guwahati	Aquifer Mapping and Management plan of Karbi Anglong district, Assam	Mophi Mili, Sc.-C, Sh. HimangshuKachari, Sc.-B
91	NER Guwahati	Aquifer Mapping and Management plan of Kokrajhar	Ms.H.V.Sophia Kay (Sc-C) , Sh. TiamenbaLongshir, AHG

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		district, Assam	
92	NER Guwahati	Aquifer Mapping and Management plan of Chirang district, Assam	Dr. L. Sardine Varay,Sc.-C, Sh. Rupam Chattaarjee, AHG
93	NER Guwahati	Aquifer Mapping and Management plan of Tinsukia district, Assam	Sh. Bibhuti Bhusan Sahoo, Sc.-C, Sh. Bipul Vishal, STA
94	NER Guwahati	Aquifer Mapping and Management plan of Karimganj district, Assam	Sh. Arijit Mitra (Sc- C), Ms. Rinku Rani Das, Sc.-B
95	NER Guwahati	Aquifer Mapping and Management plan of Bongaigaon district, Assam	Sh. C.K. Gogoi, Sc -C
96	NER Guwahati	GW Exploration report of Assam	Sh. D. Khanikar, Sc-D
97	NER Guwahati	GW Exploration report of Manipur	Sh. S.Kent (Sc - C)
98	NER Guwahati	GW Exploration report of Meghalaya	Ms. Dakshina Rabha (Sc - D)
99	NER Guwahati	GW Exploration report of Tripura	Dr.Raja Ram Purohit (Sc -D), Mrs.Ritu Orang (Sc - C), Ms.H.V.Sophia Kay (Sc-C)
100	NER Guwahati	GW chemical report of Assam	Dr.K.Radhapyari (Sc - C), Sh. Gopal Sahoo (Sc - B), Sh. Rinku Mani Barman (STA)
101	NER Guwahati	GW chemical report of Tripura	Dr.K.Radhapyari (Sc - C), Sh. Gopal Sahoo (Sc - B), Sh. Rinku Mani Barman (STA)
102	NER Guwahati	State Hydrogeological Report, Tripura	Dr.Raja Ram Purohit (Sc -D), Mrs.Ritu Orang (Sc - C), Ms.H.V.Sophia Kay (Sc-C)
103	NER Guwahati	Spring studies and management in Jampui Hills, North Tripura District.	Dr.Raja Ram Purohit (Sc -D), Mrs.Ritu Orang (Sc - C), Ms.H.V.Sophia Kay (Sc-C), Rupam Chattaraj (AHG)

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108	WCR, Ahmedabad	STATE GROUND WATER EXPLORATION REPORT OF GUJARAT STATE, UT OF DADRA & NAGAR HAVELI and DAMAN & DIU 2022 23	Dr.K.M.Nayak, Sc-B (HG)
109	WCR, Ahmedabad	STATE GEOPHYSICAL SURVEY REPORT OF GUJARAT STATE, UT OF DADRA & NAGAR HAVELI and DAMAN & DIU 2022 23	Sh N. Veerabau, Sc- B, (GP)
110	WCR, Ahmedabad	GROUND WATER YEAR BOOK OF GUJARAT STATE, UT OF DADRA & NAGAR HAVELI and DAMAN & DIU 2021-22	Dr Gajanan Ramteke, Sc-B, (HM)
111	WCR, Ahmedabad	REPORT ON DYNAMIC GROUND WATER RESOURCE OF GUJARAT STATE AS ON MARCH 2022	Sh Avinash Chandra, AHG
112	WCR, Ahmedabad	NAQUIM REPORT OF CHOTA UDEPUR DISTRICT GUJARAT	Sh Ramesh Jena, Sc-C, (HG)
113	WCR, Ahmedabad	NAQUIM REPORT OF BHARUCH DISTRICT GUJARAT	Sh Ankit Vishwakarma, Sc- C, (HG)
114	WCR, Ahmedabad	NAQUIM REPORT OF MORBI DISTRICT GUJARAT	Sh Ankit Vishwakarma, Sc- C, (HG)
115	WCR, Ahmedabad	NAQUIM REPORT OF ANAND DISTRICT GUJARAT	Sh Himesh Pandya, Sc- C, (HG)
116	WCR, Ahmedabad	NAQUIM REPORT OF JAMNAGAR DISTRICT GUJARAT	Sh Kamar Ujjam Khan, Sc-B (HG)
117	WCR, Ahmedabad	NAQUIM REPORT OF SURENDRANAGAR DISTRICT GUJARAT	Dr Subhash Singh, AHG
118	WCR, Ahmedabad	NAQUIM REPORT OF BHAVNAGAR DISTRICT GUJARAT	Sh Shah Izhar Ahmed, AHG
119	WCR, Ahmedabad	NAQUIM REPORT OF BOTAD DISTRICT GUJARAT	Sh Saddam Husain, AHG

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120	WCR, Ahmedabad	NAQUIM REPORT OF AMRELI DISTRICT GUJARAT	Sh Avinash Chandra, AHG
121	WCR, Ahmedabad	NAQUIM REPORT OF DEVBHUMI DWARKA DISTRICT GUJARAT	Sh Abdurrahman, AHG
122	WCR, Ahmedabad	NAQUIM REPORT OF VALSAD DISTRICT GUJARAT	Sh Kamar Ujjam Khan, Sc-B (HG)
123	WCR, Ahmedabad	NAQUIM REPORT OF SURAT DISTRICT GUJARAT	Sh Saddam Husain, AHG
124	WCR, Ahmedabad	NAQUIM REPORT OF DANG DISTRICT GUJARAT	Sh Shah Izhar Ahmed, AHG
125	WCR, Ahmedabad	NAQUIM REPORT OF TAPI DISTRICT GUJARAT	Sh Abdurrahman, AHG
126	WCR, Ahmedabad	NAQUIM REPORT OF NAVSARI DISTRICT GUJARAT	Dr Subhash Singh, AHG
127	WCR, Ahmedabad	NAQUIM REPORT OF UT OF DAMAN DISTRICT GUJARAT	Sh Kamar Ujjam Khan, Sc-B (HG)
128	WCR, Ahmedabad	NAQUIM REPORT OF UT OF DIU DISTRICT GUJARAT	Sh Avinash Chandra, AHG
129	WR, Jaipur	Aquifer mapping and Management Plan of Tonk district	Smt. Priya Kanwar, Sc-D(hydrogeology)
130	WR, Jaipur	Dynamic Ground Water Resources of Rajasthan-2023	Smt. Preeti Pandey, Sc-C(hydrogeology)
131	WR, Jaipur	GROUND WATER POLLUTION STUDIES IN PALI INDUSTRIAL CLUSTURE, RAJASTHAN	Ms Balinder P. Singh,Scientist D (Senior Chemist) Dr Jaipal Garg, Sc.B (Chemist), Ms Reena Borana (AHG)
132	WR, Jaipur	DISTRIBUTION OF URANIUM IN GROUND WATER OF AFFECTED DISTRICTS OF RAJASTHAN	Ms Balinder P. Singh,Scientist D (Senior Chemist) Dr Jaipal Garg, Sc.B (Chemist), Dr. R.K. Kushwaha, Sc-E
133	SER Bhubaneshwar	Ground Water Year Book, 2021	Mausumi Sahoo, Scientist-C, Purba Bera, Scientist-B

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134	SER Bhubaneshwar	Dynamic Ground Water Resources, State of Odisha (2022)	B.K.Sahoo, Scientist-E, Mausumi Sahoo, Scientist-C, Purba Bera, Scientist-B
135	SER Bhubaneshwar	Aquifer Mapping of Rayagada District	Sumita Sarkar, Scientist-D, Purba Bera, Scientist-B
136	SER Bhubaneshwar	Aquifer Mapping of Kalahandi District	S.K.Mohanty, Scientist-D, S.K.Naik, Scientist-B
137	SER Bhubaneshwar	Aquifer Mapping of Nowrangpur District	S.K.Naik, Scientist-B, Munna Sutradhar, Asst.HG
138	SER Bhubaneshwar	Aquifer Mapping of Malkangiri District	S.K.Mohanty, Scientist-D, Rajeev Tripathy, Scientist-C
139	SER Bhubaneshwar	Aquifer Mapping of Kandhamal District	Tarun Mishra, Scientist-D, Purba Bera, Scientist-B
140	SER Bhubaneshwar	Aquifer Mapping of Nayagarh District	Mausumi Sahoo, Scientist-C
141	SER Bhubaneshwar	Aquifer Mapping of Bolangir District	S.K.Mohanty, Scientist-D,
142	SER Bhubaneshwar	Aquifer Mapping of Kendrapara District	G.Prasad, Scientist-D(Retd)
143	SER Bhubaneshwar	Aquifer Mapping of Dhenkanal District	S.K.Mohanty, Scientist-D,
144	SER Bhubaneshwar	Aquifer Mapping of Nuapada District	S.K.Mohanty, Scientist-D, Rajeev Tripathy, Scientist-C
145	SER Bhubaneshwar	Aquifer Mapping of Bargarh District	Rajeev Tripathy, Scientist-C
146	SER Bhubaneshwar	89 Nos. of Basic Data Reports	
147	SER Bhubaneshwar	Angul –Talcher-Meramundali Pollution Cluster Report	B.B.Sahoo, Scientist-D (Retd), S.K.Mohanty, Scientist-D
148	SECR, Chennai	Ground Water Year Book Tamil Nadu and UT of Puducherry	Dr.K.Ravichandran,SC-D, S.Pramanayagam,SC-D, Athira R, SC-B
149	SECR, Chennai	State Hydrogeological Report of Tamil Nadu and UT of Puducherry (Under Scrutiny)	Smt. Mini Chandran, Scientist-D(HG)

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150	SECR, Chennai	State Hydrochemical Report of Tamil Nadu and UT of Puducherry (Under Scrutiny)	Dr. K. Ravichandran, SC-D (Chem) OIC,Smt. K. Padmavathi, SC-B (Chem),Smt. Bindu J. Viju, SC-B (Chem),Shri. T. Dinesh Kumar, SC-B, Shri. K. Venkatesh, STA (Chem)
151	SECR, Chennai	State Exploration Report of Tamil Nadu and UT of Puducherry (Under Scrutiny)	Shri. S. Piramanayagam, Scientist – D,Smt. R. Athira, Scientist - B, Dr. K Paramashivam, AHG
152	SECR, Chennai	Aquifer Mapping and Management plan of Pambar-Kottakaraiyar Aquifer system	Dr. M Panneer, SC-B
153	SECR, Chennai	Aquifer Mapping and Management plan of Vaippar River Basin	Dr. K Rajarajan, SC-B
154	SECR, Chennai	Aquifer Mapping and Management plan of Gundar Basin	Mrs.Mini Chandran, SC-D
155	SECR, Chennai	Aquifer Mapping and Management plan of Tambraparni Aquifer system	Dr.N Ramesh Kumar, SC-B
156	SECR, Chennai	Aquifer Mapping and Managemet plan of Kallar Aquifer system	Mrs.Mini Chandran, SC-D
157	SECR, Chennai	Aquifer Mapping and Management plan of Nambiyar Aquifer system	Dr.N Ramesh Kumar, SC-B
158	SECR, Chennai	Aquifer Mapping and Management plan of Parambikulam Aliyar Aquifer System	Dr.MPanneer,SC-B
159	SECR, Chennai	Aquifer Mapping and Management plan of Kodayar Aquifer system	Sh. V.Elanchelian,SC-D and Sh.K.Ramanand, SC-D
160	SECR, Chennai	Aquifer Mapping and Management plan of Agniyar Aquifer system	Sh.A.Ravi, SC-D

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161	KR Kerala	37 Basic data Reports of Exploratory wells	
162	KR Kerala	Aquifer Mapping and Management plan of ofPathanamthitta district	Dr. S Sakthi Murugan, Sci D
163	KR Kerala	Aquifer Mapping and Management plan of of Malappuram district	Shri. Santana Subramaniam Sci D
164	KR Kerala	Aquifer Mapping and Management plan of of Trivandrum district	Smt. Aneesha K. SciC
165	KR Kerala	Aquifer Mapping and Management plan of of Kannur district	Shri. Vijesh V.K. Sci C
166	KR Kerala	Aquifer Mapping and Management plan ofof Kasargod district	S. Singadurai, Sci C
167	KR Kerala	NAQUIM Report of Kozhikode district	Shri. Roopesh G K Sci C
168	KR Kerala	Dynamic Ground water Resource estimation of Kerala 2022	T.S. Anitha Shyam, Regional Director; Roopesh G Krishnan, Sc-C; S. Singathurai, Sc-C; Rakhi U.R, Sc-C; Saritha S, Sc-D; Anu V, Sc-B; Arun Kumar A.V, AHG; Bindhu S (Superintending Hg (GWD, Kerala), Dr. Santhosh N(Sr. Hg; GWD, Kerala); Dr.AnseenaBeegom A S (Hydrogeologist, GWD, Kerala), Santy S R (Hydrogeologist, GWD, Kerala); Bushra A (Jr. Hydrogeologist, GWD, Kerala)
169	KR Kerala	Dynamic Ground water Resource estimation of Lakshadweep 2022	M Santhana Subramani, Scientist D Vijesh V K, Scientist C Anisha K, Scientist C

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170	KR Kerala	Special study report on the flouride contamination in the Ground water of Attappadi area of the Palakkad district	Shri. Arun Kumar AHG
171	KR Kerala	Groundwater Exploration Report Kerala	Shri. Roopesh G.K Sci C, Shri. Aneesh Kumar, SciC and Shri. Anu V. Sci C
172	KR Kerala	Annual General Report CGWB Kerala Region 2021-22	Technical Section
173	SWR, Banglore	Groundwater Year book of Goa:2021-22	Jayaprakash H P (Sc-D), Deepa Gupta (Sc-B), Lubna Kouser (AHL), Hemalatha V (AHG)
174	SWR, Banglore	Groundwater Year book of Karnataka:2021-22	Jayaprakash H P (Sc-D), Deepa Gupta (Sc-B), Lubna Kouser (AHL), Hemalatha V (AHG)
175	SWR, Banglore	Ground water Resource Assessment of Karnataka-2022	Ground Water Directorate Minor Irrigation & Ground Water Development Department Govt. of Karnataka and CGWB, SWR, Bangalore
176	SWR, Banglore	Ground water Resource Assessment of Goa-2022	Water Resource Department, Govt of Goa and CGWB, SWR, Bangalore
177	SWR, Banglore	Report on Periodical Measurements of Groundwater Monitoring Stations in Goa State (For the months of August 2021, November 2021 & January 2022)	Jayaprakash H P (Sc-D), Deepa Gupta (Sc-B), Lubna Kouser (AHL), Hemalatha V (AHG)
178	SWR, Banglore	Aquifer Maps and Management Plan, Khanapur Taluk, Belagavi District, Karnataka State	A. Suresha, Sci-D
179	SWR, Banglore	Aquifer Maps and Management Plan, Belagavi Taluk, Belagavi District, Karnataka State	A. Suresha, Sci-D
180	SWR, Banglore	Aquifer Maps and Management Plan, Kollegal Taluk, Chamarajanagara District,	Dr S SVittala, Sc-B

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		Karnataka State	
181	SWR, Banglore	Aquifer Maps and Management Plan, Chincholi Taluk, Kalburgi District, Karnataka State	Dr.J.Davithuraj, Sc-C
182	SWR, Banglore	Aquifer Maps and Management Plan, Chamarajanagara Taluk, Chamarajanagara District, Karnataka State	Dr S SVittala, Sc-B
183	SWR, Banglore	Aquifer Maps and Management Plan, Channapatna Taluk, Ramanagara District, Karnataka State	S.Saritha, Sc-B
184	SWR, Banglore	Aquifer Maps and Management Plan, Bidar Taluk, Bidar District, Karnataka State	Dr.J.Davithuraj, Sc-C
185	SWR, Banglore	Aquifer Maps and Management Plan, Bangalore East Taluk, Bangalore Urban District, Karnataka State	S.Saritha, Sc-B
186	SWR, Banglore	Aquifer Maps and Management Plan, Bangalore North Taluk, Bangalore Urban District, Karnataka State	Sangeeta P. Bhattacharjee, Sc - B
187	SWR, Banglore	Aquifer Maps and Management Plan, Mysore Taluk, Mysore District, Karnataka State	Dr S SVittala, Sc-B
188	SWR, Banglore	Aquifer Maps and Management Plan, Kanakapura Taluk, Ramanagara District, Karnataka State	Rakhi U R, Sc-B
189	SWR, Banglore	Aquifer Maps and Management Plan, Hubli Taluk, Dharwad District, Karnataka State	A. Suresha, Sci-D
190	SWR, Banglore	Aquifer Maps and Management Plan, Channarayapatna Taluk,	D.Dayamalar, Sc-E

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		Hassan District, Karnataka State	
191	SWR, Banglore	Aquifer Maps and Management Plan, Bhalki Taluk, Bidar District, Karnataka State	Dr.J.Davithuraj, Sc-C
192	SWR, Banglore	Aquifer Maps and Management Plan, Malavalli Taluk, Mandya District, Karnataka State	Sangeeta P. Bhattacharjee, Sc - B
193	SWR, Banglore	Aquifer Maps and Management Plan, Bailahongal Taluk, Belagavi District, Karnataka State	Dr.J.Davithuraj, Sc-C
194	SWR, Banglore	Aquifer Maps and Management Plan, Tumkur Taluk, Tumkur District, Karnataka State	Rakhi U R, Sc-B
195	SWR, Banglore	Aquifer Maps and Management Plan, Aurad Taluk, Bidar District, Karnataka State	Dr.J.Davithuraj, Sc-C
196	SWR, Banglore	Aquifer Maps and Management Plan, Shirahatti Taluk, Gadag District, Karnataka State	Dr.J.Davithuraj, Sc-C
197	SWR, Banglore	Aquifer Maps and Management Plan, Krishnarajapete Taluk, Mandya District, Karnataka State	Deepa Gupta, Sc-B
198	SWR, Banglore	Aquifer Maps and Management Plan,Sandur Taluk, Bellary District, Karnataka State	Sangeeta P. Bhattacharjee, Sc - B
199	SWR, Banglore	Aquifer Maps and Management Plan, Siruguppa Taluk, Bellary District, Karnataka State	Sangeeta P. Bhattacharjee, Sc - B
200	SWR, Banglore	Aquifer Maps and Management Plan, Alur Taluk, Hassan District, Karnataka State	D.Anjaneyulu, AHG
201	SWR, Banglore	Aquifer Maps and Management Plan, Belur Taluk, Hassan District, Karnataka State	D.Anjaneyulu, AHG

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203	SWR, Banglore	Aquifer Maps and Management Plan, PandavapuraTaluk,Mandya District, Karnataka State	Dr.Suchetana Biswas, Sc-B
204	SWR, Banglore	Aquifer Maps and Management Plan, SrirangapatnaTaluk,Mandya District, Karnataka State	Dr.Suchetana Biswas, Sc-B
205	SWR, Banglore	Aquifer Maps and Management Plan, T Narasipura Taluk, Mysore District, Karnataka State	V.Hemalatha, AHG
206	SWR, Banglore	Aquifer Maps and Management Plan, Piriyapatna Taluk, Mysore District, Karnataka State	V.Hemalatha, AHG
207	SWR, Banglore	Aquifer Maps and Management Plan, Yelandur Taluk, Chamarajanagara District, Karnataka State	V.Hemalatha, AHG
208	SWR, Banglore	Aquifer Maps and Management Plan, Gundlupete Taluk, Chamarajanagara District, Karnataka State	V.Hemalatha, AHG
209	SWR, Banglore	Aquifer Maps and Management Plan, Byadgi Taluk, Haveri District, Karnataka State	Dr.Suchetana Biswas, Sc-B
210	SWR, Banglore	Aquifer Maps and Management Plan, Kundgol Taluk, Dharwad District, Karnataka State	A. Suresha, Sc-D
211	SWR, Banglore	Aquifer Maps and Management Plan, Nagamangala Taluk, Mandya District, Karnataka State	Deepa Gupta, Sc-B
212	SWR, Banglore	Aquifer Maps and Management Plan, Molkalamuru Taluk, Chitradurga District, Karnataka State	Dr.Suchetana Biswas, Sc-B

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214	SWR, Banglore	Aquifer Maps and Management Plan, Nanjangud Taluk, Mysuru District, Karnataka State	V.Hemalatha, AHG
215	SWR, Banglore	Aquifer Maps and Management Plan, Gubbi Taluk, TumkurDistrict, Karnataka State	N.C.Sahoo, AHG
216	SWR, Banglore	Aquifer Maps and Management Plan, Kunigal Taluk, TumkurDistrict, Karnataka State	N.C.Sahoo, AHG
217	SWR, Banglore	Aquifer Maps and Management Plan, Turuvekere Taluk, TumkurDistrict, Karnataka State	N.C.Sahoo, AHG
218	SWR, Banglore	Aquifer Maps and Management Plan, Hunsur Taluk, Mysore District, Karnataka State	Abdul Rajik Khan, AHG
219	SWR, Banglore	Aquifer Maps and Management Plan, Heggadadevanakote Taluk, Mysore District, Karnataka State	Abdul Rajik Khan, AHG
220	SWR, Banglore	Aquifer Maps and Management Plan, Hospet Taluk, Bellary District, Karnataka State	Sangeeta P. Bhattacharjee, Sc - B
221	SWR, Banglore	Aquifer Maps and Management Plan, Maddur Taluk, Mandya District, Karnataka State	S.Saritha, Sc-B
222	SWR, Banglore	Aquifer Maps and Management Plan, Hangal Taluk, Haveri District, Karnataka State	Sushanth S. Navarat, AHG
223	SWR, Banglore	Aquifer Maps and Management Plan, Savanur Taluk, Haveri District, Karnataka State	Sushanth S. Navarat, AHG
224	SWR, Banglore	Aquifer Maps and Management Plan, Shiggaon Taluk, Haveri District, Karnataka State	Sushanth S. Navarat, AHG

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226	SWR, Banglore	Aquifer Maps and Management Plan, Arkalgud Taluk, Hassan District, Karnataka State	A.Sakthivel, Sc-B
227	SWR, Banglore	Aquifer Maps and Management Plan, Holenarasipura Taluk, Hassan District, Karnataka State	A.Sakthivel, Sc-B
228	SWR, Banglore	Aquifer Maps and Management Plan, Kalburgi Taluk, Kalburgi District, Karnataka State	Bijimol Jose, Sc-C
229	SWR, Banglore	Aquifer Maps and Management Plan, Bangalore South Taluk, Bangalore Urban District, Karnataka State	Bijimol Jose, Sc-C
230	SWR, Banglore	Aquifer Maps and Management Plan, Anekal Taluk, Bangalore Urban District, Karnataka State	Bijimol Jose, Sc-C
231	SWR, Banglore	Aquifer Maps and Management Plan, Dharwad Taluk, Dharwad District, Karnataka State	Bijimol Jose, Sc-C
232	SWR, Banglore	Aquifer Maps and Management Plan, Chitapur Taluk, Kalburgi District, Karnataka State	Bijimol Jose, Sc-C
233	SWR, Banglore	Hydrogeological Investigation for Water Supply at DRDO-Electronics & Radar Development Establishment in S1 Site Ramagondanahalli& S2 Site TimmasandraVillages , Yelahanka Taluk, Bangalore Urban District.	Principal author:Damera Anjaneyulu, AHG,Co-author(s):Dr.Ananth Kumar Ars, Sc-D & Mayank Dixit, Sc-B
234	SWR, Banglore	Hydrogeological Investigation for constructing Borewells for Water Supply at ICGAE, Chicolna, Vasco, Goa State	Principal author:Dr.J.Davithuraj, Sc-C,Co-author(s):Sushanth S. Navarat, AHG & Mayank Dixit, Sc-B

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236	SWR, Banglore	Water Supply Investigation for Feasibility of Drilling Borewell at ASC Centre (South) 2 ATC Bangalore, Bangalore South Taluk, Bangalore Urban District.	Principal author:Damera Anjaneyulu, AHG,Co-author(s):Mayank Dixit, Sc-B &N.Narasihma Naidu, Sc-B
237	SWR, Banglore	Water Supply Investigation for Feasibility of Drilling Borewell at HQGNA Gomantak (Naval Area) & NOFRA-II, Vasco, Goa State (AAP: 2021-2022)	Principal author:Dr.J.Davithuraj, Sc-C,Co-author(s):Sushanth S. Navarat, AHG & Mayank Dixit, Sc-B
238	SWR, Banglore	Water Supply Investigation for Feasibility of drilling Borewell at DRDO Township, Phase-1, CV Raman Nagar, Bangalore North Taluk, Bangalore Urban District, Karnataka (AAP: 2022-2023)	Principal author:Deepa Gupta, Sc-B,Co-author(s): Mayank Dixit, Sc-B &N.Narasihma Naidu, Sc-B
239	SWR, Banglore	Water Supply Investigation for Feasibility of drilling Borewell at DRDO Township, Phase-2, CV Raman Nagar, Bangalore North Taluk, Bangalore Urban District, Karnataka (AAP: 2022-2023)	Principal author:Deepa Gupta, Sc-B,Co-author(s): Mayank Dixit, Sc-B &N.Narasihma Naidu, Sc-B
240	SWR, Banglore	Water Supply Investigation for Feasibility of Drilling Borewell at DRDO Site-2, Rajanukunte, Yelahanka, Bangalore North Taluk, Bangalore Urban District, Karnataka (AAP: 2022-2023)	Principal author:Deepa Gupta, Sc-B,Co-author(s): Mayank Dixit, Sc-B &N.Narasihma Naidu, Sc-B
241	SWR, Banglore	Water Supply Investigation for Feasibility of Drilling Borewell at Airmen Training School (ATS), Tilakwadi, Belagavi, Karnataka State (AAP: 2022-2023)	Principal author: Sushanth S. Navarat, AHG,Co-author(s):T.S.N.Murthy Sc-B & Mayank Dixit, Sc-B

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243	SR Hyderabad	Andhra Pradesh State Hydrochemical Report (Under scruitny)	Y. Satya Kumar, Sc-B , Susanta Kumar Ratha, Sc-B, Swati. D., Asst. Chem
244	SR Hyderabad	Telangana State HydrochemicalReport (Under scruitny)	Y. Satya Kumar, Sc-B , Susanta Kumar Ratha, Sc-B, Swati. D., Asst. Chem
245	SR Hyderabad	Andhra Pradesh State Hydrogeological Report (Under scruitny)	Ravi Kumar Gumma, Sc-D, Md. Sarif Khan, Sc-C
246	SR Hyderabad	Telangana State Hydrogeological Report (Under scruitny)	Rani V.R., Sc-D, Caroline Louis, Sc-C ,Resma S Pillai, Sc-C
247	SR Hyderabad	Andhra Pradesh State Geophysical Report (Under scruitny)	P. Raghavendra,Sc-B , T. Venkatagiri, Sc-B, A. Rama Devi, AGP
248	SR Hyderabad	Telangana State Geophysical Report (Under scruitny)	P. Raghavendra,Sc-B , T. Venkatagiri, Sc-B, A. Rama Devi, AGP
249	SR Hyderabad	Ground Water Year Book of Andhra Pradesh State	Bijay K Mohanta, Sc- C , Monika, Sc-C
250	SR Hyderabad	Ground Water Year Book of Telangana State	Bijay K Mohanta, Sc- C, Monika, Sc-C
251	SR Hyderabad	Ground Water Resources Estimation Report 2022 of Andhra Pradesh State	Ravi Kumar Gumma, Sc-D, L.N.Damodar,Sc-C, Md. Sarif Khan, Sc-C
252	SR Hyderabad	Ground Water Resources Estimation Report 2022 of Telangana State	Rani V.R., Sc-D, Caroline Louis, Sc-C ,Resma S Pillai, Sc-C

SNo	Region	Title of Report Published	Authors of Report
253	SR Hyderabad	Aquifer Mapping and Management plan of of Nalgonda district, Telangana state	Dr.S.S.Vittala,Sc-B
254	SR Hyderabad	Aquifer Mapping and Management plan of ofYadadriBhuvanagiri district, Telangana state	Dr.S.S.Vittala,Sc-B
255	SR Hyderabad	Aquifer Mapping and Management plan of ofSuryapet district, Telangana state	Dr.S.S.Vittala,Sc-B
256	SR Hyderabad	Aquifer Mapping and Management plan of ofMulugu district, Telangana state	Bijay Ketan Mohanta,Sc-c
257	SR Hyderabad	Aquifer Mapping and Management plan of of Jayashankar Bhupalpally district, Telangana state	Bijay Ketan Mohanta,Sc-c
258	SR Hyderabad	Aquifer Mapping and Management plan of ofHanumakonda district, Telangana state	Monika,Sc-C
259	SR Hyderabad	Aquifer Mapping and Management plan of of Warangal (Rural) district, Telangana state	Monika,Sc-C
260	SR Hyderabad	Aquifer Mapping and Management plan of ofJangaon district, Telangana state	M.S. Goutham, Sc-B
261	SR Hyderabad	Aquifer Mapping and Management plan of ofMahabubabad district, Telangana state	P. Yadaiah,Sc-B
262	SR Hyderabad	Aquifer Mapping and Management plan of of Karimnagar district, Telangana state	DibakarMohanta,Sc-D

SNo	Region	Title of Report Published	Authors of Report
263	SR Hyderabad	Aquifer Mapping and Management plan of ofPeddapalli district, Telangana state	Dr. Sudheer Kumar,Sc-D
264	SR Hyderabad	Aquifer Mapping and Management plan of ofJagityal district, Telangana state	U.V. Dhonde, Sc-C
265	SR Hyderabad	Aquifer Mapping and Management plan of of Rajanna Sircilla district, Telangana state	T. Madhav,Sc-C
266	SR Hyderabad	Aquifer Mapping and Management plan of of SPS Nellore district, Andhra Pradesh state	L.N Damodar, Sc-C
267	SR Hyderabad	Aquifer Mapping and Management plan of of Prakasam district, Andhra Pradesh state	Md.SharifKhan,Sc-C
268	SR Hyderabad	Aquifer Mapping and Management plan of of Nandyal district, Andhra Pradesh state	Atharwa S. Pawar,Sc-B
269	SR Hyderabad	Aquifer Mapping and Management plan of of Kurnool district, Andhra Pradesh state (Under scruitny)	Dr. K. Rajarajan, Sc- D
270	SR Hyderabad	Aquifer Mapping and Management plan of of Srikakulam district, Andhra Pradesh state	Nilima Patra, Asst.Hg
271	SR Hyderabad	Aquifer Mapping and Management plan of of Tirupati district, Andhra Pradesh state	Dr.SakthiMurugan,Sc-D
272	SR Hyderabad	Ground Water Information Booklet of MahabubnagarDistrict,Telangana State	Rani V.R., Sc-D

SNo	Region	Title of Report Published	Authors of Report
273	SR Hyderabad	Ground Water Information Booklet of Adilabad District,Telangana State	T. Madhav,Sc-C
274	SR Hyderabad	Ground Water Information Booklet of Nirmal District,Telangana State	T. Madhav,Sc-C
275	SR Hyderabad	Ground Water Information Booklet of Wanaparthy District,Telangana State	Swaroop Kalyan Sahoo,Asst.Hg
276	SR Hyderabad	Ground Water Information Booklet of NarayanpetDistrict,Telangana State	Dr.S.S.Vittala,Sc-B
277	SR Hyderabad	Ground Water Information Booklet of NagarkurnoolDistrict,Telangana State	Dr.S.S.Vittala,Sc-B
278	SR Hyderabad	Ground Water Information Booklet of Jogulamba Gadwal District,Telangana State	B.J. Madhusudhan,Sc-B
279	SR Hyderabad	Ground Water Information Booklet of Kumuram Bheem AsifabadDistrict,Telangana State	Resma S. Pillai,Sc-C
280	SR Hyderabad	Ground Water Information Booklet of Mancherial District District,Telangana State	Resma S. Pillai,Sc-C
281	SR Hyderabad	Ground Water Information Booklet of BhadradriKothagudem District District,Telangana State	Caroline Louis,Sc-C
282	ER Kolkata	Ground Water Year Book 2022-23 of West Bengal and Andaman & Nicobar Islands	Chirashree Mohanty, Sc.D, Dr. Shaista Khan, Sc. C, Dr.Neelamoni Burman, Sc.B
283	ER Kolkata	Report on Ground Water Resource Assessment 2022 of West Bengal	Dr. Indranil Roy, Sc.D

SNo	Region	Title of Report Published	Authors of Report
284	ER Kolkata	Report on Ground Water Resource Assessment 2022 of Sikkim	Dr. Indranil Roy, Sc.D, Anirvan Chowdhury, Sc.D, Sandip Bhowal, AHG
285	ER Kolkata	Report on Ground Water Resource Assessment 2022 of Andaman & Nicobar Islands	Dr. Indranil Roy, Sc.D, Anirvan Chowdhury, Sc.D
286	ER Kolkata	Aquifer Mapping and Management plan of West Sikkim district, Sikkim State	Ankita Bhattacharya, Sc.C& Sandip Bhowal, AHG
287	ER Kolkata	Aquifer Mapping and Management plan of parts of Paschim Medinipur &Jhargram districts, West Bengal State	S.M.Hossain, Sc.D
288	ER Kolkata	Aquifer Mapping and Management plan of Malda district, West Bengal State	Debashis Bagchi, Sc.D
289	ER Kolkata	Aquifer Mapping and Management plan of parts of Birbhum district, West Bengal State	Sonam Gautam, Sc.C
290	ER Kolkata	Aquifer Mapping and Management plan of Purulia district, West Bengal State	Anirvan Chowdhury, Sc.D, Monisha Baruah, Sc.C&Zumchilo T. Ezung, Sc.C
291	ER Kolkata	Aquifer Mapping and Management plan of Cooch Behar district, West Bengal State	Rajesh Kumar Sahoo, AHG
292	ER Kolkata	Aquifer Mapping and Management plan of parts of PurbaBardhaman& Paschim Bardhaman districts, West Bengal State	Bibhuti Bhushan Sahu, Sc.B& Tapas Kumar Sahu, AHG
293	ER Kolkata	Aquifer Mapping and Management plan of Jalpaiguri district, West Bengal State	Monisha Baruah, Sc.C
294	ER Kolkata	Aquifer Mapping and Management plan of Alipurduar	Zumchilo T. Ezung, Sc.C

SNo	Region	Title of Report Published	Authors of Report
		district, West Bengal State	
295	ER Kolkata	Aquifer Mapping and Management plan of North & Middle Andaman district, Union Territory of Andaman & Nicobar Islands.	Debashis Bagchi, Sc.D& Awadhesh Kumar, STA (HG)
296	ER Kolkata	Aquifer Mapping and Management plan of Bankura district, West Bengal State	Bibhuti Bhushan Sahu, Sc.B
297	ER, Kolkata	Aquifer Mapping and Management plan of parts of Howrah district, West Bengal State	Naseema Jamal, Sc.C
298	ER, Kolkata	Aquifer Mapping and Management plan of Uttar Dinajpur district, West Bengal State	Debashis Bagchi, Sc.D
299	NCR ,Bhopal	Dynamic Groundwater Resource Assessment 2022 Madhya Pradesh	Chittaranjan Biswal, Sc C
300	NCR ,Bhopal	State Geophysical Report, Madhya Pradesh	Dr. V Arul Prakasham Sc D, Pradip Roy Sc B, Saumya Siddhartha, STA (GP)
301	NCR ,Bhopal	State Exploration Report, Madhya Pradesh	Dr. Rakesh Singh, Sc D
302	NCR ,Bhopal	State Report on Chemical quality of Groundwater in MP	Dr. V.K Kulshrestha Sc C, Rahul Vashistha ACH, Teh Singh ACH, Jitendra Kumar STA
303	NCR ,Bhopal	Annual General Report 2021-22	Ms. Lata Udsaiya, Sc C, Ms Vaishnavi Parihar Sc B
304	NCR ,Bhopal	State Hydrogrologicalreport , Madhya Pradesh	Mrs. Anakha Ajai, Sc C

SNo	Region	Title of Report Published	Authors of Report
305	NCR ,Bhopal	Aquifer Mapping and Management plan of Seoni District MP	B. Abhishek, Sc C
306	NCR ,Bhopal	Aquifer Mapping and Management plan of Burhanpur District MP	SayelliTemburne, Sc C
307	NCR ,Bhopal	Aquifer Mapping and Management plan of Jabalpur District MP	Gargi Walia, Sc C
308	NCR ,Bhopal	Aquifer Mapping and Management plan of Rewa District MP	Nirmal Kumar Nanda, Sc B
309	NCR ,Bhopal	Aquifer Mapping and Management plan of Jhabua District MP	J. Sreemannarayana, STA
310	NCR ,Bhopal	Aquifer Mapping and Management plan of Morena District MP	Kodali Laxman Pradeep, AHG
311	NCR ,Bhopal	Aquifer Mapping and Management plan of Alirajpur District MP	Lata Udsaiya, Sc C
312	NCR ,Bhopal	Aquifer Mapping and Management plan of Katni District MP	Lata Udsaiya, Sc C, Kamlesh Ojha, Sc B
313	NCR ,Bhopal	Aquifer Mapping and Management plan of Sehore District MP	S.K Shrivastava, Sc D
314	NCR ,Bhopal	Aquifer Mapping and Management plan of Berasia block, Bhopal District MP	Lata Udsaiya, Sc C
315	NCR ,Bhopal	Aquifer Mapping and Management plan of Ashoknagar District MP	Naima Akhtar, Sc C
316	NCR ,Bhopal	Aquifer Mapping and Management plan of Harda District MP	Venkatajalapathy, AHG

SNo	Region	Title of Report Published	Authors of Report
317	NCR ,Bhopal	Aquifer Mapping and Management plan of Sheopur District MP	Sumanta Kumar Mohanta, Sc B
318	NCR ,Bhopal	Aquifer Mapping and Management plan of Sidhi District MP	Dr. K. Paramshivam, Sc B
319	NCR ,Bhopal	Aquifer Mapping and Management plan of Umaria District MP	Raj Kishore Mohanty, Sc B
320	CR, Nagpur	Aquifer Maps and Ground Water Management Plan, Ratnagiri District, Maharashtra	Nelofar, Sc. C,, Ms. SayeliTembhurne, Sc. C, Nirmal Kumar Nanda, Sc. B, Prakash C. Maharana, AHg.
321	CR, Nagpur	Aquifer Maps and Ground Water Management Plan, Sindhudurg District, Maharashtra	Sandeep D. Waghmare, Sc-C, Bhagyashree Sahoo, Sc-B
322	CR, Nagpur	Aquifer Maps and Ground Water Management Plan, Palghar District, Maharashtra	Ms. Pournima Barahate, Sc. C, Sh. Ashwin Kumar Atey, Sc. B, Sh. Amarnath, AHG.
323	CR, Nagpur	Aquifer Maps and Ground Water Management Plan, Thane District, Maharashtra	Priti D.Raut, Sc. C, Pawar Srinivas, AHg., V.Venktesam,, STA(Hg)
324	CR, Nagpur	Aquifer Maps and Ground Water Management Plan, Raigad District, Maharashtra	Priti D.Raut, Sc. C, Pawar Srinivas, AHg., V.Venktesam,, STA(Hg)
325	CR, Nagpur	Aquifer Maps and Ground Water Management Plan, Gadchiroli District, Maharashtra.	Pournima Barahate, Sc-C, Ashwin Kumar Atey, Sc-B
326	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Chandrapur District, Maharashtr	Nelofar, Sc-C, Nirmal Kumar Nanda, Sc-B
327	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Yavatmal District, Maharashtra.	Priti Raut, SC-C, Ashwin Kumar Atey, Sc-B
328	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Gondia District, Maharashtra	Nelofar Khan, Sc-C, Abhay Nivsarkar, Sc-D

SNo	Region	Title of Report Published	Authors of Report
329	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Bhandara District, Maharashtra	Nelofar Khan, Sc-C, Nirmal Kumar Nanda, Sc-B
330	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Kolhapur District, Maharashtra	Sandeep D. Waghamare, Sc-C
331	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Sangli District, Maharashtra	Sandeep D. Waghamare, Sc-C
332	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Pune District, Maharashtra.	Anu Radha Bhatia
333	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Dhule District, Maharashtra	Junaid Ahmed, Sc-C
334	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Nashik District, Maharashtra	Junaid Ahmed, Sc-C, Bhagyashree Sahoo, Sc-B
335	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Satara District, Maharashtra	Sandeep D. Waghamare, Sc-C
336	CR, Nagpur	Aquifer Maps and Ground Water Management Plan of Jalgaon District, Maharashtra	Pournima Barahate, Sc-C
337	CR, Nagpur	Status of Ground Water Levels in Maharashtra State During May 2022	Pournima Barahate, Sc-C
338	CR, Nagpur	Annual General Report 2021-22	Ms. Nelofar, Scientist-C
339	CR, Nagpur	Annual General Report 2020-21	Pournima Barahate, Sc-C
340	CR, Nagpur	Ground Water Year Book Maharashtra, 2021-22	Pournima Barahate Sc-C, Nelofar Sc-C, Amarnath (AHG), Dr. P. Kiran Kumar, Sc-B (HM), Dr.Devsharan Verma, Scientist-C (Chem), Dr. R. K. Sharma, Scientist-C (Chem), Dr. Sailee Bhange

SNo	Region	Title of Report Published	Authors of Report
			STA (Chem)
341	CR, Nagpur	Dynamic Ground Water Resources of Maharashtra 2020	Sh. Rahul R Shende, Sc-C, Smt. Nelofar, Sc-C, Ms. Anu Venkitaraman, Sc-B, Sh. Nirmal Kumar Nanda, Sc-B
342	CR, Nagpur	Dynamic Groundwater Resource Assessment 2022, Maharashtra	Sh. Kartik P.Dongre, Scientist D, , Ms.PournimaBarahate, Scientist C,Sh. Amarnath, AssttHydrogeologist,
343	CR, Nagpur	Ground Water Quality Monitoring for Arsenic in Maharashtra And U.T. of Dadra & Nagar Haveli	Dr.Devsharan Verma, Sc-C
344	CR, Nagpur	Comprehensive Report of Impact Study of Water Conservation and Recharge Interventions Being Taken Up In Convergence With MGNREGA In Warud Taluka, District – Amravati, Maharashtra	Priti D. Raut, Sc-C, K P Dongre, Sc-D
345	CR, Nagpur	Status of Ground Water Levels in Maharashtra State During May 2022	
346	CR, Nagpur	90 Nos. Basic Data Report	Pournima Barahate, Sc-C
347	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Araria district of Bihar	Shri Pankaj Kumar, Sc-D
348	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Arwal district of Bihar	Shri Singaren Sandeep Purty, Sc-C
349	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Aurangabad district of Bihar	Shri Singaren Sandeep Purty, Sc-C

SNo	Region	Title of Report Published	Authors of Report
350	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Banka district of Bihar	Shri Sanjib Chakraborty, Sc-D
351	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Bhagalpur district of Bihar	Shri Sanjib Chakraborty, Sc-D
352	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Gaya district of Bihar	Shri Pankaj Kumar, Sc-D
353	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Jamui district of Bihar	Shri Sanjib Chakraborty, Sc-D
354	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Jehanabad district of Bihar	Mrs. Sulekha Bhaya, Sc-C
355	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Kaimur district of Bihar	Shri C.K.Gogoi, Sc-C
356	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Katihar district of Bihar	Shri Singaren Sandeep Purty, Sc-C & Mrs. Sulekha Bhaya, Sc-C
357	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Khagaria (part) district of Bihar	Shri Aneesh Kumar, Sc-C
358	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Kishanganj district of Bihar	Shri Pankaj Kumar, Sc-D

SNo	Region	Title of Report Published	Authors of Report
359	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Lakhisarai (Part) district of Bihar	Shri Sanjib Chakraborty, Sc-D
360	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Madhepura district of Bihar	Mrs. Richa Gautam, Sc-B
361	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Munger (Part) district of Bihar	Shri Sanjib Chakraborty, Sc-D
362	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Nalanda district of Bihar	Mrs. Sulekha Bhaya, Sc-C
363	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Nawada district of Bihar	Shri Aneesh Kumar, Sc-C
364	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Purnia district of Bihar	Shri Singaren Sandeep Purty, Sc-C
365	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Rohtas district of Bihar	Shri Aneesh Kumar, Sc-C
366	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Saharsa district of Bihar	Shri Aneesh Kumar, Sc-C
367	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Sheikhpura district of Bihar	Mrs. Sulekha Bhaya, Sc-C

SNo	Region	Title of Report Published	Authors of Report
368	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Sheohar district of Bihar	Shri Alok Kumar Sinha, Sc-D
369	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Supaul district of Bihar	Shri Sanjib Chakraborty, Sc-D
370	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Vaishali (Part) district of Bihar	Mrs. Sonam, Sc-C
371	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Begusarai (Part) district of Bihar	Shri Pankaj Kumar, Sc-D
372	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Darbhanga district of Bihar	Mrs Siperna Nayak, AHG
373	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of East Champaran district of Bihar	Shri Singaren Sandeep Purty, Sc-C
374	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Gopalganj district of Bihar	Shri Santosh K Sen, AHG
375	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Madhubani district of Bihar	Ms. Divya Kujur, STA (HG)
376	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Muzaffarpur district of Bihar	Shri Alok Kumar Sinha, Sc-D, and Sh. Kamar Ujjam Khan, Sc-B, WCR, Ahmedabad

SNo	Region	Title of Report Published	Authors of Report
377	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Samastipur (Part) district of Bihar	Shri Pankaj Kumar, Sc-D and Sh. Bibhuti Bhusan Sahu, Sc-B, ER, Kolkata
378	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Saran (Part) district of Bihar	Sh. Alok Kumar Sinha, Sc.-D and Mrs. Sonam, Sc-C
379	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Sitamarhi district of Bihar	Shri Santosh Kumar Sen, AHG
380	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of Siwan district of Bihar	Mrs Siperna Nayak, AHG
381	MER, Patna	Aquifer Mapping and Management of Ground Water Resources of West Champaran district of Bihar	Shri Sanjib Chakraborty, Sc-D and Sh. Ankit Vishwakarma, Sc-C, WCR, Ahmedabad
382	MER, Patna	Dynamic Groundwater Resource Assessment -2022, Bihar	Shri Alok Kumar Sinha, Sc-D, Imam Hasan, Sc-B and Sh. Vishal Srivastava, Sc-B
383	MER, Patna	Ground Water Year Book (2021-22), Bihar	Shri Singaren Sandeep Purty, Sc-C
384	MER, Patna	State Geophysical Report, Bihar	Dr. Subrata Das, Sc-C and Sh. Rittik Das, Sc-B
385	MER, Patna	State Hydrochemical Report	Ms. Manashi Bhattacharya, Sc-B, Dr.Shelja Tiwari, ACH and Sh. Anurag Shukla, STA
386	MER, Patna	Basic Data Reports (22 Nos.)	
390	SUO Ranchi	Aquifer Mapping and Management plan of Bokaro district, Jharkhand	Sh. Atul Beck, Sc.-B

SNo	Region	Title of Report Published	Authors of Report
391	SUO Ranchi	Aquifer Mapping and Management plan of Chatra district, Jharkhand	Sh. Atul Beck, Sc.-B
392	SUO Ranchi	Aquifer Mapping and Management plan of Parts of Dhanbad district, Jharkhand	Sh. Atul Beck, Sc.-B
393	SUO Ranchi	Aquifer Mapping and Management plan of Koderma district, Jharkhand	Sh. Atul Beck, Sc.-B
394	SUO Ranchi	Aquifer Mapping and Management plan of Palamu district, Jharkhand	Sh. Atul Beck, Sc.-B
395	SUO Ranchi	Aquifer Mapping and Management plan of Parts of Ranchi district, Jharkhand	Sh. Atul Beck, Sc.-B
396	SUO Ranchi	Aquifer Mapping and Management plan of Deoghar district, Jharkhand	Dr.Anukaran Kujur, Sc.-B
397	SUO Ranchi	Aquifer Mapping and Management plan of Gumla district, Jharkhand	Dr.Anukaran Kujur, Sc.-B
398	SUO Ranchi	Aquifer Mapping and Management plan of Latehar district, Jharkhand	Dr.Anukaran Kujur, Sc.-B
399	SUO Ranchi	Aquifer Mapping and Management plan of Lohardaga district, Jharkhand	Dr.Anukaran Kujur, Sc.-B
400	SUO Ranchi	Aquifer Mapping and Management plan of Parts of West-Singhbhum district, Jharkhand	Sh. Prahlad Das , Sc.-C
401	SUO Ranchi	Aquifer Mapping and Management plan of Hazaribag district, Jharkhand	Smt. Sulekha Bhaya , Sc.-C and Md. Shadman, AHG

SNo	Region	Title of Report Published	Authors of Report
402	SUO Ranchi	Ground Water Year Book, Jharkhand, 2022-23	Smt. Sulekha Bhaya , Sc.-C, Miss Sapna Sakshi, Sc.-B
403	SUO Ranchi	Ground Water Resource Assessment, 2022 of Jharkhand State	Dr.Anukaran Kujur, Sc.-B, Sh. Atul Beck, Sc.-B, Dr. S. Shekhar, Sc.-D
404	SUO Ranchi	GROUND WATER EXPLORATION IN JHARKHAND, (AS ON March 2022)	Prahallad Das, Scientist-'C', Dr. Sudhanshu Shekhar, Scientist-'D', Atul Beck, Asstt.Hydrogeologist
405	SUO Ranchi	Assessment of Fluoride Contamination in Surface and Groundwater and its management in north Garhwa district, Jharkhand	Dr. Sudhanshu Shekhar, Scientist-D, Dr.Anukaran Kujur, Scientist-B, Atul Beck, AHG

CHAPTER 12 TECHCNICAL PAPER

Table 12.1 List of International Papers during 2022-23

Sr No	Author/Authors	Topic of Paper	National Journal	Status- Published/ Submitted
1	Anurag Khanna, Debasish Bagchi	Rejuvenation and Management of Springs in Uttarakhand Himalaya	Bhu-Jal News, Quaterly Journal of Central Ground Water Board, 31(1-4), pp. 25-36	Published
2	Anirudh Singh, KC Mondal, M. Veerababu, Akoju Ramadevi, N. Rao Elisela	'Composite Geoelectrical Investigation to Delineate Groundwater Feasibility in Hard Rock Area of Raipur, Chhattisgarh, India'	International Journal of Quality Control and Standards in Science and Engineering, Vol. 8, Issue 1, March 18, 2022	Published
3	Sanjay Raut and Nitin M. Labhane, Dr Bhushan R. Lamsoge	Rainwater Harvesting for Metro Rail Projects in India — Scope for Efficiency Improvement'	Journal of the Geological Society of India, volume 98, pages937–946 (2022)	Published
4	Priya Kanwar & K. P. Singh	Identification of Suitable Recharge Sites by Integrated Hydro-Geophysical Approach for Augmenting Groundwater in Reodar Block, district: Sirohi, Rajasthan.	Journal of Geological Society of India, Vol. 98, Oct.-2022, 1447-1454, Springer Publication,	
5	Beena Kouser, Anjali Bala, Omkar Verma, Prashant Manthena, Ashu Khosla, Rayees Ahmad Pir	Hydrochemistry for the assessment of groundwater quality in the Kathua region, Jammu and Kashmir, India	Applied Water Science (2022) 12:143, https://doi.org/10.1007/s13201-022-01673-9	
6	B. Abhishek, V N Negi, Sidhanta Kumar, Rayees Ahmad Pir	Aquifer characterization of Quaternary alluvium sequence of Northwest Himalayas	International Journal of Scientific and Engineering Research	

S. No.	Author/Authors	Topic of Paper	National Journal	Status-Published/ Submitted
7	Purba Bera, Sc. B, Tarun Mishra, Sc. D, CGWB, SER, Bhubaneswar.	Aquifer Disposition, Ground Water Issues and Management Strategy of Kandhamal District, Odisha	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
8	Purba Bera, Sc. B, Sumita Sarkar, Sc. D, CGWB, SER, Bhubaneswar.	Disposition of Aquifer System, Ground Water Issues and Management Strategy of Rayagada District, Odisha	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
9	T.Mishra, S.Shukla	Formulation of Artificial Recharge Plan and other water saving methods in Koraput District, Odisha	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
10	S. K. Mohanty and B. N. Dehury,CGWB, SER. Bhubaneswar	Assessment of groundwater quality and variation of fluoride concentration in shallow aquifer of Balangir district,Odisha	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
11	Sh B.N Dehury, ACH and Sh R.K. Tripathy, Sc-C, CGWB, SER. Bhubaneswar	Hydrogeological aspects and Ground water Quality of Nuapada District	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
12	Maya Jha, **Suresh Kumar, T. B. N. Singh, S. K. Srivastava**, Gajendra Kumar Azad and Shahla Yasmin	Potential health risk assessment through the consumption of arsenic-contaminated groundwater in parts of the middle Gangetic plain	Bulletin of the National Research Centre, SPRINGER	Published
13	Kshitindra Kumar Singh, Geeta Tewari, **Suresh Kumar , Ravikalyan Busa,** Aaditya Chaturvedi, Sudarshan Singh Rathore, Ranjeet Kumar Singh, Aabha Gangwar	Understanding urban groundwater pollution in the Upper Gangetic Alluvial Plains of northern India with multiple industries and their impact on drinking water quality and associated health risks	Groundwater for Sustainable Development, ELSEIVIER	Published
14	Kshitindra Kumar Singh, Geeta Tewari,Mamta Bisht, R. K. Tiwary, **Suresh Kumar**, Kiran Patni, Aabha Gangwar & Bhawana Kanyal	Hydrogeochemical characteristics and multivariate statistical approach for monitoring groundwater quality scenario in the vicinity of industrial area of western Himalaya, India	Chemistry and Ecology, TALOR & FRANSIS	Published

S. No.	Author/Authors	Topic of Paper	National Journal	Status-Published/ Submitted
15	Arun Kumar, Pascal Salaun, Suresh Kumar, Akhouri Bishwapriya, Ashok Kumar Ghosh	Assessment of disease burden in the arsenic exposed population of Chapar village of Samastipur district, Bihar, India, and related mitigation initiative	Environmental Science and Pollution Research , Springer	Published
16	Arun Kumar, Akhouri Bishwapriya, Suresh Kumar	Severe Disease Burden and the Mitigation Strategy in the Arsenic-Exposed Population of Kaliprasad Village in Bhagalpur District of Bihar, India	Biological Trace Element Research, Springer	Published
17	Sanjib Chakraborty, Indranil Roy and Suresh Kumar	Estimation of Dry-Season Fluctuation in Specific Yield using Water Budgeting Approach in Bonasuria Micro-watershed of Damodar River Basin, India	Jour. Geol. Soc. India (2022)	Published
18	Somvir Singh, Param Gautam, Tandrila Sarkar, Ajay Kumar Taloor	Characterization of the groundwater quality in Udham Singh Nagar of Kumaun Himalaya, Uttarakhand	Journal of Environmental Earth Sciences, (2022) 81:468, https://link.springer.com/article/10.1007/s12665-022-10579-3 (Impact factor: 3.119).	
19	Somvir Singh	Delineation of Mesozoic Sediments through two-dimensional imaging using the seismic refraction data of Umapur- Brahmanawada (W-E) profile, Central India	Asian Journal of Earth Sciences, ID of 28-AJES (13April 2023) (Under Review)	
20	Mausumi Sahoo, Scientist-C, B K Sahoo, Scientist-E, P Das, Scientist-C,	Groundwater Issues and Management in hard rock terrain of Nayagarh District, Odisha	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
21	Prahallad Das, Scientist-'C' & Mausumi Sahoo, Scientist-'C'	Steps to address Groundwater issues in Jagatsinghpur District, Odisha State	workshop volume on NAQUIM for the state of Odisha, Jharkhand & Bihar.	
22	S Somarendro Singh1, Suresh Chandra Kapil,Biplab Ray, Dhrubajyoti Khanikar, Dipjyoti Khound, Chao Konseng Gogoi, Himangshu Kachari, Dimpi Barpatra, Sunil Toppo,Shivam Prajapati	Success Story of Groundwater Irrigation Schemes Under Pradhan Mantri Krishi Sinchayi Yojana (PMKSY) - Har Khet Ko Paani (HKKP) A Component Of Ground Water (GW) in the North Eastern States of India (Pgae, 14-35)	Volume II Of Workshop on "Success Stories of NAQUIM Studies & PMKSY-HKKP-GW Schemes In North East and East India" : Guwahati , the 24th January 2023; Organized by CGWB,NER, Guwahatiin association with CGWB, ER, Kolkota	Published

S. No.	Author/Authors	Topic of Paper	National Journal	Status-Published/ Submitted
23	Chao Konseng Gogoi,Dhrubajyoti Khanikar,Dimpi Barpatra, S Somarendro Singh, D.J.Khound, Himangshu Kachari	Aquifer Characterization and Ground Water Flow Dynamics in Dibrugarh District, Assam (Page 104-113)	Volume II Of Workshop on "Success Stories of NAQUIM Studies & PMKSY-HKKP-GW Schemes In North East and East India" : Guwahati , the 24th January 2023; Organized by CGWB,NER Guwahatiin association with CGWB, ER, Kolkota	Published
24	Himangshu Kachari1, Dimpi Barpatra2, Dhrubajyoti Khanikar2, **S. Somarendro Singh**3, Chao Konseng Gogoi3	Aquifer characteristics of baksa district of Assam	Volume II Of Workshop on "Success Stories of NAQUIM Studies & PMKSY-HKKP-GW Schemes In North East and East India" : Guwahati , the 24th January 2023; Organized by CGWB,NER Guwahatiin association with CGWB, ER, Kolkota	Published
25	N.Varadaraj, Nelofar, S.K. Bhatnagar , P. C. Maharana	Strategic Management of Ground Water Resources in Basltic Terrain, Terna Sub-Basin, Godavari River Basin, Osmanabad District, Maharashtra	workshop on NAQUIM for the state of MP,CG,UP and Maharashtra	Submiited in RGI for Bhujal News
26	Anu V., Nelofar, Sayeli Tembhurne,	Groundwater Scenario, Delineation of Aquifer, Issues and Management of Amravati District, Maharashtra (India)	workshop on NAQUIM for the state of MP,CG,UP and Maharashtra	Submiited in RGI for Bhujal News
27	Vidyasagar, M.,Mophi Mili, Raja Babu Thammu, Dimpi Barpatra	DELINEATION OF GROUNDWATER POTENTIAL ZONES IN ALLUVIUM-COVERED HARD ROCK TERRAIN IN SONAPUR REVENUE CIRCLE, KAMRUP (M) DISTRICT, ASSAM USING VES AND HYDROGEOLOGICAL STUDIES	Volume II Of Workshop on "Success Stories of NAQUIM Studies & PMKSY-HKKP-GW Schemes In North East and East India" : Guwahati , the 24th January 2023; Organized by CGWB,NER Guwahatiin association with CGWB, ER, Kolkota	Published

S. No.	Author/Authors	Topic of Paper	National Journal	Status- Published/ Submitted
28	S.C.Kapil, Raja Babu Thammu, M. Vidyasagar	ADVANCE FEATURES OF NATURAL GAMMA RAY LOGGING TECHNIQUE IN THE GROUND WATER RESEARCH AND DEVELOPMENT- A CASE STUDY OF AHMEDABAD, MAHASENA AND RAJKOT DISTRICTS, GUJARAT	Volume II Of Workshop on "Success Stories of NAQUIM Studies & PMKSY-HKKP-GW Schemes In North East and East India" : Guwahati , the 24th January 2023; Organized by CGWB,NER Guwahatiin association with CGWB, ER, Kolkota	Published
29	Lata Udsaiya, A.K.Biswal	Delineating Aquifer Geometry, Ground Water Quality Issues and Management of Agar Malwa district, Madhya Pradesh: A case study	National Seminar on Recent Trends in Geosciences and it's application (RTGS - 2023) at Guntur, Andhra Pradesh organized by Department of Geology, Acharya Nagarjuna University.	Published
30	Lata Udsaiya, Vaishnavi Parihar, A.K.Biswal, Rose Anita Kujur	Hydrogeological and Hydrochemical Analysis of Basaltic Terrain of Madhya Pradesh - Assessment and Management Plan	workshop on NAQUIM for the state of MP,CG,UP and Maharashtra	Submitted
31	Dr.Suchetana Biswas	AQUIFER MAPPING STUDIES AT GRAM PANCHAYAT LEVEL, A CASE STUDY FROM KORATAGERE TALUK, TUMKUR DISTRICT, KARNATAKA, INDIA	Special volume on contributing Research Papers- Bhujal News	Submiited in RGI for Bhujal News
32	Lubna Kouser*, H.P. Jayaprakash, D. Dhayamalar, N. Jyothikumar	Drought assessment in NAQUIM taluks of Chickballapur district,Karnataka State, India	Special volume on contributing Research Papers- Bhujal News	Submiited in RGI for Bhujal News
33	Deepa Gupta	Findings of National Aquifer Mapping and Management Plan Studies in Krishnarajpet Taluk, Mandya District, Karnataka State	Special volume on contributing Research Papers- Bhujal News	Submiited in RGI for Bhujal News

S. No.	Author/Authors	Topic of Paper	National Journal	Status- Published/ Submitted
34	Vijesh V.K, Anisha K, Anitha Shyam T.S, M. Santhana Subramani, K. Balakrishnan	Outcomes of Aquifer Mapping in Kavaratti Island, U.T. of Lakshadweep: Hydrochemical Evolution, Aquifer Characteristics and Management Plans'	4 th National Seminar (volume 1.) Of Current Trends in Earth System Sciences (CTESS 2022-23) organised by Department of Geology, University of Kerala, Trivandrum, India on 19-21 January 2023.	Published
35	Anisha K*, Vijesh V. K., M. Santhana Subramani	Mapping and Management Plan for the Sustainability of Aquifer Systems in Thiruvananthapuram District, Kerala	Special volume on contributing Research Papers- Bhujal News	Submiited in RGI for Bhujal News
36	Priya Kanwar	Options for Sustainable Groundwater Management in Tonk District, Rajasthan, India	Current Trends in Earth System Sciences (CTESS 2022-23), ISBN 978-81-961543-0-1	
37	Aruna Saini and Priya Kanwar	Vulnerability Analysis of Ground Water in and Around MSW Disposal Site Mathuradaspura, Jaipur using Multivariate Statistical Techniques	Global Journal of Science Frontier Research: H Environment & Earth Science Volume 22 Issue 2 Version 1.0 Year 2022	

12.2 List of International Papers Papers during 2022-23 (For AGR 2022-23)

Sr No	Author/Authors	Topic of Paper	International Journal	Status-Published/ Submitted
1	Mohit Arora, Mayank Dixit	Assessment of Water Storage Changes Using Satellite Gravimetry and GLDAS observations over a part of Indus Basin, India	Water Conservation Science and Engineering	Published (2022)
2	Mayank Dixit, Mohit Arora, HP Jayprakash, Veena R. Achutha	Achievements and Significance of National Hydrograph Stations Network and 55 Years of Continuous Groundwater Level Monitoring in India	Ground Water for Sustainable Development	Submitted (under review)
3	Mayank Dixit	Covid-19 and Earthquakes: Challenges and Lessons Learned from June 2020 Mw5.3 Kachchh, NW India Earthquake	Natural Hazard	Submitted (under review)
4	Mayank Dixit, Mohit Arora, HP Jayprakash, Veena R. Achutha	Monitoring of National Hydrograph Stations in India during the Covid-19 Pandemic: Challenges and Lessons Learned	Ground Water Policy	Submitted (under review)
5	Suchetana Biswas, T.S.Anitha Shyam	Groundwater management plan for an overexploited village cluster: A case study in Chinakavajra gram panchayat, Madhugiri taluk, Tumkur district, Karnataka state, India	Workshop volume of 24th International Congress on Irrigation and Drainage, 24th ICID Congress & 73rd IEC Meeting October 2022, Adelaide, Australia	Published
6	Priya Kanwar and Kanwar P. Singh	Impact of climate change on water resources	Current Directions in Water Scarcity Research, Water Scarcity, Contamination and Management, Vol.5,	
7	Aruna Saini and Priya Kanwar	A study on impact of progressive urban expansion on groundwater quality by using graphical, statistical and WQI methods: Jaipur, Rajasthan, India	Current Directions in Water Scarcity Research, Water Scarcity, Contamination and Management, Vol.5,	
8	Sayelli Tembhurne, Priya Kanwar, Kaveri Salunke and S.K. Pareek	Interrelationship of rainfall and groundwater levels: Impact on socioeconomic conditions in Sawai Madhopur District, Rajasthan (India)	Current Directions in Water Scarcity Research, Water Scarcity, Contamination and Management, Vol.5,	

CHAPTER 13 SPECIAL STUDIES

Table 13.1 List of Demand Driven Studies/Special Studies in AAP 2022-23

Sr No	State/UT	Region	Special Studies/Collaborative Studies (AAP 2022-23)	
			Title	Collaborating Institute(s)
1	Madhya Pradesh	NCR	Title -Implementation of Outcome of NAQUIM Report for artificial recharge to Groundwater in Convergence with Rural Development Schemes in Bijawar block of Chhatarpur district.	District Administration (MGNREGA, PHE, Watershed development schemes under PMKSY, Rajiv Gandhi Mission, etc)
2	Madhya Pradesh	NCR	Implementation of Outcome of NAQUIM Report for artificial recharge to Groundwater in Convergence with Rural Development Schemes in Bagh block of Dhar district, Short Term Investigation in Bhopal district	District Administration (MGNREGA, PHE, Watershed development schemes under PMKSY, Rajiv Gandhi Mission, etc)
3	Madhya Pradesh	NCR	Implementation of Outcome of NAQUIM Report for artificial recharge to Groundwater in Convergence with Rural Development Schemes in MomanBadodiya block of Shajapur district	District Administration (MGNREGA, PHE, Watershed development schemes under PMKSY, Rajiv Gandhi Mission, etc)
4	Himachal Pradesh	NHR	Spring-shed conservation and management in and around Dharamshala MC area	NONE
5	Jammu & Kashmir	NWHR	Ground water Quality in Bari Brahmna-Samba industrial Clusters, District Samba	NA
6	Ladakh	NWHR	Hydrochemical studies in Phyang valley, Leh	NA

Sr No	State/UT	Region	Special Studies/Collaborative Studies (AAP 2022-23)	
			Title	Collaborating Institute(s)
7	Haryana	NWR	2. In situ salinity Remediation - Indo -australian Collaboration	GWC & WRD, Haryana
8	Haryana	NWR	1. Impact of paddy cultivation in Haryana	HWRA & IWRD, Haryana
9	Haryana	NWR	3. High resolution aquifer mapping and management in arid regions of NW India in collaboration with NGRI	NGRI
10	Gujarat	WCR	Coastal Salinity/ ingress extent demarcation in Navsari district	NA
11	Gujarat	WCR	Coastal Salinity/ ingress extent demarcation in Surat district	NA
12	Rajasthan	WR	1.High resolution aquifer mapping and management in arid regions of NW India in collaboration with NGRI.	NGRI
			2. Ground Water Quality in Shree Khatu Shyam Temple Area of Sikar district Rajasthan.	NA
			3.Augmentation of Water Supply through Roof Top Rainwater Harvesting to Adarsh Vidhya Mandir (AVM), Shankar Vidya Peeth, Mt. Abu, Block Abu Road, District, Sirohi, Rajasthan	CWC/WAPCOS

Sr No	State/UT	Region	Special Studies/Collaborative Studies (AAP 2022-23)	
			Title	**Collaborating Institute(s)**
			4.Ground Water Augmentation through Artificial Recharge in certain Water Stressed Areas of Jodhpur, Jaisalmer and Sikar Districts in Rajasthan under Phase-I and II	WAPCOS/CWC
13	Uttar Pradesh	NR	Study for understanding recharge dynamics of deeper aquifers in Ramganga Basin through isotope studies	GIZ & BARC
14	Uttar Pradesh	NR	Study on Uranium, Flouride Mercury contamination of Ground Water in Mathura, Hathras and Aligarh Districts, UP asper MoU between CGWB & GSI.	**GSI**
15	Uttar Pradesh	NR	Ground water flow modelling study in Yamuna - Kali sub basin (Ghaziabad &G.B.Nagar district and parts of Hapur and Bulandsahar district)	NA
16	Uttar Pradesh	NR	Ground Water Pollution study on Kali-Krishni-Hindon Drainage area.	NA
17	Delhi	SUO	Special Studies on Uranium Concentration in Parts of NCT Delhi	-
18	Uttarakhand	UR	Springshed Mapping and allied work in Pratapnagar&Bhilangna Blocks, Tehri Garhwal District and Ukimath Block, RudraprayagDistrict, Uttrakhand State	NIH
19	Andaman & Nicobar	ER	-	-
20	Sikkim	ER	-	-
21	West Bengal	ER	-	-

Sr No	State/UT	Region	Special Studies/Collaborative Studies (AAP 2022-23)	
			Title	**Collaborating Institute(s)**
22	Jharkand	MER	Assessment of fluoride contamination in surface and groundwater and its management in Jharkhand	CGWB, GSI & Drinking Water and Sanitation Dept, Jharkhand Govt. (DWSD)
23	Bihar	MER	1. Reassessment of Arsenic contamination in Patna & Bhojpur District.	**GSI**
24	Bihar	MER	2. Factors controlling Fluoride Geochemistry in Ground Water Resources of Amas Block, Gaya Dist.,Bihar	**GSI**
25	Bihar	MER	3. Uranium hotspot sampling in (i) Bhagalpur (ii) Bihar Sharif, (iii) Gopalganj (iv) Katihar (v) Madhepura, (vii) Muzaffarpur, (viii)Nawada (ix) Purnea (x) Siwan (xi) Vaishali	**GSI**
25	Bihar	MER	**Special Studies** 4. Tracing the source of Uranium in ground water resources of Biharsarif and Roh block of Nawada district of Bihar: A hydrogeochemical approach	
26	Chhattisgarh	NCCR	Geo-environmental appraisal of Fluoride contamination and assessment of toxic heavy metals in Bagbahara area, Mahasamund district, Chhattisgarh with emphasis on groundwater	**GSI**
27	Tripura	NER	Artesian zone mapping in North Tripura	None
28	Arunachal Pradesh	NER		
29	Assam	NER	Study on Uranium, Lead, Arsenic, Fluoride and Mercury Contamination of Ground Water in the industrial area of North Guwahati, Kamrup, Assam	GSI

Sr No	State/UT	Region	Special Studies/Collaborative Studies (AAP 2022-23)	
			Title	Collaborating Institute(s)
30	Maharashtra	CR	"Geogenic Fluoride Contamination in parts of Yavatmal district"	NA
31	Kerala	KR	Ground water pollution around Valapattanam industrial area, Kannur ,Kerala	NA
32	Kerala	KR	Sea water Ingress studies along Kollam Coast, Kerala	NA
33	Punjab	NWR	1. Analysis of Uranium contamination in identified hotspots in parts of Punjab and Haryana.; 2. Selenium toxicity in parts of Punjab;	NA
34	Puducherry	SECR	Sea water intrusion studies in Puducherry Region	Public Works Department, UT of Puducherry and Pondicherry Central University, UT of Puducherry
35	Puducherry	SECR	Augmentation of recharge in Cretaceous aquifers in Puducherry region	Public Works Department, UT of Puducherry and Pondicherry Central University, UT of Puducherry
36	Tamilnadu	SECR	1. Sea Water Intrusion Studies in Minjur area, Tiruvallur district	State Ground & Surface Water Resources Data Centre & Denmark Geological Survey
37	Tamilnadu	SECR	2. R&D Studies on Uranium occurrence and movement in the districts of Dharmapuri and Krishnagiri	NA
38	Tamilnadu	SECR	3. Soil Moisture Studies in different soil formation in Kancheepuram	NA
39	Tamilnadu	SECR	4. Study of ground water pollution along the course of Tamiraparni River	NA

Sr No	State/UT	Region	Special Studies/Collaborative Studies (AAP 2022-23)	
			Title	Collaborating Institute(s)
40	Andhra Pradesh	SR	Comprehensive Water Audit in Paleru and Musi sub basins, Prakasam district, AP	Ground Water & Water Audit department, Govt. of Andhra Pradesh
41	Andhra Pradesh	SR	Uranium Contamination in Ground Water in & around Tummalapalle Uranium Deposit, Vemula Mandal, YSR Kadapa district, A.P	**GSI**
42	Andhra Pradesh	SR	Sea Water Intrusion and Tackling Salinity in Ground Water in Coastal Aquifers of Andhra Pradesh State	NA
43	Andhra Pradesh	SR	Characterisation of Ground Water, Emerging contaminants and its impact on Ground Water in Krishna-Godavari Delta, Andhra Prad	NIH, Deltaic Centre, Kakinada, Andhra Pradesh
44	Telangana	SR	A pilot study on Impact Assessment of Artificial Recharge through Check Dams on ground water regime in Telangana State	State Ground water Department, Govt. of Telangana
45	Karnataka	SWR	" Collaborative study on Water Quality with special reference to Uranium with the technical guidance of CGWB"	State Groundwater Directorate, Karnataka
46	Goa	SWR	Special study:Demarcation of ground water potential in the coastal tract with special emphasis on sea water intrusion - Candolim, Calangute and Anjuna villages in Bardez taluk, Goa State	NA

CHAPTER 14 CENTRAL GROUND WATER AUTHORITY

14.0 Regulation and Control Of Ground Water Extraction

Central Ground Water Authority, constituted under Section 3 (3) of the Environment (Protection) Act, 1986 has been regulating ground water development in the country The Authority regulates in 19 States/ UTs , 15 States/ UTs. have their own Acts and 2 States are regulating under some go. The Authority issues No Objection Certificate for ground water abstraction.

- Exercise of powers under section 5 of the Environment (Protection) Act, 1986 for issuing directions and taking such measures in respect of all the matters referred to in sub-section (2) of section 3 of the said Act.
- To resort to penal provisions contained in sections 15 to 21 of the said Act.
- To regulate and control, management and development of ground water in the country and to issue necessary regulatory directions for the purpose.
- Exercise of powers under section 4 of the Environment (Protection) Act, 1986 for the appointment of officers.

Important activities of CGWA during the period mentioned are given below:

14.1 Guidelines For Ground Water Regulation

Ministry of Jal Shakti notified the revised guidelines on 24.09.2020 and for regulation and control of ground water development in the country. These guidelines have Pan India applicability. Further the guidelines has been amended on 29.03.2023 The salient features of guidelines are as follows:

- The seeking NOC is exempted for individual domestic consumers (in both rural and urban areas) drawing ground water for drinking/ domestic use, Rural drinking water supply schemes, Armed Forces and Central Armed Police Forces Establishments, Agricultural users and Micro and Small Enterprises drawing < 10 KLD of groundwater, All industries/ mining projects/ infrastructure projects drawing ground water only for drinking/ domestic purposes up to 5 Cum /day in all assessment units, Residential Apartments and Group Housing Societies: (a) For drinking water and domestic uses, drawing ground water upto 20 m3/day subject to the conditions mentioned in Para 2.0 of the guidelines. (b) Dwelling units for Economically Weaker Sections (EWS) under Government schemes.

- No new industries except those MSME based on ground water are permitted in Over-exploited assessment units.
- Expansion of existing industries except MSME located in Over-exploited assessment units involving increase in quantum of ground water abstraction is not permitted.
- Industries drawing >100 KLD of ground water in OE, Critical and Semi-critical assessment units are required to submit impact assessment reports along with the application.
- Industries drawing > 100 KLD of ground water in OE, Critical and Semi-critical assessment units are required biennial (once in two years) water auditand reduce fresh groundwater consumption by 20% over the next three years.
- No NOC is granted for groundwater extraction for water parks, theme parks, amusement parks in OE assessment units. Commercial infrastructure projects requiring ground water for drinking /domestic use shall also be covered under this category. Further, the Indicative list of location specific Infrastructure projects as per **Annexure VI** of the guidelines.
- All ground water users seeking NOC are required to pay ground water abstraction charges in Critical, Semi-critical and Safe assessment units and ground water restoration charges in OE assessment units. The charges so collected are to be utilized by States/ UTs on specific water conservation measures, specific supply/ demand side interventions.
- No charges are to be paid by users drawing saline ground water.
- Users drawing groundwater illegally are liable to pay Environmental Compensation.
- Penal provision has been kept for ensuring compliance of various conditions of NOC.
- Bulk water suppliers have also been brought in the ambit of NOC.
- Installation of Sewage Treatment Plants is mandatory for all new residential apartments/ Group Housing Societies where ground water requirement is more than 20 m3 /day. STP treated water is to be used by such users for flushing, green belt etc.
- All users seeking NOC are required to install tamper proof flow meters along with telemetry on all the ground water abstraction structures.
- In over-exploited assessment units, use of ground water for construction activity is permitted only if no treated sewage water is available within 10 km radius of the site.

14.1.1 Processing Of Applications For Grant / Renewal Of No Objection Certificate (Noc) For Ground Water Withdrawal

CGWA evaluated applications from Industries/ Infrastructure Units / Mining Projects for grant of NOC for ground water withdrawal as per provisions of the Notified guidelines dated 24.09.2020. A total **of 2735**nos. of new NOCs were **issued** and **748** nos. of NOCs **renewed** during the 2022-23. In addition, **exemption** was given in respect of **8043**Nos. **new** applications and **277 Nos renewal** cases. As compared to previous year, there has been an increase to **104%** in issuance of new NOC/

exemption . Presently, NOCs are also being issued in Over-exploited, Critical and Semi-critical areas as per revised guidelines which were earlier held up owing to observation made by Hon'ble NGT in its various orders. All such applications have since been processed and pendency cleared.

14.2 Monitoring Of Compliance of Conditions Stipulated In The Noc

CGWA has developed online self-compliance format for the project proponent to upload the compliance conditions as stipulated in the NOC and the same is being monitored by the Regional Offices for ensuring the compliance conditions by the project proponent. Mobile app has also been developed for this purpose with the provision of verification by CGWA officers.

Show-cause Notices are also being issued to units, which have been found nottohavefully complied with the NOC conditions. Orders for sealing of bore/tube wells and/ or disconnection of electricity supply through the concerned DCs/ DMs were also issued in respect of units, which did not give satisfactory replies to the show-cause notice.

While processing applications for renewal of NOC, compliance of NOC conditions by the proponents was ensured. Penalty of Rs. 1 lakh under Section 15 of Environment (Protection) Act of 1986 was imposed on the violators. NOC was issued after the compliance was completed and penalty was deposited by such offenders to the concerned Regional Office of CGWB.

14.3 47th Meeting of Central Ground Water Authority

The 47th meeting of Central Ground Water Authority (CGWA) was held on **07.06.2022 at 11:00 a.m**. The meeting will be held both online and in person. under the Chairmanship of Shri Sunil Kumar, Chairman, CGWA through Video Conferencing. Member, CGWA apprised in brief about the developments that had taken place since the last meeting of the Authority held on 08.06.2022. Regional Director, CGWA made a brief presentation on the agenda items and the following decisions were taken during the meeting.

- All the member agreed to and accorded approval to the accreditation policy of ground water professional dated 21.12.2021 and further amended on 04.02.2022.
- All the member agreed on requirement of Comprehensive Hydrogeological Report in case of mining applications drawing more than 100 KLD ground water.
- The committee agreed that there should be digital flow meter installed in common Outlet point in case of multiple wells. However, it should be ensured that no other delivery pipe should bypass the flow meter. Certain criteria should be introduced in this regard like distance between two connected wells, plot size etc.
- All those cases where ground water abstraction is upto 500 KLD and falling in OCS blocks, and upto 2000 KLD in safe blocks, a lower-level state EAC is to be constituted for approval of IAR/Modelling/Mining reports at regional level.

- Water Audit Certification course other issues related to water audit was discussed. It was agreed that a policy guidelines incorporating various provisions including certification, penal provision, delisting, renewal etc may be prepared for approval of the Board.

14.4 On Site Inspection By CGWB

On-site inspections were carried out by the Regional Offices of CGWB to check the compliance of NOCs accorded by CGWA before recommending the renewal applications to CGWA, New Delhi. Necessary show- cause notices were issued to the project proponents who have not complied with the conditions of the NOC issued by CGWA.

14.5 Developemt Of Advanced Nocap Software

Development of upgraded version of NOCAP has been initiated as eNIWARAN.

CHAPTER 15 TECHNICAL COLLABORATION

Table 15.1 List of National collaboration

Participating Country	Participating organization/University (India)	Area of Study	Nodal Officer / alternate officer	Activities
India	**Central Water Commission (MOU)**	Pan India		**For the testing of water samples in CWC labs for pathogenic/bacteriological/pesticide analysis**
India	**Central Water Commission (Joint Study)**	Area to be identified	Member N&W, Chairman of the Committee	**Sea water ingress and surface-ground water interaction in Coastal aquifers.Committee constituted. Study area to be identified**
India	**Geological Survey of India (MOU)**	Punjab, Haryana, Andhra Pradesh, Uttar Pradesh, Chhattisgarh, Jharkhand and Assam		**Study of As, U, Pb, Hg, and F in ground water**
India	**B.A.R.C, Trombay (MOU)**	Ramganga Basin (UP and Uttarakhand), Vadodara and Baruch (Gujarat)		**Environmental stable isotopes Studies, Training for chemists and Study on uranium Occurrence in Shallow Aquifer in India**

Participating Country	Participating organization/University (India)	Area of Study	Nodal Officer / alternate officer	Activities
India	**IIT, Hyderabad (MOA)**	Pan India		**To create a GIS based web portal by name 'India- Groundwater Resource Estimation System (IN-GRES) in association with IIT-Hyderabad.**
India	**Department of Earth Sciences, Indian Institute of Technology, Kanpur (MOA)**	Punjab, Haryana and Bundhelkhand Region		**Development of Groundwater Flow Models and Preparation of Aquifer Management Plans (1.) Punjab and Haryana (2.) Bundelkhand Region:Completed and report submitted**
India	**Prajapita Brahma Kumaris Ishwariya Vishwa Vidyalaya**	Pan India		**Creating Awareness for Ground Water Conservation**

Table 15.2 List of International Collaboration

Participating Country	Participating organization/University (Foreign)	Area of Study	Nodal Officer / alternate officer	Activities
Australia	CGWB, CHQ & NWR, Haryana GW Dept Australia – CSIRO	Insitu Salinity Remediation Jhaggar, Haryana	Nodal - Shri. Satish Kumar, Member (HQ) Supporting- Dr M.Senthilkumar, Scientist D	Concept note shared with Australian side.
EU	CGWB, CHQ, NR & UR, GIZ, IHE –Delft	Understanding Recharge dynamics in Ram Ganga Basin	Nodal- Shri. Satish Kumar, Member (HQ) Supporting- Sh S G Bhartariya, ,RD Shr S N Dwivedi, , Scientist D	Sampling for Carbon-14 isotopes to be taken up in November.2023. Workshop organized at Lucknow on Stable Isotope data interpretation by EU Expert Prof Jochen Wenniner, IHE Delft.
Netherlands	CGWB, CHQ & CGWB, SECR	Recharge measure to arrest Saline Water intrusion North of Chennai City Tamilnadu	Nodal – Shri. Satish Kumar, Member (HQ) , Supporting- Dr M. Senthilkumar, Scientist D	Study area finalized north of Chennai City. Identification of state department Preliminary discussion held with Chennai Metro Water & Sewerage

				Board Further, meeting with Chennai Metro Water is being planned.
Hungary	CGWB, RGI	Vulnerability studies in Karst Aquifer system of Central Chhattisgarh.	Dr Ranjan Kumar Ray, Scientist E	Study area finalized in Chhattisgarh Exchange program and joint field investigations to follow.
Morocco	CGWB, RGI	Ground Water Management	Nodal –Dr. A. Asokan, Member (E)	Activities being finalized
Israel	CGWB, CHQ & CGWB NR	Ground Water Management	Nodal – Shri. Satish Kumar, Member (HQ)	Activities are being finalized
Tanzania	CGWB, RGI	Ground Water Management	Nodal – Shri. Satish Kumar, Member (HQ) and RD, RGI	Proposal submitted for capacity building of officers of Govt. of Tanzania
Japan	CGWB,	Ground Water Management	Nodal – Shri. Satish Kumar, Member (HQ), Supporting- Dr M.Senthilkumar, Scientist D	Land subsidence due to groundwater exploitation study is being taken up Groundwater quality changes due to irrigation water

CHAPTER 16 RAJIV GANDHI NATIONAL GROUND WATER TRAINING & RESEARCH INSTITUTE

16.1 Introduction

Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) is the training wing of CGWB and functions as a `Centre of Excellence' with thenational role of capacity building of CGWB, groundwater professionals from Central and State Government Organizations, Public Sector Undertakings, Non-Governmental Organizations, Academic institutions, and other stakeholders through three-tier training -Tier I (National Level), Tier II (State Level) and Tier III (Block level) trainings.

16.2 Human Resource Development

To improve the capacity building of groundwater professionals for sustainable groundwater development & management with a participatory approach, Institute has embarked upon a **three-tier training system (Tier I, II, III)** which can create a pool of trained resource persons who would be working towards sustainability of groundwater resources across the country with community participation.

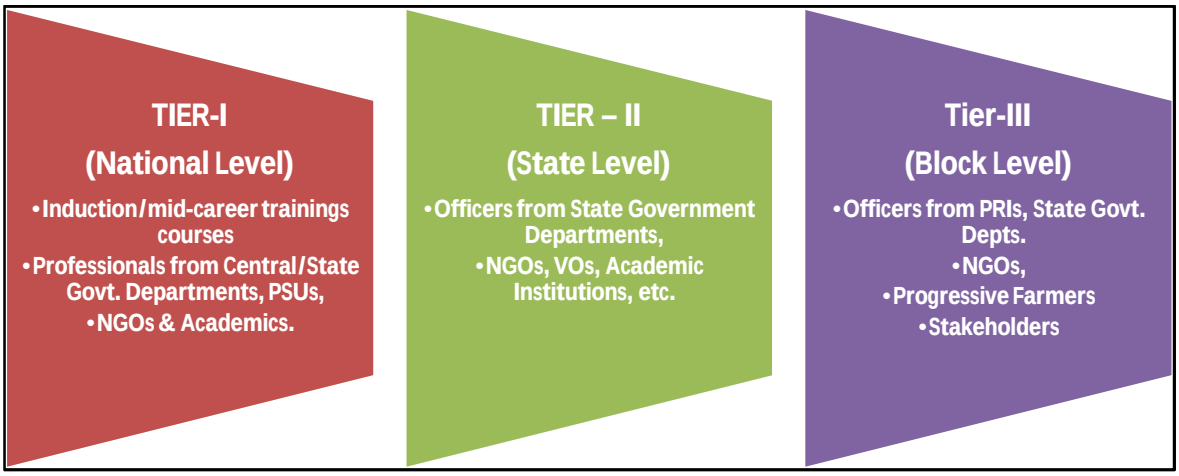


Fig 16.1: -Tier Training Program

During the AAP 2022-2023, against the target of 60 trainings, 60 National Level Training courses under Tier I have been conducted not only to cover the Mandatory training for CGWB officers and officials but also for the Central and State Govt. Organizations, Public Sector Undertakings (PSUs), Academic Institutions, NGOs, etc. working in the field of groundwater. Out of 60 tier-I trainings, Fifty Four (54) were conducted in Offline mode.

The Tier I training are categorized into three buckets: Domain-specific, Functional, and Behavioural, and are mentioned below:

Table 16.1 : Training Types and Domains

Buckets	Type of the training
Domain	• Groundwater Resource Estimation • Managed Aquifer Recharge • Groundwater Exploration in Different Terranes • Groundwater Regime Monitoring And Data Analysis • Groundwater Regulation and Management • Water Well Drilling Techniques • Ground Water Quality Contamination and Remediation • Impact of Climate Change on Water Resources • Spring shed Management • Coastal Aquifers and Island Hydrogeology • Applications of Remote Sensing and GIS • Scientific Paper Writing • Material Management
Functional	• Administration And Establishment Rules • Operation And Maintenance of Vehicles • Project Management • CGWA Guidelines & Evaluation of Reports
Behavioral	• Management Aspects and Leadership Traits • Management Principles and Practices • Mentoring Skill • Leadership Skill • Stress Management

The Tier II training at the State level is conducted to build up the required manpower, mainly at the users and stakeholders level, to make them aware of the groundwater resources, and ensure a better understanding of the aquifers on which they depend. The Tier II trainings were organized through CGWB Regional Offices and envisaged the involvement and participation of Central and State Government Organisations, NGOs, PRIs, Industries, and end-users to infuse a sense of ownership amongst the stakeholders.

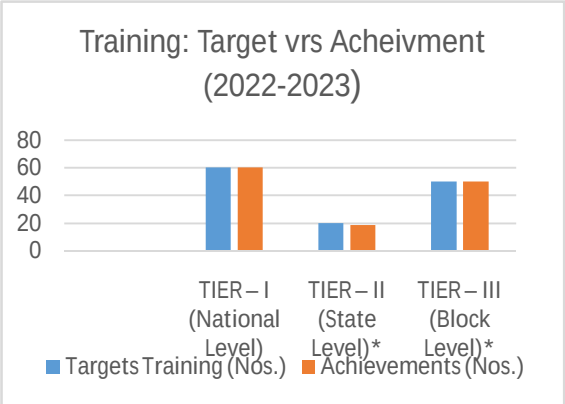
During the AAP 2022-2023, the Tier II State Level training was held on the theme **"Local Ground Water Management Issues"**. A total of 19 training courses against the target of 20 were organized in the entire country from April 2022 to March 2023, in which 544 trainees have participated including 157 women participants.

During the AAP 2022-2023, 50 Block level (Tier-III) trainings on ***"Local Ground Water-Related Issues and Participatory Ground Water Management"*** were organized to build capacity at the village/ block level in local vernaculars and thereby prepare the rural youths and the general public to function as *Jalrakshaks*. Out of 50 Tier-III programs, 01 training was organized in online mode while 49 trainings were organized in physical/offline mode. These trainings would enable them in acquiring and providing information on the status of ground groundwater at the aquifer levels and in turn assist the Panchayat Administrations in groundwater planning and sustainability of the aquifers. The course modules were designed to cover the local hydrogeological set-ups, local water-related issues, and the benefits of the participatory groundwater management approach for mitigation of water scarcity.

Table 16.2: Summary of Targets& Achievements of training courses in AAP 2022-2023

Training Courses	Targets Training (Nos.)	Targets Trainees (Nos.)	Achievements (Nos.)	No. of Participants	Male	Female
TIER – I (National Level)	60	1200	60	1251	931	320
TIER – II (State Level)*	20	400	19	544	387	157
TIER – III (Block Level)*	50	6000	50	6318	3976	2342
	130	**7600**	**129**	**8113**	**5294**	**2819**

*Tier–II and Tier – III training courses are conducted through CGWB's Regional Offices.



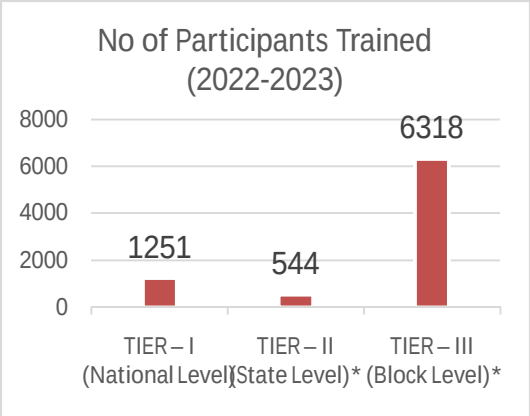


Fig 16.2: Training Details in 2022-23

16.3 Research & Development Activities

16.3.1 Research and Development Projects

Apart from regular trainings, RGNGWTRI has taken up various research projects. The project-wise progress of Research & Development Projects of RGNGWTRI, CGWB, and Raipur as on 31st March, 2023 is given below:

Table 16.3: Details of Research work in 2022-23

Sl. No	Title of the project	Collaborating Organizations	Name of faculty	Duration	Progress as on 31.03.2023
1.	Occurrence and Movement of Arsenic in groundwater, Rajanandgaon, Chhattisgarh	RGNGWTRI	Dr. Anandi Gayen, Scientist D	June 2017-March 2020	Report submitted
2.	Ground Water Flow Modelling Androth island, Lakshadweep	RGNGWTRI	Dr. AVSS Anand, Scientist D	June 2019-March 2021	Report submitted
3.	Sea water intrusion in the coastal area of the Sundarbans: a study in and around Ganga Sagar Islands, West Bengal	RGNGWTRI	Dr. Anandi Gayen, Scientist D	June 2019-March 2021	Report submitted
4.	Impact of Climate Change on Coastal Aquifers in Parts of Tamil Nadu	RGNGWTRI	Sh. Sivakumar, Scientist D	2020-2022	Report Submitted
5.	Impact of Jawaharnagar dumping site on Surface and Ground Water Quality in Mechel-Malkajgiri District, Telangana State	RGNGWTRI	Sh. J. Siddharth Kumar, Scientist D	2020-2022	Report Awaited

16.4 BHUJAL News

Bhujal News is a quarterly Journal of CGWB and in the year 2020, the publication of Bhujal News has been started by Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI), Naya Raipur.

The work assignments in CGWB are mostly field-based and the output is measured by the submission of reports based on field studies. But, because the reports have limited circulation, the results of these field studies and ideas developed thereof do not reach the scientific community. Even CGWB's Officers are deprived of this information. Bhujal News fulfills this gap and serves as a conduit between CGWB and the outside world by highlighting CGWB's contributions in the field of groundwater to the general public.

RGNGWTRI, Naya Raipur has published volume Bhujal News, Volume 32, No 1-4 in the year 2022.

It includes six papers that cover Aquifer mapping, Geophysical, Hydrogeological, and Chemical aspects of groundwater. These papers are

Table 16.4: List of Technical Papers in 2022-23

Sl. No.	Title of Paper	Author(s)
1.	Aquifer Mapping encapsulating Disposition of Aquifer system, investigating the Groundwater Quality and Sustainable Groundwater Management plan of Sendurai village, Senduraifirka, Ariyalur district, Tamil Nadu, India	Raj Kishor Mohanty and K.Rajarajan
2.	Groundwater Exploration and Drilling Problems Encountered in Basaltic Terrain of Dhar District, Madhya Pradesh	Naresh Kumar Jatav, Mahesh M. Sonkusare, Rose Anita Kujur, Kamlesh Ojha
3.	Aquifer Disposition using Surface and Sub-Surface Geophysical Methods: A case study of Lakhimpur Kheri District in parts of Ganga Plain, Uttar Pradesh, India	Shashi Kant Singh, Anirudh Singh and Rajesh Babu
4.	Inter-annual Variability of Ground Water Level and Rainfall Over Jabalpur District, India	Sarita Tiwari & Rose Anita Kujur
5.	Rooftop Rainwater Harvesting incorporating the Structural Design and Augmentation of 56 rainwater to the aquifers through Artificial Recharge at the Hostel Building of Rajiv Gandhi	Raj Kishor Mohanty, H V Sophia Kay, M Siva Kumar, Lalatendu Behera, Gargi Walia, Debajyo_ Mondal, Mohit Jain,

	National Ground Water Training and Research Institute, Nava Raipur, Chhattisgarh	Mofid Ali, Ramkishore
6.	Long-term water level trends, groundwater resources, and delineation of areas suitable for artificial recharge to groundwater in the state of Chhattisgarh, India	Mahesh M. Sonkusare, Sudarsan Sahu, Prabir K Naik, Prachi Gupta, Uddeshya Kumar

16.5 Research Papers Published by the Faculties

1. Dr. Bhushan R. Lamsoge, Scientist-E as a co-author with Sanjay Raut and Nitin M. Labhane published a research paper 'Rainwater Harvesting for Metro Rail Projects in India — Scope for Efficiency Improvement' Journal of the Geological Society of India volume 98, pages937–946 (2022), Published: 15 July 2022, Springer Link https://doi.org/10.1007/s12594-022-2098-7.

2. Dr. Sudarshan Sahu published a chapter on "Fluvio-dynamics and Groundwater system in Narmada Basin" in the Book titled: Riverine Systems Understanding the Hydrological, Hydro social and Hydro-heritage Dynamics.

16.6 Dissertation

RGNGWTRI has approved the dissertation work of the following students during 2022-23.

a) "Statistical Assessment of Climate Change Impact on Groundwater Region of Chennai Basin, Tamil Nadu" submitted by Sh Ramavath Lalu, M.Sc Geology from Dept. of Geology, School of Earth Science, Central University of Tamil Nadu. The work was supervised by Sh M Siva Kumar, Scientist-D, RGNGWTRI, Raipur.

b) "Assessment of Groundwater Quality Change Due to Climate Change in Chennai Basin, Tamil Nadu, Tamil Nadu" submitted by Smt Rosalin C, M.Sc Geology from Dept. of Geology, School of Earth Science, Central University of Tamil Nadu. The work was supervised by Sh M Siva Kumar, Scientist-D, RGNGWTRI, Raipur

16.7 Accreditation Of Groundwater Professionals

As per new Notified Guidelines of Central Ground Water Authority, all projects extracting/proposing to extract groundwater in excess of 100 m3/day in Over-exploited, Critical, and Semi-critical areas and mining projects shall have to submit mandatorily the impact assessment report of existing/ proposed groundwater withdrawal on the groundwater regime and also the report on the socio-economic impacts prepared by accredited consultants. Accreditation is a process of instilling assurance and confidence amongst all stakeholders to submit reports to acceptable standards. Accreditation is a tool to assess and evaluate the standards and quality of the work done or to be done by an

individual or an organization. The Applicants who seek permission for Water Extraction are confident in hiring the services of accredited individuals or institutions for conducting a hydrogeological survey and prepare a report on the groundwater scenario and the impact of the proposed extraction on the groundwater reservoir, which is mandatory for issuance of No Objection Certificate (NOC) from Central Ground Water Authority. RGNGWTRI is entrusted with the screening and scrutiny of applications received from individuals and Institutes for accreditation, conducting examinations, and issuance of certificates. During the year 2022-23, RGNGWTRI has conducted two rounds of Accreditation.

The details of the 'Accredited individuals / Institutes' are given in the next page:

Table 16.5 Result of the Interaction held between the Accreditation Board and Individuals / Institutions during 3rd Round of Accreditation (Validity Period: 09.06.2022 to 08.06.2025)

Name of Organisation	Individual/ Institution	IAR Without Modelling	IAR With Modelling	Mining Without Modelling	Mining With Modelling	Contact No	Email
Wasim Ahmed	Individual	Yes	No	Yes	No	7505426166, 9414373257	jazabilal@yahoo.co.in
Suriya Kumar	Individual	Yes	No	No	No	9842729655	suriyakumarsemban@gmail.com
Gouranga Charan Pati	Individual	Yes	Yes	Yes	Yes	9674181991	gcpaticgwb@gmail.com
Gajendra Singh	Individual	Yes	No	No	No	9826352755	gsthakur52@gmail.com
Ashok Kumar	Individual	Yes	No	No	No	7926870919 9427001963	ashokabi35@yahoo.com
Ifthikhar Ahmed	Individual	Yes	No	No	No	9442278601 04272431989	ifthiahmed@gmail.com
Hydro Geo Solutions	Institution	Yes	Yes	Yes	Yes	9873125140	info@hydrogeosolutions.in
Oggn E Consultancy	Institution	Yes	No	No	No	8955728433, 961007941	mktoggnec@gmail.com

Table 16.6: Result of Interaction held between the Accreditation Board and Individuals / Institutions during 4th Round of Accreditation

Validity Period: 08.02.2023 to 07.02.2026

Name of Organisation	Individual/ Institution	IAR Without Modelling	IAR With Modelling	Mining Without Modelling	Mining With Modelling	Contact no	Email
Saidul Haq	Individual	Yes	Yes	No	No	7245099349	saidul.haq@gmail.com
Central Mine Planning and Design Institute Limited (CMPDIL)	Institution	No	Yes	No	Yes	08987789044, 09479001646, 08827510510	debasis.bandopadhyay@coalindia.in, rambabu.singh@coalindia.in, rambabusingh2@gmail.com
Akshat Ground Water Consultant Services	Institution	Yes	Yes	Yes	Yes	8447557844, 9873640962	akgwcs37@gmail.com, sunilkumar_jain@rediffmail.com

16.9 Lectures Delivered Outside RGNGWTRI

1. Sh. Sivakumar, Scientist E, has delivered a guest lecture on Watershed Management at Induction level training for Dy. Collectors of Chhattisgarh state at CAA Nimora on 07.06.2022.
2. Sh. Sivakumar, Scientist E delivered a lecture/presentation on 'Watershed Management for Natural Resource', to the 1st batch of Class-II officers of CGPSC at Chhattisgarh Administrative Academy, Nimora on 29/08/2022.
3. Shri Bishwarup Mohapatra, Sc-'D', CGWB, RGI delivered a lecture as a resource person for the Tier-II training program on "Sustainable Development & Management of Ground Water Resources" during 12-14 December 2022 at Dehradun.
4. Smt. Sandhya Mohapatra, Scientist-D, RGNGWTRI delivered a lecture on "Groundwater: Issues and Challenges" during the Two-week AICTE (ATAL) sponsored faculty development program (FDP) on AI and Data Science based Emerging Trends in Earth Science organized by the Department of IT and Department of Computer Science and Engineering, NIT, Raipur on 16/11/2022.

16.10 Webinars And National Seminars Organised

1. RGNGWTRI has successfully organized a **National Webinar on Awareness Generation on Sexual Harassment of Women at Workplace** on 5th December 2022

In which all the officers/officials of CGWB have participated. Presentations were given by Advocate Smita Sarode Singalkar, High Court, Nagpur on "Awareness Generation on Sexual Harassment of Women at Workplace (Prevention, Prohibition, and Redressal) Act, 2013" and by Mrs. Archana Singh Member, Founder and President of Ambpalli Handloom and Handicraft Multi-State Coop Society Limited, on "Empowering Women through Self Awareness".

The National Webinar was coordinated by Smt Sandhya Mohapatra, Scientist D, and RGNGWTRI.

Fig 16.3 Glimpse of National Webinar

16.11 Other Activities

1. RGNGWTRI, Raipur has successfully conducted Limited Departmental Competitive Examination (LDCE) for the Post of LDC for Recruitment Year 2022 on date 21/06/2022. A total 47 MTS appeared for the written exam.
2. RGNGWTRI, Raipur has successfully conducted Skill/trade Test for the Direct Recruitment of Staff Car Driver (OG) for the year 2022 from 23rd to 25th June 2022.
3. A 4 day of international training on "Integrated Water Resource Management-India experience", in collaboration with African- Asian Rural Development Organization

(AARDO) was conducted by RGNGWTRI on Online mode from 14th to 17th November 2022. A total of 120 trainees from 27 countries participated in the training.

4. A brainstorming session on "Collaborative Groundwater study" was organized at RGNGWTRI on 29.11.2022. The meeting was attended by various institutes like NIT-Raipur, IGKV, IMD, CAA, etc. Shri Nidhish Verma, Regional Director & HOO, and Sh Sourabh Gupta, Regional Director chaired the meeting.
5. RGNGWTRI organized an interactive session on Training Need Analysis (TNA) of various groundwater-related organizations/line departments / academic institutes and stakeholders on 1.2.2023. A total 09 participants from various organizations including departments like CMPDI, GSI, NRDA, RDA, and academic institutions like CAA, NIT, IGKV, and Pandit Ravi Shankar Shukla University, participated in the interactive session and deliberated on various spheres where RGI can extend the training & expertise.
6. As per the gazette notification dated 24th September 2020, the Department of Water Resources, River Development & Ganga Rejuvenation, Ministry of Jal Shakti notified the guidelines to regulate and control groundwater extraction in India and as per the approval from the Ministry of Jal Shakti, Govt. of India, vide letter No. T-81011/53/2022-GW, Section-MOWR, dated 27.09.2022, Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI), Naya Raipur, Chhattisgarh in collaboration with National Water Academy (NWA), Pune, Maharashtra was entrusted to conduct training on Water Auditing. Hence a training Course in "Water Auditing" was scheduled from March 20th-April 26th, 2023 at RGNGWTRI, Raipur. A total of 51 no. of applications were received from all over India, out of which 25 nos of trainees were selected as per the criteria published in the notice. Sh Mahesh M Sonkusare, scientist-E was the course Director of the training.
7. RGNGWTRI has prepared six (6 nos) of e-learning modules on Atal Bhujal Yojna for uploading on the iGOT Platform. The modules were prepared by ShBiswarup Mohapatra, Scientist-D, Sh G Sreenath, Scientist-D and SmtBijimol Jose, Scientist-D.
8. The onboarding of all the training institutes under the Ministry of Jal Shakti, on the National Standards for Civil Service Training Institutions (NSCSTI) web portal created by CBC is the new activity initiated in the year 2022, and SmtBijimol Jose, Scientist-D is the nodal person for the Accreditation of RGNGWTRI.
9. The guidelines from the Ministry of Home, Department of Official Languages, have been implemented in the training Institute.

16.12 Photographs of the training courses conducted by RGNGWTRI during 2022-2023

PHOTO GALLERY INAUGURAL SESSION

PHOTO GALLERY VALEDICTORY SESSION

PHOTO GALLERY VALEDICTORY SESSION

PHOTO GALLERY CLASSROOM & COMPUTER LAB

PHOTO GALLERY CLASSROOM & COMPUTER LAB

PHOTO GALLERY FIELD

PHOTO GALLERY FIELD

PHOTO GALLERY-TRAINING

CHAPTER 17 NATIONAL HYDROLOGY PROJECT (NHP)

17.1 National Hydrology Project (NHP) is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with a total outlay of Rs. 3679.7674 Crore (Rs. 3,640 Crore for NHP and Rs. 39.7674 Crore for establishment of National Water Informatics Centre (NWIC), a repository of Nation-wide Water Resources data for a period of 08 years extending from 2016-17 to 2023-24. Out of the total outlay, 50% of the amount has to be repaid by Central Government to the Word Bank (WB) and the remaining 50% would be Central Assistance. The project will be implemented by 49 agencies involving 10 Central and 39 States agencies. Since CGWB is one of the implementing agencies of the NHP project, a budget of Rs 41.72 Crore has been allocated for the purpose.

17.2 Objective of NHP

To improve the extent, quality, and accessibility of water resources information, decision support system for floods and basin level resource assessment/ planning and to strengthen the capacity of targeted water resources professionals and management institutions in India.

The components of NHP are:

- **Component A:** Water Resources Monitoring systems
- **Component B:** Water Resources information systems
- **Component C:** Water Resources Operation and Planning systems
- **Component D:** Institutional Capacity Enhancement

NHP will improve and expand hydrology data and information systems, strengthen water resources operation and planning systems, and enhance institutional capacity for water resources management. The project will strengthen the information base and institutional capacity for effective decision making in water resources planning and operational management at the basin level across the country using the latest available technology and tools. NHP will also help in contributing to the Digital India initiative by integrating water resources information across state and central agencies.

17.3 Role of CGWB in NHP

CGWB being the Nodal Agency for Ground Water, NHP cell in all the CGWB, Regional Offices have been created for providing Technical Guidance & Support to States in the implementation of NHP.

17.4 The proposed major activities of CGWB under NHP

- **Real time monitoring of Water Level &Water Quality in coastal aquifers of Tamil Nadu & UT of Puducherry**through construction of 60 piezometers & installation of 60 DWLRs.
- **Establishment of Center of Excellence for groundwater modeling** for providing technical support to the Central/ State Agencies in Ground Water Flow Model Studies.
- **Real time monitoring of Water Level through Installation of 3400 additional DWLRs (Phase-1) in the existing Piezometers** located in the Overexploited, Critical and Semi-Critical Blocks of the Country.
- **Real time monitoring of Water Level through Installation of 1800 additional DWLRs (Phase 2) in the existing Piezometers** located in the Overexploited, Critical and Semi-Critical Blocks of the Country.
- **Capacity Building** of the officers from State / Central implementing agencies through National and International level trainings every year.
- **Awareness Campaigns** are being conducted at six places every year on state specific groundwater issues.
- Organizing **National Level Workshops** on need based Topics

17.5 The Status of various activities of CGWB under NHP is given below:

The construction of 60 Piezometers in coastal aquifers of Tamil Nadu & UT of Puducherry and Installation of 60 DWLRs for Real time monitoring of Water Level & Water Quality has been completed in the month of Oct-2021 and July-2022, respectively. Data from all the 60 Stations started sharing with India WRIS portal of NWIC of Department of Water Resources, River Development and Ganga Rejuvenation.

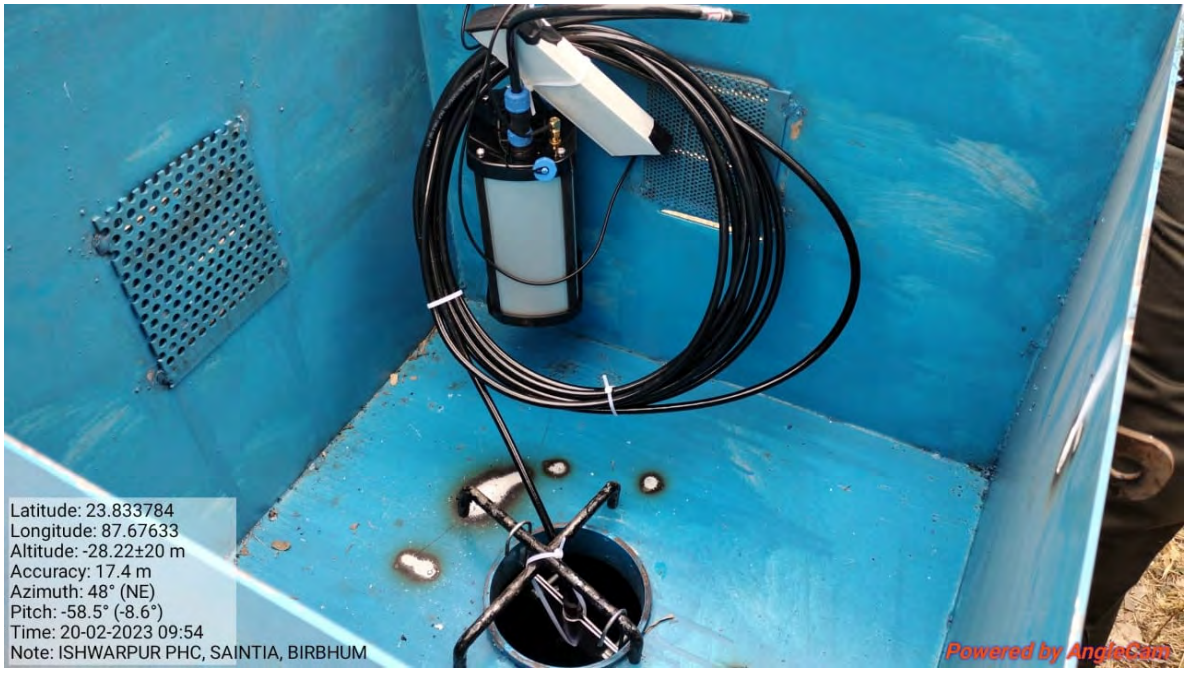
The glimpse of the activity is given in Fig-1

12 Contracts for 5200 DWLRs under Phase-1&2 was awarded to different vendors for installation of DWLRs in the existing Piezometers of CGWB located across the country for near real time monitoring of Water Level. The state-wise breakup of DWLRs is given in table below.

Table 17.1 The state-wise Breakup of DWLRs

-	Regional Office	States	3400 DWLRs	1800 DWLRs	Total DWLRs Awarded
1	2	3	4	5	6 = 4+5
1	CR, Nagpur	Maharashtra	148	29	177
2	KR, Trivandrum	Kerala	226		226
3	MER, Patna	Bihar	7	16	23
4	NCCR, Raipur	Chhattisgarh	124	144	268
5	NCR, Bhopal	Madhya Pradesh	291	100	391
6	NER, Guwahati	Arunachal Pradesh	4		4
7	NER, Guwahati	Assam	24	38	62
8	NER, Guwahati	Meghalaya	8	8	16
9	NER, Guwahati	Nagaland	5		5
10	NER, Guwahati	Tripura	7		7
11	NHR, Dharamshala	Himachal pradesh		10	10
12	NR, Lucknow	Uttar Pradesh	34	138	172
13	NWHR, Jammu	Jammu & Kashmir	14	20	34
14	NWR, Chandigarh	Chandigarh, UT	19	1	20
15	NWR, Chandigarh	Haryana	245	5	250
16	NWR, Chandigarh	Punjab	199	60	259
17	SECR, Chennai	Puducherry	7		7
18	SECR, Chennai	Tamil Nadu	468	165	633
19	SER, Bhubaneshwar	Odisha	50	52	102
20	SR, Hyderabad	Andhra Pradesh	119	161	280
21	SR, Hyderabad	Telangana	397	120	517
22	SUO, Delhi	Delhi	51	41	92
23	SUO, Ranchi	Jharkhand	5	15	20
24	SWR, Bengaluru	Goa		39	39
25	SWR, Bengaluru	Karnataka	260	134	394
26	UR, Dehradun	Uttarakhand	1	11	12
27	WCR, Ahmedabad	Diu	1		1
28	WCR, Ahmedabad	Gujarat	295	116	411
29	WR, Rajasthan	Rajasthan	304	258	562
30	ER, Kolkata	West Bengal	87	119	206
	Total		**3400**	**1800**	**5200**

The installation of DWLRs is under progress and the glimpse of the work is shown in Fig.2



- Organized 03 Nos Domain Specific Training under NHP through RGI, Raipur for 37 Participants (24 Male + 13 Female) from State/Central Departments during FY 2022-23.

- 3rd Modellers Meet was organized under NHP between 19-20th Dec 2022 in New Delhi and presented the study on "Aquifer Basin Management Plan through Numerical Modelling and Application of High Frequency Quality of Monitoring was made.

CHAPTER 18 PUBLIC INVESTMENT BOARD

18.1 Introduction of Public Investment Board (PIB)

The activities proposed under the PIB project were originally proposed as part of the Ground Water Management and Regulation Scheme for the period 2021-26 under Component I (Aquifer Mapping and Management Programme, NAQUIM) of the Scheme. Concept paper of Aquifer Mapping and Management has already been approved in the XII Plan. The same concept is being followed for completing the remaining activities. As per the recommendations of the EFC meeting of GWMR scheme held on 31st August 2021 (Minutes of meeting of EFC Annexed as Annexure, following time bound activities of similar nature are proposed to be taken up as a separate project: (i) automation of infrastructure for water level monitoring through construction of Piezometers and installation of Digital Water Level Recorders (DWLRs) (ii) high resolution aquifer mapping through heli-borne geophysical surveys and (iii) data generation for aquifer mapping through well construction Detailed Project Report for the above three activities envisaged under the Project has been prepared and is enclosed (Component I, II and III) .

I) Creation of Infrastructure for high density automated monitoring of water levels

It is proposed to create infrastructure for automated monitoring of water levels to help generate high frequency water level data in areas falling under assessment units that have been categorised under Over-exploited (OE), Critical and Semi-Critical category and other priority areas like large urban agglomerations, industrial clusters, mining areas, coastal areas, seismic zones IV and V and other such assessment units where adequate monitoring networks are not available. It is proposed to install Piezometers with Digital Water Level Recorders and telemetry in such areas for generation of high frequency water level data. The exercise would be taken up in convergence with the National Hydrology Project (NHP) and Atal Bhujal Yojana. Considering existing piezometers with CGWB and State Governments and those proposed under NHP and Atal Bhujal Yojana, additional requirement of piezometers has been estimated as 9,000. The actual requirement may vary depending upon field conditions like availability of suitable sites, security issues etc. Out of the 9,000 Piezometers, 2000 piezometers are planned to be constructed through in-house resources of CGWB under the Ground Water Management and Regulation Scheme. Construction of remaining 7000 piezometers and installation of 7,000 DWLR with telemetry is envisaged to be implemented through outsourcing under this project at

an estimated expenditure of Rs 529 Cr. High frequency water level data would form a critical input to water budgeting and planning for ground water interventions at the Panchayat level in such areas besides providing inputs in planning for source sustainability under Jal Jeevan Mission and participatory ground water management under Atal Bhujal Yojana.

II) High resolution Mapping of aquifers through heli-borne geophysical survey in nearly 3 lakh sq km area in arid parts of NW India

It is proposed to take up high resolution mapping of aquifers on 1:10,000 scale in ~3 lakh sq km area covering the arid parts of the States of Rajasthan, Gujarat, Haryana, Punjab and Himachal Pradesh using heliborne geophysical studies. The heli-borne study is aimed at providing 3D Geophysical model of the area, Geophysical Thematic maps at horizontal and vertical planes, Aquifer Geometry in 2D and 3D, Aquifer system with fresh and saline zones and suitable sites for artificial recharge. Spatial distribution of paleochannel network, if any, and its linkage with aquifer system would also be studied. The high resolution maps would help in formulating site specific plans for improving groundwater situation in the water stressed regions. An outlay of Rs 151 crores is envisaged towards conducting heliborne survey along with associated ground geophysical surveys covering 3 lakh sq km during 2022-25.

III) Data Generation for Aquifer Mapping through construction of 1135 wells in parts of 11 States of the country

National Aquifer mapping and Management Programme was initiated under the Ground Water Management and Regulation Scheme during the XII Plan (2012-17) period. Under the programme, an area of nearly 24.8 lakh sq km of the country has been identified for mapping out of which an area of 21.15 lakh sq km has already been covered upto March 2022. To meet the data generation requirement for completing aquifer mapping on 1:50,000 scale by March 2023, drilling of 1135 wells through outsourcing has been envisaged in parts of 11 States of the country. This activity is proposed to be taken up through outsourcing. The data generated through outsourcing will be used for preparation of aquifer maps and management plans. An outlay of Rs 125 Crore has been envisaged towards this activity.

18.2 PROJECT ACTIVITIES AND OUTPUTS

The outputs of this project will serve as inputs for preparing of aquifer maps and management plans under the ground water management and regulation scheme. This information, in turn will be shared with state government departments for planning sustainable ground water management.

Activity wise major outputs are given in the following table.

S.No.	Item	Measurable Outcomes
1	Construction of 7000 Piezometers and installation of DWLRs with telemetry in OE, Critical, Semi- Critical assessment units and other priority areas	Creation of infrastructure for automated monitoring of water levels in water stressed assessment units. High frequency data would also provide scientific inputs for driving corrective action by State/Central agencies based on ground water level monitoring system. High density and high frequency water level data would form a critical input to water budgeting and planning for ground water interventions at the Panchayat level in such areas and in improved assessment of groundwater resources.
2	Heli-borne surveys for high resolution mapping in ~3 lakh sq km area in arid parts of NW India	High Resolution mapping of aquifers on 1:10,000 scale in arid parts of the country will help formulate better ground water management plans. Will help provide Pin-pointed sites and depth zones for taking up appropriate management interventions for water harvesting and recharge in water stressed areas.
3	Data Generation for Aquifer Mapping through construction of Wells in 11 States	Construction of 1135 wells for data generation for completing remaining works of Aquifer Mapping on 1:50,000 scale. Improved information about aquifer disposition, quantification of available ground water resources and assessment of groundwater quality.

18.3 ESTIMATED COST

Total estimated cost of the project is Rs 805 Cr (Rupees Eight Hundred and five crores) for the entire project period i. 2022-23 to 2025-26. Year wise and activity wise proposed outlay for the project are given in the following table.

Table 1: Estimated Cost

Activity	Total Physical targets	Financial outlay (Rs in Cr)				
		2022-23	2023-24	2024-25	2025-26	Total
Construction of Piezometers	7000 no of piezometers	25	150	150	59	**384**
Installation of DWLRs with telemetry	7,000 no of DWLRs	-	21	62	62	**145**
Heli-borne surveys for high resolution mapping in 3 lakh sq km area in arid parts of NW India	~3 lakh sq km	45	45	61	-	**151**
Data Generation for Aquifer Mapping through construction of Wells in parts of 11 States	1135 number of wells	55	70	-	-	**125**
Total		**125**	**286**	**273**	**121**	**805**

CHAPTER 19 GROUND WATER MODELLING

19.1 MODELLING TALK SERIES -CENTRE OF EXCELLENCE FOR GROUND WATER MODELLING

Centre of Excellence (CoE) has been set up with its headquarters in Faridabad and four regional Hubs each at NR, Lucknow, SECR, Chennai, SER, Bhubaneswar& WCR, Ahmedabad who are further looking after the regional offices as listed below.

Regional Hub	Location	Regions
North	NR, Lucknow	NWR, Chandigarh/ NWHR, Jammu/ NHR, Dharamshala/ UR, Dehradun/ SUO, Delhi
South	SECR, Chennai	SR, Hyderabad/ SWR, Bangalore/ KR, Trivandrum
East	SER, Bhubaneswar	ER, Kolkata/ MER, Patna/ NER, Guwahati
West	WCR, Ahmedabad	WR, Jaipur/ NCR, Bhopal/ NCCR, Raipur/ CR, Nagpur

19.2 The core activities of Centre of Excellence includes

- Providing technical expertise and hands-on support on ground water modelling.
- Coordinate the activities related to Ground Water Modelling with the officers concerned and Mentors.
- Progress monitoring on Modelling assignments to CGWB officers
- Organize regular presentations for exchange of knowledge within CGWB and with other institutions/departments/organizations.
- Standardization of methods and procedures for modelling and preparation of protocols in consultation with TAMC
- Procurement, distribution and maintenance of modelling as well as various GIS software in CGWB
- Coordination with NHP modelling team.
- Inventory of availability of important Software in CHQ as well as at the regional offices of CGWB is completed.
- Groundwater modelling framework document has been prepared and provided to all the modelers of CGWB.

CHAPTER 20 PRADHAN MANTRI KRISHI SINCHAYEE YOJANA

20.1 Introduction to Pradhan Mantri Krishi Sinchayee Yojana- Har Khet Ko Pani - Ground Water (PMKSY-HKKP-GW)

The Government has formulated Pradhan Mantri Krishi Sinchayi Yojana *'Har Khet Ko Pani'* Ground Water Irrigation scheme with the vision of extending the coverage of irrigation in a focused manner. The scheme of PMKSY was approved in 2015-16 for creating additional irrigation from ground water resources. Ground water component aims utilizing ground water for irrigation purpose in areas, where ground water is sufficiently available. Further, to enhance small and marginal farmer's income in such areas by providing assured irrigation facility under the scheme. The Operational guidelines were issued by the Department of Water Resources, RD & GR, Ministry of Jal Shakti vide OM no P-15011/24/2016-SPR/2037-2105 dated July 2016. However, keeping in view of various requirements to implement the scheme and feedback received from State Governments, guidelines have been revised on 8.11.2018 and 28.05.2019. The scheme effectively launched in 2019-20 after revision of guidelines

Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti has sanctioned 13 projects under PMKSY-HKKP-GW scheme since 2019 in 10 States namely Assam, Arunachal Pradesh, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, Uttarakhand.

The total cost of these 13 ongoing projects are Rs. 978.00 Crore and Central Assistance is Rs.827.15 Crore of which **764.89** crore has already been released as on 31st March'23 by the Ministry of Jal Shakti.

Presently, nine project in Arunachal Pradesh, Phase – I & Phase - II, Assam, Phase – I & Phase - II, Manipur, Nagaland, Tripura, Phase – I, Uttrakhand and Gujarat have been completed with the help of combined effort of CGWB, DoWR, RD& GR, Ministry of Jal Shakti and concerned State Implementing AgencyAs on March 2023, 29695 wells have been constructed, 77964 Ha command area have been created and 67285 number of small & marginal farmers have been taken the benefit of PMKSY-HKKP-GW scheme so far.

The details & present status (As on 31st March 2023) of these projects are summarized in the following table:

Table 20.1 Status of Fund Allocation in last 4 years under PMKSY-HKKP-GW (as on 31 March' 23)

Fund release to the State in FY- 2019-20, 2020-21, 20221-22 and 2022-23 under PMKSY-HKKP-GW Scheme								
Sl. No	State	Projects	Project cost in crore	CA Releasd (As on 31 March'23)	CA released in 2019-20	CA released in 2020-21	CA released in 2021-22	CA released in 2022-23
1	Assam	Assam Phase-I	246.69	439.29	132.87	13	37.8	3.69
2		Assam Phase-II	292.96		0	10	242.29	0
3	Arunachal Pradesh	Arunachal Pradesh Pahse-I	45.3	79.90	24.46	10	5.985	0
4		Arunachal Pradesh Pahse-II	44.95		24.15	0	15.295	0
5	Gujarat	Gujarat	119.19	71.44	6	0	28.5	36.94
6	Nagaland	Nagaland	18.15	16.25	9.75	0	5.85	0.65
7	Manipur	Manipur	61.68	54.40	0	33.306	21.09	0
8	Mizoram	Mizoram	16.04	13.86	0	8.66	0	5.20
9	Tamil Nadu	Tamil Nadu	9.13	5.355	0	3.67	1.61	0.08
10	Tripura	Tripura Phase-I	13.31	43.63	7.15	0	2.38	0.264
11		Tripura Phase-II	48.34		0	0	26.1	7.735
12	Uttarakhand	Uttarakhand	15.89	13.72	0	1.36	12.36	0
13	Uttar Pradesh	Uttar Pradesh	46.37	26.69	16.69	0	0	10.00
	Total		**978.00**	**764.89**	**221.07**	**80.0**	**399.26**	**64.55**

Table 20.2 Status of PMKSY-HKKP-GW As on 31st March'23				
S.No	**Projects**	**Wells to be constructed Target/ Achievement Nos.**	**Project Command Target/ Achievement Ha.**	**Beneficiaries Target/ Achievement Nos.**
1	Assam-Ph-I	4779 / 4779	19116 / 19116	19643 / 19643
2	Assam Phase-II	4916/ 4916	19664/ 19532	17216/ 17200
3	Arunachal Pradesh-Ph-I	473 / 473	1785/ 1785	3350/ 3350
4	Arunachal Pradesh Ph –II	519/ 519	1957/ 1957	3633/ 3633
5	Nagaland	262/ 262	667/ 667	264/ 264
6	Tripura Ph-I	231/ 231	339/ 339	851/ 851
7	Tripura Ph-II	890/ 885	2670/ 735	1639/ 1166
8	Manipur	550/ 550	2057/ 2057	1445/ 1445
9	Mizoram	209/ 133	553/ 333.4	411/ 296
10	U.P.	14752/ 14752	36365/ 27944	15252/ 15252
11	Uttarakhand	206/ 206	1030/ 1030	1085/ 1085
12	Gujarat	1826/ 1826	1866/ 1866	1908/ 1908
13	Tamil Nadu	166/ 163	610/ 603	1233/ 1192
	Total	**29779/ 29695**	**88679/ 77964**	**67930/ 67285**

CHAPTER 21 SEMINARS/MEETINGS/WORKSHOPS/PAPER CLIPS

21.1 Workshops & Seminars

- CGWB, CR, Nagpur conducted Tier-II Training Programme on "Regional Groundwater Management issues" in Alibag, Raigad District, Maharashtra from 02.11.2022 to 04.11.2022 under the aegis of RGNGWTRI, Raipur and @ 75 years of India's Independence (Azadi Ka Amrit Mahotsav). Sh. Amol Abhiman Yadav, Additional Collector, Raigad District was the Chief Guest. Sh. Dattatray kalbhor, Agriculture Dy. Director, Raigad; Smt. Angai Salunkhe, CO, Alibag Nagar Parishad, Alibag, Raigad Distrct; Sh. B. R. Tonpe, Deputy Director, GSDA, Konkan Region and Sh. H. M. Sangnor, Sr. Geologist, GSDA, Alibag, Raigad district were present as Guest of Honors.

One day Workshop was organized by CGWB, NER Guwahati on “Success Stories of NAQUIM STUDIES and PMSKY-HKKP-GW Schemes in North East and East India” , in Guwahati on 24.01.2023. Shri Ashok Singhal, Hon’ble Minister of Housing, Urban affairs & Irrigation, Govt of Assam inaugurated the workshop. The Guest of Honour Shri Subodh Yadav, I.A.S., Joint Secretary (Admin, IC&GW), DoWR, RD & GR, Ministry Jal Shakti, Govt of India, addressed the gathering with a power point presentation on present groundwater scenario of India, National Perspective of NAQUIM and future plans. Shri M. S. Manivannan, IAS, Commissioner and Secretary Irrigation, Govt of Assam, discussed on implementation of PMKSY-HKKP-GWI schemes in the State of Assam. Shri A. K. Agarwal, Member (HQ), CGWB, Faridabad discussed on concept of NAQUIM and its objectives and scope of utilisation NAQUIM outputs. Shri Pabitra Ram Khaund, IAS, Secretary, Irrigation, Govt of Assam elaborated on the success of implementation of PMKSY-HKKP-GW scheme in the State of Assam.

21.2 Meetings/Visits by MPs/Ministers/Hon’ble Ministers

- Hon'ble Prime Minister took a Virtual Meeting to address all the new appointees in a function on 22nd November 2022 at 11:00 AM at GC, CRPF, Nagpur. Shri. Dinesh Laxman Waradi, Lab Attendant, CGWB, CR and Sh. Durga Prasad, Sc-C, CGWB,CR, Nagpur attended the function along with Sh. Thyagarajan, AO, CGWB, CR, Nagpur.

- Hon'ble Minister of Jal Shakti Sh. Gajendra Singh Shekhawat took a review meeting of Jal Jeevan Mission Scheme at Leela Palace, Chennai. Dr. A. Asokan, Regional Director, SECR, attended the meeting.
- Sh. Gajendra Singh Shekhawat, Hon'ble Minister of Jal Shakti released the report of dynamic ground water resource assessment for the country for the year 2022.
- Sh. Gajendra Singh Shekhawat, Hon'ble Ministry of Jal Shakti visited Meghalaya from 02.11.2022 to 04.11.2022 and reviewed implementation of Flagship Programmes of Govt. of India in the state.

➢ A Joint review meeting by Hon'ble Minister for Jal Shakti with Maharashtra state administration in presence of Hon'ble Chief and Deputy Minister on 14.11.2022 was conducted at Mumbai and the same was attended by Chairman, CGWB and Regional Director, CR with other officers.

- Hon'ble Minister of Jal Shakti Sh. Gajendra Singh Shekhawat took a joint review meeting with Maharashtra state administration in presence of Hon'ble Chief and Deputy Minister on 14.11.2022 at Mumbai. Chairman CGWB and Regional director, CR, Nagpur attended the meeting.

21.3 Visit of Chairman/ Members of CGWB & Other Meetings

- The 10th Meeting of Indian National Committee on Ground Water (INCGW) was held on 21.11.2022, under the Chairmanship of Chairman, CGWB.
- **The 1st Meeting of the** Ground Water Resource Estimation Committee to review and revise Ground Water Estimation Methodology 2015 (GEC- 2015) was held on 29.11.2022 under the chairmanship of Sri Sunil Kumar, Chairman, CGWB.
- The ninth meeting of the state Level Ground Water Co-ordination Committee (SGWCC) for National Aquifer mapping Programme of CGWB was held on 22.11.2022 under the Chairmanship of Secretary, Water Resources Department, Govt. of Kerala.
- A meeting of Core committee &Committee for preparation of operational guidelines for NAQUIM 2.Owas held on 23rdNovember, 2022 under the Chairmanship of Chairman, CGWB.
- Sh. Gajendra Singh Shekhawat, Hon'ble Minister of Jal Shakti convened a meeting in the conference hall of National Centre for Drinking Water, Sanitation & Quality (NCDWSQ), Joka, Kolkata, on 27.12.2022. Officers from CGWB, ER, Kolkata attended the meeting.
- Hon'ble Chief Minister Uttrakhand convened a meeting at Joshimath on 12.01.2023. Sh. Prashant Rai, Scientist- E and HOO CGWB, UR, attended the meeting.
- A team of officers from CGWB, UR and CGWB, NR visited Joshimath, Chamoli to study the land subsidence issues during 10.01.2023 to 16.01.2023.
- A state level meeting was convened by Hon'ble Chief Secretary of Uttarakhand on land subsidence issues in Joshimath, Chamoli through VC on 18.01.2023. HOO, UR attended the meeting.
- Chief Secretary, Bihar Pollution Control Board, Govt. of Bihar chaired a meeting on preparation of SOP (Standard Operating Procedure) for Regulation/ Conservation of Groundwater. Regional Director, CGWB, MER, Patna attended the meeting and provided the technical inputs.
- The 11th meeting on Indian National Committee on Ground Water (INCGW) was convened under the Chairmanship of Sh. Sunil Kumar, Chairman, CGWB, on 09.12.2022 through hybrid mode from CGWB, CHQ, Faridabad to review the R & D proposals received from various Institutions/ Universities for funding from Ministry of Jal Shakti.
- Chairman, CGWB joined the meeting virtually to review Key Performance Indicators (KPIs) of Composite Water Management Index held under the chairpersonship of Prof. Ramesh Chand, Hon'ble member NITI Aayog on 12.12.2022.
- Chairman CGWB attended the review meeting on BSR preparation for WARIMS Project review and vetting for CGWB held on 21.12.2022 under the chairmanship of Sh. Sriram Vedire, Adviser to Hon'ble Minister of Jal Shakti.

- A meeting was held between Chairman, Central Ground Water Board (CGWB), and Dy. Surveyor General, Survey of India, (SOI) on 10th February 2023 at Dehradun for working out the modalities on data sharing and collaborative studies between the organizations.
- The 12th meeting of Indian National Committee on Ground Water (INCGW) was held on 13th February,2023 under the Chairmanship of Chairman, CGWB through online mode in which Project proposals considered for R&D studies under INCGW were presented by various organizations/ institutions.
- The 10th meeting of the National Level Expert Committee (NLEC)was held under the Chairmanship of Chairman, CGWB during 1st and 2nd February ,2023 at CGWB, Bhujal Bhawan, Faridabad.
- A meeting regarding Detailing of Risk assessment of Tapi Basin under Phase II India EU water Partnership was held on 14th February 2023.
- 2nd review meeting for the project "Assess fluoride contamination in surface and ground water and its management in Garhwa district" was held on 21st February 2023. Meeting was attended by Head of Office, MER, Patna
- A meeting was convened by Deputy Collector, CEO Indore on CGWA guidelines for control and regulation of ground water extraction.
- The 4th Meeting of State Ground Water Coordination Committee (SGWCC) on Aquifer Mapping and Management was held under the Chairmanship of Additional Chief Secretary, Minor Water Resources Department, Govt. of Bihar at Vikash Bhawan, Patna, Bihar.
- Proposal of CGWB for the NAQUIM project was presented and discussed in the meeting of the Public Investment Board (PIB) held on 06th October, 2022 under the Chairmanship of Finance Secretary & Secretary (Expenditure) in Ministry of Finance, New Delhi

21.4 <u>Parliamentary Committee Visits</u>

- The committee of Parliament on Official Language under the Chairman ship of Prof Rita B Joshi held an inspection meeting with RD , CGWB, Easter Region Kolkata at Kolkata on 16.11.2022
- A meeting with Parliamentary Hindi Committee was held on 14.11.22 at Mayfair Resort Raipur. Various departments of the Central Government attended the meeting.
- Inspection Meeting of CGWB, WR, Jaipur by the Parliamentary Committee on Official Language held at Vigyan Bhawan, New Delhi on 17th October,2022
- Inspection Meeting of CGWB, Division Office, Ahmedabad by the Parliamentary Committee on Official Language held at Ahmedabad on 18th October,2022

- Parliamentary Standing Committee (PSC) on Water Resources visited Andaman and Nicobar Islands from 17.01.2023 to19.01.2023 and reviewed activities.
- Parliamentary Committee on Official Language visited CGWB, Divisional Office on 27th February,2023. Chairman CGWB attended the inspection meeting.

21.5 Press Clippings

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বিপ্লবী মেদিনীপুর টাইমস ১/১২/২০১২
ভূগর্ভস্থ জলের পুনরায় রিচার্জ বিষয়ক কর্মশালা ঝাড়গ্রামে
স্বপ্নীল মজুমদার (ঝাড়গ্রাম), ৩০ নভেম্বর : ভূগর্ভস্থ জলের পুনরায় রিচার্জ করে জলস্তর বৃদ্ধি এবং সুষ্ঠু ব্যবস্থাপনা গ্রহণ বিষয়ে বুধবার একদিনের প্রশিক্ষণ শিবির হল জামবনি ব্লকের কাপগাড়ির সেবাভারতী কলেজে। কেন্দ্রীয় সরকারের জলশক্তি মন্ত্রকের অধীনে কেন্দ্রীয় ভূমিজল বোর্ডের পূর্বক্ষেত্র কলকাতার উদ্যোগে এই প্রশিক্ষণ শিবিরের আয়োজন করা হয়। অনুষ্ঠানে উদ্বোধন করেন সেবাভারতী কলেজের অধ্যক্ষ দেবপ্রসাদ সান্ত। উপস্থিত ছিলেন কলেজের পড়ুয়া সহ অধ্যাপক-অধ্যাপিকারা। ভূগর্ভস্থ জল কিভাবে রিচার্জ হয় সে বিষয়ে আলোচনা করেন বিজ্ঞানী ইন্দ্রনীল রায়। কিভাবে ভূগর্ভস্থ জলের রিসোর্স ম্যানেজমেন্ট করা যায় সে বিষয়ে আলোচনা করেন সেবাভারতী কলেজের ভূগোলের বিভাগীয় প্রধান গুণব সান্ত। জিওফিজিক্যাল যন্ত্রের মাধ্যমে ভূগর্ভস্থ জলের সন্ধান ও চিহ্নিত করণ করার বিষয়ে আলোচনা করেন বিজ্ঞানী কে. সি. মণ্ডল। কেন্দ্রীয় ভূমিজল বোর্ডের পূর্ব ক্ষেত্রের (কলকাতা) রসায়নবিদ তথা প্রশিক্ষণের নোডাল অফিসার অতলান্ত নারায়ণ চৌধুরী বলেন, ভূগর্ভস্থ জল সম্পর্কে সাধারণ মানুষকে সচেতন করতেই এই প্রশিক্ষণ শিবির। কলেজ পড়ুয়াদের প্রশিক্ষণের মাধ্যমে নিজ নিজ এলাকায় ভূগর্ভস্থ জল বিষয়ে সচেতনতার বার্তা ছড়িয়ে দেওয়াই মূল লক্ষ্য।</td>
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<td>On the ocassion of World Water Day 2023 Celebration which was marked by workshop organized on 22.03.2023. The event was published in Aajkal newspaper.</td>
<td>Tier –III Training Programme On “Ground Water Development & Management Practices”, at Biplavi, Jamboni Block, Jhargram District, West Bengal.</td>
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মেদিনীপুর
পঁচেট পঞ্চায়েত প্রধানের পদত্যাগপত্র ঘিরে জল্পনা তুঙ্গে
৭ দিনের জেল হেফাজত ধৃত প্রাক্তন পুরপ্রধান শ্যামলের
মৎস্যজীবীদের প্রযুক্তিগত সহায়তা পূর্ব মেদিনীপুরে</td>
<td>
জল সংসাধন की संसदीय स्थायी समिति का दमण दौरा हुआ संपन्न: जल जीवन मिशन और स्वच्छ भारत मिशन की हुई समीक्षा</td>
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<td>Tier – III Training Programme On “Ground Water Development & Management Practices”, at Daspur, Ghatal Block, Paschim Medinipur District, West Bengal.</td>
<td>Media Coverage of Parliamentary Standing Committee visit in Daman on a study tour on Water Resources in Daman for the Period from 23.08.2022 to 25.08.2022. Member of the Committee and officers of the Lok Sabha Secretariat along with officers from DOWR, RD& GR, CGWB, CWC, WAPCOS, UT Of Daman, Diu, DNH present during the Study-visit program.</td>
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પાક્કું કચ્છનું છાપું

કચ્છ ભાસ્કર

સુવિચાર
પોતાની મસ્તીમાં જીવતી વ્યક્તિને કોઈની વાહ વાહની જરૂર નથી હોતી

સૂર્યોદય	આજે	સૂર્યાસ્ત
7.33 વાગ્યે	ભુજ	6.19 વાગ્યે
7.31 વાગ્યે	ગાંધીધામ	6.17 વાગ્યે

લખપત | અબડાસા | નખત્રાણા | ભુજ | માંડવી | મુન્દ્રા | અંજાર | ગાંધીધામ | ભચાઉ | રાપર

દિવ્ય ભાસ્કર | ભુજ, શુક્રવાર, 6 જાન્યુઆરી, 2023 | પોષ સુદ - 15 વિક્રમ સંવત 2079 | divyabhaskar.com

ભાસ્કર ઈન્ટરવ્યૂ કચ્છ યુનિ.માં સેન્ટ્રલ ગ્રાઉન્ડ વોટર બોર્ડનો ત્રિદિવસીય સેમિનાર, જળવૈજ્ઞાનિકોએ વરસાદી પાણી રીચાર્જની વ્યવસ્થા સૂચવી

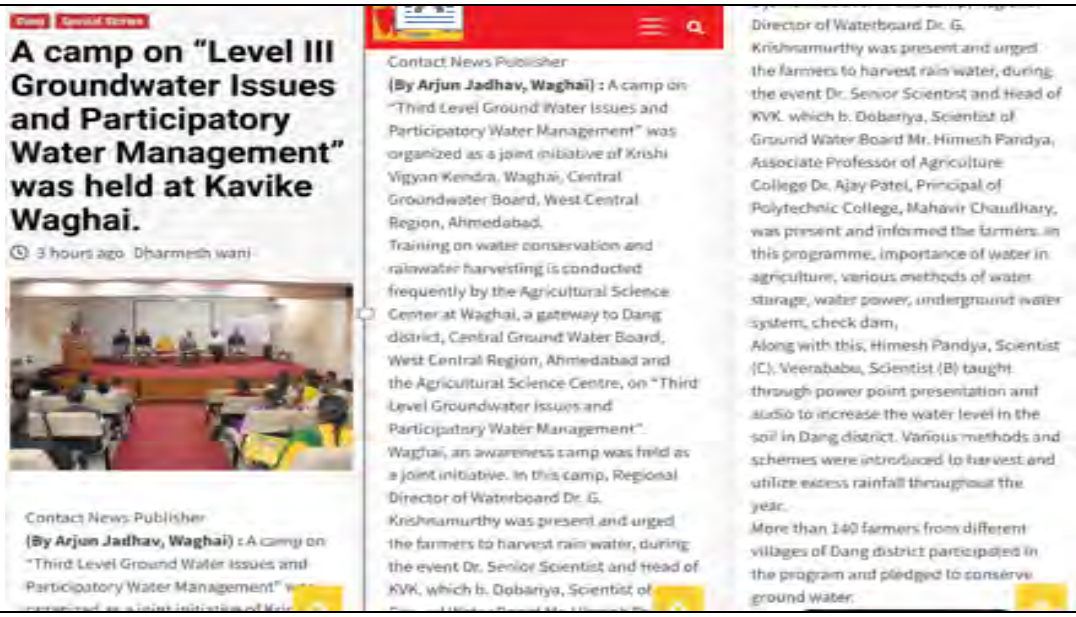
ભૂજળ માટે કચ્છ-ઉત્તર ગુજરાતમાં ચેતવું પડશે

દરિયામાં વહી જતા પાણીના લીધે જમીન અસંતુલિત થતાં તળ ખારા બન્યા અને મીઠું પાણી ખરીદવાનો સમય આવ્યો

બન્ની ભૂસ્તરીય વિરાસત:વરસાદ બાદ ની ઇકોસીસ્ટમ મજબૂત

રાજ્યમાં 189 બ્લોક્સ સુરક્ષિત :23માં 100ટકાથી વધુ

Media coverage of Tier II training program conducted under the aegis of Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) from 05/01/2023 to 07/01/2023 at KSKV University, Kachchh District, Gujarat on "Sustainable Development And Management of Groundwater Resources". A total of 35 participants actively participated in the training program.



A camp on "Level III Groundwater Issues and Participatory Water Management" was held at Kavike Waghai.

3 hours ago Dharmesh wani

Contact News Publisher

(By Arjun Jadhav, Waghai) : A camp on "Third Level Ground Water Issues and Participatory Water Management" was organized as a joint initiative of Krishi Vigyan Kendra, Waghai, Central Groundwater Board, West Central Region, Ahmedabad.
Training on water conservation and rainwater harvesting is conducted frequently by the Agricultural Science Center at Waghai, a gateway to Dang district, Central Ground Water Board, West Central Region, Ahmedabad and the Agricultural Science Centre, on "Third Level Groundwater Issues and Participatory Water Management". Waghai, an awareness camp was held as a joint initiative. In this camp, Regional Director of Waterboard Dr. G. Krishnamurthy was present and urged the farmers to harvest rain water, during the event Dr. Senior Scientist and Head of KVK, which b. Dobariya, Scientist of Ground Water Board Mr. Himesh Pandya, Associate Professor of Agriculture College Dr. Ajay Patel, Principal of Polytechnic College, Mahavir Chaudhary, was present and informed the farmers in this programme, importance of water in agriculture, various methods of water storage, water power, underground water system, check dam,
Along with this, Himesh Pandya, Scientist (C), Veerababu, Scientist (B) taught through power point presentation and audio to increase the water level in the soil in Dang district. Various methods and schemes were introduced to harvest and utilize excess rainfall throughout the year.
More than 140 farmers from different villages of Dang district participated in the program and pledged to conserve ground water.

Media coverage of Tier III training program jointly organized by CGWB and KVK Waghai in Dang District Gujarat 24 January 2023. A total of 140 people participated in the training program.

વઘઇ કૃષિ વિજ્ઞાન કેન્દ્ર ખાતે ભૂગર્ભ જળનું સંરક્ષણ શિબિરનો ૧૪૦ ખેડૂતોએ લાભ લીધો

● વાત્સલ્યમ સમાચાર
● મદન વૈષ્ણવ ડાંગ

કૃષિ વિજ્ઞાન કેન્દ્ર, વઘઈ, કેન્દ્રીય ભૂમિજળ બોર્ડ, પશ્ચિમ મધ્ય ક્ષેત્ર, અમદાવાદના સંયુક્ત ઉપક્રમે ''ત્રીજી કક્ષાનો ભૂગર્ભ પાણીને લગતા પ્રશ્નો અને સહભાગી પાણી વ્યવસ્થાપન વિષયક શિબિરનું આયોજન કરવામાં આવ્યું હતું ડાંગ જિલ્લાના પ્રવેશદ્વાર એવા વઘઈ ખાતે આવેલ કૃષિ વિજ્ઞાન કેન્દ્ર દ્વારા અવારનવાર પાણીનો બચાવ અને વરસાદી પાણીના સંગ્રહની તાલીમ યોજાતી રહે છે, ''ત્રીજા કક્ષાનો ભૂગર્ભ પાણીને લગતા પ્રશ્નો અને સહભાગી પાણી વ્યવસ્થાપન'' બાબતે કેન્દ્રીય ભૂમિજળ બોર્ડ, પશ્ચિમ મધ્ય ક્ષેત્ર, અમદાવાદ અને કૃષિ વિજ્ઞાન કેન્દ્ર, વઘઈ, સંયુક્ત ઉપક્રમે જાગૃતિ શિબિર યોજાઈ હતી. આ શિબીરમાં વોટરબોર્ડના રિજિયોનલ ડાયરેક્ટર ડો. જી. કૃષ્ણમૂર્તિ હાજર રહ્યા હતા અને તેમણે ખેડૂતોને વરસાદી પાણીના સંગ્રહ કરવાની હાંકલ કરી હતી, આ પ્રસંગ દરમિયાન કે.વી.કે ના વરિષ્ઠ વૈજ્ઞાનિક અને વડા ડો. જે બી. ડોબરીયા, ગ્રાઉન્ડ વોટર બોર્ડના વૈજ્ઞાનિક શ્રી હિમેશ પંડ્યા, કૃષિ કોલેજના સહપ્રાધ્યાપક ડો. અજય પટેલ, પોલિટેકનિક કોલેજના આચાર્યશ્રી, મહાવીર ચૌધરી, હાજર રહ્યા હતા.

Media coverage of Tier III training program jointly organized by CGWB and KVK Waghai in Dang District Gujarat 24 January 2023

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कडैया ग्राम पंचायत में ग्राम स्तरीय जलभृत सूचना एवं प्रबंधन पर सार्वजनिक विचार विमर्श कार्यक्रम का हुआ आयोजन

असली आजादी न्यूज नेटवर्क, दमण 30 जून। भारत की आजादी की 75वें साल में आजादी के अमृत महोत्सव के तहत जल शक्ति मंत्रालय, जल संसाधन, नदी विकास और गंगा संरक्षण विभाग के तहत केन्द्रीय भूमि जल बोर्ड पश्चिम मध्य क्षेत्र अहमदाबाद द्वारा दमण जिले के कडैया ग्राम पंचायत में आज ग्राम स्तरीय जलभृत सूचना एवं प्रबंधन पर सार्वजनिक विचार विमर्श कार्यक्रम का आयोजन किया गया। जिसमें केन्द्रीय भूमि जल बोर्ड अहमदाबाद के रिजनल डायरेक्टर जी. कृष्णामूर्ति एवं उनकी टीम (वैज्ञानिकों, एक्सपर्ट्स एवं सहायकों) के साथ उपस्थित होकर कडैया विस्तार के लोगों को भूमिगत जल के संचय, रखरखाव तथा जल बचाने के लिए विभिन्न तरीकों से अवगत कराया। इस दौरान उपस्थित 100 से अधिक लोगों को रजिस्ट्रेशन कीट, जल संचय से जुडे बातों को दर्शाने वाली पुस्तिका एवं मार्गदर्शिका भी दी गयी। कार्यक्रम में उपस्थित कडैया पंचायत के सरपंच शंकर पटेल ने गांव के लोगों को पानी बचाने की अपील की। कृषि विभाग के अधिकारी प्रवीण बगडा भी इस कार्यक्रम में उपस्थित रहे। ग्राम पंचायत के बोर्ड मेम्बर, पंचायत सेक्रेटरी, पंचायत स्टाफ के साथ बडी संख्या में ग्रामीण उपस्थित रहे।

Media Coverage of PIP organized in UT of Daman on 30th June 2022.

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'India is at forefront in mapping water sources'

Staff Reporter

"INDIA is at the forefront in mapping water sources and also in conducting research on availability of water in any specific area. Drinking water is now made available at all rural households", said Sunil Kumar, Chairman, Central Ground Water Board (CGWB), Faridabad. He was speaking at the one-day regional workshop on 'National aquifer mapping and management' (NAQUIM). The workshop was organised at at Chitnavis Center, on Friday.

Also present on the occasion A K Agarwal, Member, Central Ground Water Board, Member, CGWA; N Varadraj, Regional Director, Nagpur Region; and Kamalkishore Futane, Deputy Divisional Commissioner, Nagpur district.

In the first technical session Agarwal mentioned that the aquifer atlas which is a map of water resources across the country is under preparation by the Central Ground Water Board. NAQUIM is the backbone of groundwater resource management, he added.

Sunil Kumar, Chairman, Central Ground Water Board, Faridabed, performing 'Jal Puja' to inaugurate the one-day workshop on 'National aquifer mapping and management,' on Friday. Other digintaries also are seen.

Agarwal informed the gathering that the 'idea of 'Know the water sources, manage the water sources' is being carried by the local self-government bodies across the country. Committees are working at the state and local levels for the implementation of NAQUIM, he added.

N Varadraj, presented the Maharashtra aquifer study report on 'National Aquifer Mapping and Management' which is a component of the NAQUIM programme. He also stressed the workshop was jointly organised by Central Ground Water Board, Central Region, Nagpur; Central Ground Water Board North Region, Lucknow; North Central Chhattisgarh Region, Raipur; North Central Region, Bhopal under Department of Water Resources, Ministry of Jal Shakti, Government of India.

A large number of delegates from all regions participated in the one-day event.

Kartik Dongre, Senior Scientist, Central Ground Water Board, Central Region, Nagpur proposed a vote of thanks.

भूजल योजना से जुड़े सुझावों को लागू किया जाएगा

- जिलाधिकारी योगेश कुंभेजकर ने दी जानकारी

ब्यूरो | भंडारा

केंद्रीय भूमि जल बोर्ड का मध्यक्षेत्र, नागपुर द्वारा भंडारा जिले का भूजल प्रबंधन योजना का प्रारूप तैयार किया गया है। प्रारूप बोर्ड के अधिकारियों ने जिलाधिकारी योगेश कुंभेजकर को प्रस्तुत किया। भूजल योजना में दिए गए सुझाव क्रियान्वित किए जाएंगे। यह जानकारी जिलाधिकारी ने दी। इस समय जिलाधिकारी के साथ मुख्य कार्यकारी अधिकारी समीर कुर्तकोटी, जल जीवन मिशन से संबंधित सभी अधिकारी, बोर्ड के वैज्ञानिक अभय निवसरकार, निलाफर, प्रकाश महाराणा, जलापूर्ति व जल संपदा विभाग के अधिकारी उपस्थित थे। केंद्र सरकार के जल शक्ति मंत्रालय के केंद्रीय भूमि जल बोर्ड, जल संसाधन, नदी विकास व गंगा संरक्षण विभाग ने जिले का भूजल प्रबंधन प्रारूप तैयार किया है। इस प्रारूप में भंडारा जिले का तापमान, मौसम पूर्व व उसके पश्चात पानी स्तर आदि अभ्यास कर प्रारूप तैयार किया गया है। वैज्ञानिक नीलोफर ने रिपोर्ट प्रस्तुत की। भूमिजल बोर्ड के अधिकारियों से जिलाधिकारी ने जिले के भूजल प्रबंधन के बारे में चर्चा की। बोर्ड ने अपनी रिपोर्ट में सुझाई बातों पर कार्रवाई किए जाने की बात जिलाधिकारी योगेश कुंभेजकर ने कही।

भाषा भारती द्वारा हिंदीसेवियों का सत्कार

नागपुर। कृष्णमोहन मेमोरियल फाउण्डेशन, प्रयागराज द्वारा भारतीय भाषाओं के अस्तित्व एवं संघर्ष के लिए संचालित संस्था भाषा भारती ने प्रशासनिक क्षेत्र में कार्यरत दो हिंदी सेवियों का सत्कार किया। ऑफिसर्स क्लब स्थित वेलो रेस्ताँ के सभागार में एक सादे समारोह के दौरान मिनरल एक्सप्लोरेशन एण्ड कंसल्टेंसी लिमिटेड के वरिष्ठ हिंदी अधिकारी चंद्रशेखर तिवारी और केन्द्रीय भूमिजल बोर्ड के प्रभारी राजभाषा अधिकारी पूरणचन्द जिन्दुराम बाहमनिया का सत्कार किया गया।

भाषा भारती की ओर से हिंदी अधिकारीद्वय पूर्णिमा तलवारे एवं डॉ. वैशाली चिरडे ने शॉल व पुष्पगुच्छ दिया तथा सहायक निदेशकगण(राजभाषा) अनिल त्रिपाठी एवं पुरूषोत्तम साहू ने सत्कारमूर्तियों को प्रशस्ति-पत्र अर्पित किया। अरूणा वंजारी ने सरिता तिवारी को अंगवस्त्र प्रदान कर सत्कार किया। संयोजक डॉ. जय प्रकाश ने प्रशस्ति-पत्र का वाचन करते हुए उनके कार्यों का वर्णन किया। अनुवाद अधिकारी तेजवीर सिंह ने समयानुकूल शेरों-शायरी और ग़जल से सभी को प्रफुल्लित किया तो नवीन कुमार सिंह ने भोजपुरी का स्वर लगाते हुए फगुआ का राग छेड़ा और वातावरण को भावविभोर कर दिया। अनुवादक श्रीकांत राय ने सभी को धन्यवाद ज्ञापित किया। भाषा भारती के सर्वेसर्वा रवीन्द्र देवघरे और हेमंत रंगशाही ने हिंदीसेवियों को अपनी शुभकामनाएं प्रेषित की। इस अवसर पर पूर्व राजभाषा अधिकारी सी.एन. नंदेश्वर ने अंतरराष्ट्रीय मातृभाषा दिवस के बारे में में अपने विचार रखते हुए कहा कि माँ से मिलने वाली नेमतों में से एक मातृभाषा भी है जो बालक की वैचारिक क्षमता को सशक्त बनाती है और फिर सर्वांगीण विकास संभव हो पाता है। कार्यक्रम में अध्यापिका किरण त्रिपाठी, पुस्तकालय सहायक आशा प्रकाश, अपूर्वा बाहमनिया एवं प्रणय प्रकाश भी उपस्थित रहे।

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CGWB presents plan for Bhandara's ground water managememt

Collector Yogesh Kumbhejkar, Abhay Nivsarkar, Anant Jagtap and others at the meeting.

BHANDARA, Jan 27

THE plan for ground water management for Bhandara was presented by the Central Ground Water Board (CGWB) to Collector Yogesh Kumbhejkar, the other other day. The Collector has assured that administration will implement the plan and will take effective steps by duely following Board's recommendations. In a meeting held recently, Scientist Nilofar presented the plan and held elaborate discussions on it. The meeting was informed that the plan had been prepared taking into account rainfall, water-level recorded before and after monsoon, ground water table, geological conditions, availability of resources, etc.

Along with the Collector, Zilla Parishad CEO Sameer Kurtkoti, Board's scientist Abhay Nivsarkar, Prakash Maharana, Anant Jagtap and others were present in the meeting.

अमृत सरोवर एवं सीटीआर के कार्यों का उप सचिव ने किया निरीक्षण

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डिंडोरी. उप सचिव कृषि एवं किसान कल्याण मंत्रालय विभाग भारत सरकार सुशील पाल गहलोत ने मनरेगा के अमृत सरोवर, पुस्कर धरोहर, चैकडेम, स्टापडेम, निर्माण एवं सीटीआर के अन्य कार्यो का निरीक्षण किया गया। उप सचिव श्री गहलोत द्वारा जनपद पंचायत समनापुर की ग्राम पंचायत झांकी, पड़रिया-3 धनोली, जनपद पंचायत बजाग की ग्राम पंचायत बोदर, करौदा में एवं जनपद पंचायत करंजिया की ग्राम पंचायत सैलबार, बुंदेला एवं ग्राम पंचायत जाड़ामुरंग में मनरेगा योजना के अंतर्गत चल रहे निर्माण कार्यों का निरीक्षण किया एवं स्व- सहायता समूह की महिलाओं से चर्चा की एवं अमृत सरोवर स्थलों पर पौधरोपण भी किया। निरीक्षण के दौरान वैज्ञानिक कृषि एवं किसान कल्याण मंत्रालय विभाग भारत सरकार, जिला पंचायत सीईओ अरूण कुमार विश्वकर्मा, ग्रामीण यांत्रिकी सेवा डिंडोरी के कार्यपालन यंत्री डीएस बघेल, एनआरएलएम के डीपीएम मीना परते, मनरेगा के परियोजना अधिकारी ऋषिराज चढ़ार, वाटरशेड के तकनीकी अधिकारी स्टीपन इक्का, सीईओ प्रदीप कुमार बघेल, सहायक यंत्री पंकज सोलंकी, आर.जी. श्रीवास्तव सहित अन्य अधिकारी मौजूद रहे। उप सचिव ने जनपद पंचायत समनापुर की ग्राम पंचायत पड़रिया तीन में मनरेगा के अमृत सरोवर का अवलोकन किया और तालाब के लिए भूमि दान में देने वाले हितग्राही हरे सिंह धुर्वे से बात कर लोगों की भलाई के लिए दी गई ज़मीन की तारीफ की। उन्होंने तालाब में मत्स्य पालन के लिए बनाए गये यूजर ग्रुप महामाया स्व-सहायता समूह से मत्स्य पालन के सम्बन्ध में चर्चा की। उप सचिव ने तालाब के किनारे आम का पौध लगाया।

हिन्दी दैनिक **वान्टेड टाइम्स**

संक्षिप्त समाचार

जल शक्ति अभि.के तहत केन्द्रीय दल ने जल संरचनाओं का किया निरीक्षण

वान्टेड टाइम्स अनूपपुर। जल शक्ति अभियान के तहत जिले के जनपद पंचायत पुष्पराजगढ़ के विभिन्न कार्यों का जायजा केन्द्रीय दल भारत सरकार द्वारा निरीक्षण कर लिया गया, भारत सरकार के किसान कल्याण एवं कृषक विकास नई दिल्ली के उप सचिव वित्त सुशील पाल गहलोत तथा नेलोफर वैज्ञानिक सीजीडब्ल्यूबी नागपुर द्वारा 22 से 24 अगस्त तक जिले में जल शक्ति अभियान के तहत किए गए कार्यों के निरीक्षण के तहत 22 अगस्त को कलेक्ट्रेट में बैठक उपरांत जनपद पंचायत जैतहरी के ग्राम पंचायत जमुड़ी के ग्राम डिड़वापानी में निर्मित अमृत सरोवर का निरीक्षण किया गया, इसके बाद जिले के जनपद पंचायत पुष्पराजगढ़ के ग्राम पंचायत जीलंग एवं कोहका के ग्राम गोगा में निर्मित अमृत सरोवर का निरीक्षण व उपयोगकर्ता समूह के सदस्यों के चर्चा की गई, तत्पश्चात् केन्द्रीय दल के सदस्यों द्वारा ग्राम पंचायत लीला, बेनीबारी, करनपठार में निर्मित स्टॉप डेम, चेक डेम, बोल्डर वाल, गेबियन संरचना का निरीक्षण किया गया, 23 अगस्त को जनपद पंचायत पुष्पराजगढ़ के ग्राम पंचायत हर्राटोला, बरसोत एवं पोंड़की ग्राम पंचायतों के पुष्कर धरोहर कार्यों का निरीक्षण एवं उपयोगकर्ता समूहों से चर्चा की गई, इसके उपरांत ग्राम पंचायत भेजरी में पौधरोपण एवं प्रोसेसिंग यूनिट का निरीक्षण करते हुए स्वसहायता समूह के सदस्यों के साथ परिचर्चा की गई, तत्पश्चात् दल द्वारा ग्राम पंचायत किरगी में रूफ वाटर हार्वेस्टिंग का निरीक्षण किया गया, दल द्वारा जनपद पंचायत जैतहरी के ग्राम पंचायत पपरौड़ी एवं मौहरी में कैच दा रेन का निरीक्षण किया गया, निरीक्षण के दौरान ग्रामीण यांत्रिकी सेवा के कार्यपालन यंत्री ए.के. एक्का, जिला पंचायत के परियोजना अधिकारी मनरेगा रावेन्द्र पटेल, जनपद पंचायतों के मुख्य कार्यपालन अधिकारी, सहायक यंत्री, उप यंत्री मौके पर मौजूद रहे। दल द्वारा 24 अगस्त को जनपद पंचायत अनूपपुर एवं कोतमा के जल संरचनाओं के कैच द रेन कार्यों का निरीक्षण किया जावेगा।

उपयोगकर्ता समूह से किया संवाद

केंद्रीय दल ने किया अमृत सरोवर का निरीक्षण, देख कार्य

अनूपपुर » दबंग रिपोर्टर

जल शक्ति अभियान के तहत जिले के जनपद पंचायत पुष्पराजगढ़ के विभिन्न कार्यों का जायजा केन्द्रीय दल द्वारा लिया गया। केंद्र के किसान कल्याण एवं कृषक विकास के उप सचिव वित्त सुशील पाल गहलोत तथा नेलोफर वैज्ञानिक सीजीडब्ल्यूबी नागपुर द्वारा 22 से 24 अगस्त तक जिले में जल शक्ति अभियान के तहत किए गए कार्यों के निरीक्षण के तहत सोमवार को कलेक्ट्रेट में बैठक उपरांत जनपद पंचायत जैतहरी के ग्राम पंचायत जमुड़ी के ग्राम डिड़वापानी में निर्मित अमृत सरोवर का निरीक्षण किया गया। तत्पश्चात जिले के जनपद पंचायत पुष्पराजगढ़ के ग्राम पंचायत जीलंग एवं कोहका के ग्राम गोगा में निर्मित अमृत सरोवर का निरीक्षण व उपयोगकर्ता समूह के सदस्यों के चर्चा की गई। केन्द्रीय दल द्वारा ग्राम पंचायत लीला, बेनीबारी, करनपठार में निर्मित स्टॉप डेम, चेक डेम, बोल्डर वाल, गेबियन संरचना का निरीक्षण किया गया। 23 अगस्त को जनपद पंचायत पुष्पराजगढ़ के ग्राम पंचायत हर्राटोला, बरसोत एवं पोंड़की ग्राम पंचायतों के पुष्कर धरोहर कार्यों का निरीक्षण किया गया। इसके उपरांत ग्राम पंचायत भेजरी में पौधरोपण एवं प्रोसेसिंग यूनिट का निरीक्षण करते हुए स्वसहायता समूह के सदस्यों के साथ परिचर्चा की गई। तत्पश्चात दल द्वारा ग्राम पंचायत किरगी में रूफ वाटर हार्वेस्टिंग का निरीक्षण किया गया। दल द्वारा जनपद पंचायत जैतहरी के ग्राम पंचायत पपरौड़ी एवं मौहरी में कैच दा रेन का निरीक्षण किया गया।

रांची, बुधवार, 24 अगस्त 2022
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नीलकंठेश्वर डिग्री महाविद्यालय में विश्व जल दिवस मनाया गया

आजाद सिपाही संवाददाता

भुवनेश्वर। तालचेर साऊथ बलंडा स्थित नीलकंठेश्वर डिग्री महाविद्यालय में भारत सरकार के जल शक्ति मंत्रालय द्वारा विश्व जल दिवस मनाया गया। संयुक्त राष्ट्र प्रदत प्रेरणा के अनुसार जल और स्वच्छता के मुद्दों पर बदलती परिस्थितियों समस्याओं के कार्रवाई शीर्षक को लेकर आलोचना चक्र का आयोजन किया गया था। अध्यक्ष डॉ. बलभद्र जैना की अध्यक्षता में कार्यक्रम को तालचेर तहसीलदार डॉ रोजालिन बेहेरा और अंगुल संयुक्त शाखा समबाय बैंक के अध्यक्ष तथा तालचेर विधायक के प्रतिनिधि श्री संतोष कुमार प्रधान ने उद्घाटन किया।

कार्यक्रम के पहला चरण में महाविद्यालय के रसायन विज्ञान विभाग के मुख्य डॉ मधुमिता राउत राय, अंग्रेजी विभाग के मुख्य शुभदर्शिनी महापात्र और अर्थशास्त्र विभाग के प्रमुख और महाविद्यालय के अध्यक्ष डॉ बलभद्र जेना ने भूतल जल और परिमल के बारे में वक्तव्य रखे।

कार्यक्रम के दूसरे भाग में सेंट्रल भूजल बोर्ड भुवनेश्वर शाखा के वरिष्ठ वैज्ञानिक (डी) सुधांशु कुमार महांति, वरिष्ठ वैज्ञानिक (सी) भूजल के विनय कुमार बेहेरा, भूजल प्रबंधन वरिष्ठ रसायनज्ञ बिरंची नारायण देहुरी, विश्व जल दिवस पर अंगुल के वरिष्ठ रसायनज्ञ सिनोस पीके की भूजल गुणवत्ता की स्थिति संदर्भ में आलोचना की। उद्घाटनी कार्यक्रम को अध्यापक एवं प्रशासनिक अधिकारी सुमंत मांझी, और तकनीकी विभाग के कार्यक्रम का संचालन अध्यापिका शुभदर्शिनी महापात्र ने किया।

Celebration of WORLD WATER DAY,2023 on the theme 'Accelerating Change' on 22nd March,2023 at Nilkantheswar Degree Mahavidyalaya, Talcher, District-Angul, Odisha

Orissa TODAY
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Training camp held on groundwater development and management methods

Bargarh, (TNB): One-day training program on groundwater development and management methods has been held at Biju Pattanaik Town Hall Bargarh under the leadership of Gandhi National Groundwater Training and Research Institute Raipur, Government of India. Scientist Sudhanshu Kumar Mohanty of Central Geology Board Bhubaneswar participated in the program and highlighted the need for training camps. In this program Jaylal Shahish Executive Engineer of Agriculture Department, Rabi Narayan Senapati Asst. Director of Agriculture, Sudshanshu Shekhar Nanda Agriculture District Officer Bhatli and Gobind Chandra Panda Assistant Horticulture Officer Bargarh participated in the program and expressed their thought.

Scientists Rajeev Kumar Tripathy, Sudhanshu Kumar Mohanty, Biranchi Narayan Dehuri and Satyam Shukla were also present in the program and trained hundreds of farmers of Bargarh district through power point and field exhibition. In this program, all the assistant agricultural workers of the Agriculture Department of Bargarh cooperated and made the program a grand success.

One day Tier – III training programme on Ground Water Development and Management organized at Biju Pattanaik Town Hall, Bargarh, Dist-Bargarh on 24th December, 2022

କନ୍ଧମାଳ ଜିଲାରେ ଭୁଜଳ ପରିଚାଳନା ସମ୍ପର୍କିତ ବୈଠକ

ଫୁଲବାଣୀ -28/07(ସୁନୀଲ ପଟନାୟକ)କେନ୍ଦ୍ରୀୟ ଭୁଜଳ ପରିଷଦ ଭୁବନେଶ୍ୱର ଅନୁକୁଲ୍ୟରେ ଫୁଲବାଣୀ ଠାରେ ଜିଲା ପରିଷଦ ସମ୍ମିଳନୀ ଗୃହ ଠାରେ ଗୁରୁବାର ଏକ ଦିନିକିଆ ଭୁ ଜଳ ପରିଚାଳନା ସମ୍ପର୍କିତ ଜନବାର୍ତ୍ତାଳାପ କାର୍ଯ୍ୟକ୍ରମ ଅନୁଷ୍ଠିତ ହୋଇଥିଲା ଜିଲା ମୁଖ୍ୟ କୃଷି ଅଧିକାରୀ ବସନ୍ତ କୁମାର ପାଣିଗ୍ରାହୀ ଙ୍କ ସଭାପତିତ୍ୱ ରେ ଅନୁଷ୍ଠିତ ହୋଇଥିଲା ।ଅନ୍ୟମାନଙ୍କ ମଧ୍ୟରେ ପ୍ରକଳ୍ପ ନିର୍ଦ୍ଦେଶକ ଜଳ ଛାୟା ବିଭାଗ ପ୍ରଶାନ୍ତ ତ୍ରିପାଠୀ, ଉପ ପ୍ରକଳ୍ପ ନିର୍ଦ୍ଦେଶକ, ଆତ୍ମା, ଅନିଲ ପ୍ରହରାଜ, ପ୍ରମୁଖ ଯୋଗଦେଇ ଭୁଜଳର ସୁପରିଚାଳନା ସମ୍ପର୍କରେ ଆଲୋଚନା କରିଥିଲେ ।ସେହିପରି ଅନ୍ୟମାନଙ୍କ ମଧ୍ୟରେ କେନ୍ଦ୍ରୀୟ ଭୁଜଳ ପରିଷଦ ଭୁବେନଶ୍ୱର ବୈଜ୍ଞାନିକ ସୁଧାଂଶୁ ମହାନ୍ତି, ତରୁଣ ମିଶ୍ର, ପୂର୍ବ ବେରା, ମୁନ୍ନା ସୂତ୍ରଧର, ଯୋଗ ଦେଇ ଭୁଜଳ ର ସ୍ଥିତି, ଓପରିଚାଳନା ସମ୍ପର୍କରେ ସୁଚିନ୍ତିତ ମତ ବ୍ୟକ୍ତ କରିଥିଲେ ।ଏହି କାର୍ଯ୍ୟକ୍ରମ ରେ ସୁଧାଂଶୁ ମହାନ୍ତିଧନ୍ୟବାଦ ଅର୍ପଣ କରିଥିବା ବେଳେ ଅମିୟ ନାୟକ କୃଷି ଅଧିଦର୍ଶକ ସହଯୋଗ କରିଥିଲେ ।ଏଥିରେ ଜିଲ୍ଲାର ବିଭିନ୍ନ ଅଂଚଳରୁ ଚାଷୀ ମାନେ ଯୋଗ ଦେଇଥିଲେ ।

PIP at CDAO Office, Kandhamal, District- Kandhamal on 28.07.2022 on Local Ground Water Development and Management.

GOOD RESULT CAN BE ACHIEVED IF GROUNDWATER AUGMENTATION IS CARRIED OUT IN A SCIENTIFIC MANNER : ADDL COLLECTOR PRATHIMA SINGH

Medak, April 29 (Capital Information) Additional Collector, Local Bodies, Pratima Singh said that good results can be achieved by advancing in a scientific manner with a strategic plan for groundwater augmentation. Scientists Mahadev and Vittal presented the report to the district administration through a power point presentation on Friday at the video conference hall of the Collectorate, Central Ground Water Board, Central Hydroelectric Division, Hyderabad South, on aquifer maps (reservoir maps) and management plans for Medak district.

The Central Groundwater Team studied the results of various projects like Mission Kakatiya, Drip, Drip Irrigation, Check Dams, Clarification Tanks, Water Sheds, Rainwater Harvesting, Farm Ponds undertaken in Medak District and expressed their views on various activities yet to be undertaken for groundwater recharge. Provided. She said groundwater was being questioned in seven mondal in the district.

She said that groundwater would be improved if activities were carried out in this scientific manner. On the occasion, Additional Collector Pratima Singh said that in order to provide fresh water to future generations, everyone should be aware of water conservation. She is going to drive and catch every drop of water. It was suggested to conserve rain water, manholes in every house and roof water in schools. In agriculture, methods like drip and drip irrigation should be adopted.

Similarly, the construction of check dams and manholes will increase the groundwater level so that there will be no problems for cultivable water and drinking water. She pointed out that there are many resources available for groundwater augmentation and that everyone should be made partners in the process by looking at what works can improve groundwater. Advised the authorities to create awareness among the people especially at the village level.

Under the Water Power Program in the District, the DRDO Prathima Singh directed the authorities to send out success stories and photos on the progress made so far. Zilla Parishad CEO Shailesh, DRDO Srinivas, DFO were present on the occasion. Ravi Prasad,, D.P.O. Tharun Kumar, Irrigation Department EE Srinivas, Municipal Commissioner Srihari and other Department officials were present.

Sharing of NAQUIM report and presentation on NAQUIM Outcomes of Medak District,Telangana state on 29.04.2023

Increase the number of workers in NREGS

DISTRICT COLLECTOR P. UDAY KUMAR

DECCAN NEWS SERVICE
NAGARKURNOOL

District Collector P. Uday Kumar instructed the MPDOs to increase the number of laborers in NREGS . He held a review meeting with MPDOs, APOs, DRDA Additional Project Officers at the Collectorate

Conference Hall on Tuesday Speaking on the occasion, the Collector said, "We want to ensure that at least 200 laborers in each village work in the EGS ." It is suggested that ORS packets be distributed to workers in the workplace in view of the intensity of the Summer . The Workers should be encouraged to come for the work .

He asked the village meetings to approve the resolution and carry out the work only for the benefit of the farmers and the villagers.

According to the new software, 260 types of work can be done under the employment guarantee. They want to create new jobs to increase the number of workers.

The Collector also expressed displeasure over the MPDOs in the mandals where the number of laborers is less. He warned that action should not be taken if the number of laborers is not increased.Similarly, the Collector directed the MPDOs to immediately address the loan applications received from various welfare departments.

Earlier, the Collector had directed the authorities to take steps to increase groundwater from time to time under the Jalashakti Abhiyan to prevent groundwater depletion in the district.

The first district level coordination meeting was held in the Collector Chamber. On this occasion, the Collector said that through the Departments of Irrigation, Ground Water, DRDA, the ground water in any village should be collected and accurate data on how many feet deep and how deep the water should come when the bore wells are drilled and in coordination with the Central Water Energy Campaign Department officials in any village.

Want to make sure it needs to be increased. Accordingly, he said, works like percolation tank and soak pits to augment groundwater through employment would help in intensification.

The collector unveiled groundwater annual report book prepared in collaboration with various line departments .PD DRDA Narsing Rao, Additional PD Rajeshwari, APDs, Chandrasekhar Srinivasan, DRDA staff, District Groundwater Officer Ramadevi, Central Groundwater Board Scientist Dr.S.S.Mittal , GM Industries Hanumanthu Nayak, E.D. SC Corporation Ramlal, T. Mathews,M.Linguswamy participated.

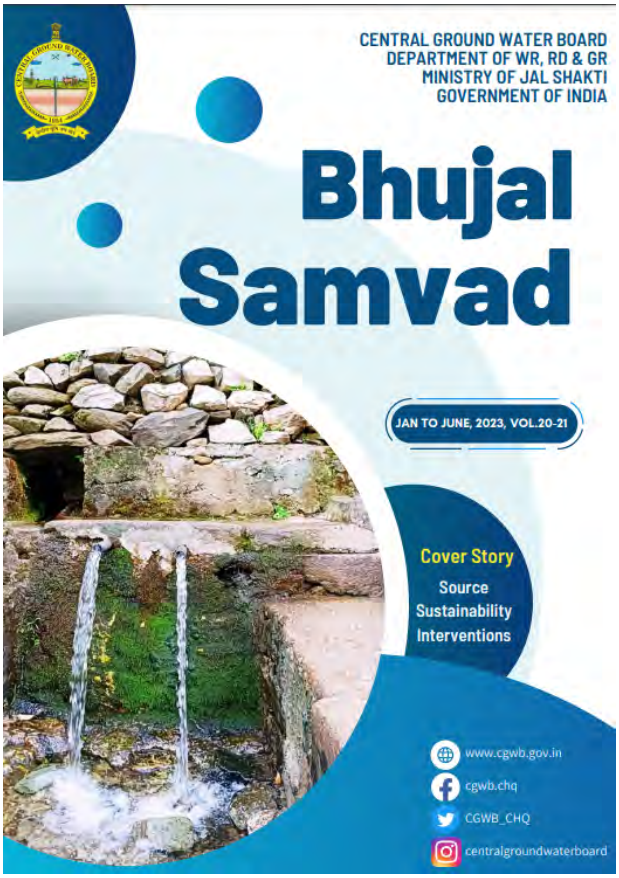
District Ground Water Coordination Committee meeting held at District Collectorate with District Collector Shri. P. Uday Kumar,IAS on 10.05.2022.

CHAPTER 22 BHUJAL SAMVAD

22.1 INTRODUCTION

BhujalSamvad, a quarterly publication by the Central Ground Water Board, was inaugurated in January 2018. Since then, it has seen the release of 21 editions until June 2023, all of which are available on the official CGWB website. The magazine is divided into several distinct segments, including the Cover Story, In Focus, NAQUIM Success Stories, Reports, Pathshala, and Social Media Highlights and Collectables.

The intention of the "Cover Story" is to spotlight significant and prominent endeavors undertaken by the Board. 'In Focus' captures the primary activities and accomplishments of CGWB in each quarter, providing readers with a snapshot of its major achievements. The "Report" and "Pathshala" segments aim to educate our diverse readership about various hydrogeological subjects.



A dedicated section of the magazine is dedicated to the triumphs of the NAQUIM initiative, which is a pivotal undertaking of CGWB. This exclusive section serves as a platform to share the success stories related to NAQUIM.

22.2 Azadi Ka Amrit Mahotsva (AKAM)

"Azadi Ka Amrit Mahotsav" mirrors India's socio-cultural, political and economic progress. The official journey of "Azadi ka Amrit Mahotsav" commencedfrom12th March, 2021with a 75 week countdown to our 75th anniversary of Independence, and shall end after a year on 15th August, 2023. The Department has prepared programmes for all of the 75 weeks to celebrate this Mahotsav at large scale by using resources of various wings, organizations, institutions and attached and subordinate offices of this Department spread all over India. Apart from the programmes mentioned in the calendar for 75 weeks.

22.3 Public Interaction Programme:

Under Azadi ka Amrit Mahotsva, Public Interaction Programmes were organized to sensitize people in respect of NAQUIM outcomes and water related aspects in all States.

Details are as follows:

Table 22.1 List of PIPs conducted during the current year (2022-23)

		PIPs conducted during the current year (2022-23) including those in Worlds Water Day		
Sl No.	**State/UT**	**PIP**	**Total No of**	**Number of**
1	Andhra Pradesh	6	545	158
2	Bihar	20	1999	681
3	Chandigarh	1	17	5
4	Chhattisgarh	12	755	264
5	Goa	0	0	0
6	Gujarat & UT of	14	1340	499
7	Haryana	6	694	300
8	Himachal Pradesh	7	703	273
9	Jammu & Kashmir	11	618	204
10	Ladakh	0	0	0
11	Jharkhand	9	873	573
12	Karnataka	11	1033	304
13	Kerala	7	662	357
14	Madhya Pradesh	12	1354	427
15	Maharashtra	13	946	337
16	NER States	20	1956	941
17	Odisha	13	1560	438
18	Punjab	14	1892	991
19	Rajasthan	14	1843	594
20	SUO- Delhi	3	224	74
21	Tamil Nadu & U.T of	13	1591	764
22	Telangana	9	955	273
23	Uttar Pradesh	21	2588	800
24	Uttarakhand	6	759	391
25	West Bengal	10	816	387
	Total	**252**	**25723**	**10035**

		PIPs conducted during 2018 to 31st March 2022 A			PIPs conducted during the current year (2022-23) including those in Worlds Water Day B			Cumulative Number of PIPs A+B		
Sl No.	State/UT	PIP conducted	Total No of people sensitized	Number of women sensitized	PIP conducted	Total No of people sensitized	Number of women sensitized	PIP conducted	Total No of people sensitized	Number of women sensitized
1	Andhra Pradesh	26	1641	319	6	545	158	32	2186	477
2	Bihar	44	4142	1428	20	1999	681	64	6141	2109
3	Chandigarh	4	322	128	1	17	5	5	339	133
4	Chhattisgarh	58	4744	1677	12	755	264	70	5499	1941
5	Goa	2	97	31	0	0	0	2	97	31
6	Gujarat & UT of Daman & Diu	51	3189	706	14	1340	499	65	4529	1205
7	Haryana	32	2652	1107	6	694	300	38	3346	1407
8	Himachal Pradesh	40	3280	1613	7	703	273	47	3983	1886
9	Jammu & Kashmir	17	856	124	11	618	204	28	1474	328
10	Ladakh	2	41	1	0	0	0	2	41	1
11	Jharkhand	40	3055	1526	9	873	573	49	3928	2099
12	Karnataka	75	5629	1939	11	1033	304	86	6662	2243
13	Kerala	55	4586	3106	7	662	357	62	5248	3463
14	Madhya Pradesh	55	5173	1005	12	1354	427	67	6527	1432

		PIPs conducted during 2018 to 31st March 2022 A			PIPs conducted during the current year (2022-23) including those in Worlds Water Day B			Cumulative Number of PIPs A+B		
Sl No.	**State/UT**	**PIP conducted**	**Total No of people sensitized**	**Number of women sensitized**	**PIP conducted**	**Total No of people sensitized**	**Number of women sensitized**	**PIP conducted**	**Total No of people sensitized**	**Number of women sensitized**
15	Maharashtra	60	6093	1661	13	946	337	73	7039	1998
16	NER States	82	4744	2115	20	1956	941	102	6700	3056
17	Odisha	73	7308	3091	13	1560	438	86	8868	3529
18	Punjab	20	3148	1262	14	1892	991	34	5040	2253
19	Rajasthan	48	3400	329	14	1843	594	62	5243	923
20	SUO- Delhi	14	923	302	3	224	74	17	1147	376
21	Tamil Nadu & U.T of Puducherry	69	7750	4398	13	1591	764	82	9341	5162
22	Telangana	52	2916	868	9	955	273	61	3871	1141
23	Uttar Pradesh	100	9691	3329	21	2588	800	121	12279	4129
24	Uttarakhand	22	1572	354	6	759	391	28	2331	745
25	West Bengal	58	5538	1656	10	816	387	68	6354	2043
	Total	**1099**	**92490**	**34075**	**252**	**25723**	**10035**	**1351**	**118213**	**44110**

23 .0 VIGILANCE

7 disciplinary cases were under investigation which were brought forward from the year 2022. Out of these 7 cases, 5 inquiries have been completed and disposed off and 2 are under investigation. 1 chargesheet has been issued during the year 2022-23 which is under investigation. In addition to above, 1 case is being investigated by CBI, 4 cases have been sent to Ministry for the preliminary inquiry and 1 is under preliminary investigation in CGWB.

24.0 RTI

The opening balance of RTI applications of CHQ as on 01.04.2022 was 46 Requests and 07 Appeals. During the year 2022-2023, 283 RTI requests and 68 appeals were received. Total RTI requests and RTI appeals disposed of were 329 and 75 respectively. Details are given in table.

Table 24.1 Status Of RTI Applications For Year 2022-23 At Chq

Received during the year (including cases transferred from other public authorities)	Opening balance at CHQ as on 01.04.2022	Decisions where requests/appeals rejected
Requests: 329	Requests: 46	Requests: NIL
Appeals: 75	Appeals: 07	Appeals: NIL

CHAPTER 25 BUDGET AND EXPENDITURE

25.0 Budget and Expenditure

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
Demand No. 62					PLAN (GWM & R)	
					2702 (REVENUE)	
Major Head: 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION					**(Figure in lacs)**	
Sub Head: 16-GROUND WATER MANAGEMENT & REGULATION –01 (Ext.)02(Dom.)						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
1	2		5	6	7	8
16.02 Dom. Supp. (GWM&R) PLAN						
16.02.03 - समयोपरिभत्ता -OTA	1.00	0.00	0.32	**0.00**	0.32	-0.32
16.02.06 - चिकित्सा - M/Treat.	350.00	250.00	242.88	**-1.19**	241.69	8.31
16.02.11 - यात्राव्यय - D.T.E.	4000.00	3500.00	2958.67	**1.85**	2960.52	539.48
16.02.12 - विदेशयात्राव्यय - F.T.E.	50.00	0.00	0.53	**0.00**	0.53	-0.53
16.02.13 - कार्यालयव्यय - O.E.	1500.00	1698.00	1594.76	**0.24**	1595.00	103.00
16.02.14 - किराया, दरएवंकर - R.R.T.	500.00	500.00	413.34	**0.00**	413.34	86.66
16.02.16 - प्रकाशन - Publications	2.00	2.00	1.74	**0.00**	1.74	0.26
16.02.20 - O.A.E.	200.00	400.00	315.76	**4.89**	320.65	79.35
16.02.24 - पेट्रोल / डीजल -P.O.L.	2000.00	2000.00	1740.41	**17.96**	1758.37	241.63
16.02.27 - माइनरकार्य - Min/Works	400.00	400.00	237.94	**0.00**	237.94	162.06
16.02.28 - वृत्तिकसेवाएँ - P.S.	500.00	500.00	301.71	**0.60**	302.31	197.69
16.02.30 - Other Contractual Services	600.00	500.00	401.83	**0.73**	402.56	97.44
16.02.33 - Subsidies	0.00	0.00	0.00	**0.00**	0.00	0.00

16.02.43 - उचंतस्टाक - S/Stock	1600.00	1700.00	351.17	**1006.91**	1358.08	341.92
16.02.50 - अन्यप्रभार - O. C.	50.00	0.00	0.00	**0.00**	0.00	0.00
16.99.13 - OE (Information Technology)	200.00	200.00	134.76	**0.00**	134.76	65.24
Total Dom Support (PLAN)GWM&R	**11953.00**	**11650.00**	**8695.82**	**1031.99**	**9727.81**	**1922.19**
						PLAN (GWM & R)
Demand No. 62						4702 (CAPITAL)
Major Head: 4702-MINOR IRRI-02-005-Capital Nature Expenditure (GWM & R)						
Sub Head: 06.01-GROUND WATER MANAGEMENT & REGULATION						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
4702.00.800.06 (Capital Nature Expenditure) GWM&R, Dom. Supp. (Scheme) PLAN						
1	2		5	6	7	8
06.01.51 - मोटरवाइकल - M.V.	701.00	550.00	210.41	**12.28**	222.69	327.31
06.01.52 - मशीनरीएवंउपस्कर - M & E	3500.00	550.00	221.74	**57.97**	279.71	270.29
06.01.53 - मुख्यकार्य - M/Works	20996.00	18400.00	5178.85	**2549.43**	7728.28	10671.72
Total Capital Nature Expenditure	**25197.00**	**19500.00**	**5611.00**	**2619.68**	**8230.68**	**11269.32**
Grant Total (Revenue + Capital)	**37150.00**	**31150.00**	**14306.82**	**3651.67**	**17958.49**	**13191.51**

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
						Tribal Sub Plan (TSP)
Demand No. 62						
Major Head: 01.796 Tribal Sub Plan (Rewvnue)MINOR IRRI-00-445-GROUND WATER INVESTIGATION						
Sub Head: 16-GROUND WATER MANAGEMENT & REGULATION –02.01 (Ext.)02(Dom.)						
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						**(Fig. in lacs)**
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
1	2		5	6	7	8
02.01 Dom.Supp.(GWM&R) PLAN						
16.02.06 - चिकित्सा - M/Treat.	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.11 -यात्राव्यय - D.T.E.	450.00	450.00	388.88	**3.35**	392.23	57.77

16.02.12 - विदेशयात्राव्यय -F.T.E.	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.13 -कार्यालयव्यय -O.E.	200.00	200.00	176.63	**-0.16**	176.47	23.53
16.02.14 -किराया, दरएवंकर - R.R.T.	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.16 -प्रकाशन - Pub.	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.20 -ओ.ए.ई -O.A.E.	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.24 - पेट्रोल / डीजल - P.O.L.	200.00	200.00	163.60	**0.01**	163.61	36.39
16.02.27 -माइनरकार्य - Min/Works	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.28 -वृत्तिकसेवाएँ - P.S.	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.30 -Other Contractual Services	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.43 -उचंतस्टाक - S/Stock	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Dom Support (PLAN)	**850.00**	**850.00**	**729.11**	**3.20**	**732.31**	**117.69**
Total Revenue (GWMR + TSP)	**12803.00**	**12500.00**	**9424.93**	**1035.19**	**10460.12**	**2039.88**

						00.796- CAPITAL
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
Demand No. 62						Tribal Sub Plan
Major Head:00.796-Tribal Sub Plan CAPITAL OUTLAY-00.112 GROUND WATER (M.I.)						
Sub Head: 01.01-Tribal Sub Plan (Capital)						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
00.796- CAPITAL						
1	2		5	6	7	8
01.01.51 -मोटरवाहन -Motor Vehicle	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.52 -मशीनरीएवंउपस्कर -M & E	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.53 -मुख्यकार्य -M/Works	1000.00	1000.00	89.08	**-9.36**	79.72	920.28
Total: 00.796- CAPITAL	**1000.00**	**1000.00**	**89.08**	**-9.36**	**79.72**	**920.28**
Total CAPITAL (GWMR + TSP)	**26197.00**	**20500.00**	**5700.08**	**2610.32**	**8310.40**	**12189.60**
Grant Total TSP (Revnue+Capital)	**1850.00**	**1850.00**	**818.19**	**-6.16**	**812.03**	**1037.97**
Grant Total (TSP + GWMR)	**39000.00**	**33000.00**	**15125.01**	**3645.51**	**18770.52**	**14229.48**

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
Demand No. 62		**4702 - (Capital)**			DEDUCTED RECOVERY	
केन्द्गीयभूमिजलबोर्ड (जलशक्तिमंत्रालय)					ISSUE TO WORK & OTHER CREDITS	
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
Major Head: 4702-MINOR IRRI-00-800-Capital Nature Expenditure (GWM & R) (PLAN)						
Sub Head: 06 ISSUE TO WORK & OTHER CREDITS						
						(Fig. in lacs)
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
1	2		5	6	7	8
06.01.70 -Issue to Work	1500.00	1500.00	1489.40	-294.54	1194.86	305.14
06.02.70 -Other Suspense Charges	0.00	0.00	0.00	0.00	0.00	0.00
Total Recoveries	**1500.00**	**1500.00**	**1489.40**	**-294.54**	**1194.86**	**305.14**
NET CGWB (GWM&R)	**35650.00**	**29650.00**	**12817.42**	**3946.21**	**16763.63**	**12886.37**
NET CGWB (GWM&R + TSP)	**37500.00**	**31500.00**	**13635.61**	**3940.05**	**17575.66**	**13924.34**

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
						Non Scheme (Non Plan)
Demand No. 62					Establishment Expenditure 2702 (Revenue)	
Major Head: Establishment Expenditure 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION						
Sub Head: 01-CGWB-01.01-H/QRS.						**(Fig.in lacs)**
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
1	2		5	6	7	8
16.02 Domestic Support N.PLAN						
01.01.01 - वेतन -Salary	25900.00	26989.00	26892.32	**45.59**	26937.91	51.09
01.01.02 -मजदूरी -Wages	250.00	200.00	138.17	**0.46**	138.63	61.37
01.01.03 -समयोपरिभत्ता -OTA	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.06 -चिकित्सा -M/Treat.	194.00	150.00	133.37	**0.59**	133.96	16.04
01.01.11 -यात्राव्यय -D.T.E.	250.00	200.00	144.43	**1.44**	145.87	54.13

01.01.12 -विदेशयात्राव्यय -F.T.E.	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.13 -कार्यालयव्यय -O.E.	100.00	100.00	85.81	**0.09**	85.90	14.10
01.01.14 -किराया, दरएवंकर - R.R.T.	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.16 -प्रकाशन -Pub.	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.20 -ओ.ए.ई -O.A.E.	10.00	10.00	9.61	**-0.01**	9.60	0.40
01.01.24- पेट्रोल / डीजल -P.O.L.	9.00	9.00	0.00	**0.00**	0.00	9.00
01.01.26 - ऐड. एवंपब्लिस्ट -Adv. &Publi.	30.00	35.00	26.77	**-0.36**	26.41	8.59
01.01.27 - माइनरकार्य -Min. Works	50.00	50.00	38.85	**0.00**	38.85	11.15
01.01.28 -वृत्तिकसेवाएँ -P.S.	25.00	25.00	0.91	**0.00**	0.91	24.09
01.01.30 -Other Contractual Services	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.50 -अन्यप्रभार -O. C	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Dom Support (N.PLAN)	**26818.00**	**27768.00**	**27470.24**	**47.80**	**27518.04**	**249.96**

Demand No. 62				4702 (CAPITAL)		
Major Head: 4702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION (Establishment Expenditure)						
Sub Head: 05. 01-CGWB-01.01-H/QRS.						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
4702.00.800.05 (Capital Nature Expenditure) CGWB Head Quarter (NON PLAN)						
1	2		5	6	7	8
05.01.51 -मोटरवाइकल -M.V.	0.00	0.00	0.00	**0.00**	0.00	0.00
05.01.52 -मशीनरीएवंउपस्कर -M & E	0.00	0.00	0.00	**0.00**	0.00	0.00
05.01.53 -मुख्यकार्य -M/ Works	1200.00	1200.00	545.49	**70.93**	616.42	583.58
Total Capital Nature Expenditure	**1200.00**	**1200.00**	**545.49**	**70.93**	**616.42**	**583.58**
Grant Total (Non Scheme)	**28018.00**	**28968.00**	**28015.73**	**118.73**	**28134.46**	**833.54**

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
Swachhta Action Plan				Non Scheme (Non Plan)		
Demand No. 62				Establishment Expenditure 2702 (Revenue)		
Major Head: Establishment Expenditure 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION						
Sub Head: 01-CGWB-01.01-H/QRS.						
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
01.96 Swachhta Action Plan						
01.96.13 - कार्यालयव्यय -O.E.	2.00	2.00	1.75	**0.00**	1.75	0.25
Total	**2.00**	**2.00**	**1.75**	**0.00**	**1.75**	**0.25**
Total (Estt. Exp. + SAP)	**28020.00**	**28970.00**	**28017.48**	**118.73**	**28136.21**	**-116.21**

Information Technology				Non Scheme (Non Plan)		
Demand No. 62				Establishment Expenditure 2702 (Revenue)		
Major Head: Establishment Expenditure 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION						
Sub Head: 01-CGWB-01.01-H/QRS.						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
01.99 Information Technology						
01.99.13 -कार्यालयव्यय -O.E.	180.00	130.00	123.93	**0.00**	123.93	6.07
Total (Information Technology)	**180.00**	**130.00**	**123.93**	**0.00**	**123.93**	**6.07**
Total (Estt. Exp. + SAP + IT)	**28200.00**	**29100.00**	**28141.41**	**118.73**	**28260.14**	**839.86**

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
						Non Scheme : R.G.N.G.W.T.R.I (Training)
Demand No. 62						2702 (REVENUE)
Major Head: 2702 : Minor Irrigation, 02 : Ground Water, 800 : Other Expenditure (Non Scheme)						
Sub Head: 03 : Rajiv Gandhi National Ground Water Training and Research Institute. Detailed Head : 00 (Default)						
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						**(Figure in lacs)**
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving

				Rco. In PAO		
1	2		5	6	7	8
03.00 : Domestic Support (RGNGWTRI)						
03.00.06 - चिकित्सा -M/Treat.	13.00	8.00	5.17	**0.01**	5.18	2.82
03.00.11 - यात्राव्यय -D.T.E.	15.00	15.00	14.89	**0.00**	14.89	0.11
03.00.12 -विदेशयात्राव्यय -F.T.E.	0.00	0.00	0.00	**0.00**	0.00	0.00
03.00.13 - कार्यालयव्यय -O.E.	180.00	130.00	127.10	**0.00**	127.10	2.90
03.00.14 -किराया, दरएवंकर - R.R.T.	0.00	0.00	0.00	**0.00**	0.00	0.00
03.00.16 - प्रकाशन -Pub.	3.00	1.00	0.00	**0.00**	0.00	1.00
03.00.24 - पेट्रोल/डीजल -P.O.L.	10.00	7.00	6.34	**0.00**	6.34	0.66
03.00.30 - Other Contractual Services	0.00	0.00	0.00	**0.00**	0.00	0.00
03.00.28 - वृत्तिकसेवाएँ -P.S.	129.00	128.00	124.58	**2.83**	127.41	0.59
Total Dom Support (RGI)	**350.00**	**289.00**	**278.08**	**2.84**	**280.92**	**8.08**
						R.G.N.G.W.T.R.I (Training)
Demand No. 62						4702 (CAPITAL)
Major Head: 4702-06.01 HRD/Capacity Building (PLAN)						
Sub Head: 07.01-RGN. G/W TRG AND RESURCH INSTITUTE.						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
4702 Capital Nature Expenditure RGNT &RI (PLAN)						
1	2		5	6	7	8
07.01.51 -मोटरवाइकल - Motor Vehicle.	0.00	0.00	0.00	**0.00**	0.00	0.00
07.01.52 -मशीनरीएवंउपस्कर - M & E	0.00	11.00	0.00	**0.00**	0.00	11.00
07.01.53 -मुख्यकार्य - M/Works	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Capital Nature Expenditure	**0.00**	**11.00**	**0.00**	**0.00**	**0.00**	**11.00**
Grant Total (RGI) PLAN	**350.00**	**300.00**	**278.08**	**2.84**	**280.92**	**19.08**

Hydrology Project-Phase-III						
Demand No. 62			**2701**	REVENUE		
Major Head: 2701-.80.004.08.01(Ext.Supp.) & 08.02 (Dom. Supp) National Hydrology Project-Phase-III **(PLAN)**						
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
08.01 - Ext. Support NHP-III 1	2	3	4	5	6	7
08.01.11 - यात्राव्यय - D.T.E.	2.50	1.00	0.00	**0.00**	0.00	1.00
08.02.12 - विदेशयात्राव्यय - F.T.E.	54.00	3.77	3.76	**0.00**	3.76	0.01
08.01.13 - कार्यालयव्यय - O.E.	4.00	6.25	0.00	**0.00**	0.00	6.25
08.01.20 - Other AdminstativeExpe.	5.00	2.75	2.71	**0.00**	2.71	0.04
08.01.27 - Minor/Works	0.00	0.00	0.00	**0.00**	0.00	0.00
08.01.28 - Professional Services	17.00	17.20	16.78	**0.00**	16.78	0.42
08.01.30 - Other Contratual Services	2.00	0.24	0.24	**0.00**	0.24	0.00
TOTAL (Ext. Support) NHP	**84.50**	**31.21**	**23.49**	**0.00**	**23.49**	**7.72**
08.02 Dom. Support (NHP)-III						
08.02.11 - D.T.E.	2.50	1.00	0.00	**0.00**	0.00	1.00
08.02.12 - विदेशयात्राव्यय - F.T.E.	54.00	3.77	3.76	**0.00**	3.76	0.01
08.02.13 - कार्यालयव्यय - O.E.	4.00	6.25	0.00	**0.00**	0.00	6.25
08.02.20 - Other AdminstativeExpe.	5.00	0.00	0.00	**0.00**	0.00	0.00
08.02.27 -Minor/Works	0.00	0.00	0.00	**0.00**	0.00	0.00
08.02.28 - Professional Services	0.00	0.00	0.00	**0.00**	0.00	0.00
08.02.30 - Other Contratual Services	2.00	0.24	0.24	**0.00**	0.24	0.00
TOTAL (Dom. Support) NHP	**67.50**	**11.26**	**4.00**	**0.00**	**4.00**	**7.26**
TOTAL (Ext. Sup. + Dom. Support) NHP	**152.00**	**42.47**	**27.49**	**0.00**	**27.49**	**14.98**

Demand No. 62	4701 CAPITAL				Hydrology Project-Phase-III	
Major Head: 4701-.80.004.08.01 (Ext. Supp) & 08.02 (Dom. Supp)Hydrology Project-Phase-III (PLAN)						
Sub Head: 06 : National Hyydrology Project – 01 (Ext. Support) & 02 (Dom. Support)						
4701. 80.004.04.02 (Capital Nature Expenditure) National Hydrology Project (PLAN)						
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
4701. 08.01 Ext. Supp. NHP (Capital)						
08.01.51 - M V	0.00	0.00	0.00	**0.00**	0.00	0.00
08.01.52 - M & E	1065.50	250.54	174.62	**0.00**	174.62	75.92
08.01.53 - Major/Works	0.00	0.00	0.00	**0.00**	0.00	0.00
08.01.60 - Other Capital Expenditure	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Capital Nature (Ext. Supp.)	**1065.50**	**250.54**	**174.62**	**0.00**	**174.62**	**75.92**
4701. 08.02 Dom. Supp. NHP (Capital)						
08.02.51 - M V	0.00	0.00	0.00	**0.00**	0.00	0.00
08.02.52 - M & E	1065.50	250.54	174.62	**0.00**	174.62	75.92
08.02.53 - Major/Works	0.00	0.00	0.00	**0.00**	0.00	0.00
08.02.60 - Other Capital Expenditure	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Capital Nature (Dom. Supp.)	**1065.50**	**250.54**	**174.62**	**0.00**	**174.62**	**75.92**
Total Capital (Ext. Sup. + Dom.Sup.)	**2131.00**	**501.08**	**349.24**	**0.00**	**349.24**	**151.84**
G.Total (Revenue + Capital) (NHP-III)	**2283.00**	**543.55**	**376.73**	**0.00**	**376.73**	**166.82**

जलशक्तिमंत्रालय, जलसंसाधन, नदीविकासऔरगंगासंरक्षणविभाग, केन्द्रीयभूमिजलबोर्ड, भूजलभवन, एन.एच. -4, फरीदाबाद।						
Demand No. 62						
केन्द्गीयभूमिजलबोर्ड (जलसंसाधनमंत्रालय)				(IEC) ACTIVITIES (PLAN)		
Supplemently Monthly Expenditure Report For the Month of March, 2022-2023						
Major Head: 2701-MINOR IRRI-02-005-GROUND WATER INVESTIGATION (PLAN)						
Sub Head : 20 MASS AWARENESS ACTIVITIES (IEC) (PLAN)						
						(Figure in lacs)
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2022-23	Revised Final RE F.Y. 2022-23	Prog. Exp. upto the Pre. Month of 03/2023	Expend. during the Month of 03/2023 after Rco. In PAO	**Progr. Exp. upto the Month of 03/2023**	Variation (+)Excess (-)Saving
1	2		5	6	7	8
20.01.13 - Office Expenses	0.00	0.00	0.00	**0.00**	0.00	0.00
20.01.20 - Other Administrative Expenses	0.00	0.00	0.00	**0.00**	0.00	0.00
20.01.26 - Advertising & Publicity	0.00	0.00	0.00	**0.00**	0.00	0.00
20.01.28 - Professional Services	0.00	0.00	0.00	**0.00**	0.00	0.00
20.01.31 - Grant-in-aid-General	0.00	0.00	0.00	**0.00**	0.00	0.00
Total (IEC) PLAN	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

CHAPTER 26

राज भाषा हिन्दी का प्रचार-प्रसार एवं कार्यान्वयन

- केंद्रीय भूमि जल बोर्ड और इसके सभी अधीन स्थक्षेत्रीय, प्रभागीय और राज्यए क कार्यालयों में वर्ष 2022-2023 के दौरान राज भाषा हिन्दी के कार्यान्वयन और प्रचार – प्रसार के सक्रिय प्रयास जारी रखेगए। केंद्रीय भूमि जल बोर्ड राज भाषा हिंदी के प्रगामी प्रयोग व कार्यान्वयन के लिए प्रतिब द्धहै। राज भाषा विभाग द्वारा जारी वार्षिक कार्यक्रम के अनुसार निर्धा रितलक्ष्य को प्राप्त करने के लिए बोर्ड सत तप्रयत्नशील है। राज भाषा विभाग, गृहमंत्रालय, जल शक्ति मंत्रालय और नगर राज भाषा कार्या न्वयन समिति से समय – समय पर प्राप्त अनुदेशों और दिशा निर्देशों के पूर्ण अनुपाल नसुनिश्चित करने तथा संघकीरा जभाषा नीति के प्रावधा नोंके कार्या न्वयन और वार्षिक कार्यक्रममें निर्धारि तलक्ष्यों को प्राप्त कर ने के उद्देश्य से सभी अधीन स्थकार्यालयों और मुख्यालय के सभी अनुभागों को इससे अवगत कराया गया।

- सरकारी काम काज में हिंदी के प्रयोग को बढ़ाने के लिएनियमित रूप से विभिन्न परिपत्र, आदेश और अपील जारी किए गए। कार्यालय में हिन्दी में टिप्पण और आलेखन को बढ़ावा देने के उद्देश्य से राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम 2022-2023 की प्रति, द्वि भाषी मान कमसौ दे, प्रशासनिक शब्दावली आदि सभी अधिकारियों और कर्मचारियों को उपलब्ध कराये गए। सभी कम्प्यूटरों पर हिन्दी में काम करने की सुविधा प्रदान करने हेतु यूनिकोड सुविधा उपलब्ध कराई गयी।

- राजभाषा हिन्दी के प्रभावी प्रचार-प्रसार और कुशल कार्या न्वयन के लिए राजभाषा विभाग, गृहमंत्राल यद्वा राजारी समस्त प्रोत्साहन योजनाओं यथा मूल रूप से हिन्दी में टिप्पण–आलेखन, श्रुत लेखन, हिन्दी में टंकण आदि योजनाओं को कार्यालय में लागू किया गया है तथा अधिक से अधिक अधिकारी और कर्मचारी उत्साह पूर्वक इन योजनाओं में भाग ले रहे हैं

- राजभाषा हिन्दी के प्रभावी कार्या न्वयन और इसकी प्रगति की मॉनिटरिंग के लिए बोर्ड और इसके समस्त अधीनस्थ कार्यालयों में राजभाषा का र्यान्वयन समिति का गठन किया गया है। इस समिति की बैठकें नियमित रूप से प्त्येक तिमाही में आयोजित की जाती हैं। इन बैठकों में राजभाषा विभाग, गृह मंत्राल यद्वारा जारी वार्षिक कार्यक्रम में निर्धारित लक्ष्यों की प्राप्ति हेतु विभिन्न मुद्दों परविचार-विमर्शद्वारा कार्य नीति तैयार की जाती है। इन बैठकों में लिए गए निर्णयों पर प्राथमिकता के आधार पर अनुवर्ती कार्रवाई की जाती है।

- गृहमंत्राल यद्वराजारीदिशा निर्देश के अनुसार सितंबर माह में दिनांक 14.09.2022 से 28.09.2022 के दौरान बोर्ड और इसके समस्त अधीन स्थकार्यालयों में हिन्दी दिवस, हिन्दी सप्ताह, हिन्दी पखवाड़ा का आयोजन किया गया। इस दौरान अधिकारियों और कर्मचारियों के लिए कई प्रतियोगिताओं जैसे हिन्दी निबंध प्रतियोगिता, टिप्पण-आलेखन, टंकण, वाद-विवाद, प्रश्न मंच आदि का आयोजन किया गया। सभी अधिकारियों और कर्मचारियों ने इन प्रतियोगिताओं में बढ़-चढ़ कर भागलिया।

- भारत सरकार की राजभाषा नीति के अनुसरण में राजभाषा अधिनियम, 1963 की धारा3(3) के तहत सभी दस्तावेज हिंदी और अंग्रेजी दोनों में तैयार किए जाते हैं। राजभाषा नियम, 1976 के नियम 5 के तहत हिन्दी में प्राप्त सभी पत्रों के उत्तर अनिवार्य रूप से हिन्दी में ही दिये जा रहे हैं। वार्षिक कार्यक्रम में निर्धारित लक्ष्यों को ध्यान में रखते हुए कार्यालय में जांच बिन्दुस्थापित किए गए हैं, ताकि इस दिशा में किए जा रहे कार्यों की मॉनिटरिंग की जा सके।

- राजभाषा विभाग, गृहमंत्रालय द्वारा समय-समय पर जारी दिशा निर्देशों के अनुपालन में केंद्रीय भूमि जल बोर्ड, मुख्यालय और इसके अधीन स्थकार्यालयों में नियमित रूप से प्रत्येक तिमाही मे हिन्दी कार्यशालाओं का आयोजन किया जा रहा है।

- माननीय संसदीय राजभाषा समिति की दूसरी उप समिति द्वारा दिनांक 26.09.2022 को केंद्रीय भूमि जलबोर्ड, केरल क्षेत्र, तिरुवनन्तपुरम; दिनांक 17.10.2022 को केंद्रीय भूमि जल बोर्ड, पश्चिमी क्षेत्र, जयपुर; दिनांक 18.10.2022 को केंद्रीय भूमि जल बोर्ड, प्रभाग।, अहमदाबाद;

दिनांक 14.11.2022 को केंद्रीय भूमि जल बोर्ड, उत्तर मध्य छत्तीसगढ़ क्षेत्र, रायपुर; दिनांक 15.11.2022 को केंद्रीय भूमि जल बोर्ड, प्रभागXIII, रायपुर; दिनांक 16.11.2022 को केंद्रीय भूमि जल बोर्ड, पूर्वीक्षेत्र, कोलकाता; दिनांक 03.01.2023 को केंद्रीय भूमि जल बोर्ड, प्रभागV, रांची; दिनांक 05.01.2023 को केंद्रीय भूमि जल बोर्ड, उत्तर पश्चिमी क्षेत्र, चंडीगढ़; दिनांक 19.01.2023 को केंद्रीय भूमि जल बोर्ड, राज्यएकक कार्यालय, पुणे और दिनांक 27.02.2023 को केंद्रीय भूमि जल बोर्ड, प्रभाग-XI, जोधपुर का राजभाषायी निरीक्षण किया गया। समिति द्वारा कार्यालयों में राजभाषा के प्रचार-प्रसार और इसके का र्यान्वयन की दिशा में किए जार हे कार्यों पर संतोष प्रकट किया गया।

- केंद्रीय भूमि जल बोर्ड के सभी अधीन स्थ कार्यालयों में राजभाषा का र्यान्वयन की समुचि त मॉनिटरिंग के लिए सभी कार्यालयों के राजभाषा प्रभारियों की एक सूची तैयार की गयी है और इसे केंद्रीय भूमि जल बोर्ड की वेब साइ टप रउपलब्ध करा याग या है।

- केंद्रीय भूमि जल बोर्ड हिंदी के प्रगामी प्रयोग व कार्यान्वयन के लिए प्रतिबद्ध है।राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम के अनुसार निर्धारित लक्ष्य को प्राप्त करने के लिए बोर्ड सत तप्रयत्नशील है

माननीय संसदीय राजभाषा समिति द्वारा दिनांक 26.09.2022 को
केंद्रीय भूमि जल बोर्ड, केरल क्षेत्र तिरुवनन्तपुरम का निरीक्षण

माननीय संसदीय राजभाषा समिति द्वारा दिनांक 19.01.2023 को
केंद्रीय भूमि जल बोर्ड, राज्यएकक कार्यालय, पुणे का निरीक्षण

माननीय संसदीय राजभाषा समिति द्वारा दिनांक 17.10.2022 को

केंद्रीय भूमि जल बोर्ड, पश्चिमी क्षेत्र, जयपुर का निरीक्षण

माननीय संसदीय राजभाषा समिति द्वारा दिनांक 1810.2022 को

केंद्रीय भूमि जल बोर्ड, प्रभाग-I, अहमदाबाद का निरीक्षण

माननीय संसदीय राजभाषा समिति द्वारा दिनांक 26.09.2022 को
केंद्रीय भूमि जल बोर्ड, केरल क्षेत्र तिरुवनन्तपुरम का निरीक्षण

Chapter 27 GALLERY

Workshop on Springshed Management and Rejuvenation organised by NEHARI, Brahmaputra Board, Guwahati in association with CGWB & NIH at Guwahati.

JalShaktiAbhiyan:CatchTheRain 2022 field visit to Rewa District, MP under the supervision of Sh R L Meena, Deputy Secretary, Ministry of Telecommunication (CNO) &Ms Rakiya, Scientist, CGWB (Technical Officer).

CGWB in Jaipur Expo 2022 (22-24 Sept 2022)- Popularising rainwater harvesting and artificial recharge. We thank Dr Kirodi Lal Meena, Hon'ble MP, Rajya Sabha for visiting the stall

Inaugration of One day Workshop on "Success Stories of NAQUIM studies & PMKSY-HKKP-GW Schemes in Guwahati, Assam.

दिनांक 18.10.2022 को प्रो. @RitaBJoshi जी की अध्यक्षता में संसदीय राजभाषा समिति ने अहमदाबाद में केंद्रीय भूमि जल बोर्ड प्रभाग -1, अहमदाबाद कार्यालय के साथ निरीक्षण बैठक की।

Students visited and interacted regarding various models exhibited by officers of Central Ground Water Board, Jaipur during Vision Rajasthan 2022 Expo Sirohi.

NAQUIM Report sharing meeting held in the conference hall, DM Office, South 24 Parganas under the Chairmanship of Shri Sumit Gupta, IAS, DM, South 24 Parganas on 23.02.2023. Siyad N, IAS, ADM (Gen) and Sankha Santra WBCS ADM (Dev) was also present in the meeting. In the meeting officers from CGWB, ER, shared the output of NAQUIM findings with the District Authority along with the DLA members.

DM/ DC presentation completed for Alipurduar district. Meetting was chaired by Surendra Kumar Meena, IAS and attended by ADM (admin), ADM(DL&LRO), EE PHED, EE WRIDD, Director Agriculture, CMO, DFO Buxa Tiger Reserve,EE I&WD, Sr. Geologist ,SWID.

Officers from CGWB, ER, Kolkata, attended a meeting with the Secretary, Drinking Water & Sanitation, GoI, held in the Conference Hall of National Centre for Drinking Water, Sanitation & Quality (NCDWSQ), Joka, Kolkata, on 05.11.2022.

On the ocassion of World Water Day 2023 celebration which was marked by workshop organized on 22.03.2023. The event and talk given by the Regional Director, Dr. Anadi Gayen was covered by doordarshan.	On the ocassion of World Water Day 2023 celebration which was marked by workshop organized on 22.03.2023. The event and talk given by the Regional Director, Dr. Anadi Gayen was telecasted on doordarshan.

Scientists of CGWB, NR, Lucknow conducted Pumping Test at Bhadeicha site, block- Sursa, district- Hardoi. SDT of 3 hours, APT of 1000 minutes at constant discharge of 2028 lpm

A meeting with honourable Minister of Jal Shakti Sh. Gajendra Singh Shekhawat during 'SUJALAM-International Water Conference' held at Ujjain district, Madhya Pradesh.

A team of Officers from CGWB CHQ, WCR, NCR attending a meeting of Second Mission of the EU expert for the works related to risk assessment in Tapi River basin.

Officers of CGWB, Bhopal conducted Pumping test at Koundiya Exploratory Well site, Gogawa Block, Khargone district,Madhya Pradesh. During 1000 minutes duration pumping test, discharge is about 6.1 lps with a drawdown of 2.55 m

Preliminary Yield Test (PYT) conducted by officer of CGWB, SUO, Jodhpur at Navalgarh, Jodhpur. Discharge obtained is 168 Lpm

Tier-III Training with hydrogeological field trip on the occasion of "World Water Day" on theme "Ground Water Management Through Participatory Approach and Local Ground Water Issues" at Govt. Degree College, Paukhal, Tehri Garhwal.(UR Dehradun)

Dr. M. Senthil Kumar, CGWB Faridabad gave a presentation on Groundwater Scenario of India, management initiatives, NAQUIM, Groundwater Resources Augmentation and community led initiatives in India at 2nd ECSWG meeting at Gandhinagar, Gujarat.

Contributors

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भारत सरकार

GOVERNMENT OF INDIA

जल शक्ति मंत्रालय,

MINISTRY OF JAL SHAKTI

जल संसाधन, नदी विकास और गंगा संरक्षण विभाग

DEPARTMENT OF WATER RESOURCES, RIVER DEVELOPMENT & GANGA REJUVENATION

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